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A novel method for the design of induction heating work coils

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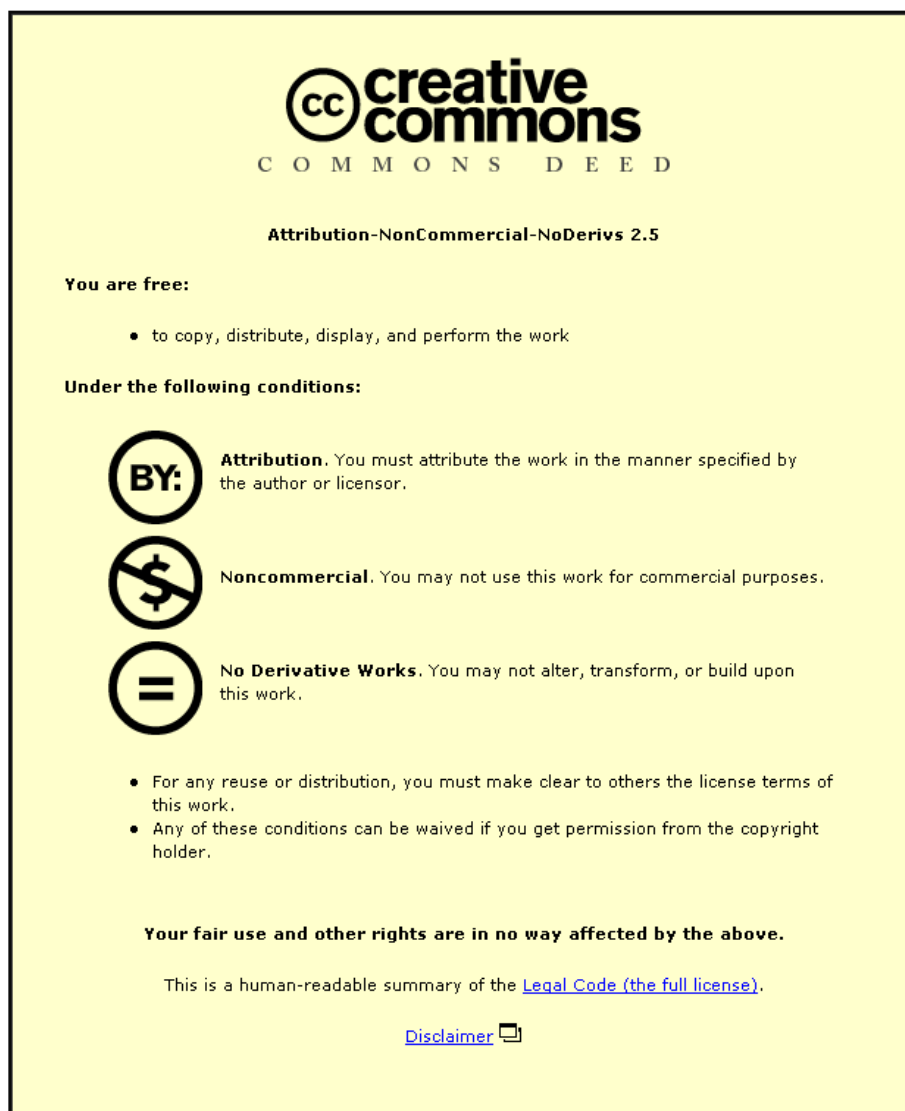
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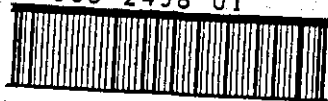
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A NOVEL METHOD FOR THE DESIGN OF
INDUCTION HEATING WORK COILS

by

ALI KAMIL MAKKI AL-SHAIKHLI

A DOCTORAL THESIS

Submitted in partial fulfilment of the requirements
for the award of Ph.D. of the Loughborough
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Supervisor: Dr L Hobson

Department of Electronic and Electrical Engineering

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Synopsis

Induction billet heating is a well established industrial process for preheating prior to forging, rolling and extrusion. In many cases the application of induction heating techniques has been hindered by the inability of existing methods of work coil design to easily produce a design which will give a non-uniform power density along the surface of a workpiece.

Existing methods of work coil design almost universally assume that the load is subjected to a uniform magnetic field, and hence it is very difficult to adapt these methods of design to tackle problems of non-uniform power density distributions.

In order to simplify the problem of work coil design for a non-uniform power density along a workpiece a more flexible approach was required. In this project a novel design technique is described in which the surface current density distribution produced by a single conductor has been investigated and equations governing this distribution established. The superposition principle was then used to predict the power density distribution produced by a number of conductors. The method was then successfully applied to typical industrial loads of different materials and shapes. The simplicity of this method means that it can be used on desk top computers and even to some programmable hand calculators. The investigations have also included the effect of workpiece material, size and shape, the existence of lamination packs and the current distribution within the work coil conductor itself.

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LIST OF PRINCIPAL SYMBOLS

ϕ	: Magnetic flux (Wb)
R_m	: Reluctance (A/Wb)
N	: Number of turns
I	: Current (A)
V	: Voltage (V)
$\omega = 2\pi f$	: Angular velocity (rad/s)
f	: Frequency (Hz)
$\mu = \mu_o \mu_r$	: Absolute permeability (H/m)
μ_o	: Free-space permeability (H/m)
μ_r	: Relative permeability
Z	: Impedance (Ω)
R	: Resistance (Ω)
X	: Reactance (Ω)
r	: Radius (m)
l	: Length (m)
k	: Empirical factor-equation (A2.13)
k_r	: Empirical factor-equation (A2.10)
k_c	: Empirical factor-equation (2.30)
k_e	: Empirical factor-equation (2.31)
P	: Dimensionless constant-equation (A1.36)
Q	: Dimensionless constant-equation (A1.37)
$\delta = \sqrt{\frac{\rho}{\pi f \mu_o \mu_r}}$	: Penetration depth (m)
ρ	: Resistivity (Ωm)
VA	: Volt-ampere (VA)
FA	: Constant of Frohlich formula-equation (2.32)
FB	: Constant of Frohlich formula-equation (2.32)
FC	: Constant of Frohlich formula-equation (2.32)
$P.F.$	: Power factor

η	: Efficiency (%)
H	: Magnetic field strength (A/m)
H*	: Conjugate of magnetic field strength
E	: Electric field strength (V/m)
J	: Current density (A/m^2)
J*	: Conjugate of current density
a	: Conductor radius (m)
h	: Distance from the centre of the conductor to the surface of the load (m)
z	: Distance along the load measured from a point directly beneath the conductor (m)
h_e	: Effective height (m)
A	: Current per unit length (A/m)
PD	: Power density (W/m^2)
T	: Temperature ($^{\circ}C$)
d	: Coil pitch (m)
α	: Dimensionless constant - Fig. 5.8
β	: Dimensionless constant - Fig. 5.9

Suffix

r	: Return
c	: Coil
t	: Total
w	: Workpiece
g	: Air gap

Other symbols are defined as they occur.

CHAPTER 1

INTRODUCTION

Induction heating was first used in industry to melt metals but now deals efficiently with many other applications in the general field of metal heating, tube welding, brazing, soldering, surface hardening, and through heating for rolling, forging and extrusion.

The assembly of the basic induction heater is shown in Fig. 1.1.

The principle of induction heating is illustrated by Fig. 1.2. A water cooled copper coil with thermally insulated lining surrounds the metal to be heated. The coil is supplied with an alternating current, thus creating a pulsating magnetic field which interacts with the billet and induces eddy currents within it. The eddy currents in turn produce heat as a result of the Joule heating or $I^2 R$ losses.

The amount of heat generated in the workpiece is dependent upon the physical properties of the billet material, and the magnitude of the induced current, which is proportional to the work coil turns and current.

Induction heating has a number of inherent advantages over its fuel fired competitive processes. Very fast heating rates can be achieved and the ease of control and the ability to repeat the working cycle make induction heating ideal for a fully automated process. When hot metal is not required the heater can be switched off thus eliminating

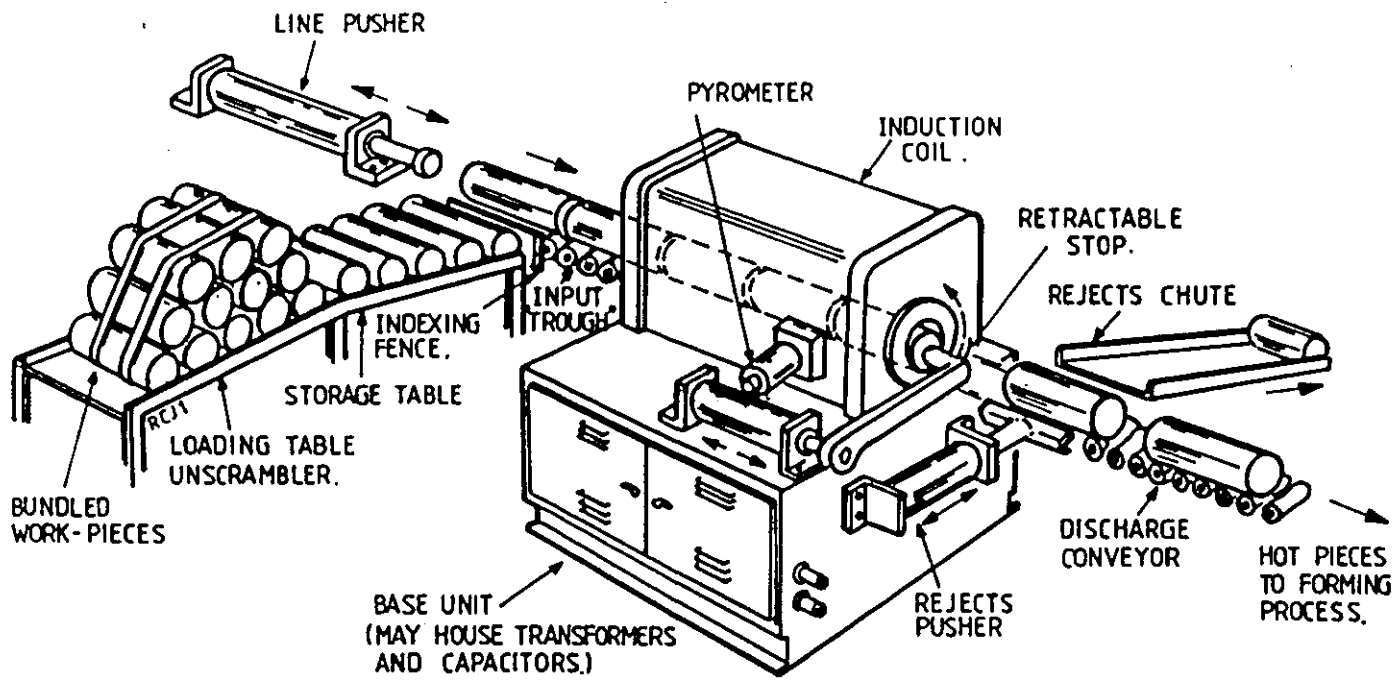


Fig. 1.1 Basic induction through heater

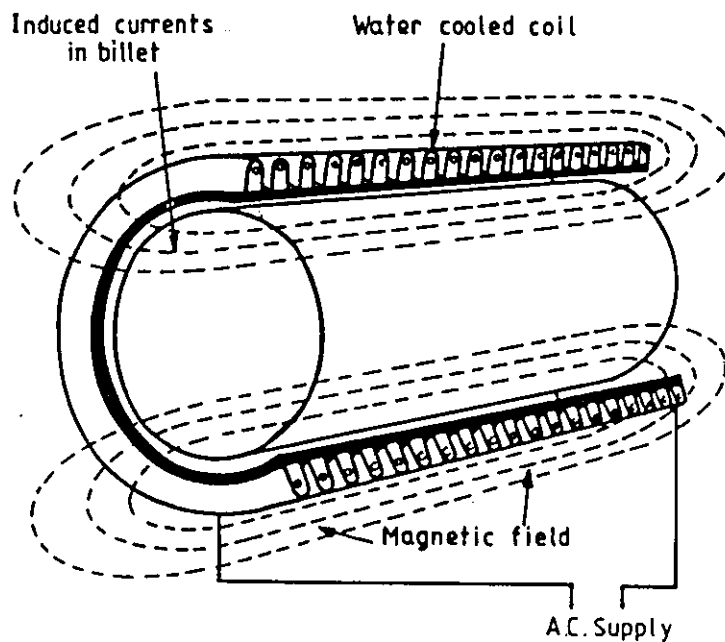


Fig. 1.2 Induction billet heating

virtually all standby losses. Induction heating equipment usually provides a clean and relatively pleasant working environment with little extraneous heating and a low noise level.

These process advantages have ensured that induction heating has achieved a substantial market share of the heating installations within certain parts of the metal forming industry [1.1]. Other areas however, have so far remained virtually untouched, especially those requiring a non-uniform surface power density along the length of the workpiece such as off-the-bar forging or the extrusion of aluminium.

Existing methods of work coil design almost universally assume that the load is subjected to a uniform magnetic field, and hence it is very difficult to adapt these methods of design to tackle problems of non-uniform power density distributions. In industrial practice the design of induction billet heaters is largely based on the equivalent circuit method devised by Baker [1.2, 1.3] and developed by Reichert [1.4] and Vaughan and Williamson [1.5, 1.6]. The method assumes a uniform magnetic field strength along the length of the workpiece and many empirical factors are incorporated to take into account such things as a shortness of coil and spacing between turns. Numerical solutions using finite difference, finite element and mutually coupled circuit techniques are in existence [1.7-1.9] but they also assume a uniform magnetic field strength and hence a surface power density along the length of the workpiece. They require specialist knowledge and powerful computational facilities not normally available to UK induction heating manufacturers.

In order to deal realistically with problems involving non-uniform surface power density along the length of a workpiece, a more flexible approach is required. A novel technique is presented in which the surface current density distribution produced by a single conductor has been investigated and equations governing this distribution established. The superposition principle to find the power density produced by a number of conductors has been verified and a work coil design to produce a particular power density distribution in a billet has been carried out. The method is very simple and can be used on some programmable hand calculators.

CHAPTER 2

METHODS OF ANALYSIS FOR INDUCTION HEATING

WORK COILS

Industrial designers of induction heating work coils depend largely on relatively simple equivalent circuit techniques supplemented by empirical data accumulated over many years. On the other hand, academics and other research workers have developed highly sophisticated numerical techniques to solve particular problems.

This chapter explains the principles behind each form of work coil design.

2.1 The Equivalent Circuit Method

The basic assembly of an induction billet heater consists of a water cooled copper coil surrounding a metallic workpiece. A relatively large air gap between the coil and the workpiece is required so as to permit the free movement of the billet through the heater.

In addition to this, thermal insulation is provided so that the radiated heat losses from the billet to the coil can be reduced.

The magnetic flux produced by the work coil, ϕ_t , has alternative parallel paths through either the workpiece ϕ_w , the coil ϕ_c or the air gap ϕ_g , as shown in Fig. 2.1. These paths have the following respective

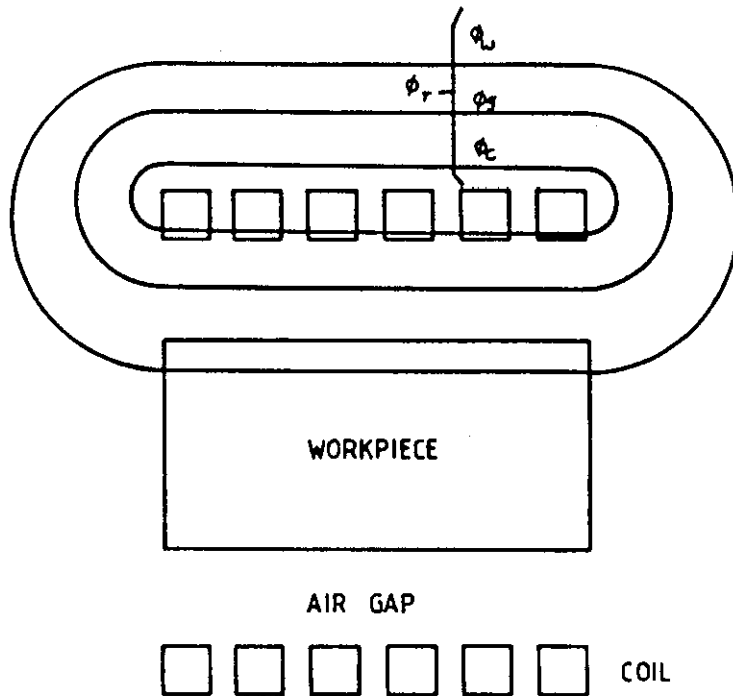


Fig. 2.1 The flux paths in induction heating system

magnetic reluctances R_{mw} , R_{mc} and R_{mg} . The return flux ϕ_r has a reluctance of R_{mr} . The effect of ϕ_c on the coil exterior is usually ignored [2.1, 2.2] because it is numerically far less than ϕ_r . The effect of ϕ_c on the coil interior is represented by R_{mc} . The assembly total reluctance equals R_{mr} in series with the parallel connection of R_{mw} , R_{mg} and R_{mc} . The position of R_{mc} relative to R_{mr} does not alter the performance of the magnetic circuit seriously because, the coil reluctance is generally far higher than the external reluctance. The magnetic circuit of Fig. 2.2 is formulated to simplify the calculations.

An expression for the total ampere-turn requirement is derived using the reluctances of each of the magnetic circuits and Ampere's law:

$$\begin{aligned}
 N_c I_c &= \phi_t R_{mt} \\
 &= \phi_t \left[\frac{R_{mc} (R_{mr} + \frac{R_{mg} R_{mw}}{R_{mg} + R_{mw}})}{R_{mc} + R_{mr} + \frac{R_{mg} R_{mw}}{R_{mg} + R_{mw}}} \right] \quad (2.1)
 \end{aligned}$$

The work coil voltage, V_c , is related to the total flux by Faraday's law:

$$V_c = j\omega N_c \phi_t \quad (2.2)$$

Hence an expression for the total circuit impedance Z can be obtained from

$$Z = \frac{V_c}{I_c} \quad (2.3)$$

The electrical equivalent circuit is, therefore, derived and shown in Fig. 2.3, where

$$X_r = N_c^2 \frac{\omega}{R_{mr}} \quad (2.4)$$

$$X_g = N_c^2 \frac{\omega}{R_{mg}} \quad (2.5)$$

$$Z_w = jN_c^2 \frac{\omega}{R_{mw}} \quad (2.6)$$

$$Z_c = jN_c^2 \frac{\omega}{R_{mc}} \quad (2.7)$$

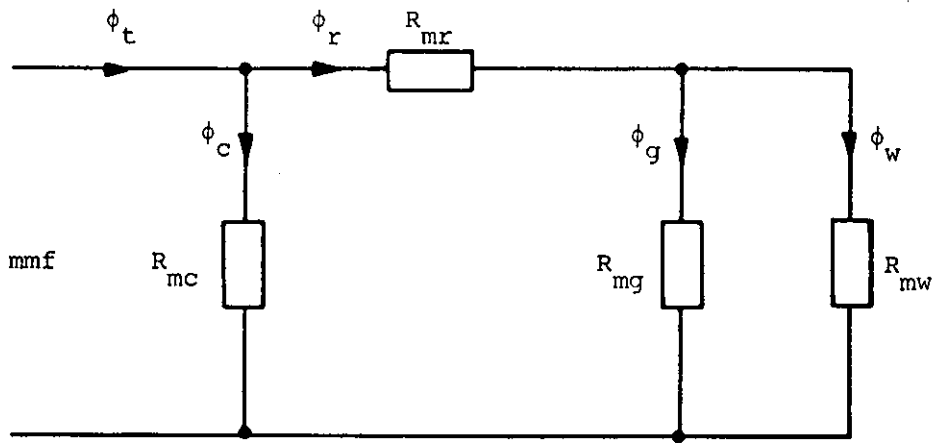


Fig. 2.2 The equivalent magnetic circuit

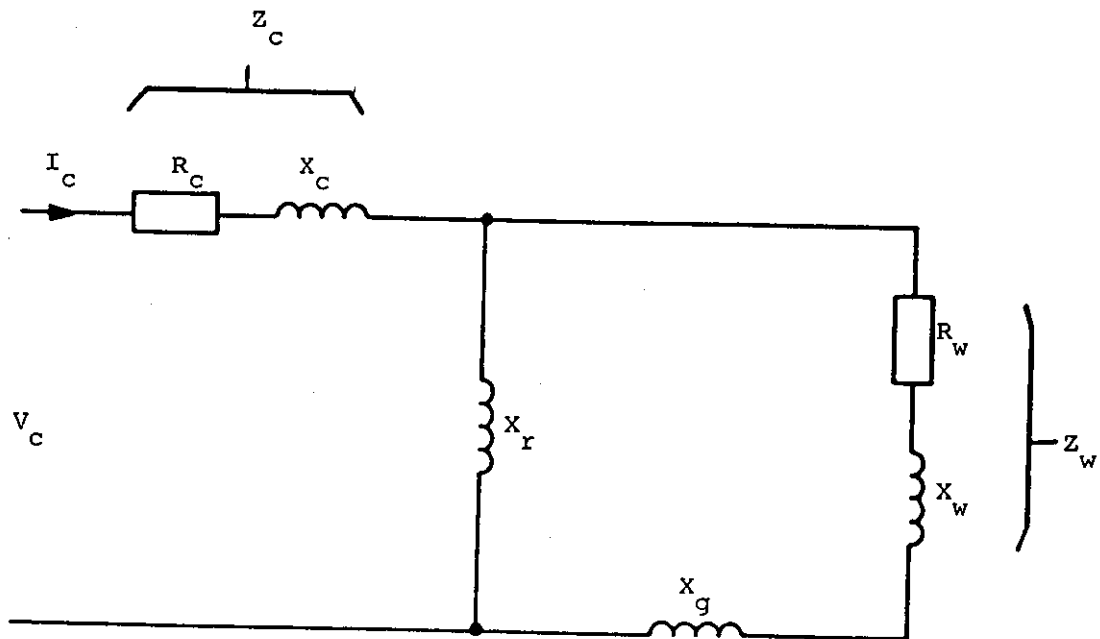


Fig. 2.3 The equivalent electrical circuit

The reluctances are derived in detail in appendices 1 and 2 and are given by:

$$R_{mr} = \frac{2(0.45+k)}{\mu_o \pi r_c} \quad (2.8)$$

$$R_{mg} = \frac{l_c}{\mu_o \pi (r_c^2 - r_w^2)} \quad (2.9)$$

$$R_{mw} = \frac{l_c}{\mu_o \mu_r \pi r_w^2 (P-jQ)} \quad (2.10)$$

$$R_{mc} = \frac{l_c}{\mu_o \pi k_r \delta r_c (1-j)} \quad (2.11)$$

The components of the electrical circuit are derived by substituting equations (2.8-2.11) respectively into equations (2.4-2.7) so as to give:

$$X_r = K_R \frac{r_c}{2(0.45+k)} \quad (2.12)$$

$$X_g = K_R \frac{(r_c^2 - r_w^2)}{l_c} \quad (2.13)$$

$$Z_w = R_w + jX_w = K_R \frac{\mu_r r_w^2}{l_w} (Q + jP) \quad (2.14)$$

$$Z_c = R_c + jX_c = K_R \frac{k \delta r_c}{l_c} (1 + j) \quad (2.15)$$

$$\text{where } K_R = \omega \mu_o \pi N_c^2 \quad (2.16)$$

The equivalent circuit of Fig. 2.3 can be solved by circuit analysis to give work coil turns, the current carrying capacity and the efficiency and power factor. As the coil number of turns is not known, the circuit components are calculated for a single turn coil. In these calculations the components of the single turn coil are represented by adding the suffix s to the symbol.

The single turn circuit impedance, Z_s , is:

$$Z_s = Z_{cs} + \frac{jX_{rs}[jX_{gs} + Z_{ws}]}{jX_{rs} + jX_{gs} + Z_{ws}} \quad (2.17)$$

Z_s is of a complex value;

$$Z_s = R_s + jX_s \quad (2.18)$$

its absolute value is;

$$|Z_s| = \sqrt{R_s^2 + X_s^2} \quad (2.19)$$

The coil efficiency is;

$$\eta = \frac{R_{ws}}{R_s} \quad (2.20)$$

and the power factor is;

$$\text{P.F.} = \frac{R_s}{|Z_s|} \quad (2.21)$$

The volt-ampere, VA, of the coil can be calculated from the power input to the coil P_c :

$$VA = \frac{P_c}{P.F.} \quad (2.22)$$

The voltage of the single turn coil V_c , can be calculated from the volt-ampere as follows:

$$V_c = Z_c I_c \quad (2.23)$$

multiply by the voltage and divide by N_c^2 ,

$$\frac{V_c^2}{N_c^2} = V_c I_c \frac{Z_c}{N_c^2} \quad (2.24)$$

take the square root

$$\therefore V_{cs} = \sqrt{(VA)Z_s} \quad (2.25)$$

As the coil voltage V_c is known, then the number of turns is;

$$N_c = \frac{V_c}{V_{cs}} \quad (2.26)$$

Since the number of turns is now derived, the impedance of the coil is;

$$Z = Z_s N_c^2 \quad (2.27)$$

and the coil current is

$$I_c = \frac{V_c}{Z} \quad (2.28)$$

2.1.1 Comments on the Equivalent Circuit

In general, the aforementioned equivalent circuit method is adopted widely by many researchers. The literature, however, does tend to differ in this domain.

Reichert [2.2] expressed the factor k_r as:

$$k_r = \frac{0.92}{\sqrt{k_c}} \quad (2.29)$$

$$\text{where } k_c = \frac{N_c t_c}{l_c} \quad (2.30)$$

t_c is the coil conductor width

while Baker [2.1] suggested that $1.5 \geq k_r \geq 1.0$ with a typical value of 1.15. Instead of taking k_r , Vaughan and Williamson [2.3, 2.4] used the space factor, which is approximately the inverse of k_r [2.5].

According to Baker [2.1] any single layer coil can be classified as being either a short or a long coil. The equivalent circuits are shown in Figs. 2.4 and 2.5 and the difference between them is the shunt reactance X_r . The justification of the position of this reactance in the circuit can be considered insufficient. Its value was given as:

$$X_r = \frac{\omega \mu_0 \pi r_c^2 N_c^2}{0.9 k_e} \quad (2.31)$$

where k_e is an empirical factor with a typical value of unity.

The above expression is similar to that of equation (2.12) where $k = 0$.

Reichert [2.2] adopted the equivalent circuit shown in Fig. 2.3.

while Vaughan and Williamson [2.3, 2.4] used the circuit shown in Fig. 2.5. The methods of Baker and Vaughan and Williamson were discussed in detail and worked examples were supplied in reference [2.5].

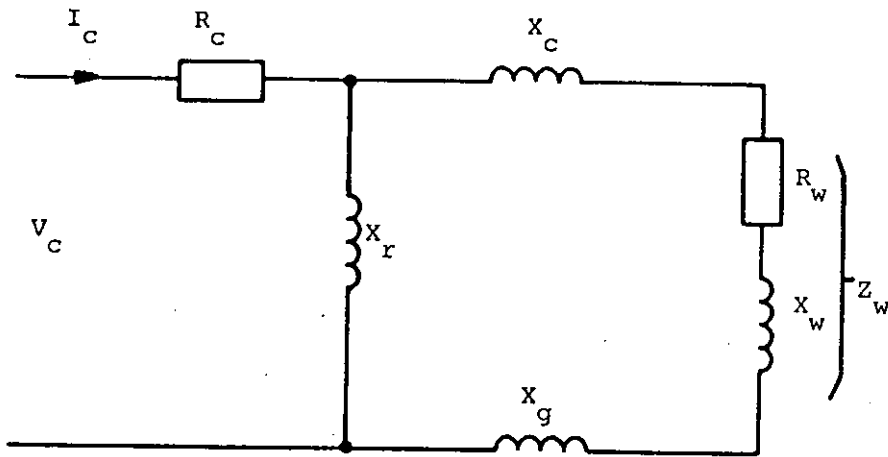


Fig. 2.4 The equivalent circuit of the short coil

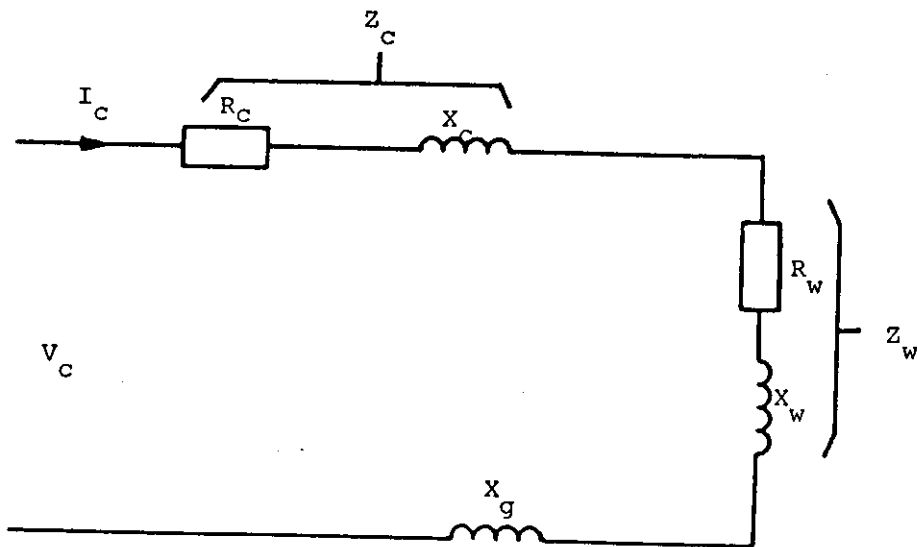


Fig. 2.5 The equivalent circuit of the long coil

In the equivalent circuit method, the workpiece material is assumed to be uniform and the calculation is based on a single value for the resistivity and the permeability. The resistivity is a function of temperature, while the ferromagnetic material permeability, below the Curie point, is a function of the magnetic field strength and can be reasonably represented by Frohlich formula:

$$\mu = \frac{FA}{FB + H} + FC \quad (2.32)$$

The equivalent circuit method suffers from three main imperfections:

1. The use of empirical data. The values of these factors are not necessarily known for every application.
2. The assumption of uniform material properties is incorrect, as the resistivity is a function of the temperature, and the permeability is a function of both temperature and magnetic field strength.
3. The assumption of uniform magnetic field along the length of the workpiece hinders any application of this method where non-uniform power densities must be induced in the load.

2.2 Numerical Methods of Work Coil Design

Three distinct numerical approaches have been applied to induction heating problems: finite difference, finite element and mutually coupled circuit methods. In the finite difference and finite element methods the workpiece is divided into subregions and the relevant non-linear

partial differential equations are replaced by a set of algebraic equations to be solved in each subregion by means of iterative procedures. The mutually coupled circuit method [2.6] divides the system into a number of subconductors mutually coupled with each other and applies Kirchoff's Laws to obtain a system of linear equations describing the problem which in turn are solved by procedures similar to those used in other numerical methods. The application of these numerical methods to induction heating problems will now be discussed.

2.2.1 Mutually Coupled Circuit Method

This method was developed to solve problems in induction heating and melting applications by Kolbe and Reiss [2.6] and later developed by Dudley and Burke [2.7]. The region to be investigated is subdivided into subconductors and using known expressions [2.8] the resistance and self inductance of each subconductor, together with the mutual inductances between each subconductor are determined. The expressions are usually presented in the form of an impedance matrix and the application of Kirchoff's Laws leads to a system of linear equations with complex coefficients of the form,

$$[Z] [I] = [V]$$

where $[Z]$ is the square matrix comprising all of the coil and billet self and mutual impedances, $[I]$ is the column matrix of unknown coil and billet segment currents and $[V]$ is the column matrix of driving voltages. A more detailed illustration of the use of the mutually coupled method is given in Appendix 3.

The mutually coupled circuit method can be used to determine the induced currents, power distribution and mechanical forces produced by any two dimensional, linear and axisymmetric induction heating problem. The limitations of the method are that magnetic materials cannot be included in the problem geometry and that the number of elements into which the conductors can be divided is limited by the size of the matrix which may be solved. This number is of the order of 300 unknowns for the IBM 3033 computer [2.9] which has a memory capability of approximately one megabyte. Difficulties also arise when dealing with curved surfaces and when the penetration depth is small in comparison with the load dimensions. The number of unknowns can partly be reduced by finely subdividing only the regions of greatest interest, typically between a distance 0.1 to 0.33 times the skin depth δ [2.9] from the surface and using larger subdivisions elsewhere. This, of course, requires more complicated software.

Kolbe and Reiss [2.6] employed the mutually coupled circuit method in order to determine:

1. Current density and power distributions in an inductively heated load of rotational symmetry.
2. Temperature distribution as a function of space and time.

These problems were firstly treated separately and then coupled together by the temperature dependence of the electrical material's values. This technique is characterized by calculating the current density distribution for initial temperature distribution. Once this has been determined one can then calculate the temperature

distribution at the time $t = \Delta t$, and with it the new material's values. The derivation of the current density distribution for the time $t = 2\Delta t$ and the corresponding materials values are then possible. This sequence of calculations is continued until the desired temperature or heating time is reached.

The application of this procedure to a specific case in the paper showed that the assumed inductor arrangement is unsuitable for the intended objective. This, of course, requires a re-run of the program; this is an expensive operation and requires the use of a powerful computer, even though the thermal conductivity of the material was ignored.

Dudley and Burke [2.7] used the mutually coupled circuit method to determine the current distribution in three induction heating configurations. The first is a coil and billet where the coil current was assumed to be distributed uniformly along the coil length; see Appendix 3. The second case is that of a sheet coil and billet; the coil current in this case was assumed to be uniform. This required the single turn and the billet to be divided into a number of segments, all of which are coupled to each other. This, of course, increases the number of equations to be solved. The third case involves a coil without a workpiece, where each turn is to be divided into a number of segments.

This method has been used in the solution of other problems and the various formulations differ, primarily in terms of the expressions used to represent the circuit elements. These expressions are usually complicated and the calculations very lengthy; see Appendix 3.

2.2.2 Finite Element Method

The finite element method has become an important and practical numerical analysis tool. It has found application in almost all areas of engineering and applied mathematics.

In this method, the field region is subdivided into a finite number of discrete sized subregions or finite elements. The unknown quantities at each element are presented by suitable interpolation functions that contain the node values of each element.

From the mathematical point of view the finite element method is based on integral formulations. By way of comparison the older finite difference methods are usually based on differential formulations. Finite element integral formulations are obtained by two different procedures: variational formulations and weighted residual formulations. Both of these techniques generate the final assembly of algebraic equations that must be solved for the unknown nodal parameters.

The earliest mathematical formulations for finite element models were based on variational techniques [2.10-2.13]. Variational techniques still are very important [2.14, 2.15] in developing elements and in solving practical problems. Variational models usually involve finding the nodal parameters that yield a stationary (maximum or minimum) value of a specific integral relation known as a functional.

The solution that yields a stationary value of the integral functional and satisfies the boundary conditions is equivalent to the solution of an associated differential equation, known as the Euler equation. If the functional is known, then it is relatively easy to find the corresponding Euler equation. Most engineering and physical problems are initially defined in terms of a differential equation. The finite element method requires an integral formulation. Thus, one must search for the functional whose Euler equation corresponds to the given differential equation (and boundary conditions). Unfortunately, this is generally a difficult, or impossible task. Therefore, there is increasing emphasis on the various weighted residual techniques [2.16] that can generate an integral formulation directly from the original differential equations.

The generation of finite element models by the utilization of weighted residual techniques is a relatively recent development. The weighted residual method starts by the assumption of an approximate solution for the governing differential equation and avoids the often tedious search for a mathematically equivalent variational statement. This method leads to a set of algebraic equations that can be solved for the unknown coefficients in the approximate solution. The variational and the weighted residual techniques, both, are not easily applied by any one other than the specialist numerical analyst. A more detailed illustration of the use of the finite element method is given in Appendix 4.

In conclusion, it must be borne in mind, that when employing the finite element methods to solve induction heating problems, one must [2.9]

- (i) Include an "exterior element" to represent the region outside the coil
- (ii) Use a sparse matrix solution routine

Typically, induction heating problems involving the computation of the electromagnetic and thermal fields at 500 nodes requires approximately 0.75 megabytes of computer memory [2.9] even when using a sparse matrix solution routine. Problems involving 500 to 2000 unknowns can be run at a reasonable cost on an IBM 3033 which compares very favourably with the mutually coupled circuit method mentioned previously.

Nemoto and Tabuchi [2.15] used the finite element method to analyse induction heating systems in two computer programs. The first was to solve Maxwell's equations so as to determine the eddy current losses which were employed in the second program as heat sources to predict the temperature distribution in the workpiece. The reduction of the computing time necessitated the representation of the relations among the elements by first degree equations. On the other hand, the accuracy derived from these equations will be minimal.

When applying these programs to a particular problem, the heat sources and the material properties were assumed to be constant. In practice, they change simultaneously and the principal advantage of the finite

element method, until now overlooked, is the facility to take this change partially into account. Moreover, this method can not be considered a trial and error procedure, as it was recommended by Nemoto and Tabuchi, because of the high cost of running the program.

Sabonnadiere [2.17] employed a weighted residual technique into a finite element program in order to predict the distributions of the flux and the power in an induction heating system during the heating cycle. The boundary conditions and the elements physical properties were required as an input data to the program. This program, like other numerical methods programs, is characterized by a lengthy running time. Recent communication with Sabonnadiere suggests that the work is still in progress.

2.2.3 Finite differences

This was the first numerical technique applied to induction heating problems [2.18] and is still in use today. The technique provides a relatively straightforward means of formulating and solving two dimensional problems.

The basis of the finite difference method is the conversion of the governing differential equations of the problem into a set of algebraic equations by the use of the Taylor series approximations. The equations are solved at every point on a grid constructed over the required space domain. The mathematical operations are performed

not upon continuous functions but rather in terms of equations about one discrete point. By this approach operations such as differentiation and integration may be reduced to simple arithmetic forms and can then be conveniently solved using digital computers.

Finite difference equations are derived by approximating the partial or total derivatives in a differential equation with algebraic expression, usually based on Taylor series expansion of the unknown function. Given the function $\phi(x)$, the value of the function $\phi(x + \Delta x)$ and $\phi(x - \Delta x)$ can be calculated from:

$$\phi(x + \Delta x) = \phi(x) + \Delta x \frac{d\phi}{dx} + \frac{1}{2}(\Delta x)^2 \frac{d^2\phi}{dx^2} + \frac{1}{6}(\Delta x)^3 \frac{d^3\phi}{dx^3} + \dots \quad (2.33a)$$

$$\phi(x - \Delta x) = \phi(x) - \Delta x \frac{d\phi}{dx} + \frac{1}{2}(\Delta x)^2 \frac{d^2\phi}{dx^2} - \frac{1}{6}(\Delta x)^3 \frac{d^3\phi}{dx^3} + \dots \quad (2.33b)$$

By omitting terms from $(\Delta x)^4$ onwards the simple expression for $\frac{d^2\phi}{dx^2}$ is obtained by adding equations (2.33a and b)

$$\frac{d^2\phi}{dx^2} = \frac{1}{(\Delta x)^2} [\phi(x + \Delta x) - 2\phi(x) + \phi(x - \Delta x)] \quad (2.34)$$

In the same way the following approximations can be obtained for the first derivative by ignoring $(\Delta x)^2$ and above

$$\frac{d\phi}{dx} = \frac{1}{\Delta x} [\phi(x + \Delta x) - \phi(x)] \quad (2.35a)$$

$$\frac{d\phi}{dx} = \frac{1}{\Delta x} [\phi(x) - \phi(x - \Delta x)] \quad (2.35b)$$

Equations (2.35a and b) are known respectively as the forward and backward difference approximation. The central difference approximation is defined as the average of the two given expressions, i.e.

$$\frac{d\phi}{dx} = \frac{1}{2\Delta x} [\phi(x + \Delta x) - \phi(x - \Delta x)] \quad (2.35c)$$

If ϕ is a function of two independent variables x and y , subdivide the x - y plane into sets of equal rectangles of sides Δx and Δy , as shown in Fig. 2.6, and let the coordinates (x, y) of the representative mesh point p be;

$$x = i\Delta x \text{ and } y = j\Delta y$$

where i and j are integers.

Denote the value of ϕ at p by;

$$\phi_p = \phi(i\Delta x, j\Delta y) = \phi_{i,j}$$

Then by equation (2.34)

$$\begin{aligned} \left(\frac{\partial^2 \phi}{\partial x^2}\right)_p &= \left(\frac{\partial^2 \phi}{\partial x^2}\right)_{i,j} = \frac{1}{(\Delta x)^2} \{ \phi[(i+1)\Delta x, j\Delta y] - 2\phi[i\Delta x, j\Delta y] \\ &\quad + \phi[(i-1)\Delta x, j\Delta y] \} \end{aligned} \quad (2.36)$$

i.e.

$$\left(\frac{\partial^2 \phi}{\partial x^2}\right)_{i,j} = \frac{1}{(\Delta x)^2} [\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}] \quad (2.37)$$

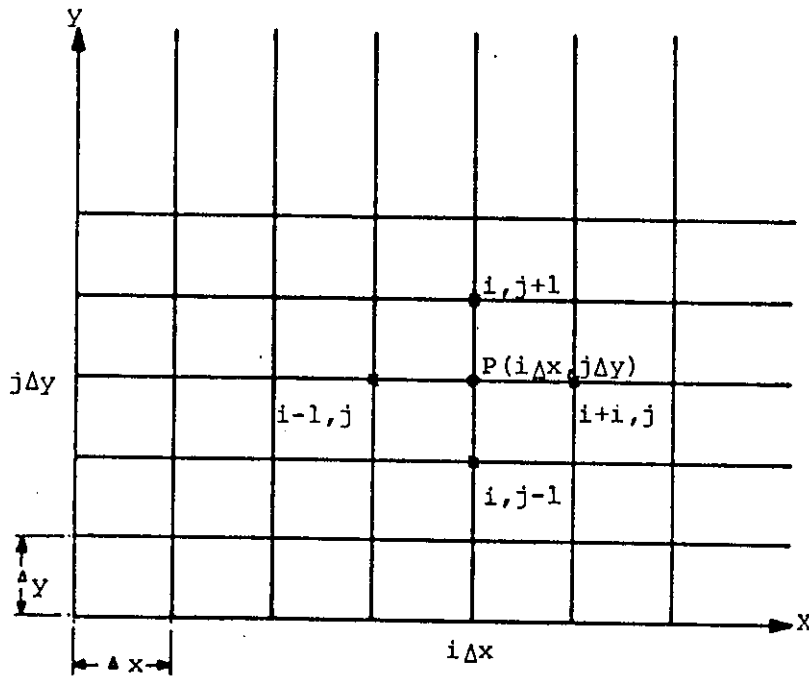


Fig. 2.6 Finite difference computation grid

$$\text{Similarly, } \left(\frac{\partial^2 \phi}{\partial y^2} \right)_{i,j} = \frac{1}{(\Delta y)^2} [\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}] \quad (2.38)$$

With this notation the forward-difference approximation for

$\frac{\partial \phi}{\partial y}$ at p is

$$\frac{\partial \phi}{\partial y} = \frac{1}{\Delta y} [\phi_{i,j+1} - \phi_{i,j}] \quad (2.39)$$

Reichert [2.19] developed a finite difference computer program to determine the field and the temperature distribution in an inductively heated workpiece. The program takes the variation of the physical properties during the heating cycle into account. A considerable data input base is one of this program's basic requirements and it is in this area that the program can create problems for the inexperienced user. Also, the computer used was not sufficient to the program which meant sacrificing in the accuracy.

Finally, the program's running time was too long. This might not be considered too important a factor, if the sole purpose of the exercise was to examine an existing system. However, if the aim is to find an optimum design criteria which require several re-runs of the program, this factor does become more significant.

Gibson [2.20] developed a finite difference computer program to determine the temperature profiles of a metal workpiece heated by induction or other heating methods. The program was complicated by the fact that it could be used for a wide range of applications and with various conditions, such as constant power or current and other such constraints. The changes in the workpiece's permeability were taken into account, by employing a functional representation based on Fröhlich formula, rather than a large amount of data, as in reference [2.19]. This does, of course, reduce the user's work load.

In general, the main features of the numerical methods are:

1. Over the heating cycle time span, the change in the physical properties of the workpiece can be taken into account. This, of course, increases the running time and a greater memory facility is essential.
2. The program's main objective is to analyse existing configurations. The optimum design criteria can be obtained by a trial and error approach, which necessitates several re-runs of the program. This is an expensive method, as the program is, usually, quite long.

3. The problem of non-uniform power density induced along the workpiece has not been tackled as there is currently no relevant research on this subject.

2.3 Conclusion

For many years, very simple models, based on the electrical equivalent circuit, were adopted together with empirical data to design induction heating equipment. During the last decade, various research workers have begun to employ numerical techniques to aid the design and development process of induction heating systems.

Almost all of the existing work coil design methods are based on the assumption that the workpiece is subjected to a uniform magnetic field. It is, therefore, not easy to employ these methods, when designing a coil capable of inducing a non-uniform power density along the load. These methods, too, either assume a uniform workpiece material and require the use of empirical data as in the equivalent circuit method or they need considerable computational facilities when adopting numerical techniques.

CHAPTER 3

INITIAL INVESTIGATIONS

Any attempt to develop or improve a design method for a particular piece of equipment naturally requires a study of the existing methods of analysis of that equipment. This applies to induction heating work coils as well as to other devices.

The aim is to gain experience in the use of work coil design methods, to assess their limitations and to examine the possibility of introducing any improvement.

This chapter describes initial investigations into the equivalent circuit and the numerical methods. The study of the equivalent circuit method includes the examination of the empirical factors' effects, the reactance of the external flux path and the nonuniformity of the workpiece.

This method has, also, been improved by developing an equivalent circuit of a multilayer load of differing physical properties so as to take the nonuniformity of the workpiece into account.

The use of the finite difference method, a numerical technique, has been investigated to assess the typical limitations and difficulties

which occur, when solving induction heating problems with these methods.

The finite difference method was implemented to predict the magnetic field and power distributions in an inductively heated workpiece.

3.1 Investigations into the Equivalent Circuit Method

Three computer programs were developed to assess the influence of the external reactance, the empirical factors and the nonuniformity of the workpiece physical properties. These programs "EQUIV", "SHORT" and "LONG" are based on the equivalent circuits of Figs. 2.3, 2.4 and 2.5 respectively.

The calculations of the program "EQUIV" is as was shown in Section 2.1. The calculations of the programs "SHORT" and "LONG" are compiled according to Baker's equations [3.1], see Section 2.1.1.

3.1.1 Uniform Load

The previously mentioned computer programs were developed in the PRIME 400 system and employed to assess the effect of the external reactance and the empirical factors on the design and performance of a coil to heat a uniform load.

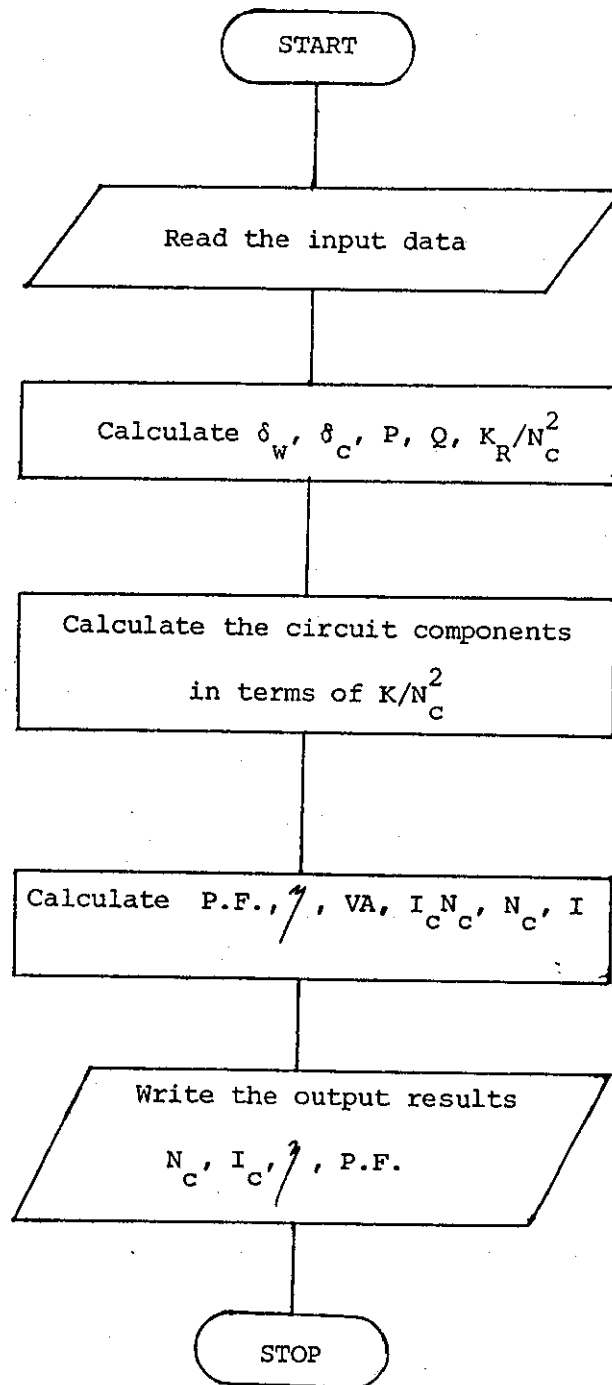
A file entitled "BESSEL" was developed independently to evaluate the Bessel functions. The polynomial expressions of these functions were obtained from reference [3.2]. This file was inserted into each of the three programs so as to obtain the evaluations of the Bessel functions required for the calculations.

The program "EQUIV" is based on the equivalent circuit of Fig. 2.3 and the expressions for the components of the circuit are those given in Section 2.1.

The program "SHORT" is based on Baker's method [3.1] for the design of a short coil, Fig. 2.4. The expressions for the components of the equivalent circuit, are those given by Baker and mentioned in the previous chapter.

The program "LONG" is similar to SHORT with the absence of the external reactance - that is the equivalent circuit of long coil, Fig. 2.5.

The three computer programs differ only on the position of the external reactance X_r . The flow chart of them is identical and is given below.



The Flow Chart of the Programs

The three computer programs were employed to assess the effects of:

1. The external flux path reactance
2. The empirical factors

Three cases were considered:

1. A nonmagnetic load with 50 Hz power supply
2. A nonmagnetic load with 3 kHz power supply
3. A magnetic load with 50 Hz power supply.

The following values of the parameters were used in the calculations:

Coil power	= 200 kW
Coil voltage	= 440 V
Load diameter	= 0.08 m
Coil diameter	= 0.14 m
Coil length = Load length	= 0.5 m
Coil resistivity	= $2 \times 10^{-8} \Omega\text{m}$
Coil relative permeability	= 1.0

The values used for each of the following factors are as follows unless its effect was examined

$$k_r = 1.15 \quad k_e = 1.0 \quad k_c = 0.8 \quad k = 0.0$$

The nonmagnetic load was assumed to be aluminium with surface temperature of about 600°C and 20°C at the centre. This corresponds

to resistivities of $10 \times 10^{-8} \Omega\text{m}$ and $2 \times 10^{-8} \Omega\text{m}$ respectively [3.3]. An average value of $\rho_w = 6 \times 10^{-8} \Omega\text{m}$ was adopted.

The magnetic load was assumed to be EN3 mild steel with a surface temperature of 600°C and a centre temperature of 20°C . This corresponds to resistivities of $75 \times 10^{-8} \Omega\text{m}$ and $15 \times 10^{-8} \Omega\text{m}$ respectively [3.3].

The permeability can be calculated from Frohlich formula [3.4]

with the following constants: $AF = 2.08$, $BF = 1380$ and $CF = 1.2566 \times 10^{-6}$. These will give a maximum relative permeability of 1200 at zero magnetic field and a minimum of 1 at very high magnetic field. Average values of $\rho_w = 45 \times 10^{-8} \Omega\text{m}$ and $\mu_{rw} = 600$ were employed in the calculations.

The results obtained from the three computer programs are shown in Tables (3.1-3.7).

Although the numerical values of the results apply to particular cases, they do show the main features of the relevant methods.

Table 3.1 shows that when dealing with a nonmagnetic load the three circuits give values which are all within 10% of each other for the required coil turns and current. The difference in the estimated performance of the system i.e. efficiency and power factor is more than 15%. The long coil, Fig. 2.5, tends to produce better results than the short coil, Fig. 2.4, when compared with the circuit of Fig. 2.3 "EQUIV".

	Nonmagnetic Load, 50 Hz				Nonmagnetic Load, 3 kHz				Ferromagnetic Load, 50 Hz			
	EQUIV	SHORT	LONG	MAX % DIFF	EQUIV	SHORT	LONG	MAX % DIFF	EQUIV	SHORT	LONG	MAX % DIFF
N_c	151	156	145	7.6	10	11	10	10	93	65	62	49
I_c	1709	1636	1678	4.5	8401	8189	8280	2.6	674	1589	674	136
η	40	35	40.4	15.4	46.2	41.3	45.5	11.9	96.5	89.4	98.3	10
P.F.	.266	.28	.27	5.3	.054	.056	.055	3.7	.675	.286	.674	136

Table 3.1: Results from the three different circuits with optimum values of the empirical factors

	Short Coil			Long Coil		
	$k_r = 1.0$	$k_r = 1.5$	Max % Diff	$k_r = 1.0$	$k_r = 1.5$	Max % Diff
N_c	153	161	5.2	143	149	4.2
I_c	1742	1449	20.2	1773	1507	17.7
η	38.3	29	32.1	43.8	34.2	28.1
P.F.	0.26	0.31	19.2	.256	0.302	18

Table 3.2: The effect of k_r on a nonmagnetic load at 50 Hz

	Short Coil			Long Coil		
	$k_r = 1.0$	$k_r = 1.5$	Max % Diff	$k_r = 1.0$	$k_r = 1.5$	Max % Diff
N_c	10	12	20	10	11	11
I_c	8832	7014	25.9	8876	7171	23.8
η	44.8	35	28	49	39	25.6
P.F.	0.051	0.065	27.5	0.051	0.063	23.5

Table 3.3: The effect of k_r on a nonmagnetic load at 3 kHz

	Short Coil			Long Coil		
	$k_r = 1.0$	$k_r = 1.5$	Max % Diff	$k_r = 1.0$	$k_r = 1.5$	Max % Diff
N_c	64	66	3.13	62	62	0
I_c	1607	1549	3.75	674	674	0
η	90.7	86.5	4.86	98.5	97.7	0.8
P.F.	0.28	0.294	5.0	0.674	0.674	0

Table 3.4: The effect of k_r on a ferromagnetic load at 50 Hz

	Nonmagnetic Load at 50 Hz			Nonmagnetic Load at 3 kHz			Ferromagnetic Load at 50 Hz		
	$k_e = 0.8$	$k_e = 1.2$	%Diff	$k_e = 0.8$	$k_e = 1.2$	%Diff	$k_e = 0.8$	$k_e = 1.2$	%Diff
N_c	153	158	3.3	10	11	10	63	66	4.8
I_c	1646	1626	1.2	8212	8165	0.6	1418	1747	23.1
η	36	34	5.9	42.2	40.5	4.2	92	86	6.5
P.F.	.276	.28	1.5	.0554	.0557	0.5	0.32	0.26	23.1

Table 3.5: The effect of k_e

	Nonmagnetic Load at 50 Hz			Nonmagnetic Load at 3 kHz			Ferromagnetic Load at 50 Hz		
	$k = -.22$	$k = 0.1$	%Diff	$k = -.22$	$k = 0.1$	%Diff	$k = -.22$	$k = 0.1$	%Diff
N_c	146	153	4.8	10	11	10	79	98	24
I_c	1742	1694	2.8	8572	8325	3.0	674	674	0
η	41.1	39.5	4.0	47.3	45.7	3.5	97.4	96.1	1.4
P.F.	0.261	0.268	2.7	0.053	.055	3.8	.674	.675	0.2

Table 3.6: The effect of k

	Nonmagnetic load at 50 Hz			Nonmagnetic load at 3 kHz			Ferromagnetic load at 50 Hz		
	$k_c=0.7$	$k_c=1.0$	% Diff	$k_c=0.7$	$k_c=1.0$	%Diff	$k_c=0.7$	$k_c=1.0$	%Diff
N_c	152	149	2.0	11	10	10	93	93	0
I_c	1663	1787	7.5	8117	8878	9.4	674	674	0
η	38.4	42.7	11.2	44.5	49	10.1	96.3	96.9	0
P.F.	0.274	0.254	7.9	.056	.051	9.8	.675	.675	0

Table 3.7 The effect of k_c

The magnetic load results obtained from the three circuits have diverse characteristics. The number of turns calculated by the short and long coil circuits are equal, but a different number is produced by the circuit of Fig. 2.3 "EQUIV". The required current calculated by the former 2 circuits again equal in both cases, but different in the latter case of the short coil circuit.

The effect of the empirical factor k_r , which allows for the spacing between the turns, on the short and long circuits is shown in Tables 3.2-3.4. The effect on magnetic materials, for the case under consideration, is very small when adopting the short coil circuit and negligible when adopting the long coil circuit. However, the difference increases with the frequency for nonmagnetic materials and

it becomes more noticeable when applying the short coil circuit. The estimation error in the number of turns can be anything up to 20%, while in the current can reach 26% and the efficiency 32%.

Changing the value of k_e , the external reactance factor, from 0.8 to 1.2 does not affect the results by more than 10% except when estimating the current for magnetic materials when they will vary by 23% (Table 3.5).

The effect of the factor k , which is for the return flux, on the circuit of Fig. 2.3 "EQUIV" is shown on Table 3.6. When this constant was varied between the maximum and minimum values (0.1 and -0.22), the results did not fluctuate by more than 10%, apart from when estimating the number of turns for a magnetic load, in this case the variation was 24%.

Table 3.7 shows k_c , which is for the spacing between turns, did not influence the magnetic load results. The effect on the case of the nonmagnetic load was within 10% on the number of turns and current, and to 11% on the efficiency.

The above work showed that the three circuits may give different answers. The discrepancy between these answers depends on the case under consideration and there is no general rule to assess these differences. Also, the empirical factors have an influence on the results produced by these circuits, and here again, this influence depends on the case

under consideration and cannot be assessed in general terms, hence it is difficult to the inexperienced person to choose the right circuit and the right values of the empirical factors suitable for a given configuration.

The other limitation of the equivalent circuit method is that it does not take the nonuniformity of the workpiece into account. This is to be discussed in the next section.

3.1.2 Nonuniform Load

The workpiece, during the induction heating cycle is not uniform, as the resistivity of any part is a function of the local temperature, while the permeability depends on the magnetic field strength in that part. The load can be considered as consisting of a number of concentric cylinders, each one of different physical properties. This approach improves the equivalent circuit method by overcoming one of its imperfections.

Fig. 3.1 shows a typical billet heating application, whereby a coil of radius r_c surrounds a billet of radius r_w . The billet is shown as being divided into n concentric cylinders.

Following the procedure outlined in Section 2., the magnetic and electrical equivalent circuits are shown in Figs. 3.2 and 3.3

respectively. The derivation of X_r , X_g and Z_c are as in Section 2.7 and they are represented below:

$$X_r = K_R \frac{r_c}{2(0.45 + k)} \quad (3.1)$$

$$X_g = K_R \frac{(r_c^2 - r_w^2)}{l_c} \quad (3.2)$$

Reichert's expression [3.5] for Z_c will be drawn upon to represent the coil impedance

$$Z_c = K_R \frac{0.92}{\sqrt{k_c}} \frac{\delta_c r_c}{l_c} (1 + j) \quad (3.3)$$

$$\text{where } K_R = \omega \mu_o \pi N_c^2 \quad (3.4)$$

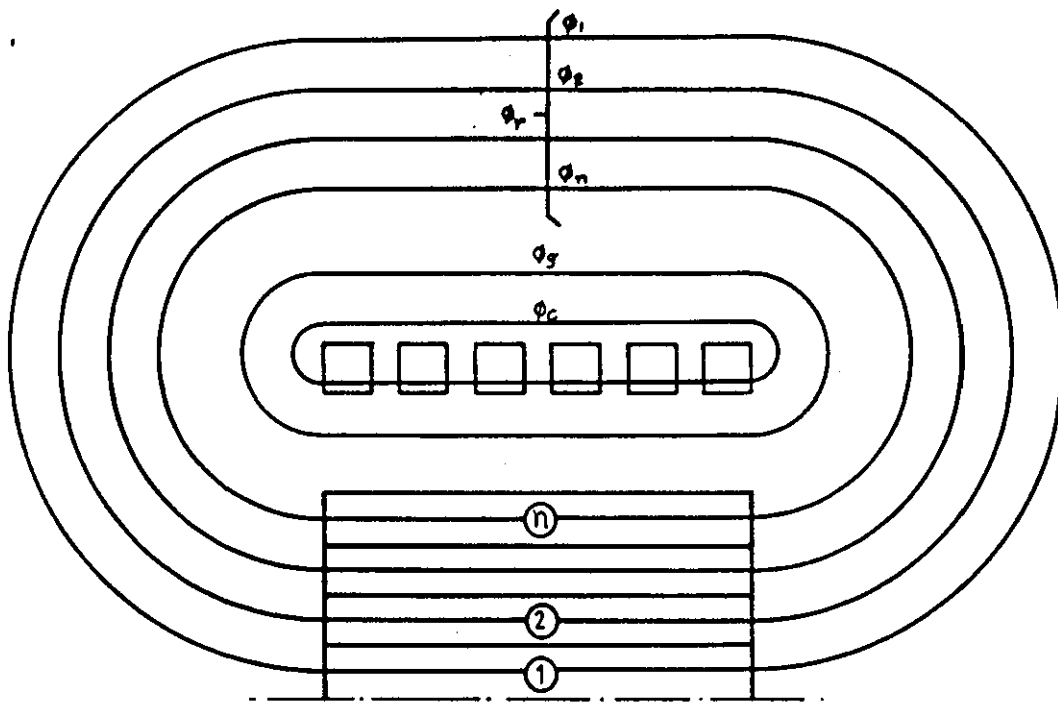


Fig. 3.1 The flux paths in a load of n concentric cylinders

The impedance of the composite load can be determined by employing Maxwell's equations and the complex Poynting vector. The total complex power S consumed by a cylinder of radius r and length ℓ_w is:

$$S_r = (E(r) \times H^*(r)) \cdot 2\pi r \ell_w \quad (3.5)$$

where $E(r)$ and $H^*(r)$ are the r.m.s. values of the electric field strength and the conjugate of the magnetic field strength on r respectively. $E(r)$ is a function of the current density $J(r)$ and the resistivity ρ_r at radius r ;

$$E(r) = -\rho_r J(r) \quad (3.6)$$

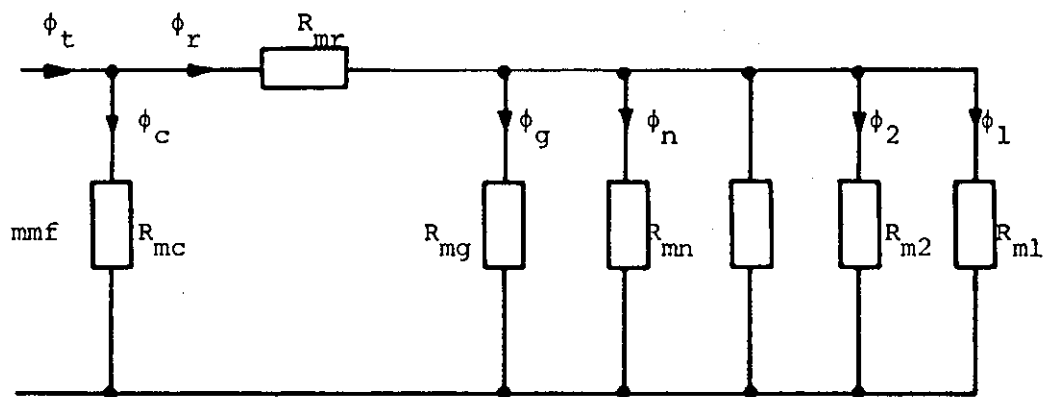


Fig. 3.2 The magnetic circuit of a coil-multilayer workpiece

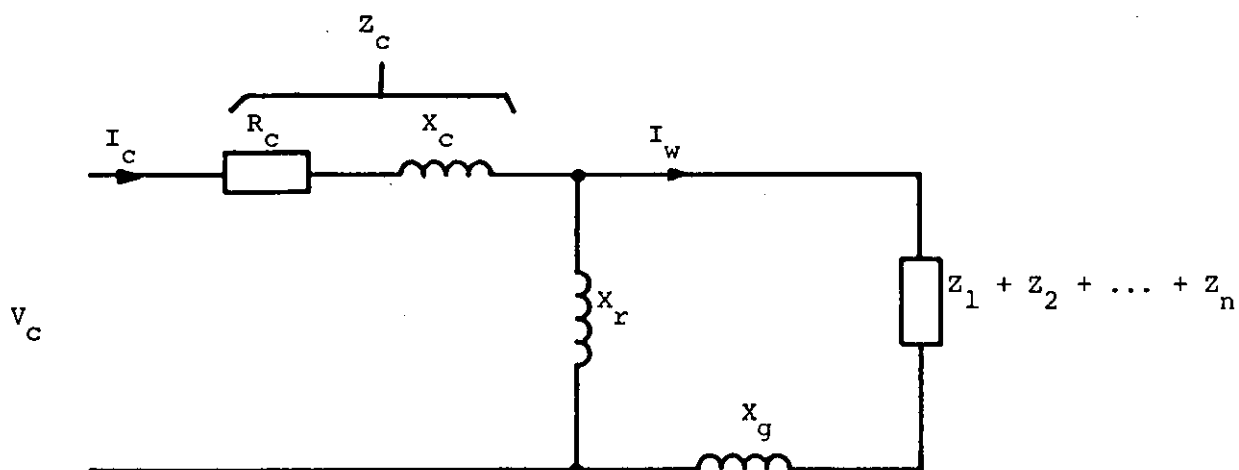


Fig. 3.3 The equivalent electrical circuit of a coil-multilayer workpiece

Since the electric and magnetic field strengths are perpendicular to each other; equation (3.5) can be rewritten as;

$$S_r = -2\pi r \ell_w \rho_r J(r) H^*(r) \quad (3.7)$$

The apparent power consumed within the cylinder of radius r , according to Fig. 3.3, is;

$$S_r = Z_r |I_w|^2 \quad (3.8)$$

where Z_r is the impedance at radius r . If the coil is a long solenoid then;

$$I_w = N_c I_c = H_o \ell_w \quad (3.9)$$

$$\therefore |I_w|^2 = H_o H_o^* \ell_w^2 \quad (3.10)$$

where H_o is the magnetic field strength and H_o^* is the magnetic field strength's conjugate on the workpiece surface.

Substitute equation (3.10) into equation (3.8) and equate it with equation (3.7) so as to yield

$$Z_r = \frac{-2\pi r \rho_r}{\ell_w} \cdot \frac{J(r)}{H_o} \cdot \frac{H^*(r)}{H_o^*} \quad (3.11)$$

The cylinder's magnetic field strength and it's current density can be evaluated by solving Maxwell's equations, see Appendix 1, and they are:

$$H(r) = C \cdot I_0(\sqrt{2j} \frac{r}{\delta}) + B \cdot K_0(\sqrt{2j} \frac{r}{\delta}) \quad (3.12)$$

$$J(r) = \frac{\sqrt{2}}{\delta} [C \cdot I_0'(\sqrt{2j} \frac{r}{\delta}) + B \cdot K_0'(\sqrt{2j} \frac{r}{\delta})] \quad (3.13)$$

where C and B are constants.

The solution in the inner cylinder, $r_1 \geq r \geq 0$, which consists of a homogenous material is:

$$H_1(r) = C_1 \cdot I_0(\sqrt{2j} \frac{r}{\delta_1}) \quad (3.14)$$

$$J_1(r) = \frac{\sqrt{2}}{\delta_1} C_1 \cdot I_0'(\sqrt{2j} \frac{r}{\delta_1}) \quad (3.15)$$

In the other cylinders, B is not necessarily zero, hence the complete solution is essential.

In the second cylinder, $r_2 \geq r \geq r_1$:

$$H_2(r) = C_2 \cdot I_0(\sqrt{2j} \frac{r}{\delta_2}) + C_3 \cdot K_0(\sqrt{2j} \frac{r}{\delta_2}) \quad (3.16)$$

$$J_2(r) = \frac{\sqrt{2}}{\delta_2} [C_2 \cdot I_0'(\sqrt{2j} \frac{r}{\delta_2}) + C_3 \cdot K_0'(\sqrt{2j} \frac{r}{\delta_2})] \quad (3.17)$$

In the third cylinder, $r_3 \geq r \geq r_2$:

$$H_3(r) = C_4 \cdot I_0(\sqrt{2j} \frac{r}{\delta_3}) + C_5 \cdot K_0(\sqrt{2j} \frac{r}{\delta_3}) \quad (3.18)$$

$$J_3(r) = \frac{\sqrt{2}}{\delta_3} \left[C_4 \cdot I'_0 \left(\sqrt{2j} \frac{r}{\delta_3} \right) + C_5 \cdot K'_0 \left(\sqrt{2j} \frac{r}{\delta_3} \right) \right] \quad (3.19)$$

where δ_1 , δ_2 and δ_3 are the skin depths in the 1st, 2nd and 3rd cylinders respectively, and this continues to be the case until the outer cylinder is reached.

$$H_n(r) = C_{2n-2} \cdot I_0 \left(\sqrt{2j} \frac{r}{\delta_n} \right) + C_{2n-1} \cdot K_0 \left(\sqrt{2j} \frac{r}{\delta_n} \right) \quad (3.20)$$

$$J_n(r) = \frac{\sqrt{2}}{\delta_n} \left[C_{2n-2} \cdot I'_0 \left(\sqrt{2j} \frac{r}{\delta_n} \right) + C_{2n-1} \cdot K'_0 \left(\sqrt{2j} \frac{r}{\delta_n} \right) \right] \quad (3.21)$$

The constants C_1 to C_{2n-1} can be obtained by considering the boundary conditions at the junctions between the concentric cylinders and also at the junctions between the surface of the billet and the air gap.

It is known that the magnetic and electric field strengths are both continuous across the boundary. Once having defined H_0 as the magnetic field strength at the surface of the billet, it is possible to determine (C_1/H_0) to (C_n/H_0) , and therefore to solve for $H(r)/H_0$ and $J(r)/H_0$ too. By substitution into equation (3.11), the impedances can be found. The following method of solution can be applied for any number of concentric cylinders. For simplicity consider a two layers load.

At the billet surface:

$$r = r_w \quad (3.22)$$

$$\text{and } H(r) = H_0 \quad (3.23)$$

$$\text{i.e. } H_0 = C_2 \cdot I_0 \left(\sqrt{2j} \frac{r_w}{\delta_2} \right) + C_3 \cdot K_0 \left(\sqrt{2j} \frac{r_w}{\delta_2} \right) \quad (3.24)$$

At the boundary between the two layers:

$$r = r_1 \quad (3.25)$$

$$\text{and } H_1(r) = H_2(r) \quad (3.26)$$

$$\rho_1 J_1(r) = \rho_2 J_2(r) \quad (3.27)$$

$$\text{i.e. } c_1 \cdot I_0(\sqrt{2j} \frac{r_1}{\delta_1}) = c_2 \cdot I_0(\sqrt{2j} \frac{r_1}{\delta_2}) + c_3 \cdot K_0(\sqrt{2j} \frac{r_1}{\delta_2}) \quad (3.28)$$

and

$$\begin{aligned} \frac{\rho_1 \sqrt{2}}{\delta_1} \cdot c_1 \cdot I'_0(\sqrt{2j} \frac{r_1}{\delta_1}) &= \frac{\rho_2 \sqrt{2}}{\delta_2} [c_2 \cdot I'_0(\sqrt{2j} \frac{r_1}{\delta_2}) \\ &+ c_3 \cdot K'_0(\sqrt{2j} \frac{r_1}{\delta_2})] \end{aligned} \quad (3.29)$$

From equations (3.24), (3.28) and (3.29) it is possible to solve for (C_1/H_0) , (C_2/H_0) and (C_3/H_0) .

The results are quoted below.

Defining,

$$x_{i,j} = \sqrt{2j} \frac{r_j}{\delta_i} \quad (3.30)$$

$$\text{and } \lambda = \sqrt{\frac{\rho_1}{\rho_2}} \quad (3.31)$$

then

$$\frac{C_1}{H_0} = \frac{K'_0(x_{2,1}) \cdot I_0(x_{1,1}) - \lambda \cdot K_0(x_{2,1}) \cdot I'_0(x_{1,1})}{\Delta} \quad (3.32)$$

$$\frac{C_2}{H_0} = \frac{\lambda \cdot I_0(x_{2,1}) \cdot I'_0(x_{1,1}) - I'_0(x_{2,1}) \cdot I_0(x_{1,1})}{\Delta} \quad (3.33)$$

$$\frac{C_3}{H_0} = \frac{K'_0(x_{2,1}) \cdot I_0(x_{2,1}) - K_0(x_{2,1}) \cdot I'_0(x_{2,1})}{\Delta} \quad (3.34)$$

where

$$\begin{aligned} \Delta = & I_0(x_{2,2}) \cdot [K'_0(x_{2,1}) \cdot I_0(x_{1,1}) - \lambda \cdot K_0(x_{2,1}) \cdot I'_0(x_{1,1})] \\ & + K_0(x_{2,2}) \cdot [\lambda \cdot I_0(x_{2,1}) \cdot I'_0(x_{1,1}) - I_0(x_{1,1}) \cdot I'_0(x_{2,1})] \end{aligned} \quad (3.35)$$

At this point all the electric circuit components were defined. The specifications of the coil can be found by solving the circuit following the method outlined previously in Section 2.7.

The computer program "EQUIV" was improved to cater for this derivation. The program's objective is to design and predict the performance of a work coil suitable to heat a multilayer cylindrical workpiece and each cylinder's material is not necessarily identical. The equivalent circuit was that shown in Fig. 3.3 and the calculation was as stated previously. If concentric cylinders of similar physical properties were to be considered, the equivalent circuit would resemble Fig. 2.3;- that being a uniform load.

In the program, the workpiece was divided into n concentric cylinders, where $n \leq 100$. The resistivity of the inner (1) and outer (n) cylinders, correspond to minimum and maximum temperatures to be supplied by the user. The resistivity of any other

cylinder, say x , is then calculated from the linear interpolation of these resistivities.

$$\rho_x = \rho_{x-1} + \frac{\rho_n - \rho_1}{n} \quad (3.36)$$

Frohlich formula, equation (3.37), was employed to calculate the permeabilities of a ferromagnetic workpiece, as a function of the magnetic field strength in each cylinder.

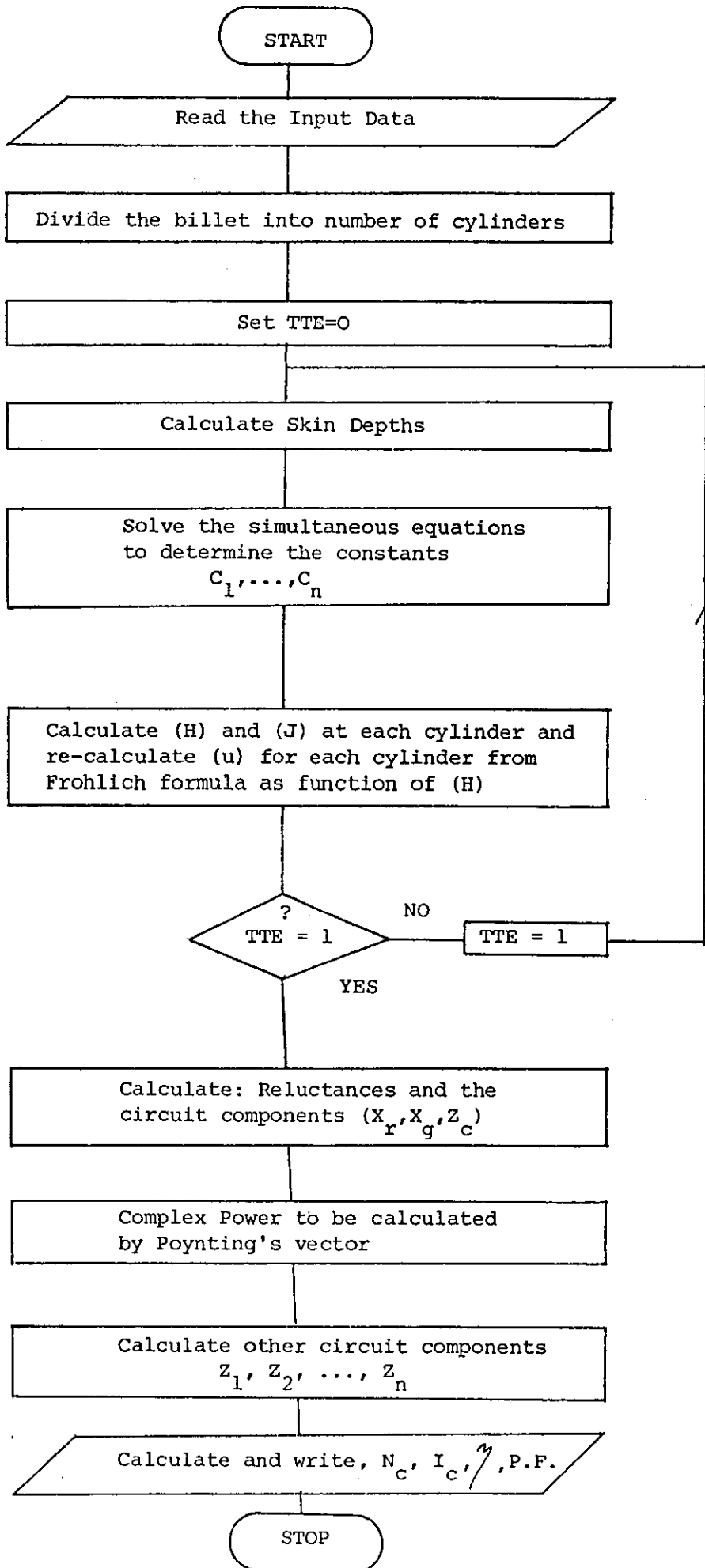
$$u_x = \frac{AF}{BF + H_x} + CF \quad (3.37)$$

The appropriate workpiece material constants AF , BF and CF should be provided.

The Gaussian elimination method was adopted for the solution of the constants $C_1, C_2, \dots, C_n$.

The flow chart of this program is shown below.

The parameters employed in the calculations were identical to those in Section 3.1.1. The resistivities of the aluminium load were taken to be: $10 \times 10^{-8} \Omega m$ and $2 \times 10^{-8} \Omega m$ on the surface and the centre respectively. The resistivities of the mild steel were $75 \times 10^{-8} \Omega m$ and $15 \times 10^{-8} \Omega m$ on the surface and the centre respectively while the permeability was represented by Frohlich formula with the constants which were supplied in Section 3.1.1. The operating frequency was assumed to be 50 Hz in every case. The results are shown in Tables 3.8 and 3.9 below.



The flow chart of "EQUIV"

number of output parameter cylinders	1	2	5	8	% Diff between 1 and 8
N_c	151	167	143	123	23
I_c	1709	1390	1865	2360	28
γ	40	51.7	36.5	24.0	67
P.F.	0.266	0.327	0.244	0.193	38

Table 3.8: The effect of dividing the nonmagnetic load into a number of concentric cylinders

number of output parameter cylinders	1	2	5	8	% Diff between 1 and 8
N_c	93	84	84	67	39
I_c	674	663	663	833	19
γ	96.5	97.2	97.2	97.2	0.7
P.F.	0.675	0.686	0.686	0.546	24

Table 3.9: The effect of dividing the ferromagnetic load into a number of concentric cylinders

The above tables showed the differences between a uniform load and, when dividing it into 2, 5 and 8 layers. If by increasing the number of concentric cylinders it was assumed that the similarities between the equivalent circuit and the actual situation increased, then if one were to ignore the nonuniformity of the load the results might be greatly distorted. This effect can be as much as 23% of the number of turns required to heat a nonmagnetic load and up to 39% when the load is a ferromagnetic material. The estimating performance of the coil also varies: the difference in the efficiency is up to 67% and the power factor difference is up to 38%. Hence the correct choice of the physical properties values is very important.

3.2 Investigations into Numerical Techniques

The adoption of the numerical techniques by many research workers necessitates one gaining experience in their use in the induction heating domain. It is only then, that their limitations and the difficulties associated with them can be assessed. It was impossible to investigate all the three numerical methods reviewed in the previous chapter, as this would require much time and effort. The time and effort are better to be devoted towards developing a method to design a coil, which is capable of inducing a nonuniform power density along the load.

One numerical technique, the finite difference method, was implemented to determine the distributions of the magnetic field and the power in a uniform long rectangular workpiece, subjected to a uniform magnetic

field. The finite difference method was chosen because it is well documented when compared with the more recent finite element and mutually coupled circuit methods. The finite difference method is sufficient when a solution is required to this particular problem, as both the uniformity of the material and the simple geometry of the region do not require the more complicated finite element or mutually coupled circuit methods.

A computer program was developed to determine the distributions of the magnetic field and the power in a rectangular billet of nonmagnetic material. It is a simple task to amend the program, if one wishes to consider magnetic material with constant permeability. The program was written for any user, as one simply needs to run the program and answer the required questions.

The program was based, mainly, on the computer program "BIEEDY" [3.4]. This piece of research has been selected because it is well documented and explained.

The derivation of the necessary magnetic field equation and the equivalent finite difference equation, together with a description of the computer program is supplied below:

3.2.1 The Finite Difference Technique

The eddy current in a conducting medium is governed by Maxwell's equations:

$$\oint E \cdot dl = \int_s \left(- \frac{\partial B}{\partial t} \right) ds \quad (3.38)$$

$$\nabla \times H = J_c + J_d \quad (3.39)$$

where J_c and J_d are the conduction and displacement current densities respectively and B is the magnetic flux density. In induction heating J_d can be ignored, as it is far less than J_c , since

$$\frac{J_c}{J_d} = \frac{1}{\rho \epsilon \omega} \quad (3.40)$$

where ϵ is the permittivity of the material. Equation (3.39) becomes:

$$\nabla \times H = J_c \quad (3.41)$$

$$\text{As: } E = \rho J \quad (3.42)$$

$$\text{and } B = \mu H \quad (3.43)$$

Substitute equation (3.41) into equation (3.42) then substitute the resultant equation together with equation (3.43) into equation (3.38) so as to yield:

$$\int \rho (\nabla \times H) \cdot dl = \int_s - \frac{\partial}{\partial t} (\mu H) ds \quad (3.44)$$

If the direction of the applied magnetic field is along the length of the workpiece H_z , as it is in billet heating, then

$$H_y = H_x = 0 \quad (3.45)$$

$$\text{and } H = H_z \quad (3.46)$$

The curl of H is:

$$\nabla \times H = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) i + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) j + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) k \quad (3.47)$$

As the change of the magnetic field occurs only within the depth of the workpiece, then

$$\frac{\partial H}{\partial z} = \frac{\partial H}{\partial y} = 0 \quad (3.48)$$

$$\therefore \nabla \times H = - \frac{\partial H}{\partial x} j \quad (3.49)$$

Substituting equation (3.49) into equation (3.44), yields

$$\rho_p \frac{\partial H}{\partial x} dl = \int_s \frac{\partial}{\partial t} (\mu H) ds \quad (3.50)$$

As the magnetic field is sinusoidal, $\frac{\partial}{\partial t}$ can be replaced by $j\omega$:

$$\rho_p \frac{\partial H}{\partial x} dl = \int_s j\omega \mu H ds \quad (3.51)$$

The magnitude of the current density J , according to equations (3.41) and (3.49) is:

$$J = \frac{\partial H}{\partial x} \quad (3.52)$$

The average volume power density (W/m^3) at any point [3.3] when sinusoidal magnetic field is in operation is:

$$P_v = \frac{1}{2} \rho J J^* \quad (3.53)$$

A program "slab" has been written to predict the magnetic field and power density distributions in a homogeneous non-magnetic rectangular workpiece subjected to a uniform surface magnetic field strength. The program was written in Fortran 77 and to reduce the computing time only one quarter of the cross-section was considered. The region is divided into MX by MY meshes. The quarter section will then have $(MX + 1)$ by $(MY + 1)$ nodes, Fig. 3.4. In finite difference method it is essential that a good mesh is selected as this can affect the results obtained to considerable degree. Where the magnetic field changes rapidly the mesh nodes should be close together, i.e. near the surface. On the other hand, where the magnetic field is changing more gradually the nodes need to be spaced further apart, i.e. towards the centre of the load.

The surface and line integrals of equation (3.51) are taken over the region about the i, j node extending halfway to the neighbouring nodes of the finite difference mesh in Fig. 3.5.

If the load is of non-uniform physical properties then the particular resistivity and permeability of each mesh should be included in the finite difference equation. In the case under consideration the load is of uniform material. Following Gibson [3.4] who considered the more general case of non-uniform load, the finite difference equation for a uniform load will be

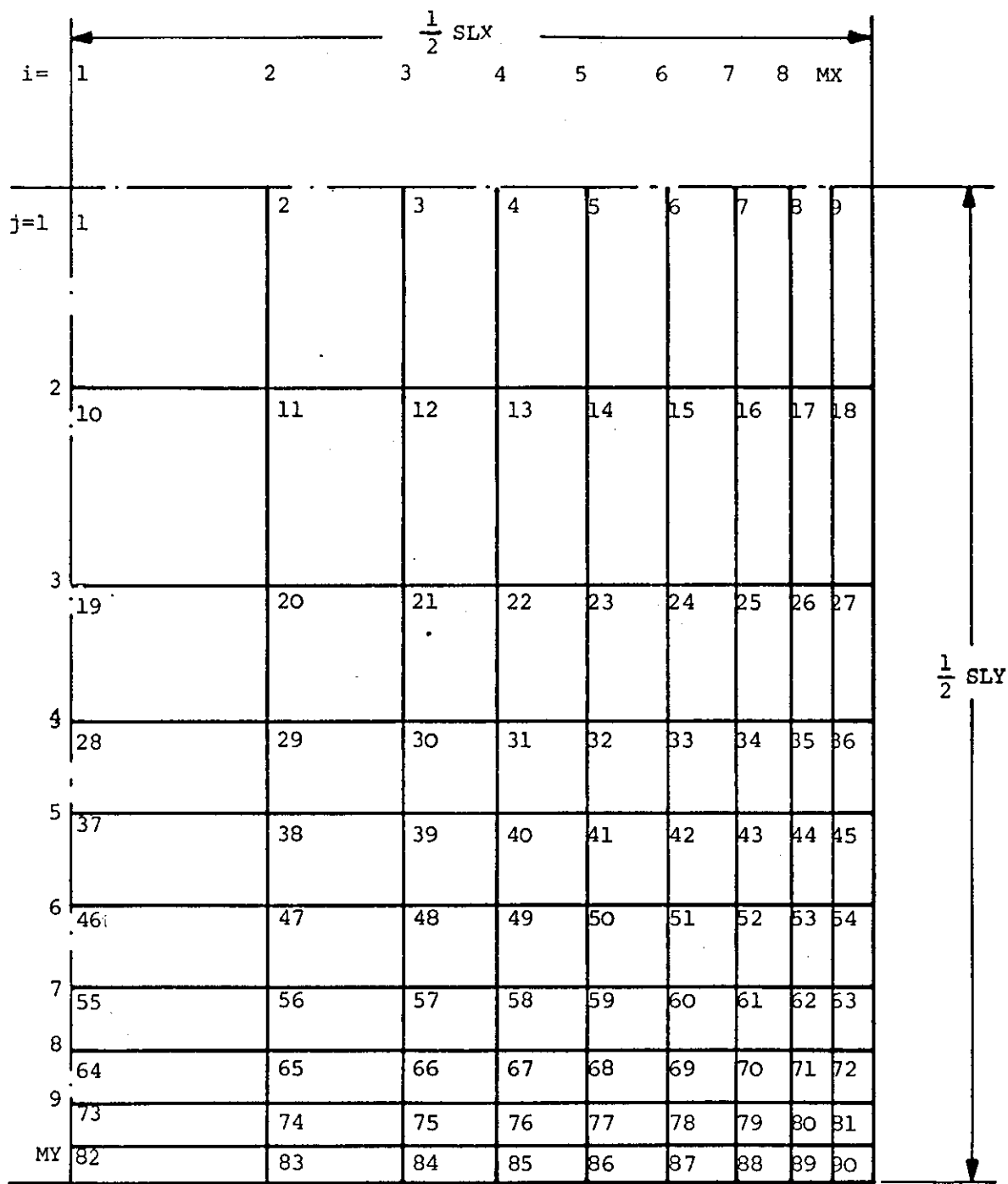


Fig. 3.4 Finite difference mesh for rectangular billet

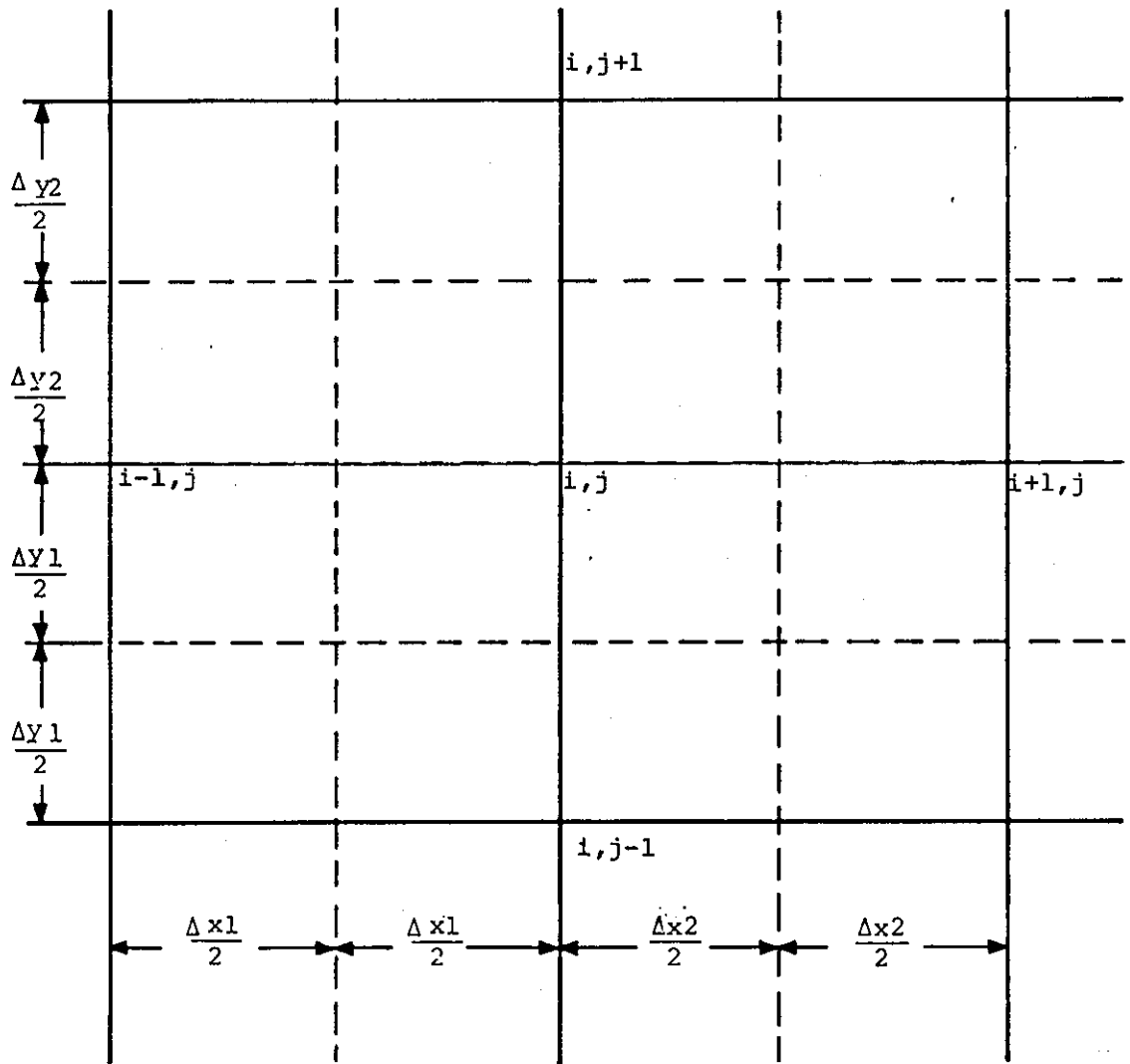


Fig. 3.5 Finite difference mesh for magnetic field equation

$$\begin{aligned}
& \rho \left\{ \left[\left(\frac{H_{i,j+1} - H_{i,j}}{\Delta y_2} \right) \left(\frac{\Delta x_1 + \Delta x_2}{2} \right) \right] + \left[\left(\frac{H_{i,j-1} - H_{i,j}}{\Delta y_1} \right) \left(\frac{\Delta x_1 + \Delta x_2}{2} \right) \right] \right. \\
& + \left. \left[\left(\frac{H_{i-1,j} - H_{i,j}}{\Delta x_1} \right) \left(\frac{\Delta y_1 + \Delta y_2}{2} \right) \right] + \left[\left(\frac{H_{i+1,j} - H_{i,j}}{\Delta x_2} \right) \left(\frac{\Delta y_1 + \Delta y_2}{2} \right) \right] \right\} \\
& = \frac{1}{4} j \omega \mu_o \mu_r H_k \left[(\Delta y_2 \cdot \Delta x_1) + (\Delta y_2 \cdot \Delta x_2) + (\Delta y_1 \cdot \Delta x_2) + (\Delta y_1 \cdot \Delta x_1) \right] \quad (3.54)
\end{aligned}$$

The boundary conditions are

$$\text{as } x = 0 \quad \frac{\partial H}{\partial x} = 0$$

$$\text{as } y = 0 \quad \frac{\partial H}{\partial y} = 0$$

$$\text{as } x = \frac{1}{2} SLX \quad H = H_o$$

$$\text{as } y = \frac{1}{2} SLY \quad H = H_o$$

Equation (3.54) was applied to each node of the grid and the set of the algebraic equations are written in matrix form to be solved by Gaussian elimination technique (Subroutine GAUSS). As a result of this the magnetic field strength at each node of the quarter section load is obtained. These will be used to calculate the power density within each mesh, the power generated at each node and the total power induced in the load which is four times the total power in the quarter of the load.

To illustrate how the power density over a mesh element is calculated in subroutine (SETUP) consider the mesh shown in Fig. 3.6.

$$J_x = \frac{H_2 - H_1}{dx} \quad (3.55)$$

$$J_x + \frac{\partial J_x}{\partial y} dy = \frac{H_{11} - H_{10}}{dx} \quad (3.56)$$

$$\text{similarly, } J_y = \frac{H_{10} - H_1}{dy} \quad (3.57)$$

$$\text{and } J_y + \frac{\partial J_y}{\partial x} dx = \frac{H_{11} - H_2}{dy} \quad (3.58)$$

where H is the magnetic field strength at each specific node and J is the current density. dx and dy are the horizontal and vertical mesh spacings for the mesh shown.

$$\text{Average } J_x = \text{Av}J_x = \frac{J_x + \left[J_x + \frac{\partial J_x}{\partial y} dy \right]}{2} \quad (3.59)$$

$$= \frac{H_2 - H_1 + H_{11} - H_{10}}{2dx} \quad (3.60)$$

$$\text{Average } J_y = \text{Av}J_y = \frac{J_y + \left[J_y + \frac{\partial J_y}{\partial x} dx \right]}{2} \quad (3.61)$$

$$= \frac{H_{10} - H_1 + H_{11} - H_2}{2dy} \quad (3.62)$$

The volume power density p (W/m^3) was calculated by the expression (3.63).

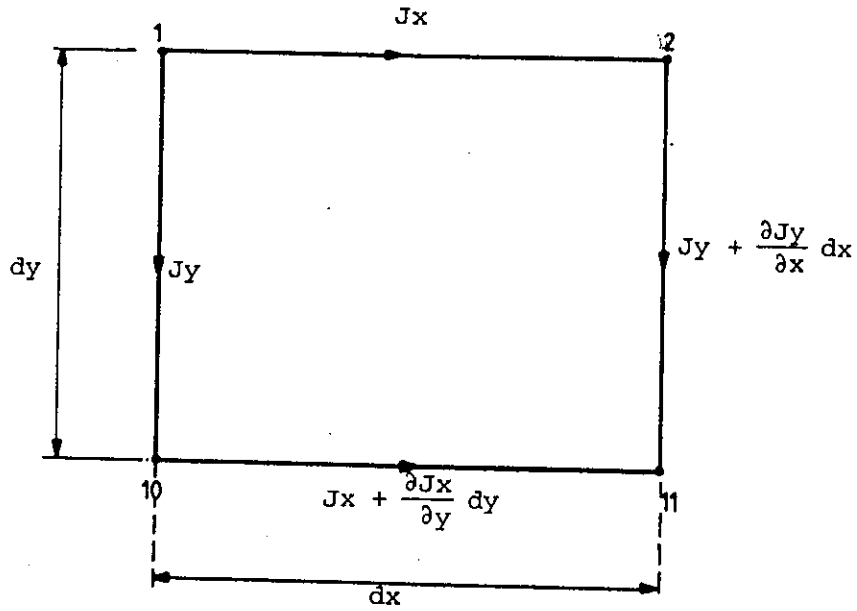


Fig. 3.6 Calculation of the Average Power Density Over a Mesh

$$P = \frac{1}{8dx \, dy} \rho \, dx \, dy \left[2 \, \text{Av} \, J_y \, 2 \, \text{Av} \, J_y^* + 2 \, \text{Av} \, J_x \, 2 \, \text{Av} \, J_x^* \right] \quad (3.63)$$

$$= \frac{1}{8} \, \rho \, \frac{4 \, dx \, dy}{dx \, dy} \left[\text{Av} \, J_y \, \text{Av} \, J_y^* + \text{Av} \, J_x \, \text{Av} \, J_x^* \right] \quad (3.64)$$

$$= \frac{1}{2} \, \rho \left[\text{Av} \, J_y \, \text{Av} \, J_y^* + \text{Av} \, J_x \, \text{Av} \, J_x^* \right] \quad (3.65)$$

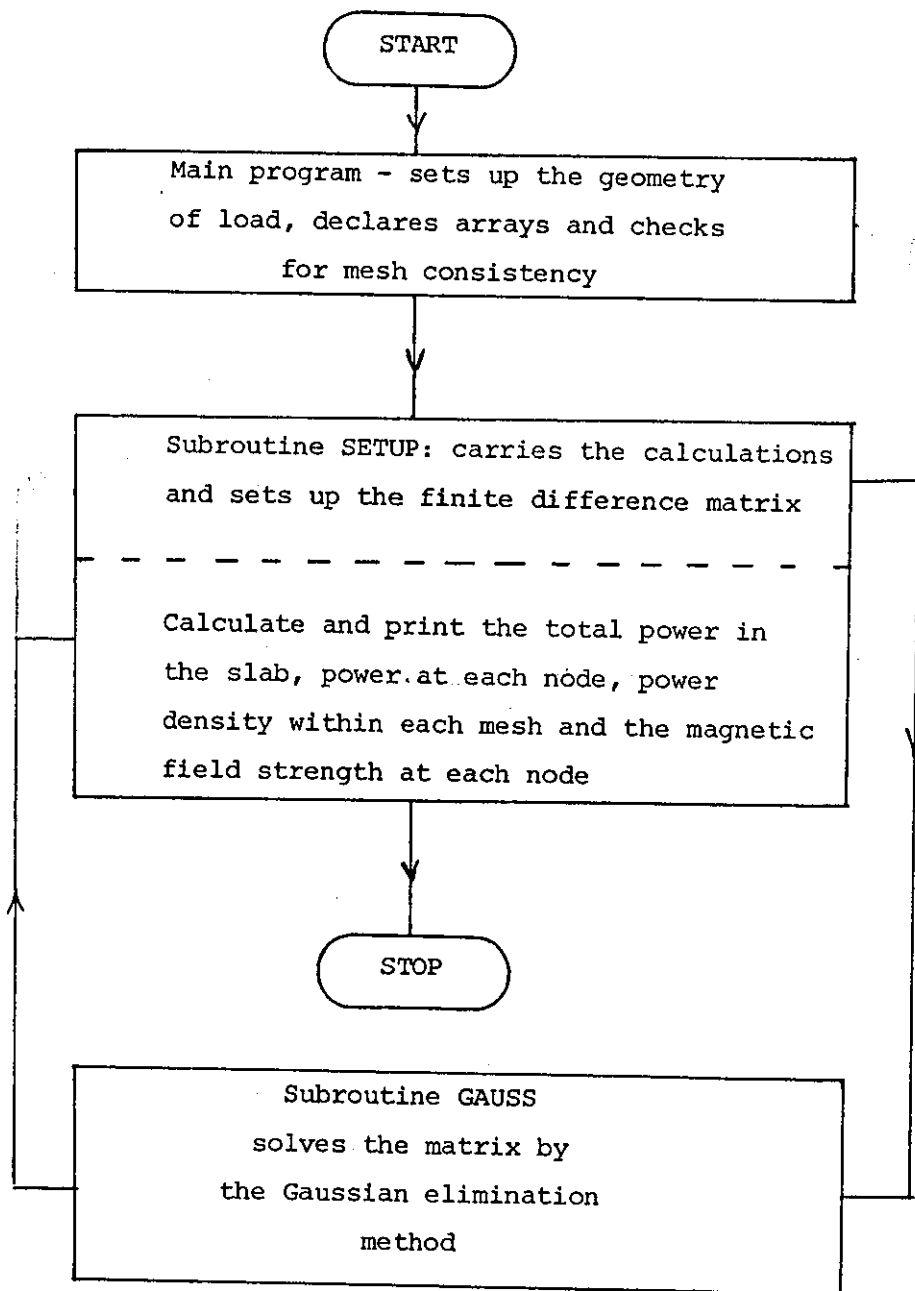
$$\therefore P = \frac{1}{2} \, \rho \, J \, J^* \quad \text{W/m}^3 \quad (3.66)$$

The program consists of four sections:

1. The main program:- sets up the geometry of the load, declares the arrays and checks for mesh consistency.
2. The subroutine (SETUP):- carries on the calculations and sets up the finite difference matrix.
3. The subroutine (GAUSS):- a call is made to this subroutine by (SETUP). The subroutine (GAUSS) solves the matrix by the Gaussian elimination method. The program then returns to the subroutine SETUP.
4. After the program has returned to SETUP, the subroutine calculates and prints the power density within each mesh, the power generated at each node and the total power induced in the load. The subroutine also goes on to print the magnetic field strength at each node.

The user has the choice of printing only the values of the outputs stated above or to print all the results of calculations.

The flow chart of the program is shown below.



The flow chart of "slab"

The magnetic field and volume power density distribution within an aluminium workpiece of cross-section 500 mm by 150 mm subjected to a surface magnetic field strength of 10^5 Am^{-1} at a frequency of 50 Hz are shown in Figs. 3.7 and 3.8.

The program was relatively simple to implement and gave satisfactory results when applied to a simple two dimensional problem.

3.3 Conclusions and Recommendations

The design and performance of a coil obtained by adopting the equivalent circuit method depend largely on the values of the empirical factors employed in the calculation; the wrong choice of these factors could affect the results quite considerably. The assumption of a uniform workpiece material can also lead to incorrect results which depend largely on the resistivity and permeability values.

An accurate knowledge of the power density and temperature distribution in an inductively heated load is of great importance. Analytical expressions are available for simple one dimensional geometries such as a cylinder or a slab [3.3] but for the solution of even the simplest two dimensional geometries numerical techniques are required.

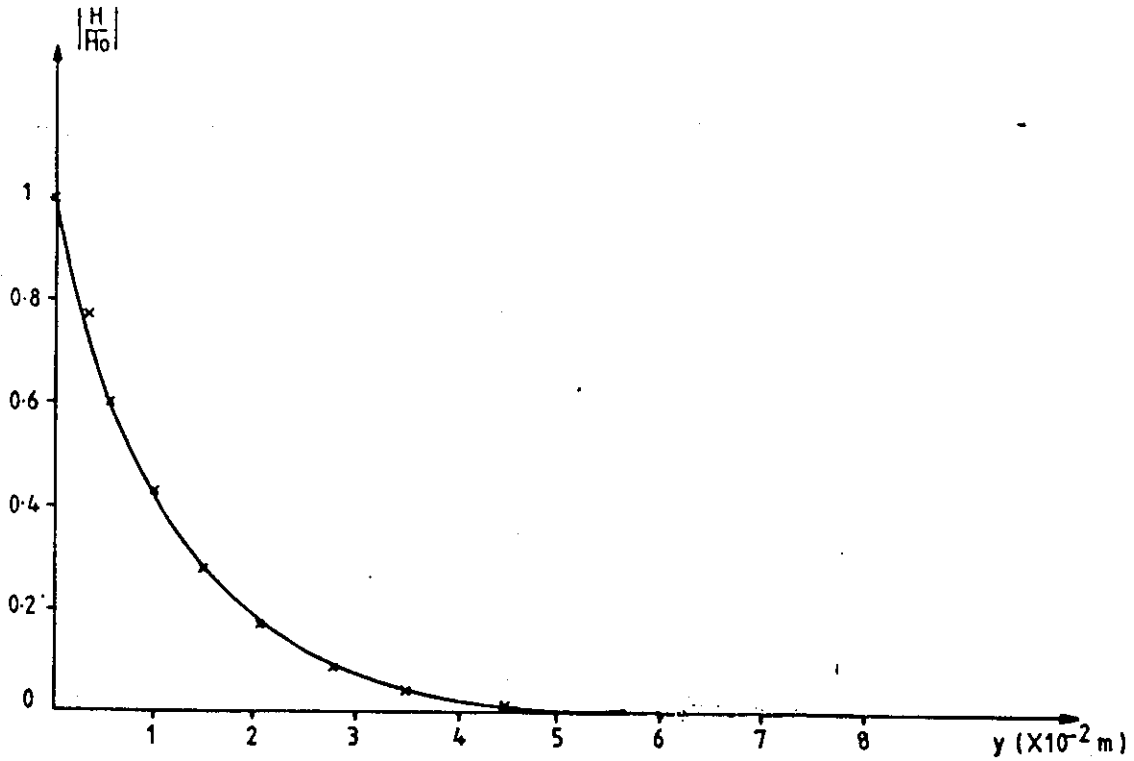


Fig. 3.7 Variation of magnetic field distribution (as a fraction of the surface field) with depth y along the centre line of the cross-sectional face.

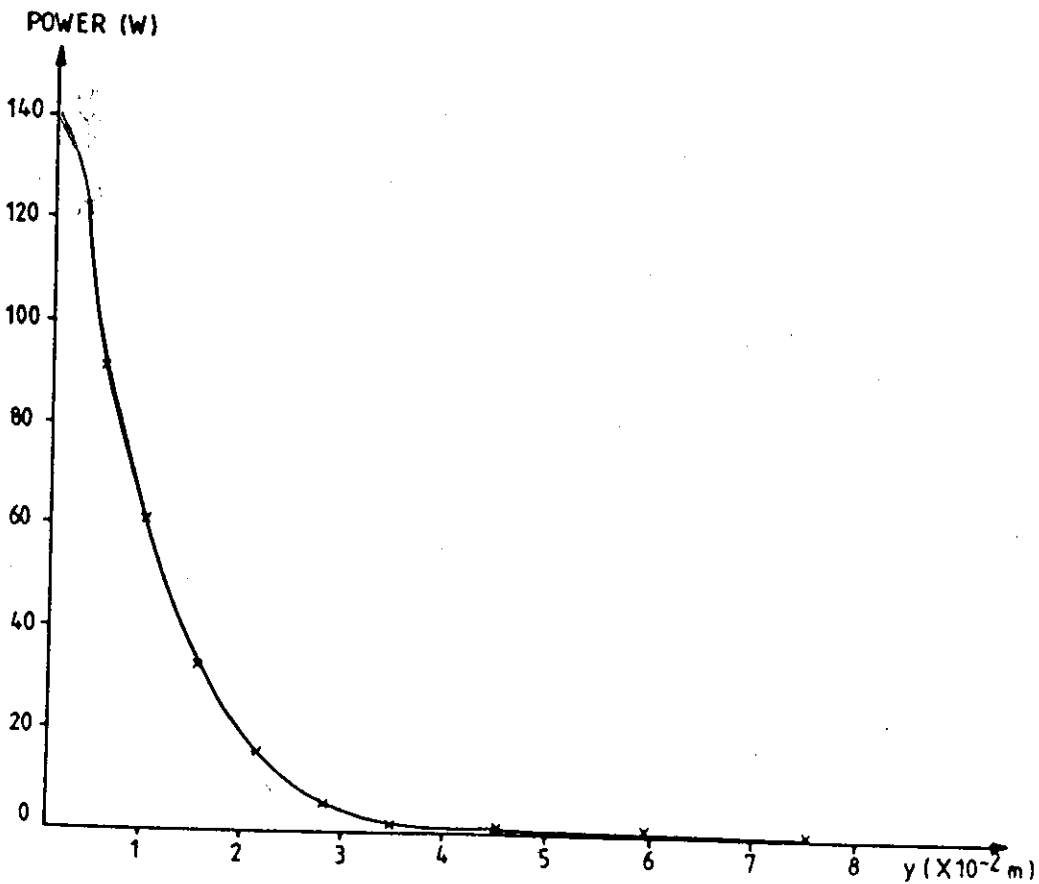


Fig. 3.8 Power distribution down the centre line of the cross-sectional face

The finite difference grid must be suited to the problem. The program should, therefore, be amended each time the dimensions of the load, its physical properties or the frequency change.

Induction billet heating is a long-standing process in the metal forming industry. There are areas, however, in which induction heating has not achieved the market share that perhaps one might expect e.g. off-the-bar forging or the extrusion of aluminium. In both applications the ability to create a non-uniform surface power density along the billet or bar is required. Existing methods of work coil design which are based on the assumption of an infinitely long solenoid and/or the use of a great amount of empirical data are not easily adapted to the design of non-uniform surface power densities. Further investigations to develop a new technique for work coil design are required which can be adapted to solve problems requiring a non-uniform surface power density. This technique should be based on the fact that the coil consists of a number of individual turns. The distribution of the power density induced on the load can be determined by applying the superposition principle.

CHAPTER 4

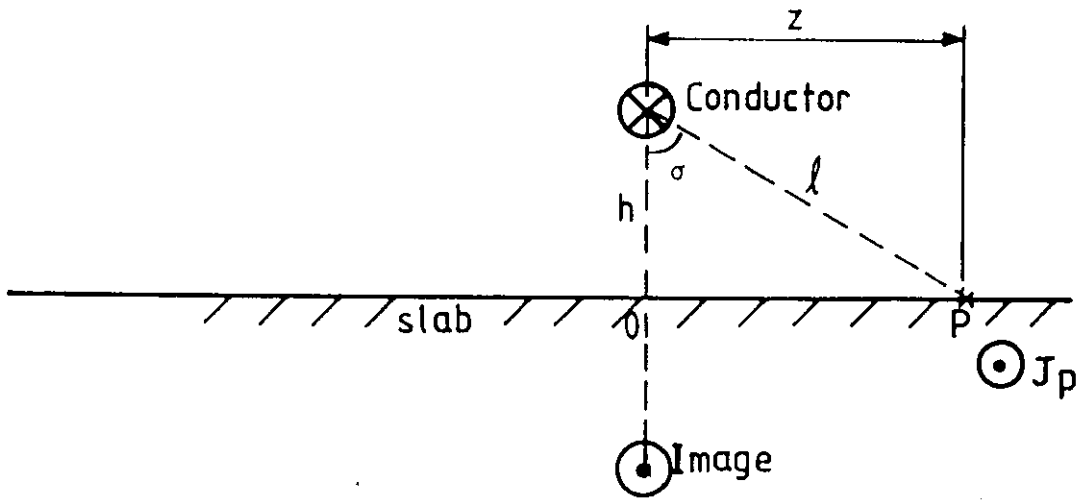
THE SUPERPOSITION THEORY

The objective is to develop a flexible work coil design method which will enable the design of a coil capable of producing a nonuniform surface power density along a workpiece. The achievement of this aim necessitated the abandonment of the assumption whereby a uniform magnetic field is produced by the coil and the adoption of a new approach, based on the superposition principles.

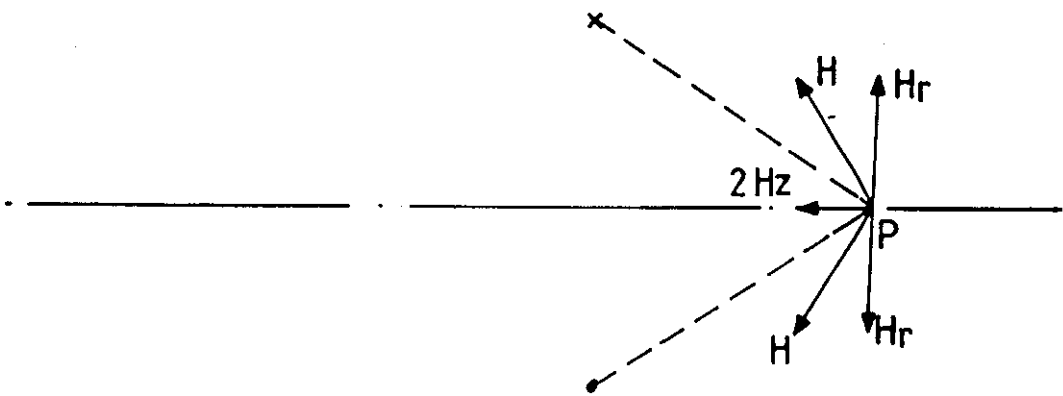
In order to deal realistically with applications requiring nonuniform surface power density to be induced along the length of the load, an equation has been derived for the surface magnetic field distribution produced by a single conductor. Then the superposition principle has been applied to determine the power density produced by a number of conductors. The theoretical derivation will be shown in the ensuing chapter.

4.1 The Derivation of the Superposition Theory

Fig. 4.1 shows a filament carrying current I at a distance h from the surface of a semi-infinite metal slab. Given that the slab is a good conductor, the magnetic field strength at any point P along the surface of the slab can be calculated by assuming an imaginary filament within the slab at a distance h from the surface and carrying current $-I$. Both the conductor and the image will produce a magnetic field H on the surface, according to Ampere's law, equal to



(a)



(b)

Fig.4.1 (a) Conductor near a semi - infinite slab

(b) The magnetic field intensity on point P .

$$H = \frac{I}{2\pi\sqrt{h^2 + z^2}} \quad (4.1)$$

where z is the distance of point P along the load measured from a point directly beneath the conductor, see Fig. 4.1(a).

The axial components of the conductor's magnetic field strength H_z and its image will aid each other, while the radial components H_r will cancel each other. The resultant field on point P will be:

$$H_P = 2H_z \quad (4.2)$$

$$\text{but } H_z = H \cos \sigma \quad (4.3)$$

$$\therefore H_P = 2 \frac{I}{2\pi\sqrt{h^2 + z^2}} \cdot \frac{h}{\sqrt{h^2 + z^2}} \quad (4.4)$$

$$\therefore H_P = \frac{I}{\pi} \frac{h}{h^2 + z^2} \quad (4.5)$$

It can be shown [4.1] that the magnetic field intensity at the surface of the slab is equal to the linear current density, A , induced per unit length of the slab

$$\text{i.e. } A_P = H_P \quad (4.6)$$

$$\therefore A_P = \frac{I}{\pi} \frac{h}{h^2 + z^2} \quad (4.7)$$

If the filament is replaced by a circular conductor of radius a and the current penetration depth into the slab and the conductor is small compared with the height h and the thickness of the slab, the case

becomes similar to the electrostatic case of charged cylinder near a metal plate [4.2]. The solution of this problem [4.3, 4.4] results on re-writing equation (4.7) in the following form:

$$A_P = \frac{I}{\pi} \cdot \frac{\sqrt{h^2 - a^2}}{h^2 - a^2 + z^2} \quad (4.8)$$

This equation can be written in general form as:

$$A_P = \frac{I}{\pi} \cdot \frac{h_e^2}{h_e^2 + z^2} \quad (4.9)$$

Where the effective height, h_e , for a circular conductor is:

$$h_e = \sqrt{h^2 - a^2} \quad (4.10)$$

According to Davies and Simpson [4.1] it is valid to treat a circular conductor around a cylinder as a line current at a distance from a flat plane and the above derivation may be applied to a single turn coil surrounding a cylinder, provided that the distance between the conductor and the cylinder, h , is:

1. Small in comparison with the cylinder radius and,
2. Large in comparison with the current penetration depth in the cylinder.

Consider next, the case illustrated in Fig. 4.2 which comprises of N identical conductors parallel to each other and parallel to a metal slab at a distance h from its surface. Each conductor carries a current I amperes.

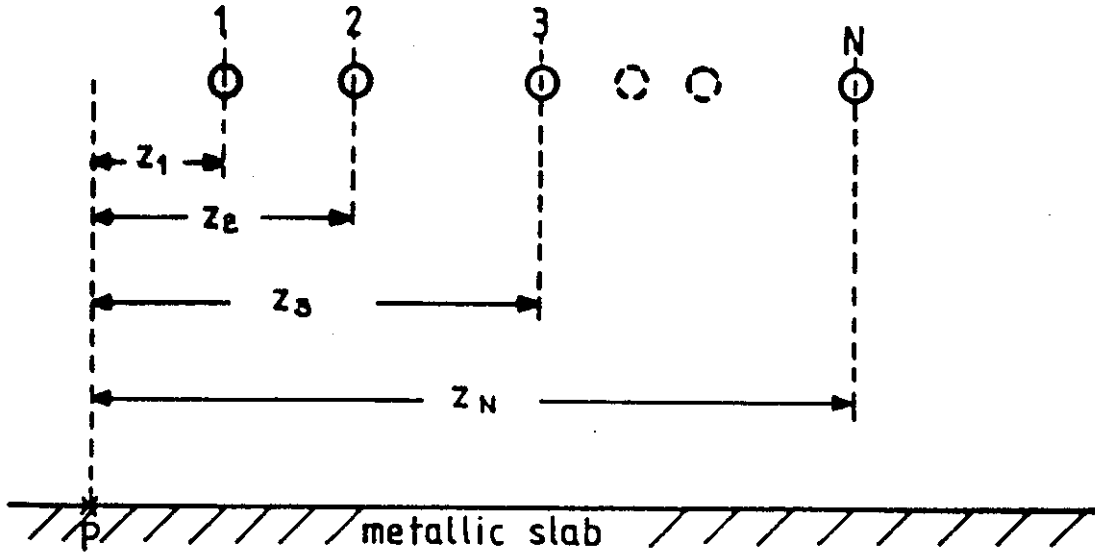


Fig4.2 Number of identical conductors at a specific air gap from a metallic load .

The current per unit length A induced by these conductors at a point P on the slab, according to superposition principle, is the sum of the current per unit length induced by each individual conductor.

This can be written in mathematical form as:

$$A_P = A_{P1} + A_{P2} + A_{P3} + \dots + A_{PN} \quad (4.11)$$

Where A_{P1} , A_{P2} , A_{P3} , ..., A_{PN} are the current per unit length induced at point P from conductors 1, 2, 3, ..., N respectively.

If the performance of each conductor complies with equation (4.9)

then

$$A_P = \frac{I}{\pi} \left[\frac{h_e}{h_e^2 + z_1^2} + \frac{h_e}{h_e^2 + z_2^2} + \dots + \frac{h_e}{h_e^2 + z_N^2} \right] \quad (4.12)$$

If the distances between the conductors are constant and equal to d , i.e. a uniform coil, then

$$\left. \begin{aligned} z_1 &= |z_1| \\ z_2 &= |z_1 + d| \\ z_3 &= |z_1 + 2d| \\ z_N &= |z_1 + (N-1)d| \end{aligned} \right\} \quad (4.13)$$

$$\text{and } A_P = \frac{I}{\pi} h_e \left[\frac{1}{h_e^2 + z_1^2} + \frac{1}{h_e^2 + (z_1 + d)^2} + \dots + \frac{1}{h_e^2 + [z_1 + (N-1)d]^2} \right] \quad (4.14)$$

$$\therefore A_P = \frac{I h_e}{\pi} \sum_{X=0}^{X=N-1} \frac{1}{h_e^2 + (z_1 + Xd)^2} \quad (4.15)$$

In induction heating, it is more convenient to deal with surface power density PD which is a function of the surface current density J . The latter is given for a semi-infinite slab at point P by [4.1] as:

$$J_P = \frac{\sqrt{2}}{\delta} H_P \exp j \left(\omega t + \frac{\pi}{4} \right) \quad (4.16)$$

$$\text{i.e. } J_P = \frac{\sqrt{2}}{\delta} H_P \left\{ \cos \left(\omega t + \frac{\pi}{4} \right) + j \sin \left(\omega t + \frac{\pi}{4} \right) \right\} \quad (4.17)$$

$$\therefore |J_P| = \frac{\sqrt{2}}{\delta} |H_P| \quad (4.18)$$

where δ is the penetration depth and is given by:

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_o \mu_r}} \quad (4.19)$$

Substitute equation (4.6) into equation (4.18) and rearrange:

$$A_P = \frac{\delta}{\sqrt{2}} J_P \quad (4.20)$$

The power density at point P is given by [4.1]:

$$(PD)_P = \frac{1}{2} \rho \delta J_P^2 \quad (4.21)$$

Equation (4.21) gives the surface power density at any particular point P on the load as a function of the surface current density at that point.

4.2 Verification of the Superposition Theory

Before applying the superposition theory to any particular problem it must be verified. The theory is to be assessed in two successive stages. The first stage is to investigate the accuracy of the theoretical derivation. If this is successful; the second stage is to examine this theory as a work coil design technique.

In order to verify the theoretical derivation, the results from the superposition theory were checked with the equations given by Callaghan and Maslen [4.5] for the case of large cylinder. The agreement was very good and the difference did not exceed 2%. Callaghan and Maslen have checked their equations thoroughly with practical measurements.

The next stage in testing the superposition theory, after having verified the theoretical derivation, is to check if it can be used as a work coil design technique. This is to be done by predicting the coil number of turns required for a given specification.

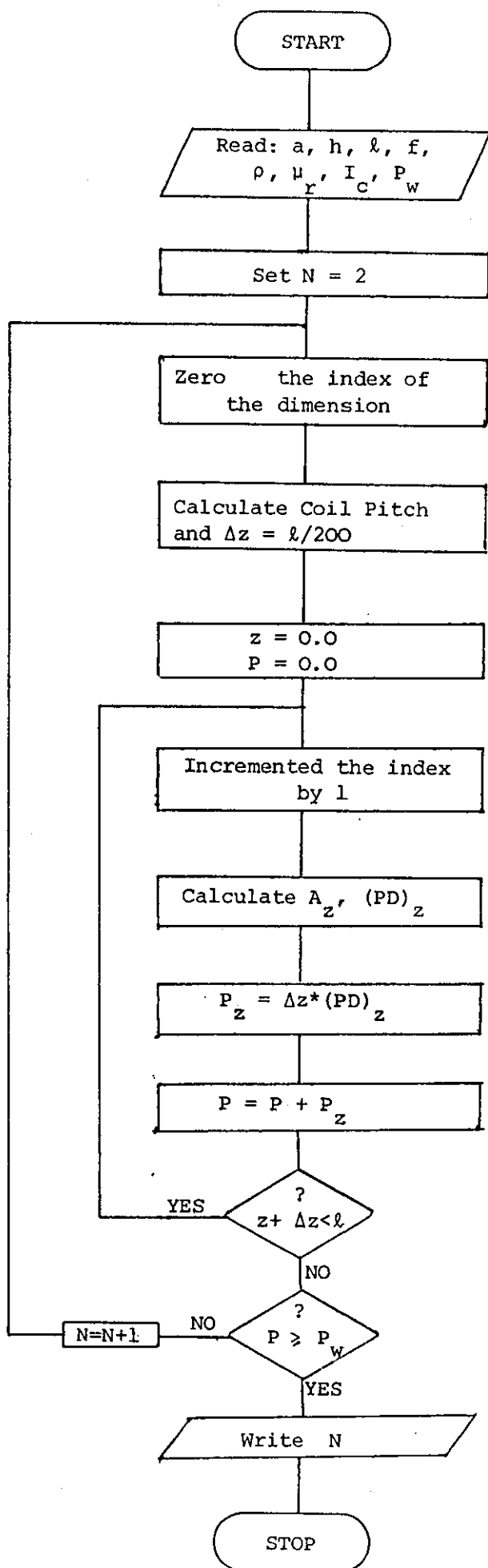
A computer program "PROXIMITY", based on the superposition principles, was developed to calculate the number of turns, in a uniform coil, required to produce a pre-determined power per unit width of the load.

As a result of adopting this program, the number of turns required for a large cylindrical load and high frequency nearly equals that obtained from the equivalent circuit method. The two results are within 10% difference from each other.

4.3 The Use of the Superposition Theory

Having verified the superposition theory, a computer program "W-P-POWER" was developed, based on equations (4.15), (4.20) and (4.21), to calculate and plot the current per unit length and the surface power density distributions induced on the load. The coil was assumed to be of uniform pitch and the load was assumed to be of nonuniform resistivity; so that the change in the physical properties can be taken into account.

The surface power density induced at any point P on a nonmagnetic or linear magnetic load was given by equation (4.21) which can be



The Flow Chart of the Program "PROXIMITY"

rewritten, by substituting equation (4.20) as:

$$(PD)_P = \frac{\rho}{\delta} A_P^2 = \sqrt{\frac{\rho \mu_o \mu_r \omega}{2}} A^2 \quad (4.22)$$

The power density induced on the load is a function of the resistivity which is not necessarily uniform along the load if the temperature is not uniform. To examine the applicability of the superposition theory to tackle the problem of the nonuniform load; two types of calculations were included in the program.

First, consider a scanning system of a billet moving inside an induction coil. When the billet is inside the coil, it's starting edge is at maximum temperature and the other end remains at minimum temperature. The temperatures of the rest of the load ranges between these two limits. A gradual change of the resistivity along the load was assumed by dividing the load into (X) equal divisions and taking the change in resistivity $\Delta\rho$ to be

$$\Delta\rho = \frac{\rho_f - \rho_i}{X - 1} \quad (4.23)$$

where ρ_i = the resistivity at minimum temperature

ρ_f = the resistivity at maximum temperature

and the resistivities of the individual segments will be:

The resistivity of first segment = $\rho_1 = \rho_i$

The resistivity of second segment = $\rho_2 = \rho_i + \Delta\rho$

The resistivity of nth segment = $\rho_n = \rho_i + (n-1)\Delta\rho \quad (4.24)$

The penetration depth of the nth segment is

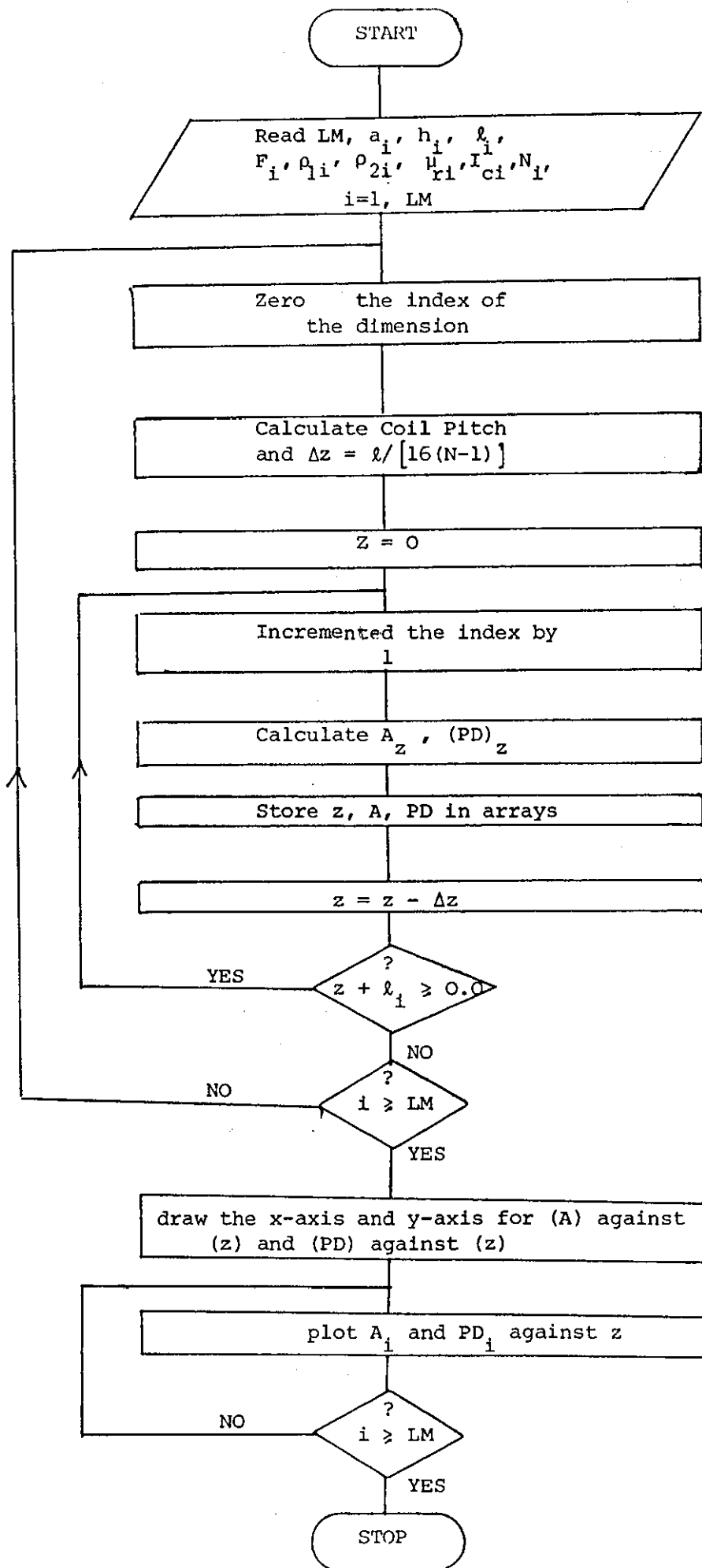
$$\delta_n = \sqrt{\frac{\rho_i + (n-1)\Delta\rho}{\pi f \mu_o \mu_r}} \quad (4.25)$$

Since the current per unit length induced at any point in the load can be calculated by the superposition theory, then the surface power density of any segment can be found from equation (4.26) below

$$(PD)_n = \frac{[\rho_i + (n-1)\Delta\rho]}{\delta_n} A_n^2 \quad (4.26)$$

Second, consider a billet heated by the "single-shot" technique. The billet, stationary inside the coil is subjected to a constant magnetic field. At first the temperature is at a minimum, as are the resistivity and the induced power density, the latter being a function of temperature, the resistivity and the power density will gradually increase until they reach a maximum. This case was included in the program and the result is to be shown in a number of curves representing different values of the resistivity.

The computer program "W-P-POWER" was used to study different cases and the results are shown in graphs which relate to theoretical situations only and therefore do not, as yet, represent a practical application. They are presented, simply to show the different parameter's effects. The flow chart of the program "W-P-POWER" is shown below.



The Flow Chart of the Program "W-P-POWER"

Figs. 4.3 and 4.4 show that increasing the number of turns will increase the uniformity, the magnitude of the current per unit length and the power density induced on the load. To be more precise, the increase in the number of turns from 15 to 30 turns, doubles the current per unit length induced in the load and increases the power density by a factor of 4. The reason for the increase in the power density by this factor being that it is a function of the current squared. This also applies to the coil of 45 turns, where the power density is increased by a factor of 9 when compared with the coil of 15 turns. The above result agrees with the fact that the current induced in the load is a function of the coil current multiplied by the number of turns. If the number of turns is increased, then the coil pitch decreases thus causing it to resemble a current sheet which will result in a very uniform power density induced in the load.

Figs. 4.5 and 4.6 were plotted, so as to show the effect of changing the air gap. Decreasing the air gap increases the current per unit length and the power density induced on the load. However, it decreases the uniformity as can be seen from the upper curves on these graphs.

Although the distributions of the current per unit length and the power density induced in the load due to single conductors are functions of the conductor diameter; this diameter, in the case of multiturn coil is not an important factor in the control of the amplitude of the current per unit length and the power density induced in the load; this can be read off Figs. 4.7 and 4.8. The curves represent 3 very

similar coils differing only in the size of the conductor. The air gap was taken to be constant in all three coils ($g = h - a = 30\text{mm}$).

Although the size of the conductor is not important when considering the induced power, the thinner the conductor the more the non-uniformity of the induced power, and the less the efficiency. This is a result of the increase in the losses.

The effect of the resistivity in the power density is shown in Figs. 4.10 and 4.11, while the current per unit length induced from the coil is constant: this is illustrated on Fig. 4.9. Fig. 4.10 shows the change of the local resistivity as against time, while Fig. 4.11 shows the axial change of resistivity. The power density induced in the load is a function of the square root of its resistivity; see Figs. 4.10 and 4.11.

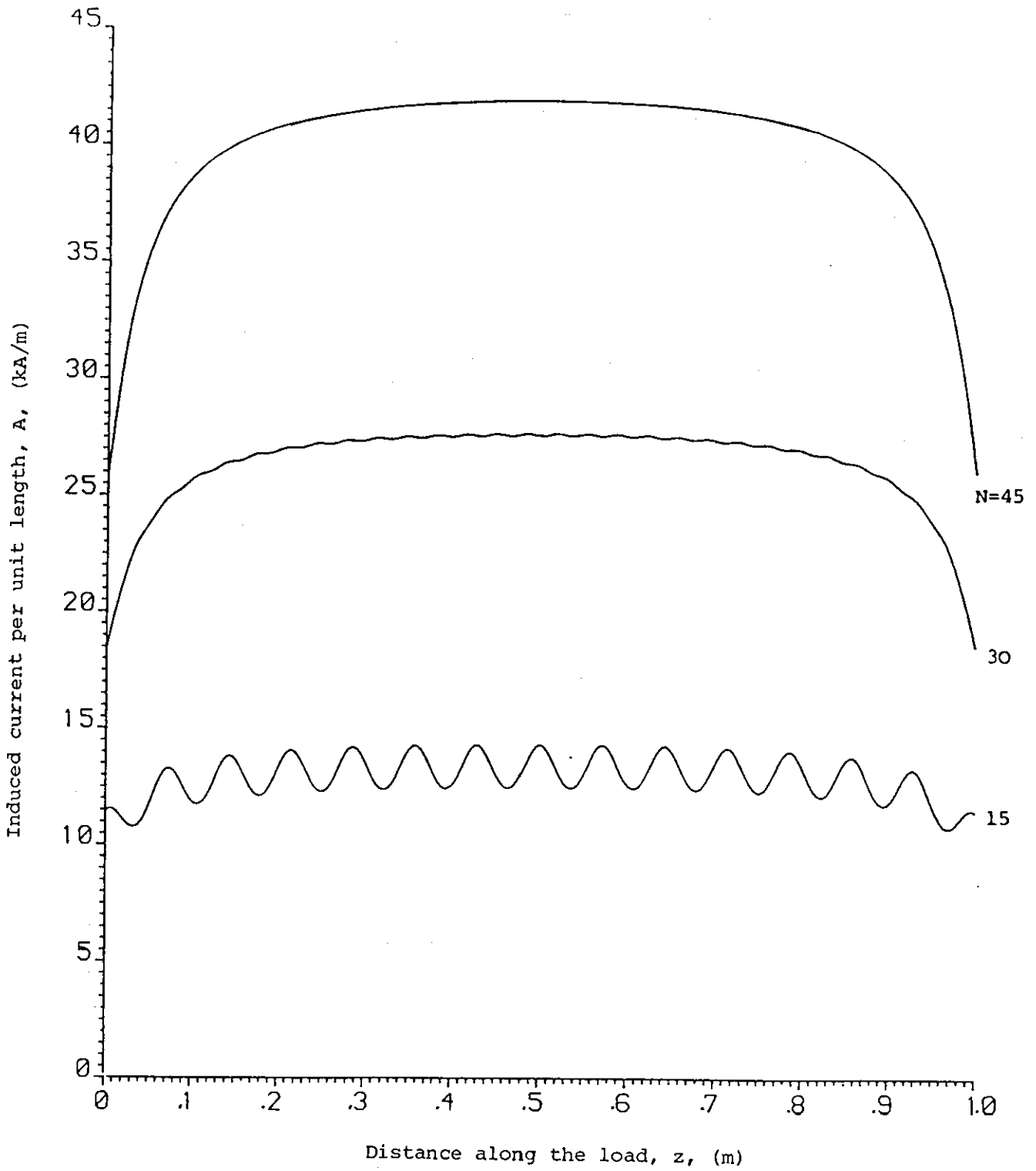


Fig. 4.3 Current per unit length along the load due to 1000A flowing through coils of different number of turns, N , at $h = 40\text{mm}$ and the conductor radius is $a = 10\text{mm}$.

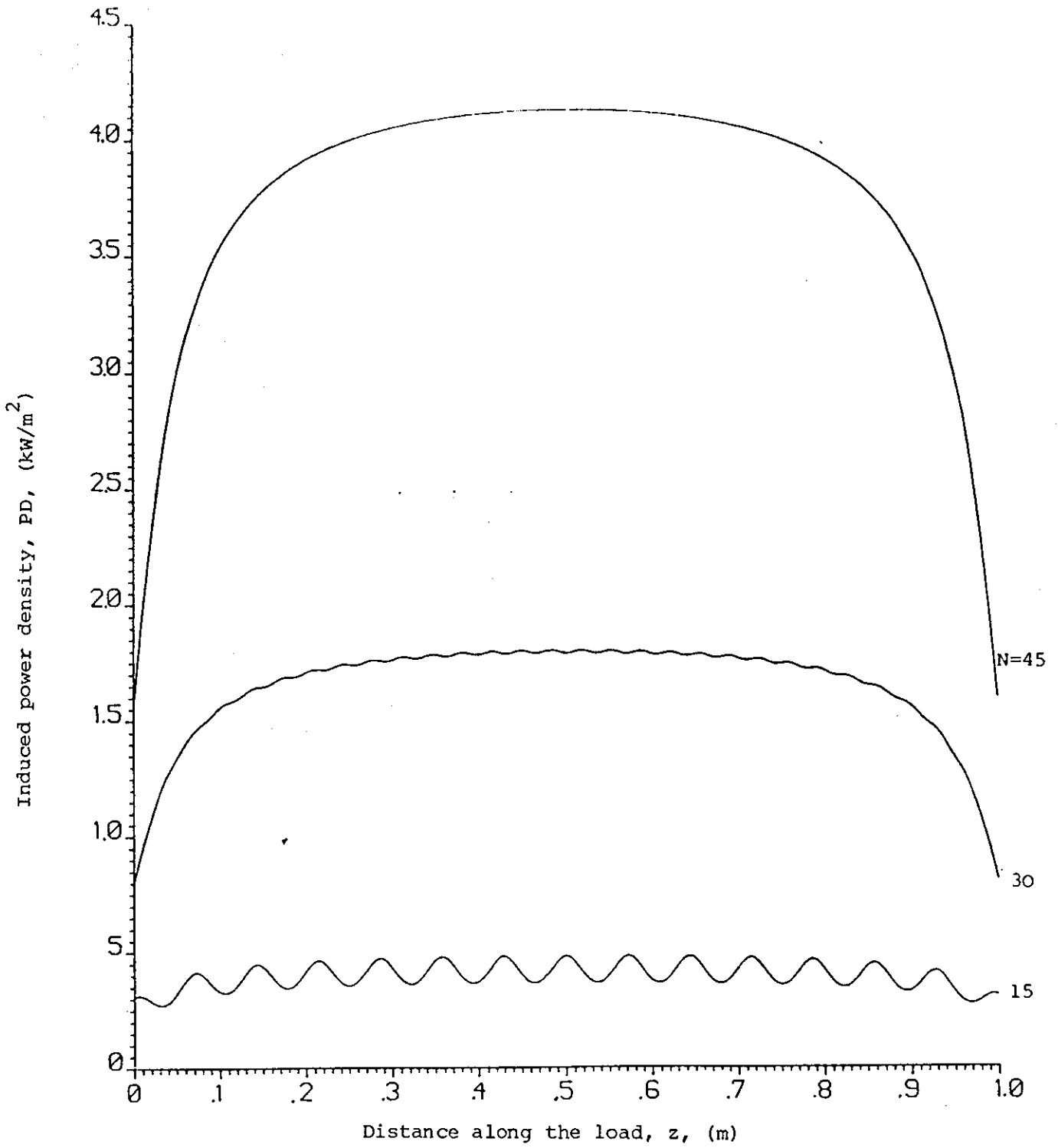


Fig. 4.4 Power density distribution along a nonmagnetic load of $2.8 \times 10^{-8} \Omega\text{m}$ resistivity due to the coils mentioned in Fig. 4.3, the frequency is 50 Hz.

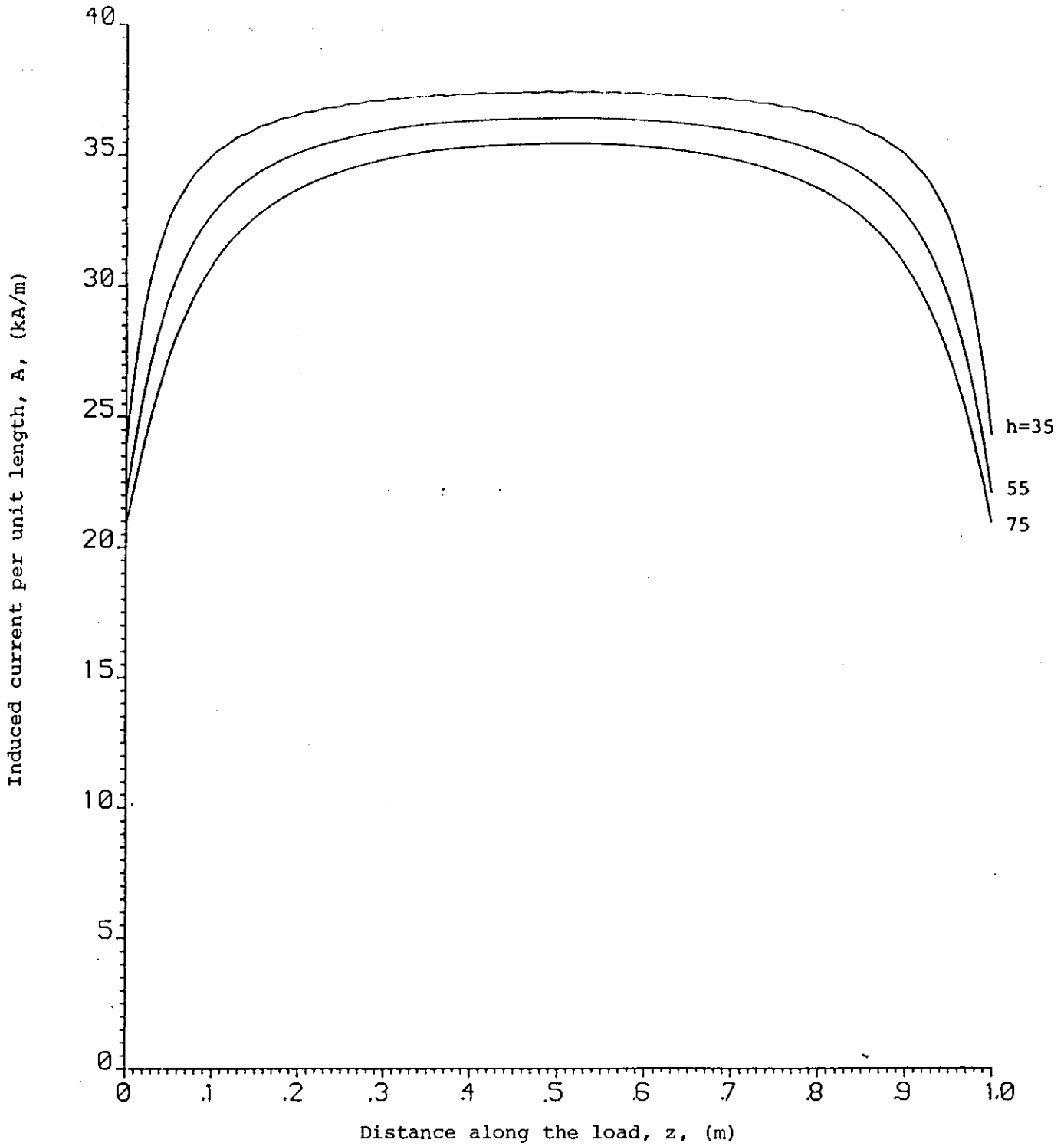


Fig. 4.5 Current per unit length along the load due to 1000A flowing through 40 turns coils at different heights, h , and the conductor radius is $a = 10\text{mm}$.

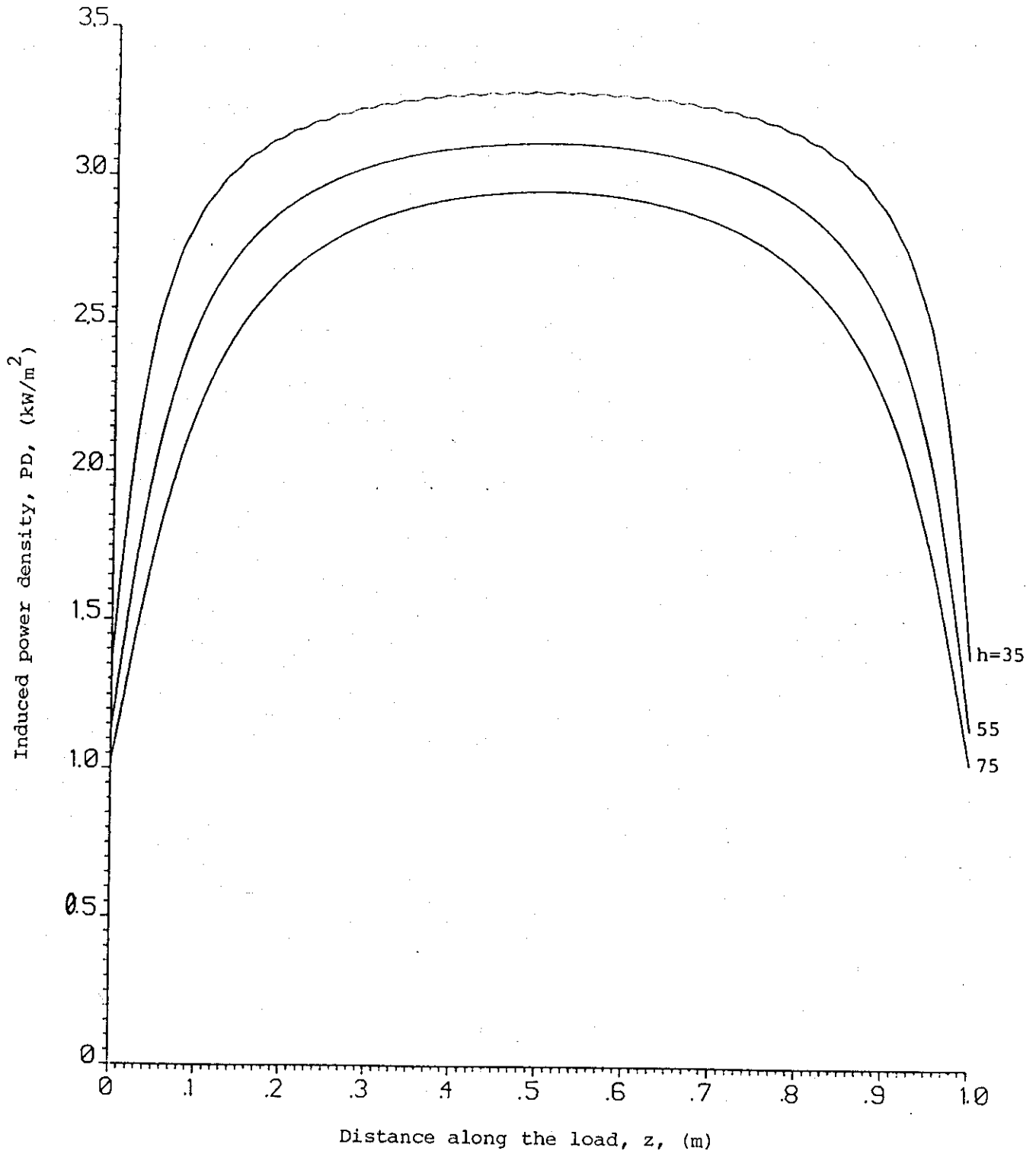


Fig. 4.6 Power density distribution along a nonmagnetic load of $2.8 \times 10^{-8} \Omega\text{m}$ resistivity due to the coils mentioned in Fig. 4.5, the frequency is 50 Hz.

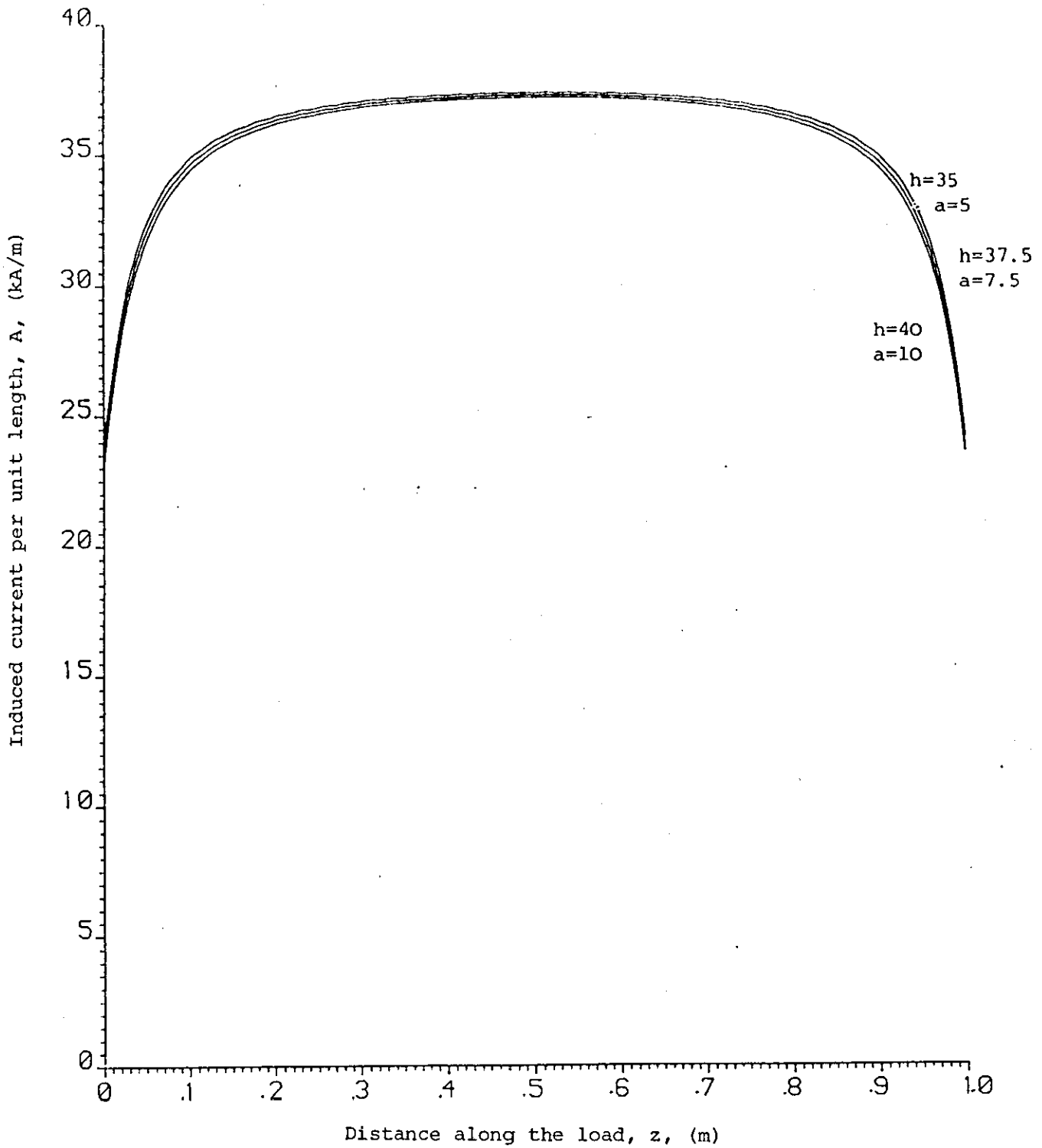


Fig. 4.7 Current per unit length along the load due to 1000A flowing through 40 turns coils of different conductor radii, a , at air gap of 30mm.

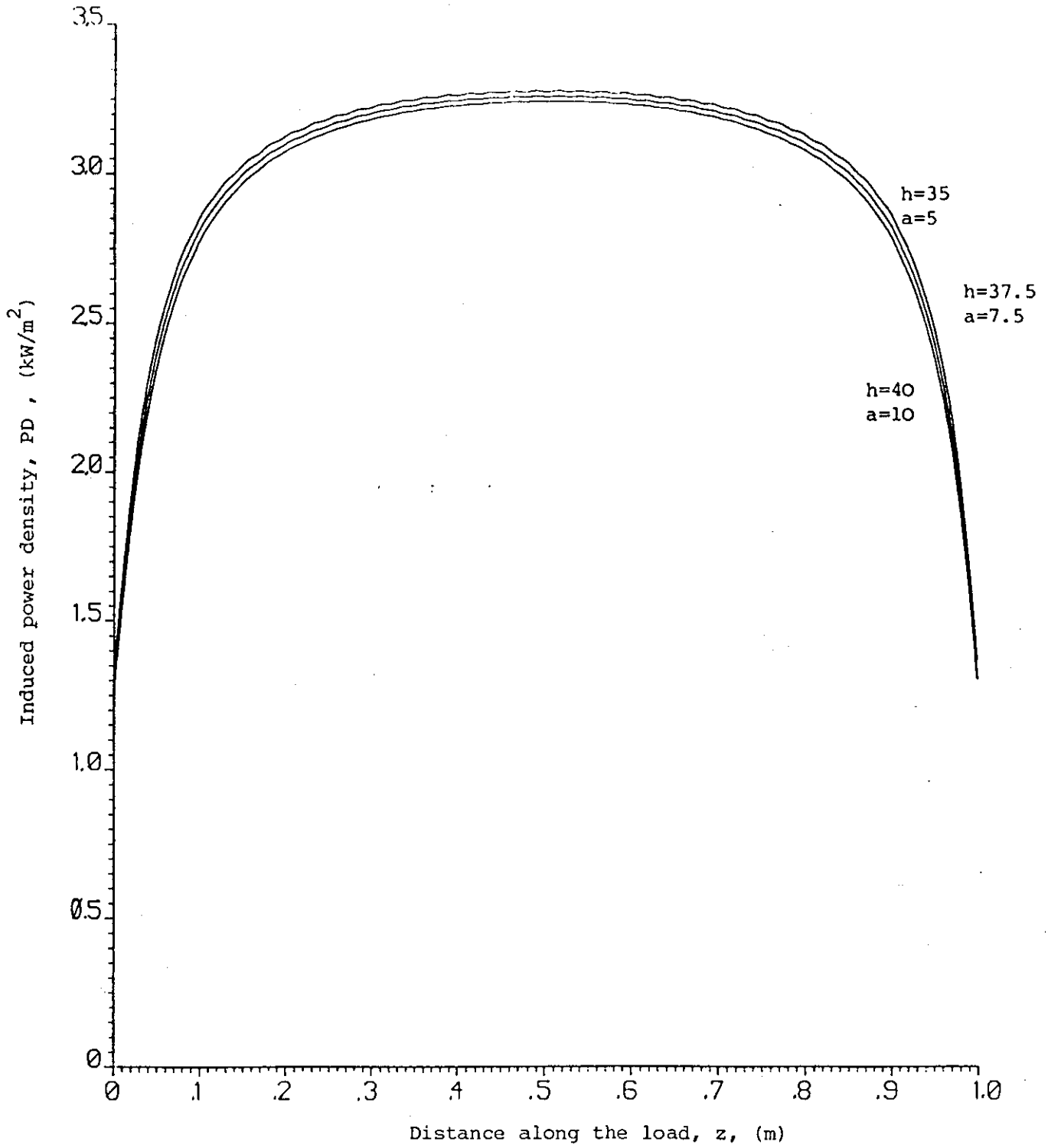


Fig. 4.8 Power density distribution along a nonmagnetic load of $2.8 \times 10^{-8} \Omega\text{m}$ resistivity due to the coils mentioned in Fig. 4.7, the frequency is 50 Hz.

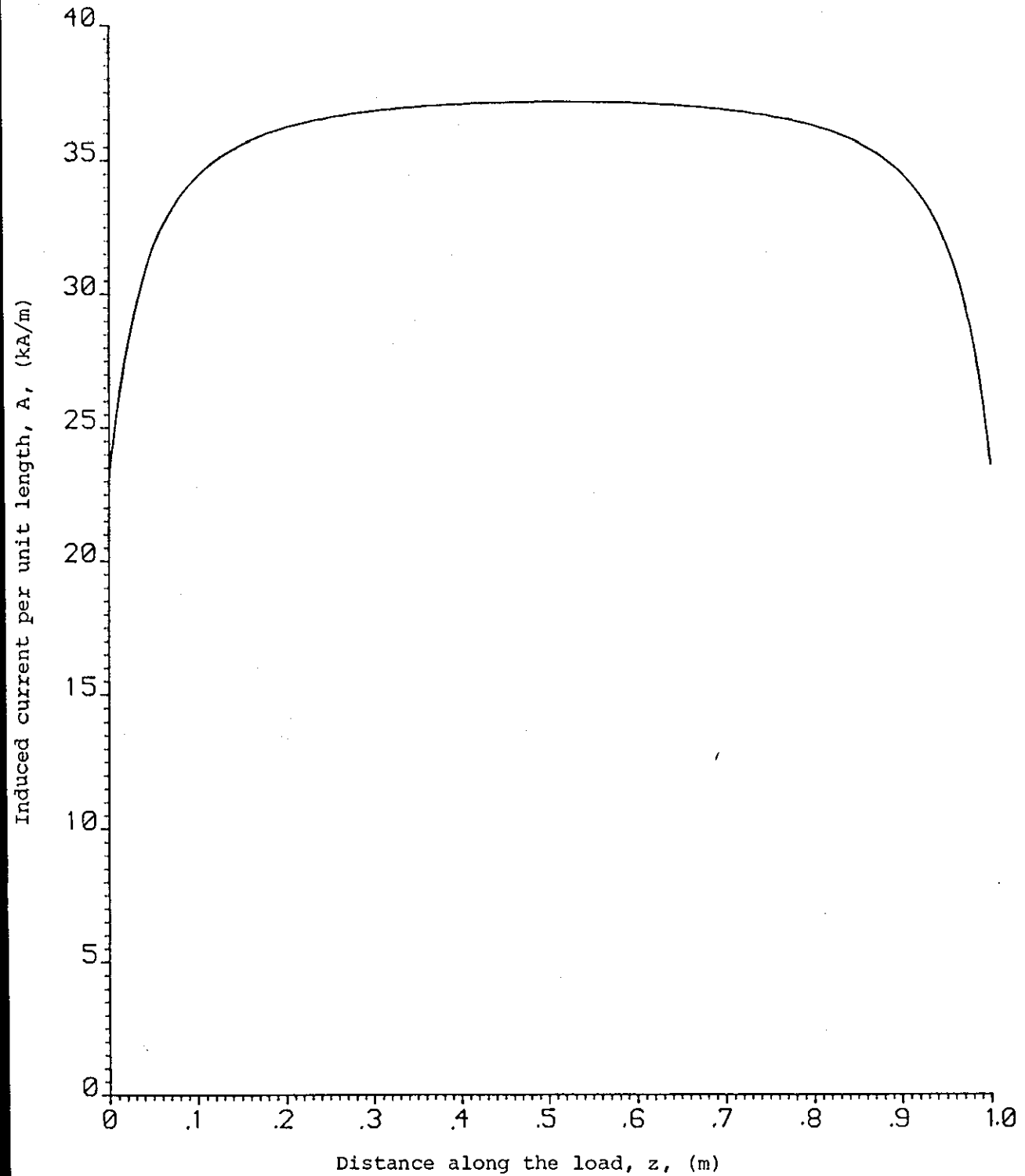


Fig. 4.9 Current per unit length along the load due to 1000A flowing through 40 turns coil of 10mm conductor radius at $h = 40\text{mm}$.

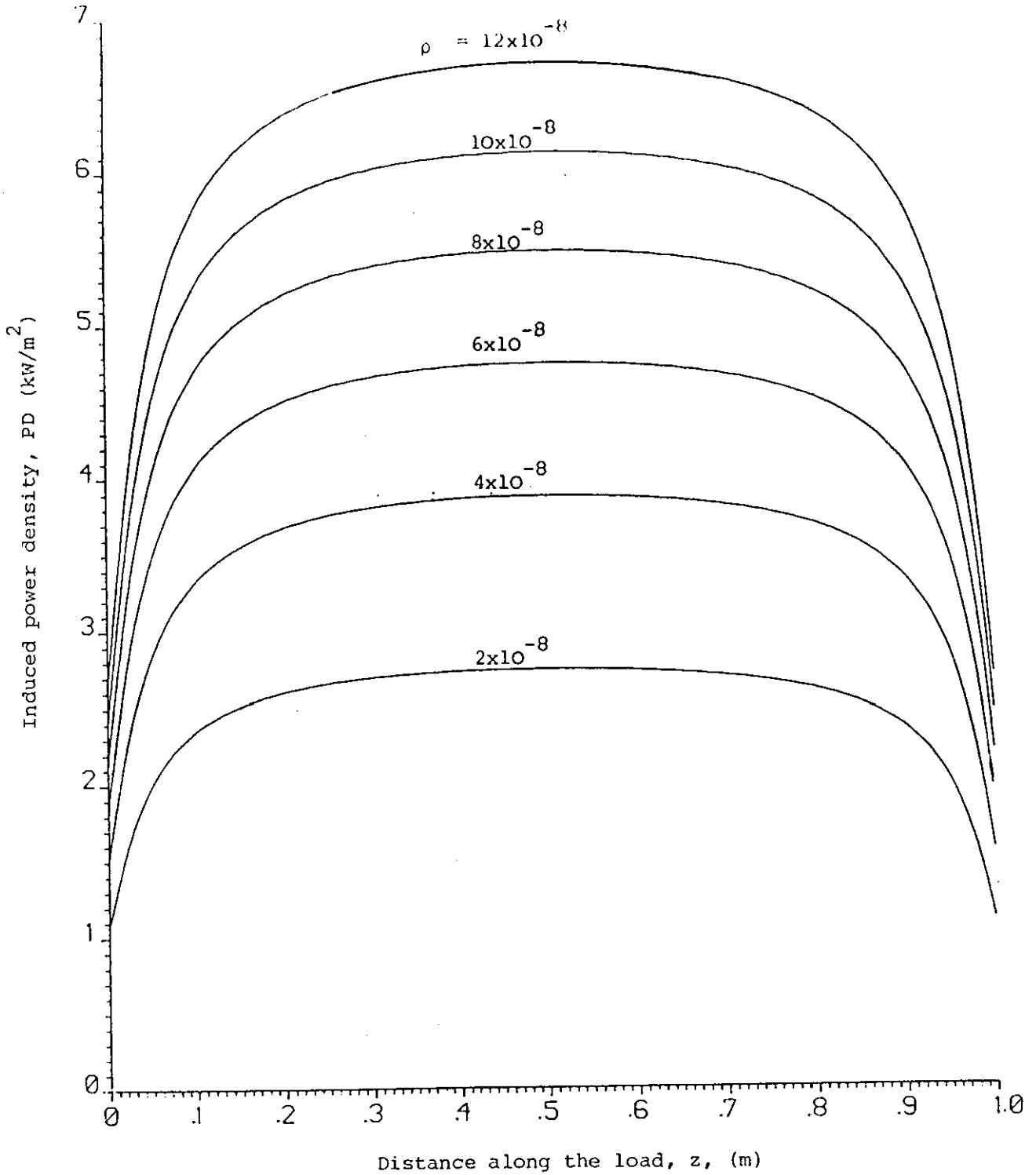


Fig. 4.10 Power density distribution along a nonmagnetic load of resistivity varies radially from $2 \times 10^{-8} \Omega\text{m}$ to $12 \times 10^{-8} \Omega\text{m}$, due to the coil mentioned in Fig. 4.9, the frequency is 50 Hz.

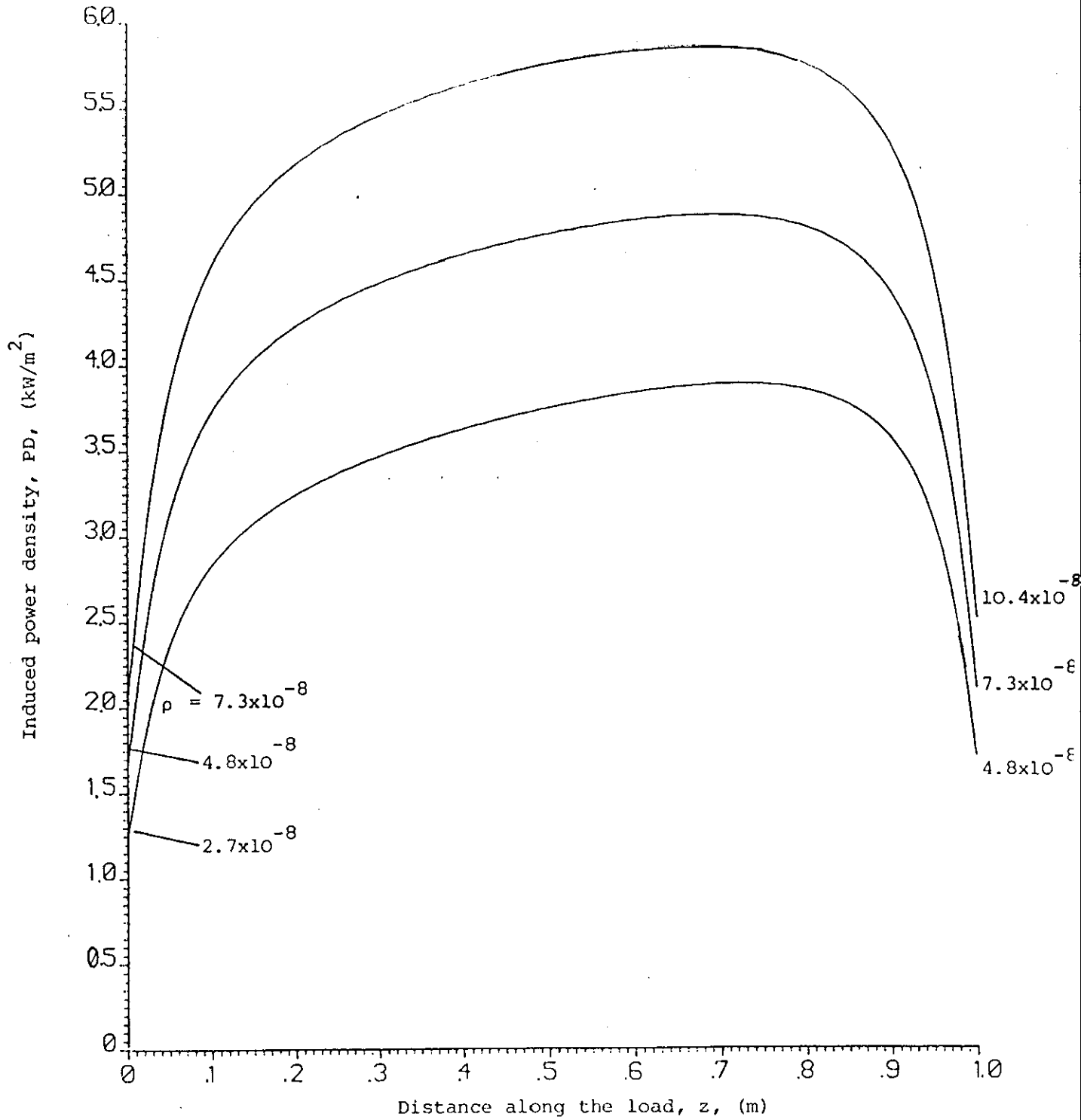


Fig. 4.11 Power density distribution along a nonmagnetic load of resistivity varies axially from $2.7 \times 10^{-8} \Omega\text{m}$ to $10.4 \times 10^{-8} \Omega\text{m}$ due to the coil mentioned in Fig. 4.9, the frequency is 50 Hz.

4.4 Discussion and Conclusions

An equation has been derived for the distribution of the induced current along the load length due to a current flowing in a nearby single conductor. The adoption of the superposition principle on this equation resulted in the development of a new theory for the design of induction heating work coils.

This theory does not assume a uniform magnetic field produced by the coil and, also, it does take the conductor diameter and the spacing between the turns into account.

The superposition theory was correlated with known equations [4.5] and the agreement was very good. This proves that the former is based on reliable theoretical work and reasonable assumptions.

This theory was used to investigate the effect of the load non-uniform physical properties and the influence of each parameter in the power density induced in the load. The computer program neither needed large memory nor required long running time.

The application of this theory showed that the uniformity of the power density induced on the load is a function of the coil number of turns, coil pitch and the air gap. The magnitude of the power density depends largely on the number of turns, the air gap and, of course, the physical properties of the load. The nonuniformity of the resistivity along

the load, due to the difference in the temperature, will result in a nonuniform power density induced in the workpiece.

The application of the superposition theory to a number of cases and it's ability to deal with different situations shows it's flexibility and encourages one to examine it practically.

CHAPTER 5

THE SUPERPOSITION THEORY - PRACTICAL EXAMINATION

The next logical step in the quest for a flexible coil design method, after having derived and verified the superposition theory theoretically, is the experimental examination of this technique; so that the practical problems can be established.

The objective of this chapter is to inspect the applicability of the superposition theory at different conditions similar to those which occur in induction heating applications. To achieve this it was necessary to divide the work into two parts. The first is to test the method when there is no ferromagnetic materials in the system which would complicate the solution due to permeability effects. The investigation concerning the effects of ferromagnetic materials was then carried out in the second part of the investigation.

5.1 Preliminary Investigations

The derivation shown in Chapter 4 was for filamentary conductors where the air gap and the spacings between the turns were large in comparison with the conductor diameter. In induction heating these distances are relatively small which can cause a non-uniform distribution of current across the conductor.

The aim of this section is to establish the difference between the theoretical case of filamentary conductors and the actual situation where there is inter-conductor effect and also to examine the superposition technique practically.

A series of experiments have been executed on an aluminium slab subjected to magnetic fields induced from mains frequency current carrying conductors. The first part of the experiments was to study the case of single conductors, of different shapes and sizes, above the aluminium slab; and the second was the examination of the superposition method by employing a number of conductors instead of a single one.

5.1.1 Single Conductor

The experimental rig of Fig. 5.1 was employed to study the influence of a single conductor above an aluminium slab. The circuit diagram is shown in Fig. 5.2.

Before starting the experiments, different precautions should be considered. Since the experiments involve high current, care should be taken to reduce the effect of the high magnetic field on the different parts of the system. This has been done by shielding the meters with mild steel sheets of 3.2 mm thickness. The current carrying cables were positioned at the greatest possible distance from the load.

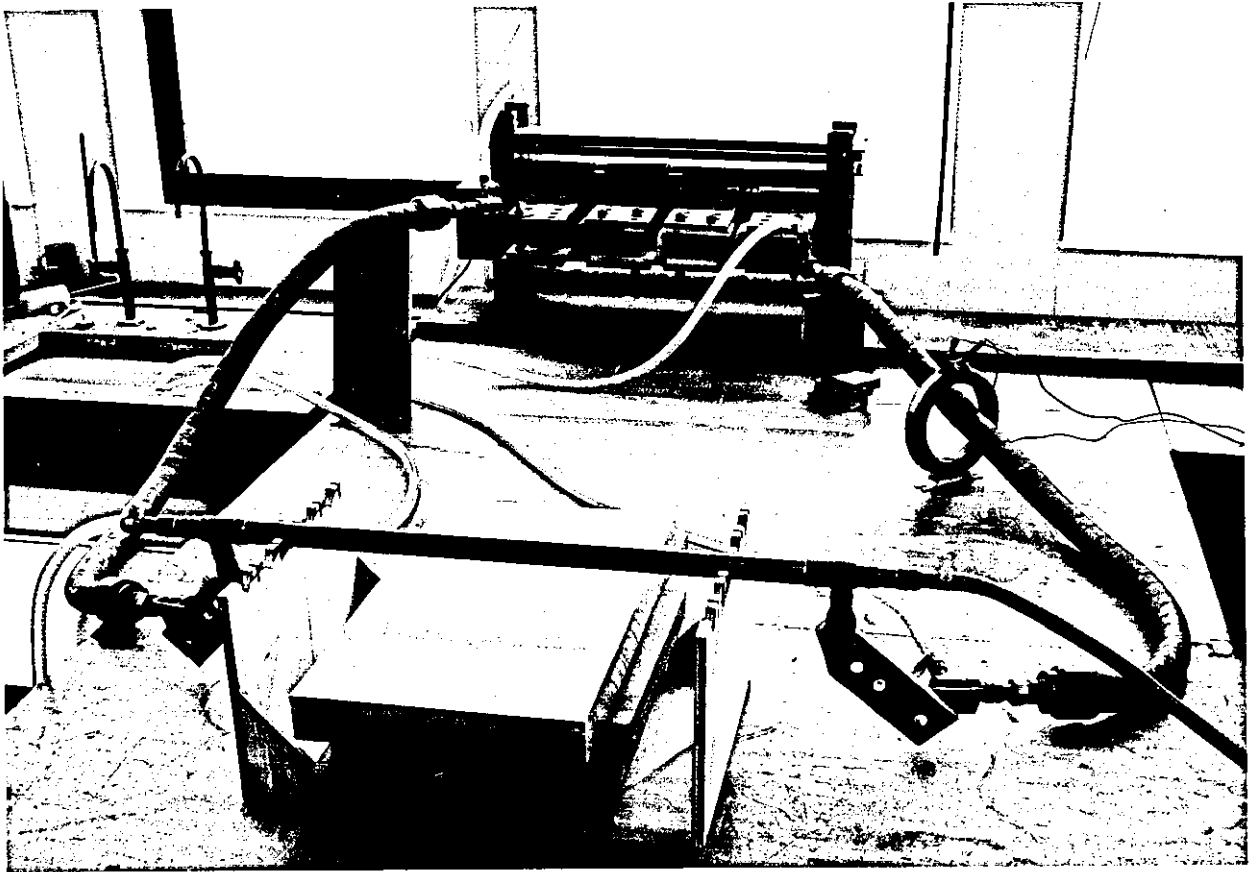


Fig. 5.1 The Experimental Rig

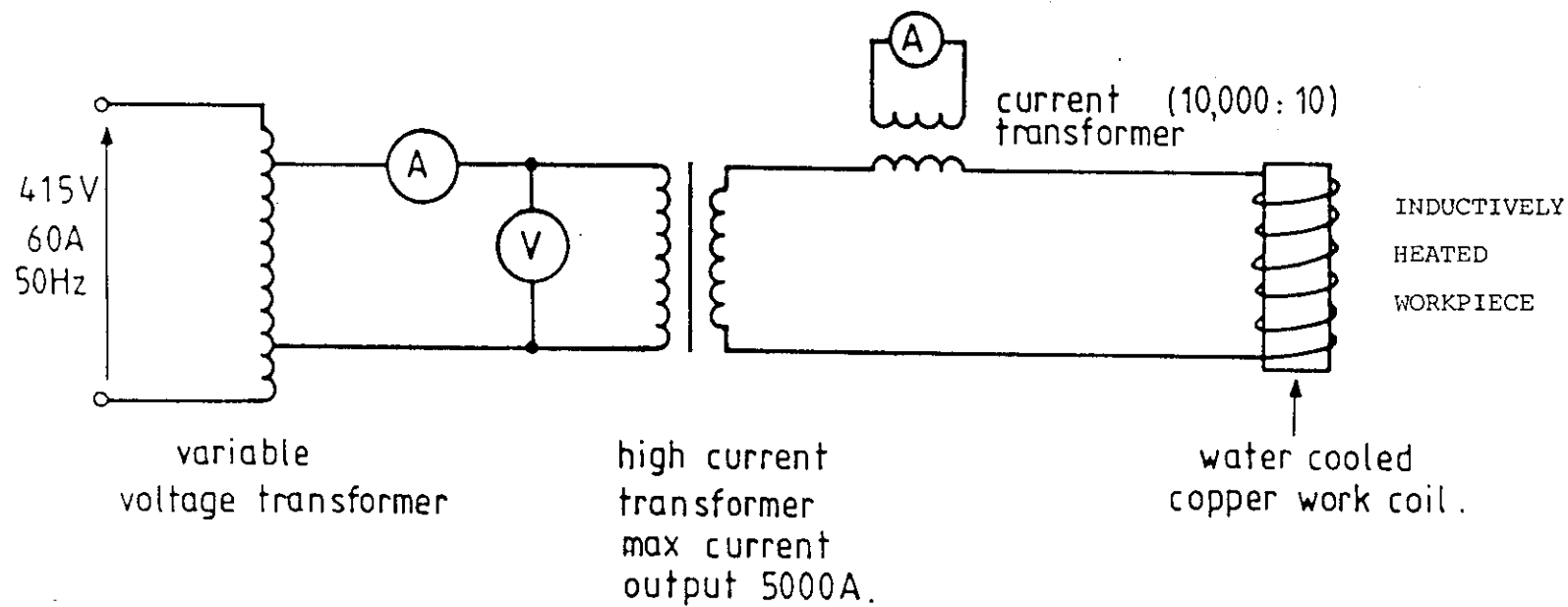


Fig. 5.2 The Circuit Diagram

The conductor was placed over the centre of the load. If this was not the case the induced current would concentrate on a point on the load and not the point directly below the conductor. This characteristic is due to the end effect. That is the concentration of the magnetic field lines near the edge of the load.

The function of the wooden frame, Fig. 5.1, was to hold stable the various conductors, parallel above the workpiece. It was able to hold conductors of various sizes at different set heights.

Four conductors were used to carry the current above the slab. The first three were circular water cooled copper conductors of 1mm wall thickness, 800mm length and external diameters of 6, 15 and 28mm. These conductors will be referred to as "circular" conductors. The fourth conductor was of 450mm length, made of solid rectangular copper brazed to a rectangular water cooled copper tube, as illustrated in Figs. 5.3 and 5.4. It will be referred to as "rectangular" conductor.

Those conductors were chosen because the induction heating work coil is usually made of water cooled copper conductors, either of circular or rectangular cross section. The rectangular conductor is used in a mains frequency induction billet heating coil. The solid part is greater than the penetration depth of copper at 50 Hz, which is about 10mm.

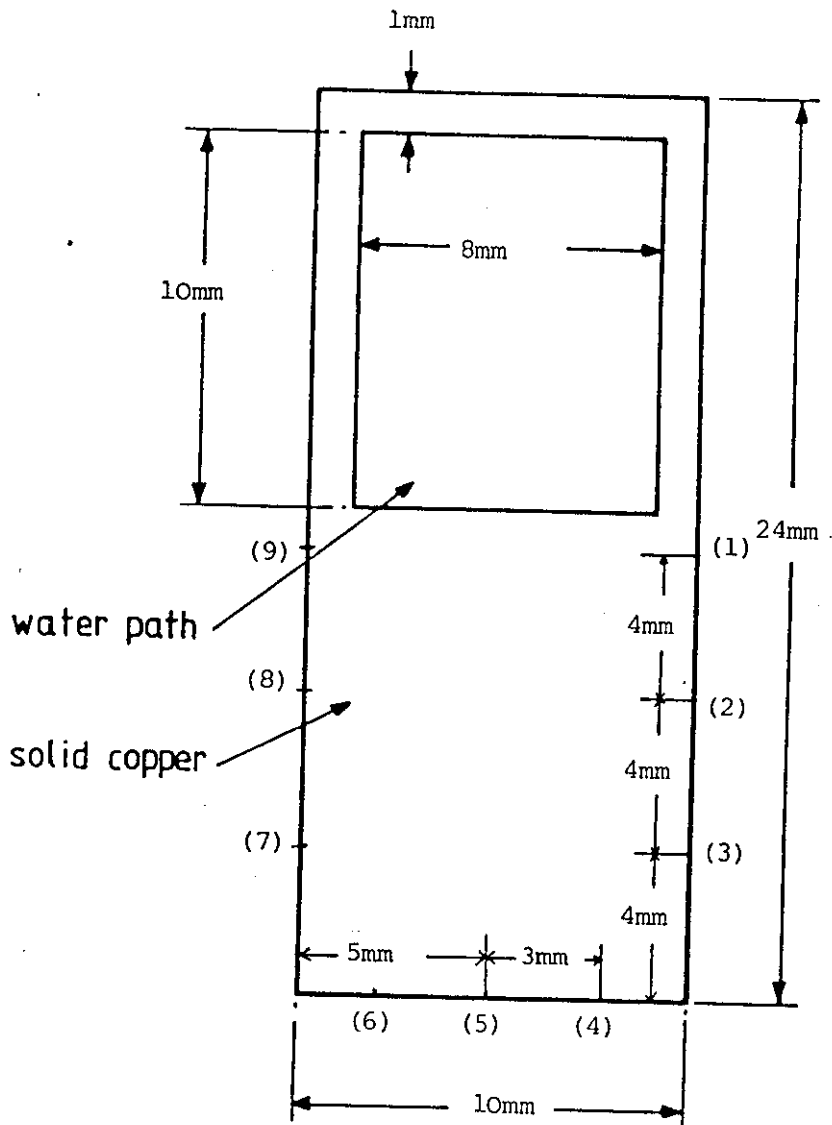


Fig. 5.3 Rectangular conductor used in mains frequency induction heating.

The number between brackets () is the probe number.

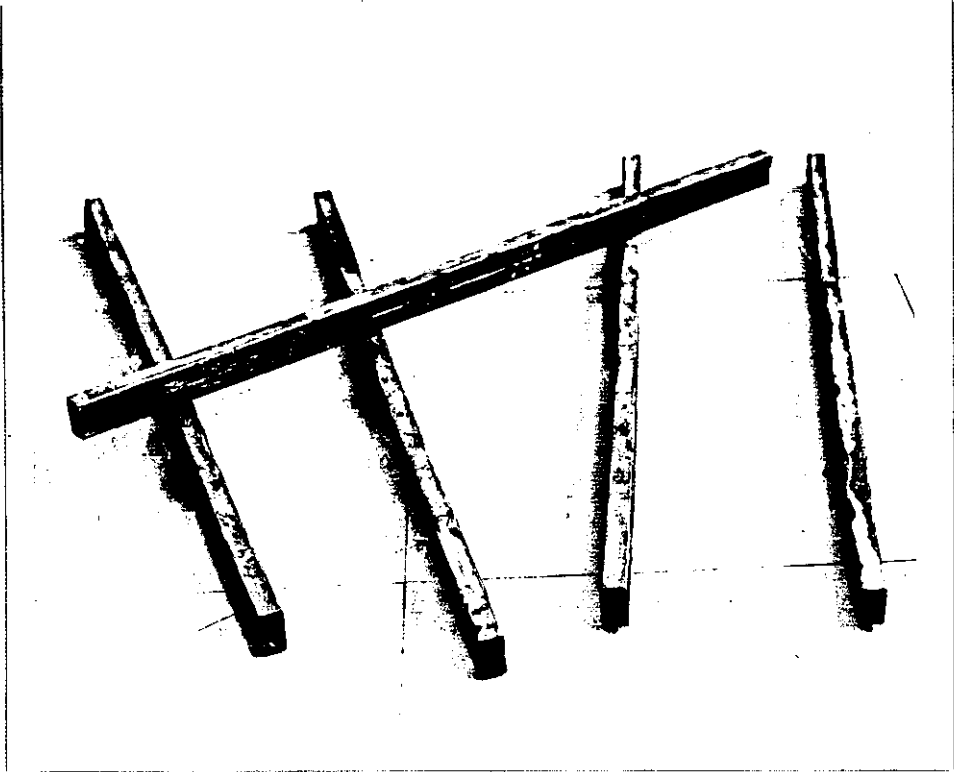


Fig. 5.4 Rectangular Conductors

An aluminium slab of (500 x 300 x 50.8)mm dimensions was employed as a workpiece. When operating at mains frequency the thickness of this slab was greater than 4 times the skin depth. This load satisfied two factors: firstly it was of nonmagnetic material so that there was no worry about the change in permeability at this early stage of the practical work; and secondly the thickness of this slab was more than 4 times the skin depth, hence the current beyond this depth is very small. The current penetration into the load is approximately

$$J_x = J_s e^{-\frac{x}{\delta}}$$

where J_s is the surface current density,

and J_x is the current density at depth x .

If $x = 4\delta$ then $J_x = 0.018 J_s$. i.e. J_x is small.

A high current should be fed through the conductor, if a measurable eddy current is to be obtained. The high current necessary for the experiments was supplied by a transformer capable of supplying 5000A. Its primary was connected to a 415V, single phase regulating transformer. Water cooled heavy duty cables, capable of carrying the required current, connected the conductor across the secondary of the transformer. A current transformer of 10000:10A and digital multi-meter, with current shunt were adopted to measure the conductor current.

The surface current density induced on the load was measured by the current density probe shown in Fig. 5.5. The theory, construction and characteristics of filament type current density probes are described by Burke and Alden [5.1], see also Appendix 5. A perspex frame containing 2 brass pins was the type of probe suitable for use on the surface of the aluminium slab. The distance between the pins was 30.5mm and each one was connected to an insulated constantan thin wire (0.1mm diameter), which was kept close to the surface. The two wires were twisted together in the centre and the other ends were connected across a voltmeter. By attaching this probe to the surface of the load, the current density at that point could be measured.

Care was taken while constructing this probe as the constantan wire must be extremely close to the surface of the load, but no contact should occur in any part other than the two pins. A coaxial wire connected the probe to the voltmeter so as to reduce pick-up.

The current density on the surface of the conductor was also measured to assess the effect of the conductor current flow on the distribution of the current density induced on the load. The probe was of 30mm length, constructed from constantan wires. The two constantan wires were laid along the conductor length and their ends were cleaned and connected to the conductor. Seven probes positioned at 30° intervals were constructed around the 28mm diameter circular conductor.

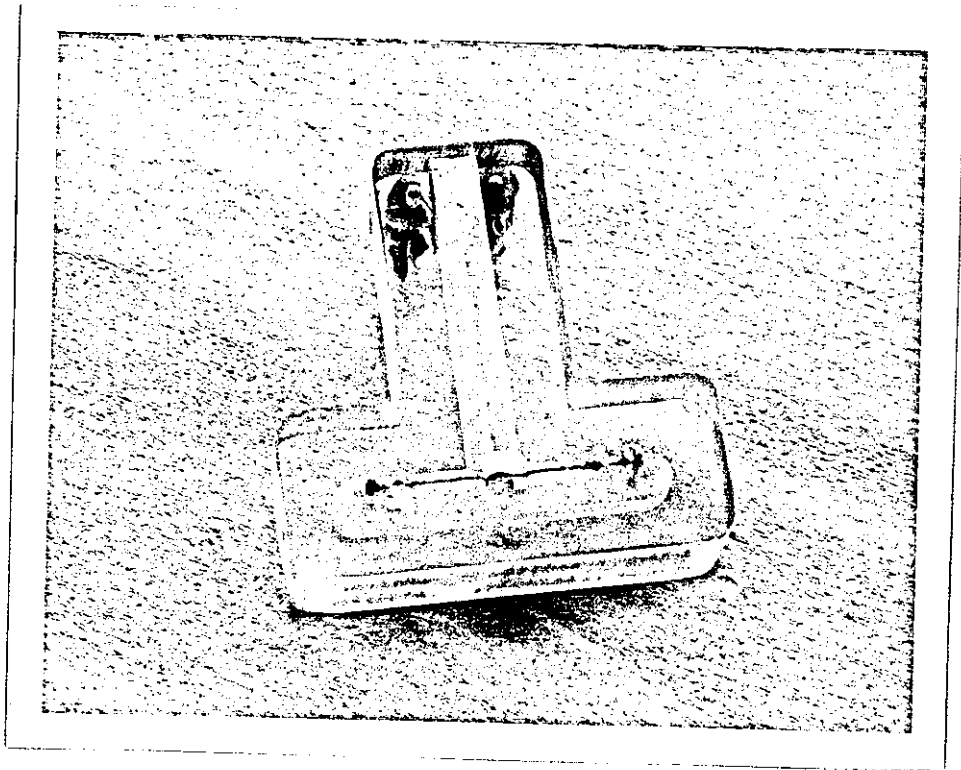


Fig. 5.5 The probe for measuring the induced current density on the slab.

On the rectangular conductor; nine probes were used and they were positioned as depicted in Fig. 5.3.

In order to obtain adequate resolution the probe must satisfy two criteria: firstly, (J) and (ρ) must not change over the probe length; secondly, the diameter of the filament must be small compared to the skin depth. The first condition was satisfied as the load was uniform and the current's direction was known and the second too was satisfied by using probes of 0.1mm diameter wire when the skin depth was 12mm. The surface current density J_s is a function of the voltage across the probe V , the resistivity ρ and the probe length l as:

$$J_s = \frac{V}{\rho l} \quad (5.1)$$

The experiments were carried out on a conductor which was placed at a fixed distance h from the aluminium slab. A mains frequency current of 1000A was used.

The current density distribution on the 28mm circular conductor and the rectangular conductor was measured by the attached probes when there was aluminium slab beneath it and when there was no load. The results are shown in Figs. 5.6 and 5.7. It can be seen that the distribution was uniform on the circular conductor, this was because the thickness of the conductor was much smaller than the penetration depth. The current distribution on the rectangular conductor was non-

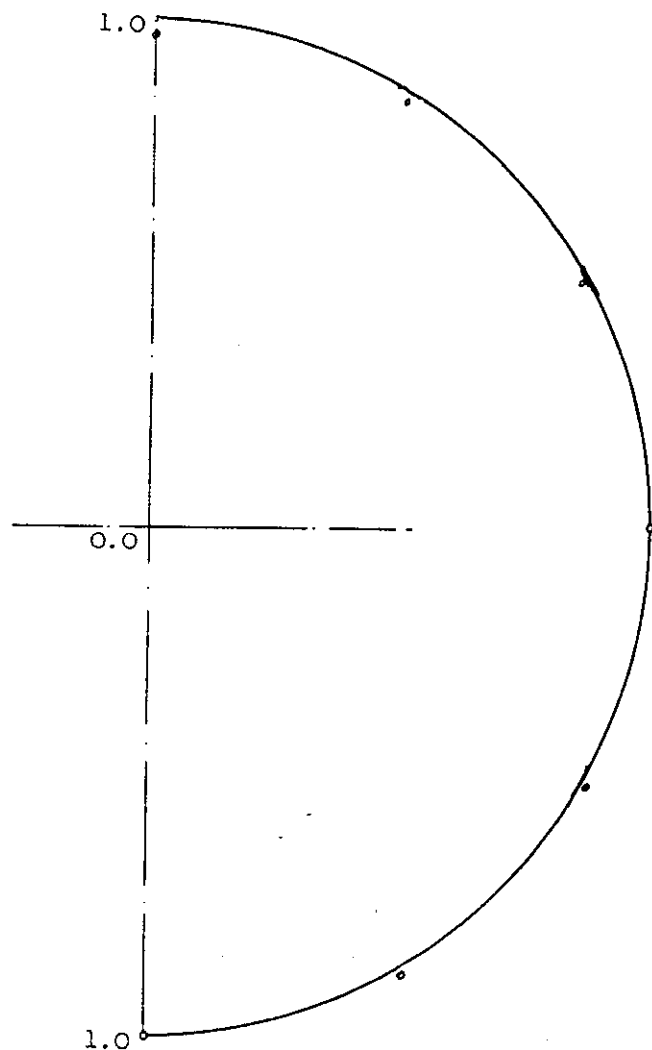


Fig. 5.6 Normalised surface current density distribution on circular conductor for no load and with load at $h = 37\text{mm}$. The graph is in polar coordinate. $a = 14\text{mm}$,
 $I = 1000\text{A}$,

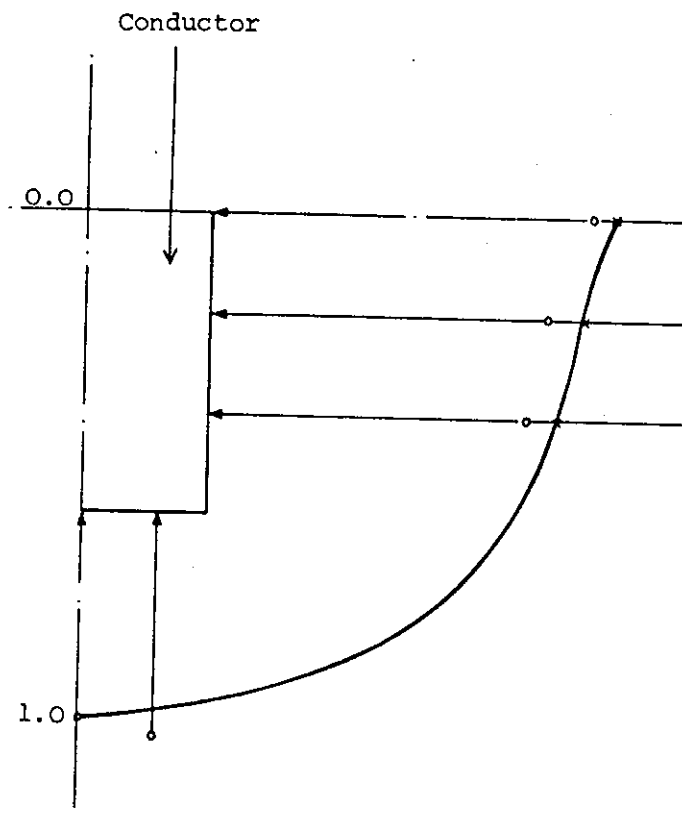


Fig. 5.7 Normalised surface current density distribution on the solid part of the rectangular conductor. The distances from 0.0 show the current density at the probe shown by the arrow

~~**~~ no load o with aluminium slab
at g = 23mm

uniform and the existence of the aluminium slab increased the concentration of the current on the lower surface of the conductor. This was due to the proximity effect on the conductor itself. That is the variation of the current distribution of the conductor because of the existence of the load. The calculation was based on constant resistivity as the conductors were water cooled and the temperature was constant at 20°C. The surface current density was calculated from equation (5.1).

To deduce the distribution of the surface power density induced on the load, a current density probe, Fig. 5.5, was attached to the surface of the slab at different points along the z direction. The z direction was perpendicular to the conductor, and the probe was parallel to the conductor, that being the direction of the induced current. The object of this exercise was to measure the surface current density induced on the slab; hence the surface power density distribution was determined by using equation (4.21).

The practical measurements of the current density induced on the slab showed that the "theoretical" equation (4.9) of a filamentary conductor cannot be applied practically.

$$A = \frac{I}{\pi} \cdot \frac{h_e}{h_e^2 + z^2} \quad (4.9)$$

Many experiments have been carried out in order to find a more suitable equation. It was found, empirically, that equation (4.9)

should be amended to the "practical" equation (5.2) below,

$$A = \frac{\beta I}{\pi} \cdot \frac{h_e}{h_e^2 + \frac{\alpha}{\cos \sigma} z^2} \quad (5.2)$$

where α and β are functions of h_e , see Figs. 5.8 and 5.9, and σ is the angle between the line joining the conductor and point P and the perpendicular line from the conductor to the slab, see Fig. 4.1(a). It will be shown later on in the study that equation (5.2) can be applied also to an aluminium cylinder of radius equal to the thickness of the slab.

Graphs of the measured power density and those calculated from the "theoretical" equation (4.9) and the "practical" equation (5.2) are shown in Figs. 5.10 — 5.21 for the circular conductors. Figs. 5.22-5.25, for the rectangular conductors, do not contain curves for the theoretical equation as this equation was derived for circular conductors only. The equivalent height of the rectangular conductor was found to be equal to the air gap g .

The theoretical power density was calculated by employing equations (4.9), (4.20) and (4.21). The calculation of the practical power density was based on equations (5.2), (4.20) and (4.21). The measured power density was determined from the measured surface current density and equation (4.21). The calculation of the practical power density was carried out on a program "ALI" developed for a

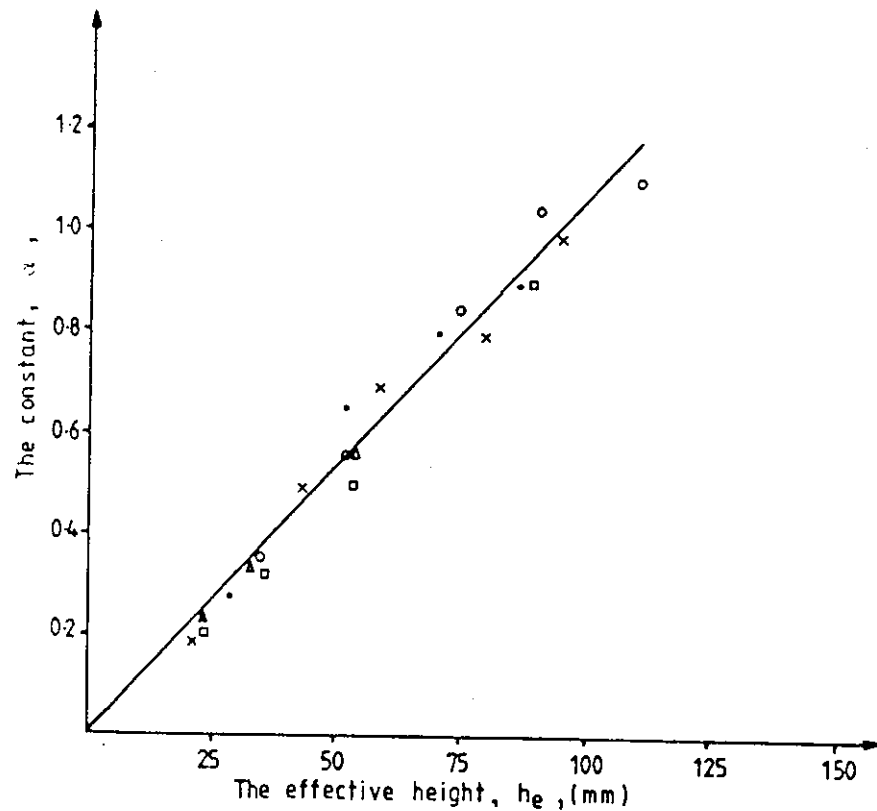


Fig. 5.8 The variation of the constant (α) with the effective height (h_e)

- Rectangular conductor
 - × Circular conductor of 6 mm diameter
 - Circular conductor of 15 mm diameter
 - Circular conductor of 28 mm diameter
 - △ Circular conductor of 6 mm diameter around the cylinder
- } Above aluminium slab

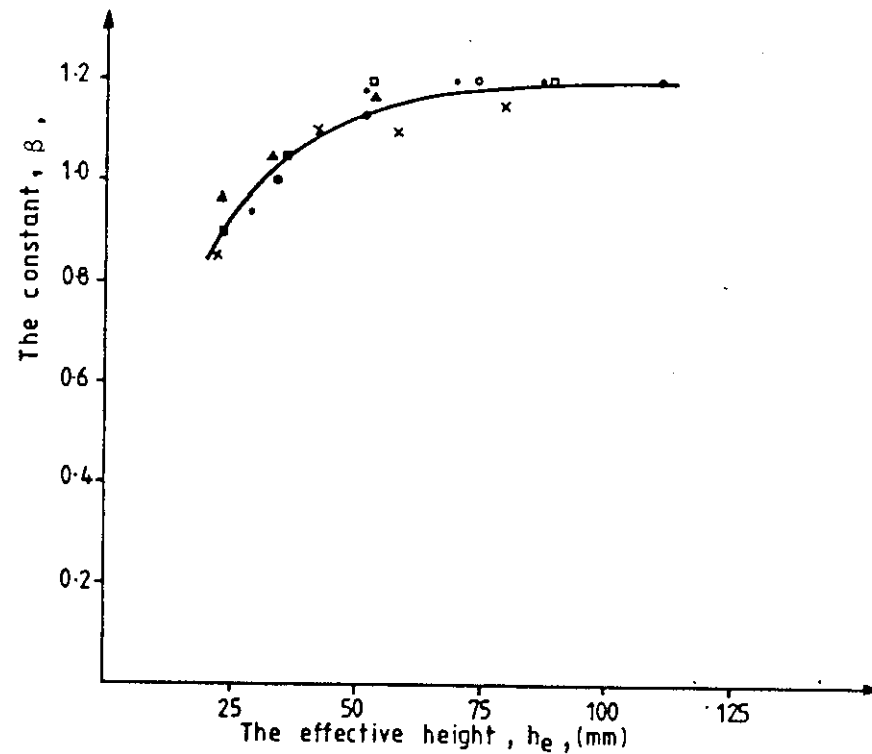


Fig. 5.9 The variation of the constant (β) with the effective height (h_e)

- Rectangular conductor
 - × Circular conductor of 6 mm diameter
 - Circular conductor of 15 mm diameter
 - Circular conductor of 28 mm diameter
 - △ Circular conductor of 6 mm diameter around the cylinder
- } Above aluminium slab

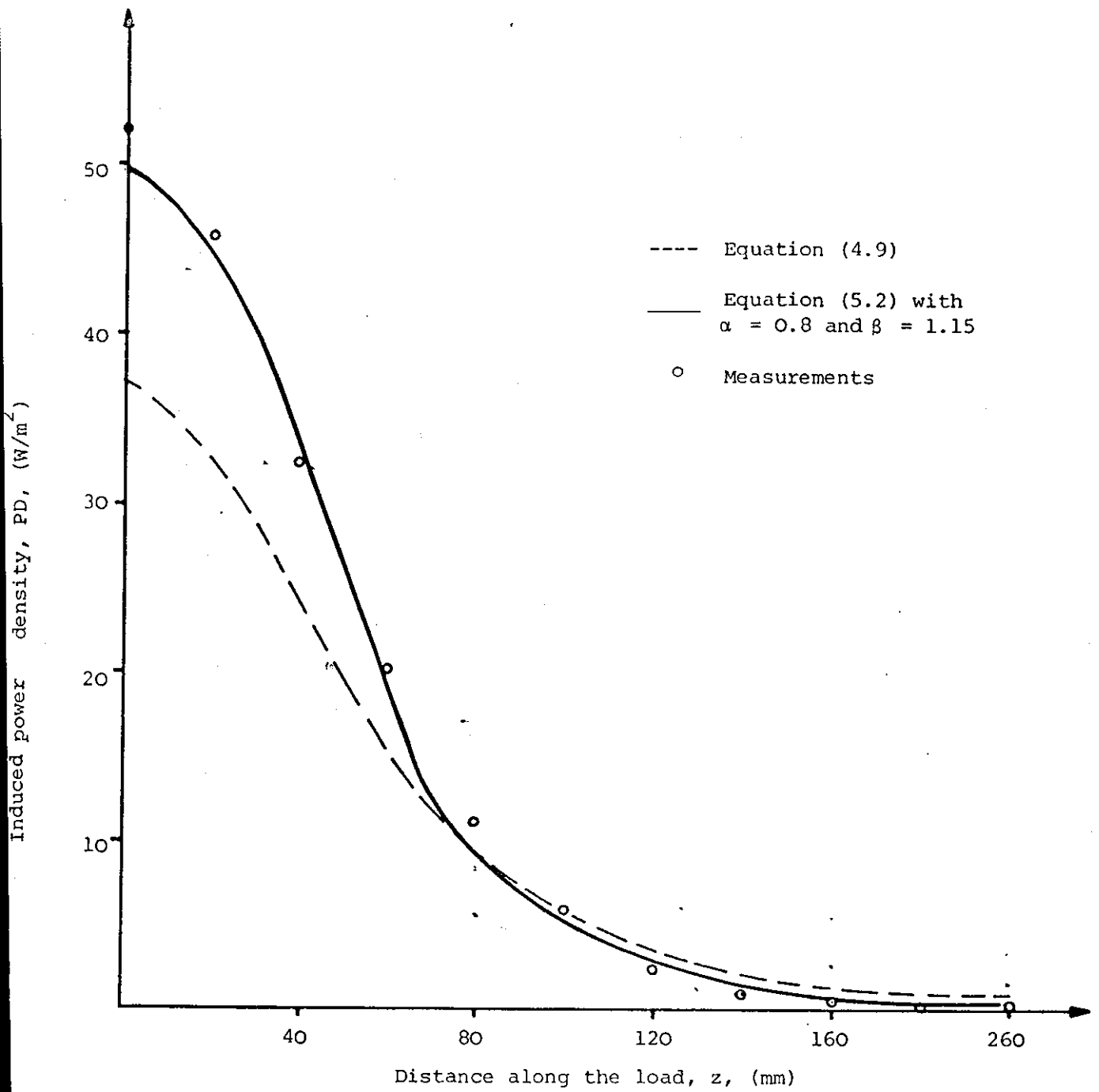


Fig. 5.10 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 6mm diameter at $h = 80\text{mm}$

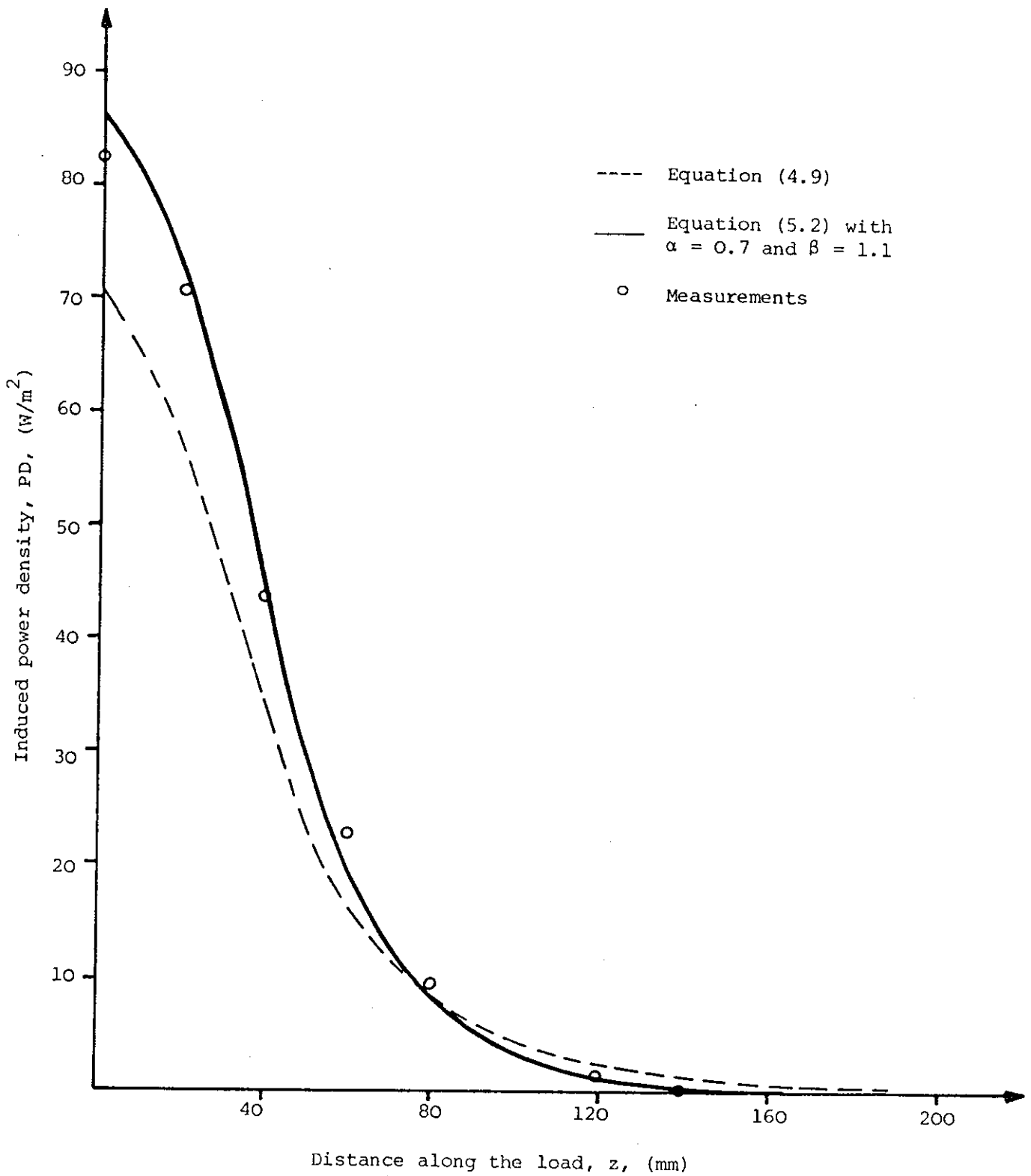


Fig. 5.11 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 6mm diameter at $h = 58\text{mm}$

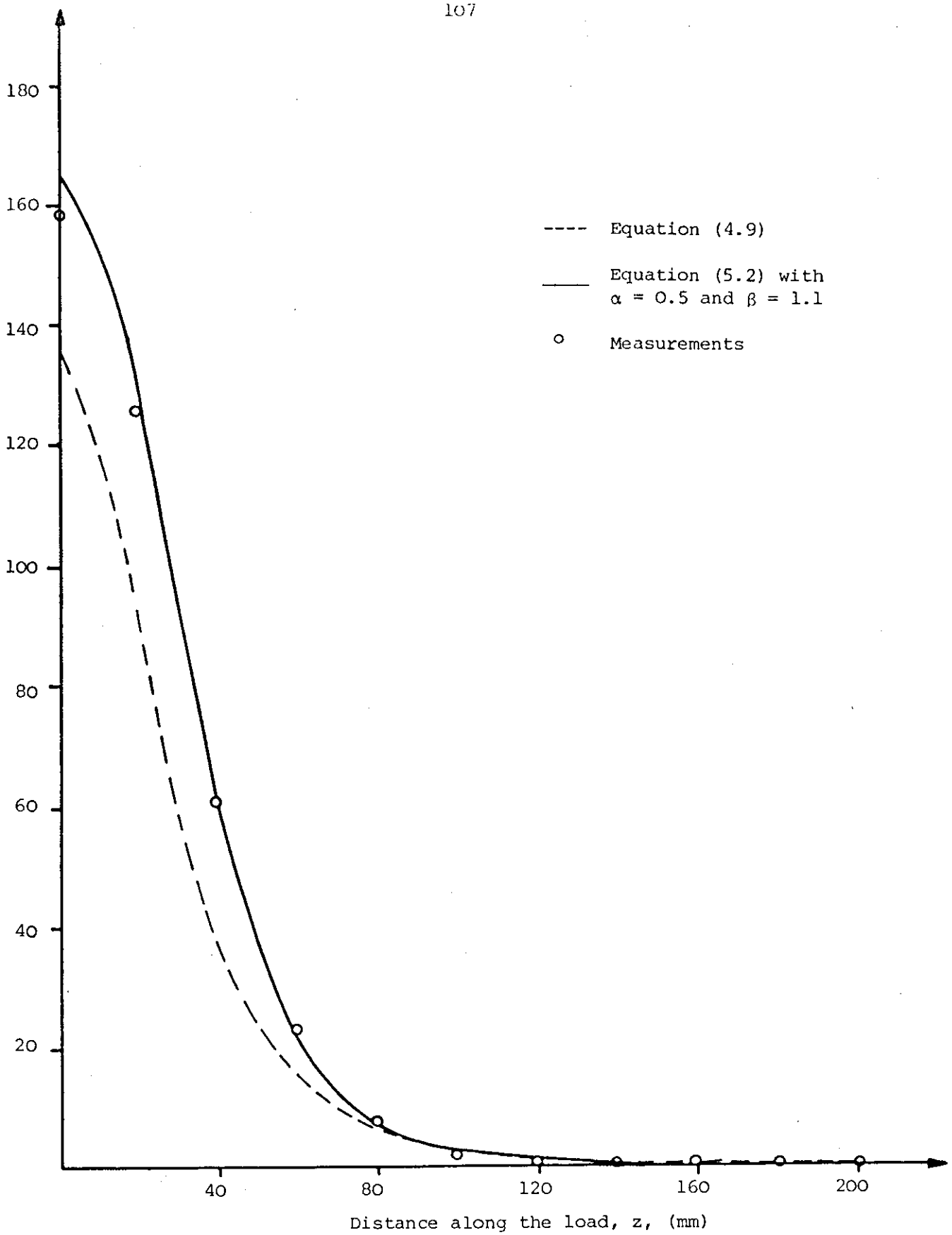


Fig. 5.12 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 6mm diameter at $h = 42\text{mm}$

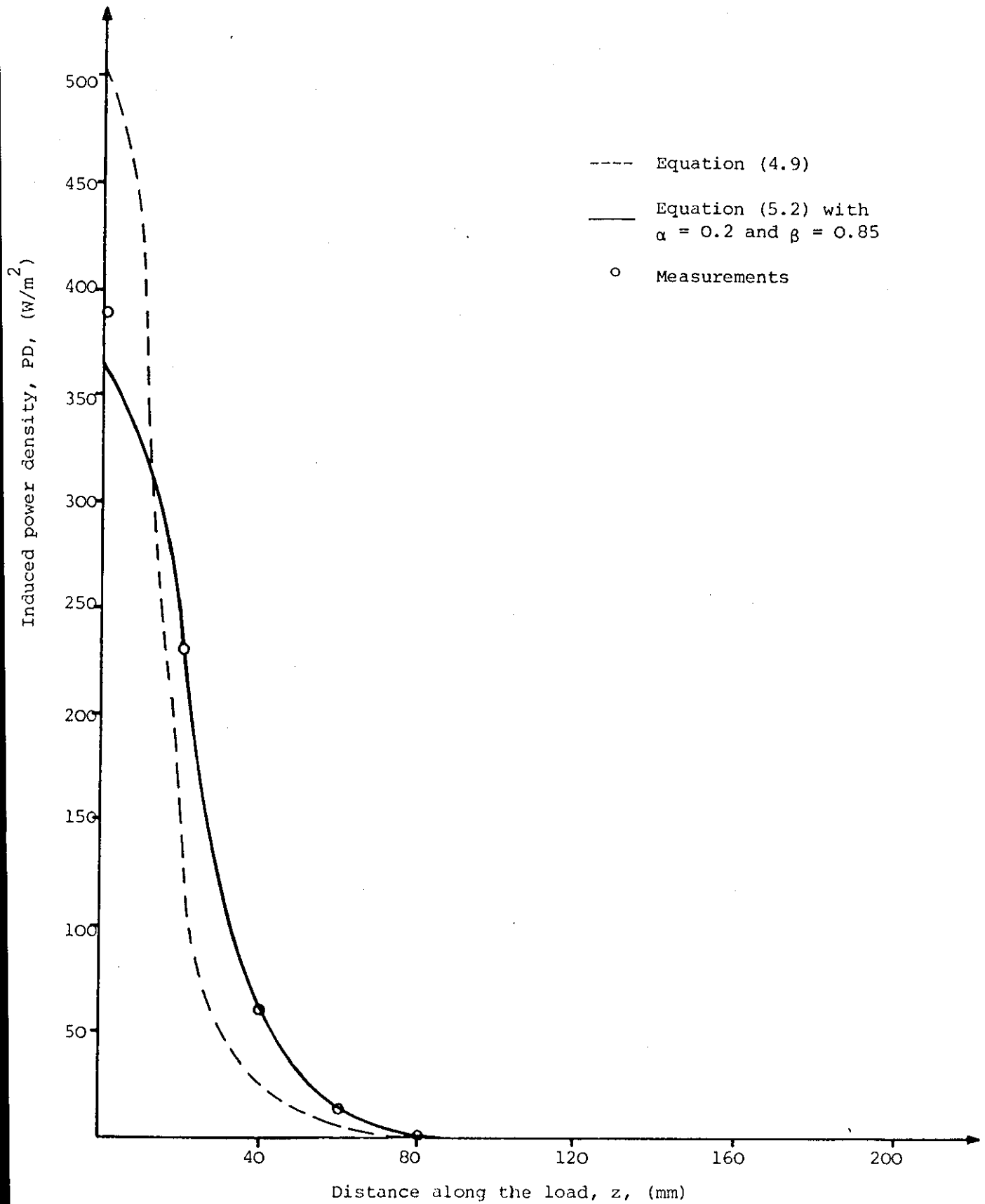


Fig. 5.13 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 6mm diameter at $h = 22\text{mm}$

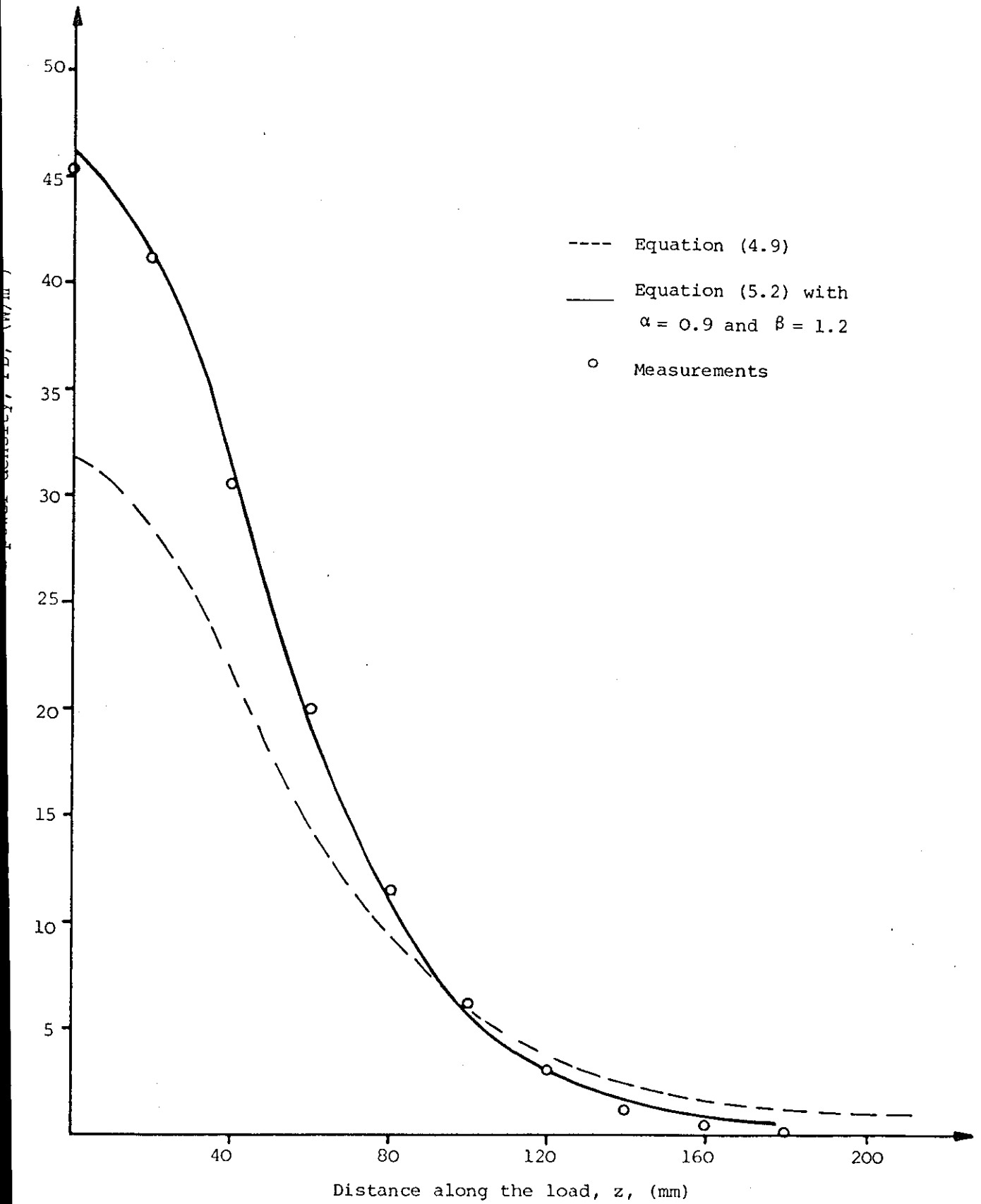


Fig. 5.14 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 15mm diameter at $h = 87\text{mm}$

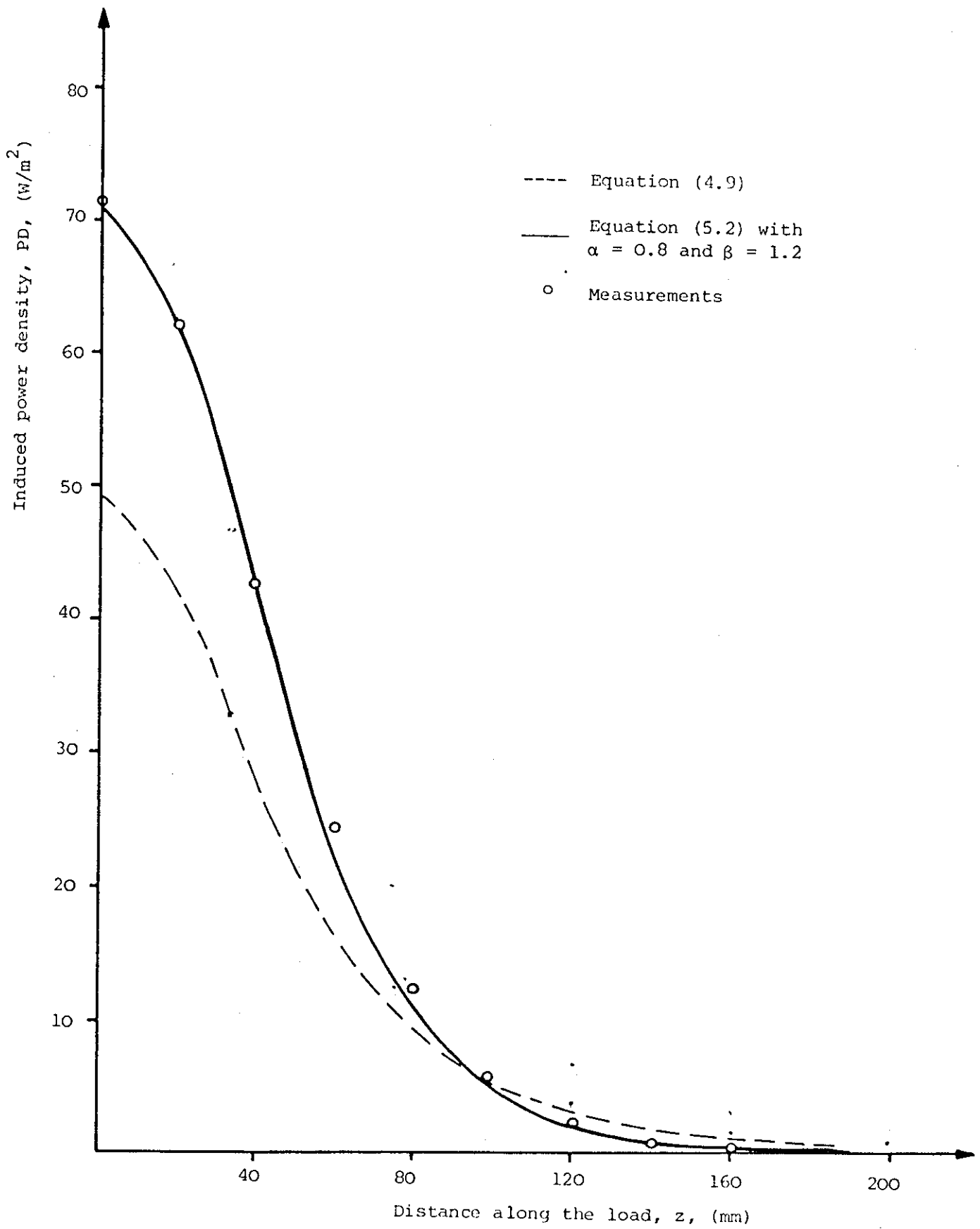


Fig. 5.15 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 15mm at $h = 70\text{mm}$

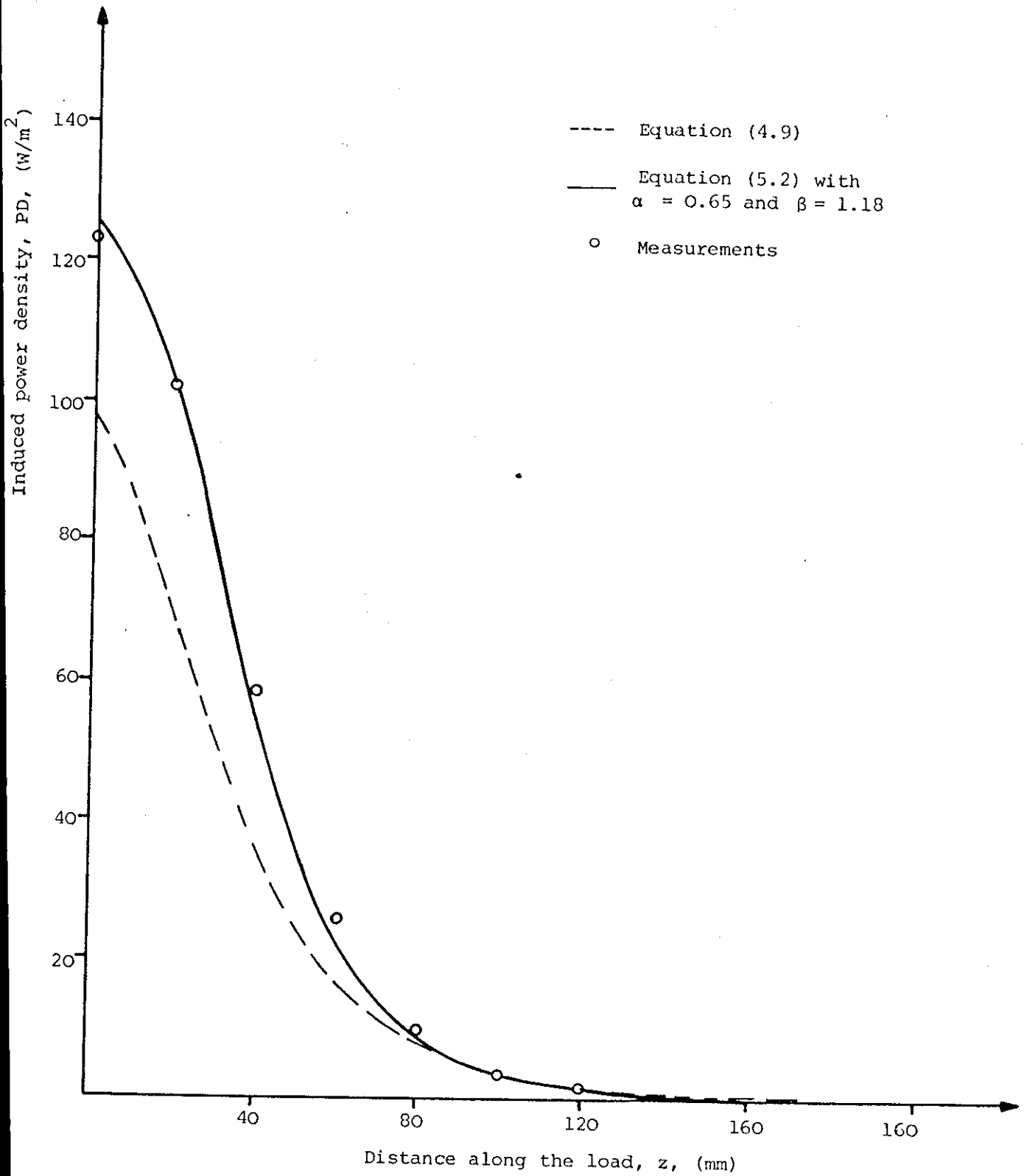


Fig. 5.16 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 15mm diameter at $h = 52\text{mm}$

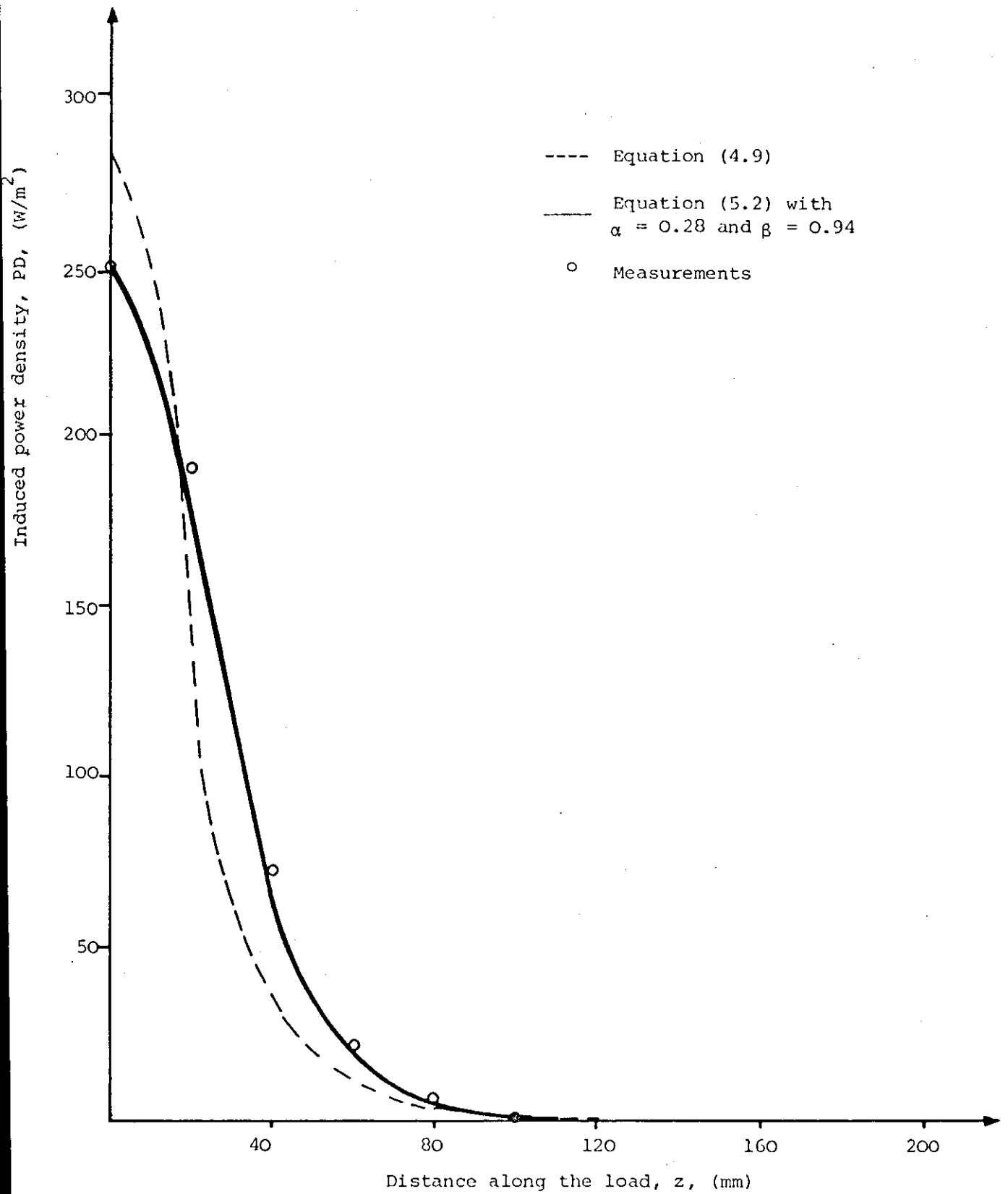


Fig. 5.17 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 15mm diameter at $h = 30\text{mm}$

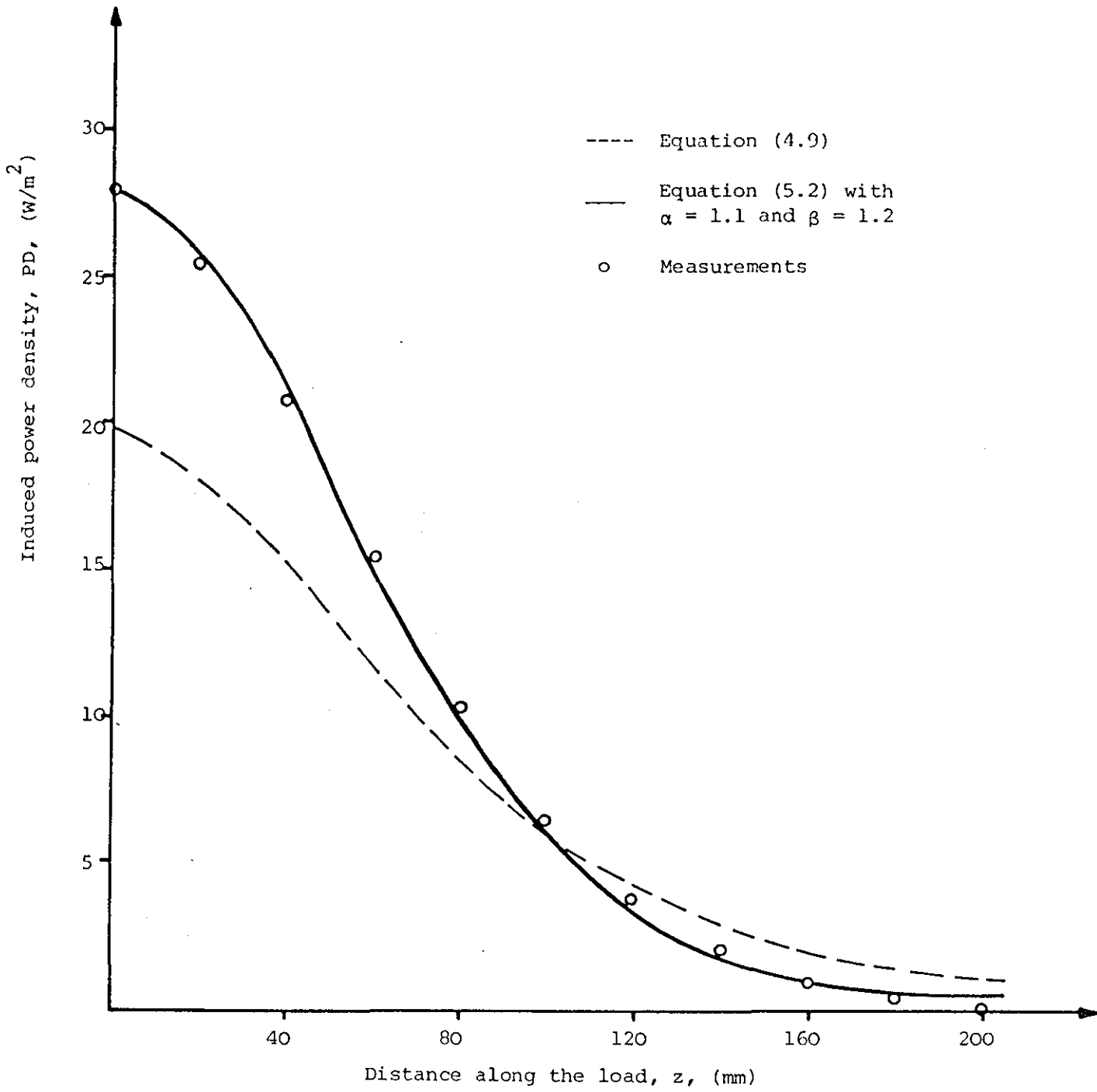


Fig. 5.18 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 112\text{mm}$

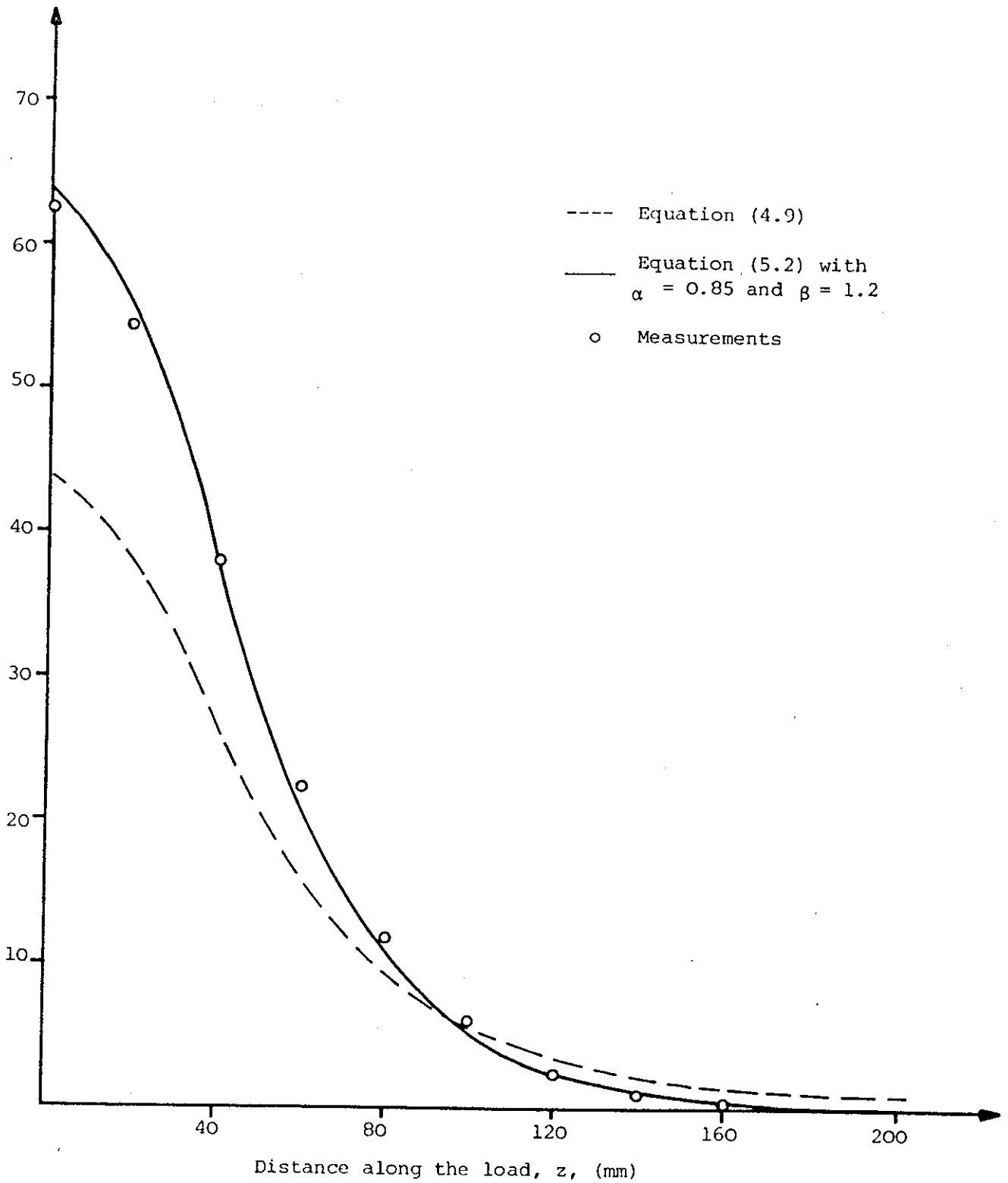


Fig. 5.19 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 75\text{mm}$

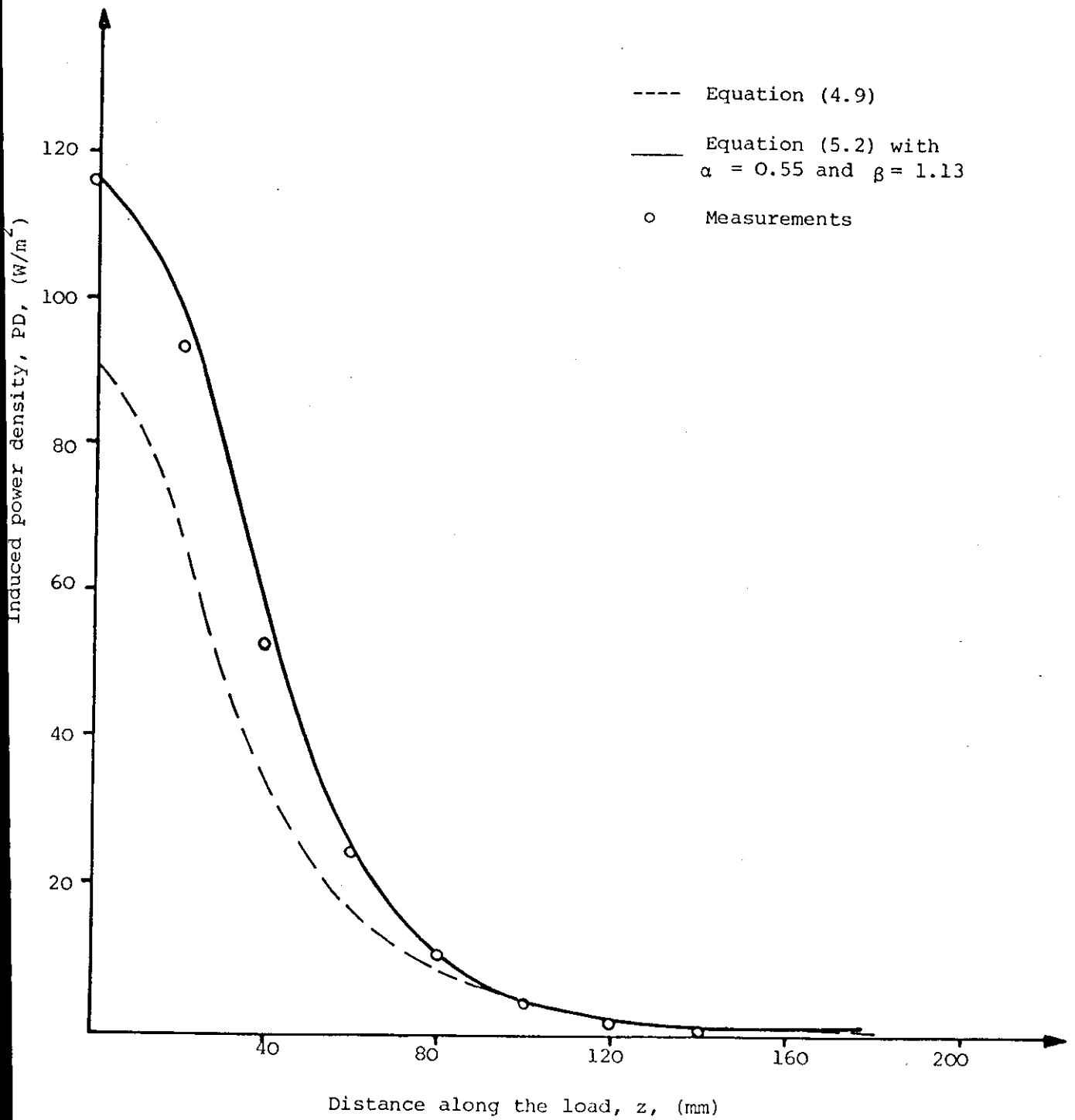


Fig. 5.20 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 54\text{mm}$

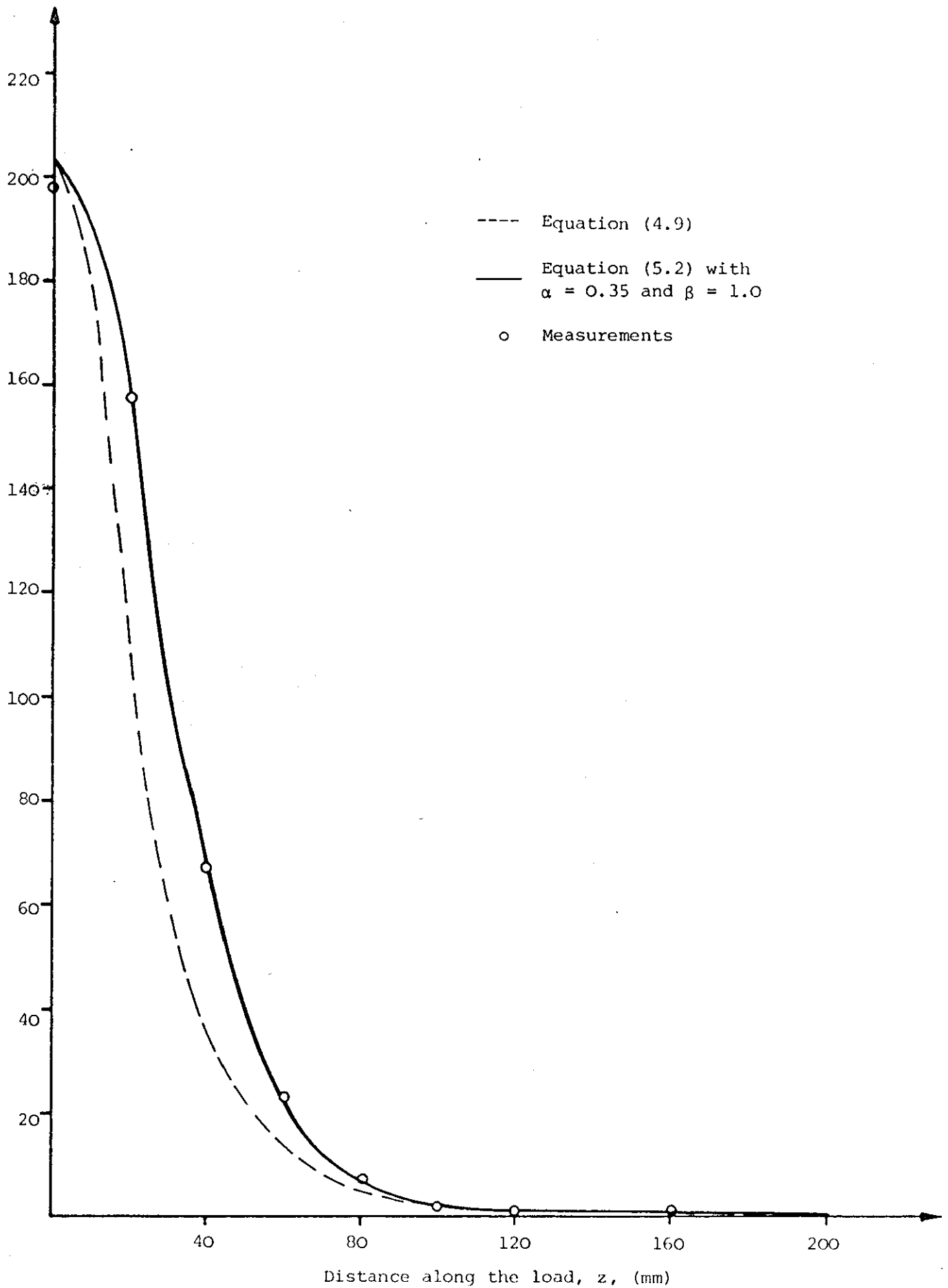


Fig. 5.21 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 37\text{mm}$

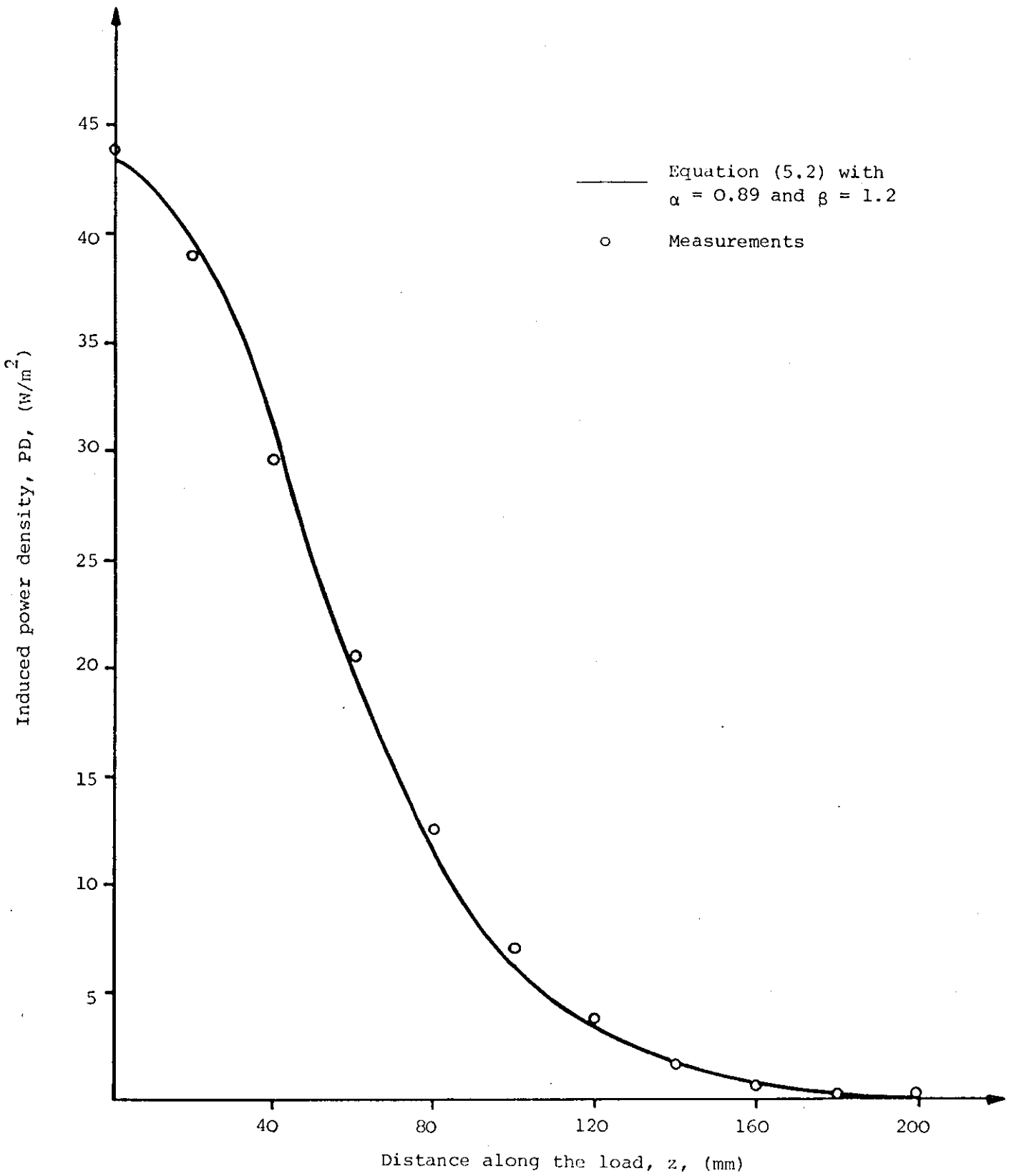


Fig. 5.22 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 89\text{mm}$

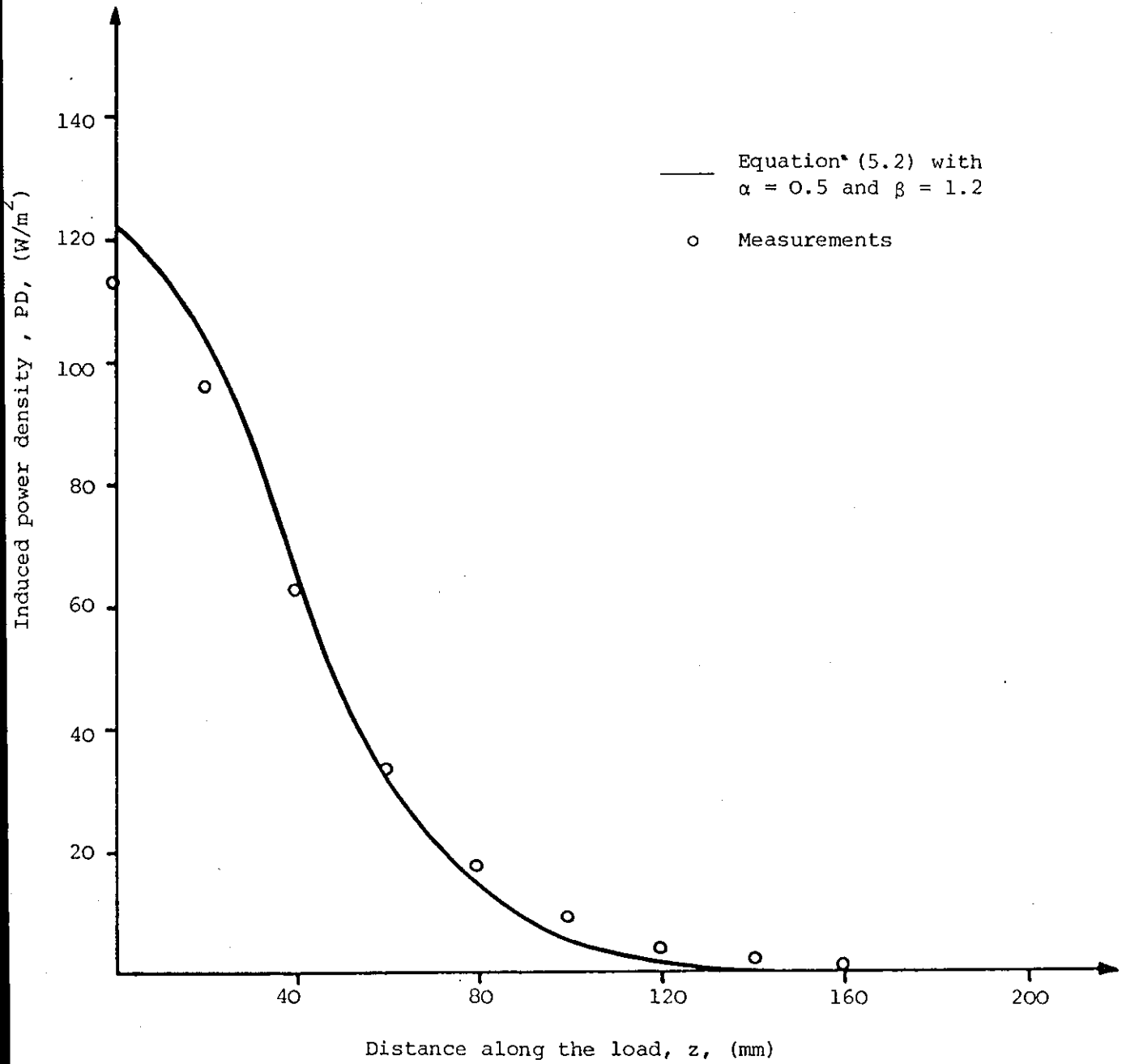


Fig. 5.23 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 53\text{mm}$

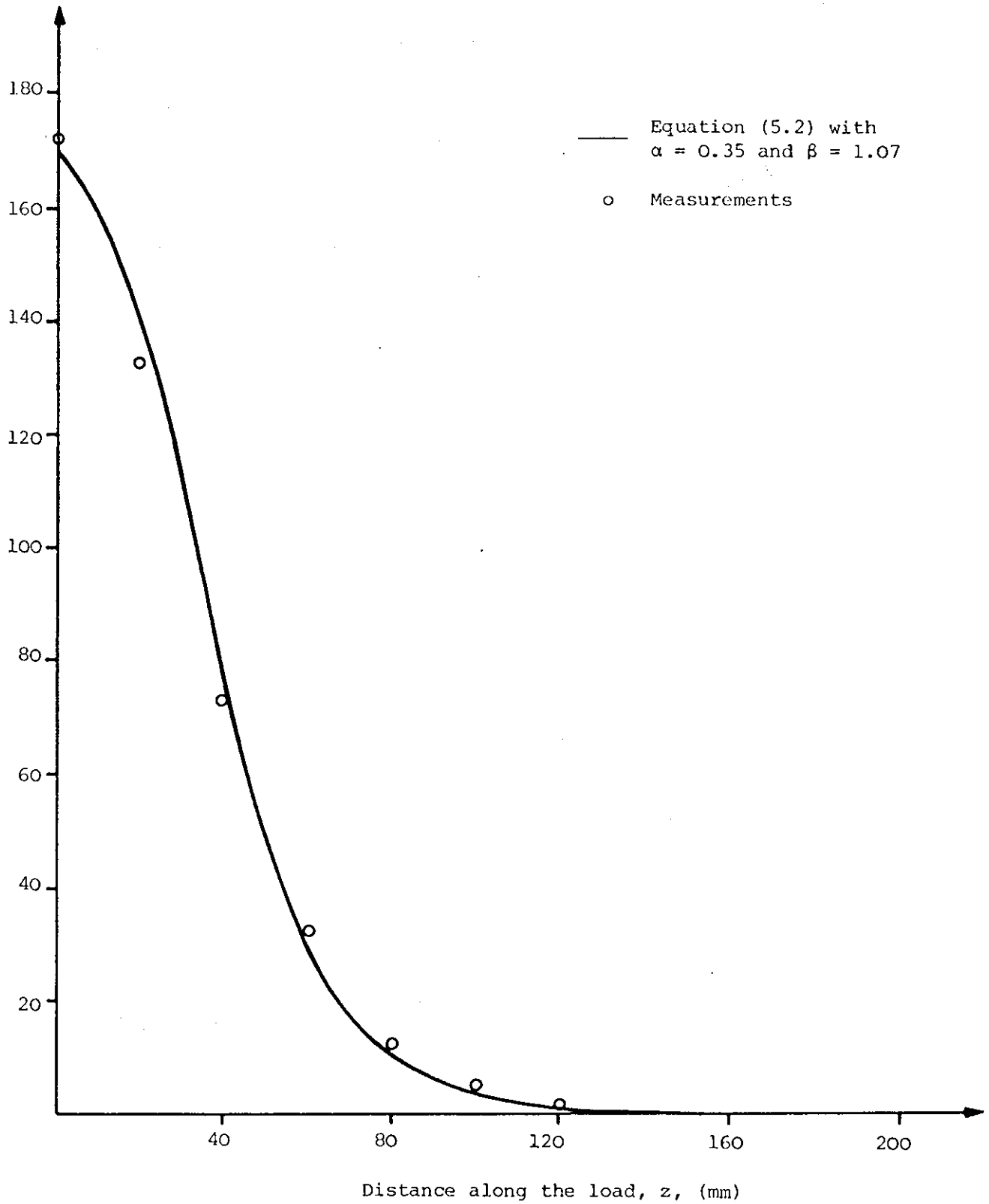


Fig. 5.24 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$

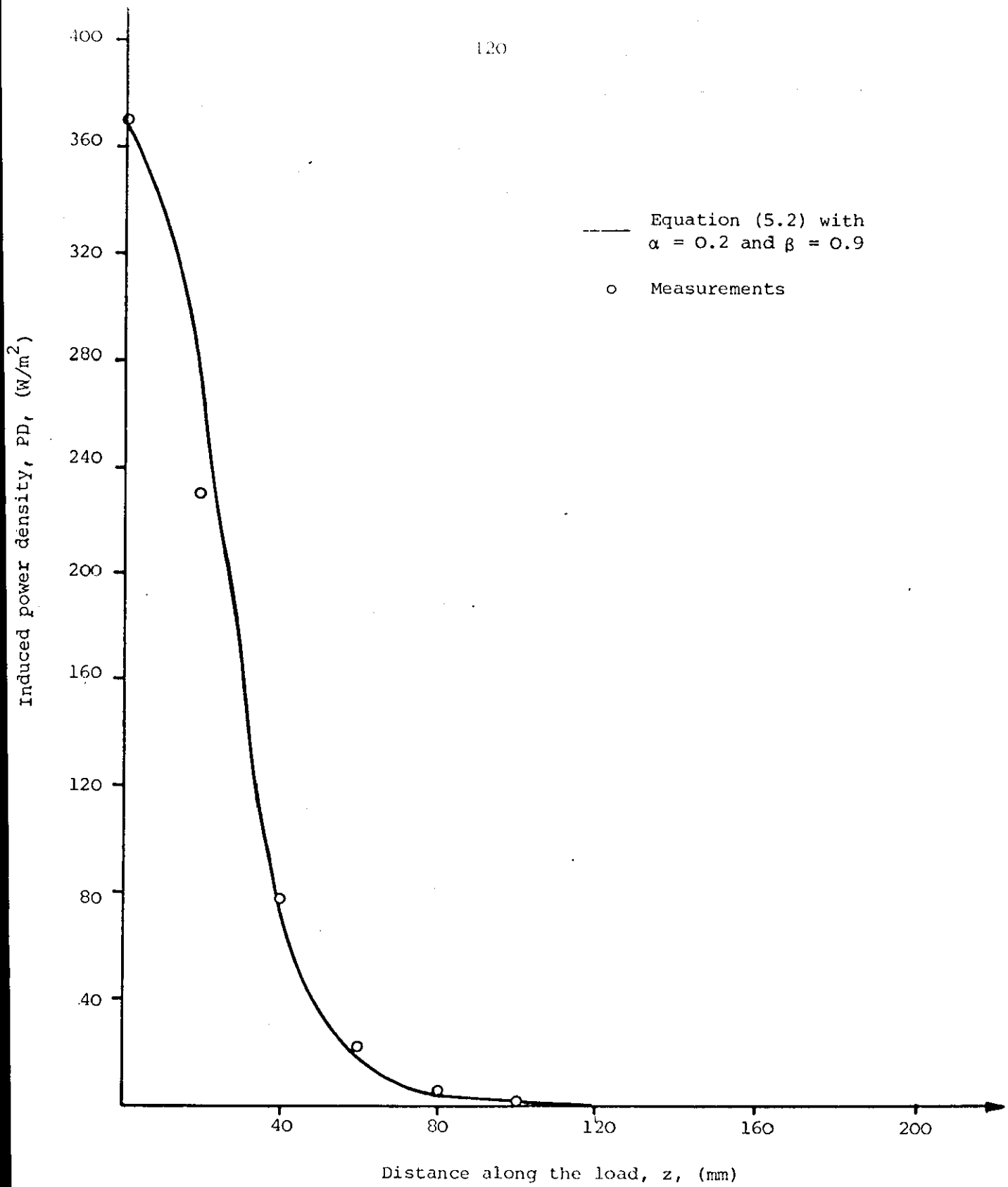


Fig. 5.25 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 23\text{mm}$

hand calculator. In all cases the resistivity of the aluminium load was taken as $2.824 \times 10^{-8} \Omega\text{m}$ which is corresponding to a temperature of 20°C [5.2]. This was the temperature of the aluminium slab during the experiments as the magnetic field was relatively low and it did not heat the slab.

The power density distribution due to a single conductor, Figs. 5.10-5.21, showed that the "theoretical" equation (4.9) of a filamentary conductor does not represent the practical situation accurately. The difference between the measurements and the values of the power density predicted by the theoretical equation (4.9) was up to 32%. Many experiments have been done to amend the theoretical equation (4.9) to the "practical" equation (5.2). Although the three correction factors were found experimentally, they can be calculated theoretically. The factor $\cos\sigma$ is a function of the distance z :

$$\cos\sigma = \frac{h}{\sqrt{h^2 + z^2}} \quad (5.3)$$

The constants α and β are functions of the effective height h_e . α is a linear function of h_e and it was found to be a straight line represented by:

$$\alpha = 10^{-2} h_e \quad (5.4)$$

The application of the computer routine (EO2ACF) was found to produce the equation of the other constant:

$$\begin{aligned} \beta = & -(2.642 \times 10^{-1}) + (7.736 \times 10^{-2})h_e - (1.539 \times 10^{-3})h_e^2 \\ & + (1.31 \times 10^{-5})h_e^3 - (3.968 \times 10^{-8})h_e^4 \end{aligned} \quad (5.5)$$

where h_e is in (mm). α and β are shown in Figs. 5.8 and 5.9 respectively. It should be mentioned here that in the case of an aluminium cylinder of radius equal to the thickness of the aluminium slab, the same constants can be applied. This will be shown later on in the study. The practical equation (5.2) predicts the current per unit length and can be applied within a range of 100% to 10% of the maximum induced power density, the error within this band being less than 10%. The practical equation applies to both conductors. The effective height of the circular conductor is:

$$h_e = \sqrt{h^2 - a^2}$$

It was found, experimentally, that the rectangular conductor can be replaced by a filament on its lower surface i.e. $h = g$ and $a = 0.0$.

The effect of the aluminium slab on the current distribution of the circular conductor, within the range of the experiments, is very small and can therefore be ignored. The current distribution

on the rectangular conductor has altered, due to the existence of the aluminium slab. This is the reason for the increase in the current density on the lower surface as shown in Fig. 5.7.

The surface current distribution of the conductor, which differs for both conductors, see Figs. 5.6 and 5.7, does not affect the applicability of equation (5.2), i.e. the induced power density distribution is a function of the total conductor current. This case greatly simplifies the application of this equation.

5.1.2 Number of Conductors

Before recommending the superposition theory as a coil design technique, it must be verified experimentally. It was decided to examine this theory in two stages. The first involved the conductor current to be approximately uniform and the proximity effect between the conductors to be at a minimum. The second stage is to examine the superposition technique in a more realistic context, that is when the current flows nonuniformly through the conductor, and there is a proximity effect between the conductors.

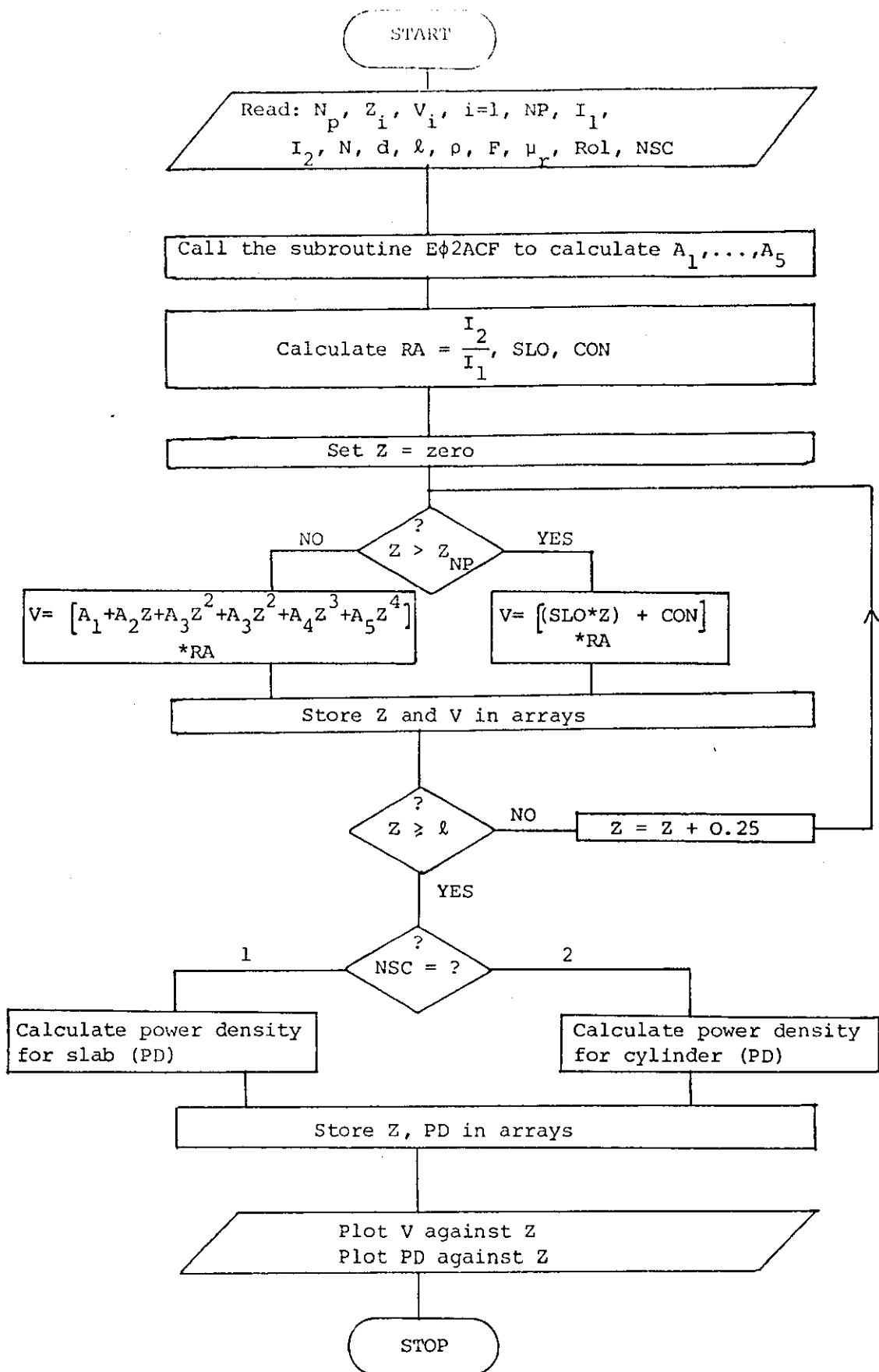
The experimental rig and the circuit diagram were identical to those of Figs. 5.1 and 5.2 respectively, but instead of using a single conductor 10 conductors were introduced in the study.

The practical applicability of the superposition theory with uniform conductor current was tested by taking a single circular conductor of 6mm diameter and situating it above the aluminium slab at various air gaps (19-77)mm. A current of 1000A flowed through this conductor, and the induced voltage distribution on the load was then measured. These voltages which are functions of the surface current density, see equation (5.1), were fed into a computer program "W-SC-FIT" to calculate the most accurate curve, representing the induced voltage $V(z)$ as a function of the distance (z) along the load in an equation which takes the following form:

$$V(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 \quad (5.6)$$

where a_0 , a_1 , a_2 , a_3 and a_4 are constants.

The above equation was then used to calculate the voltage from each individual conductor at each point on the surface of the load, and then they were summated to derive the predicted voltages (V) at different points along the load. If the load was a slab then the power density distribution on the load would be calculated from equations (5.1) and (4.21). The equation suitable for the cylindrical load will be dealt with in another chapter. The induced power densities along the load have now been calculated and the results can be presented in graphical form. The flow chart of this program is shown below.



The flow chart of the program "W-SC-FIT"

Then five similar conductors were connected in parallel above the slab, at identical air gaps to those which were applied for the single conductor. The coil pitch (d) between these conductors was 60mm, so that the proximity effect between them would be small. A total current of 3000A flowed through them, and the induced voltage distribution on the load was then measured. The current per conductor was 600A which was less than that for the single conductor. This is because it was not necessary to draw maximum current of 5000A from the transformer as 3000A would produce measurable eddy current on the load. Although many experiments have been done; only one graph is given, Fig. 5.26, to prove the applicability of the theory by showing the measured induced surface power density and that calculated from the superposition theory. The differences between the two values remain within an acceptable experimental error of 5%. The conductors were of a relatively small diameter and the proximity effect between them was small. In this case the current was distributed uniformly through the conductors. However, further on in this study it will be seen that this is not important criterion and that the superposition method can be applied even when the conductors are in close proximity to each other, and when the current flows nonuniformly. This Fig., also shows the nonuniformity of the power density along the load. This correlates with the theoretical results of Chapter 4.

To examine the superposition method under similar conditions to those in induction heating applications; that is a nonuniform flow of current through the conductors, groups of 5 and 10 rectangular

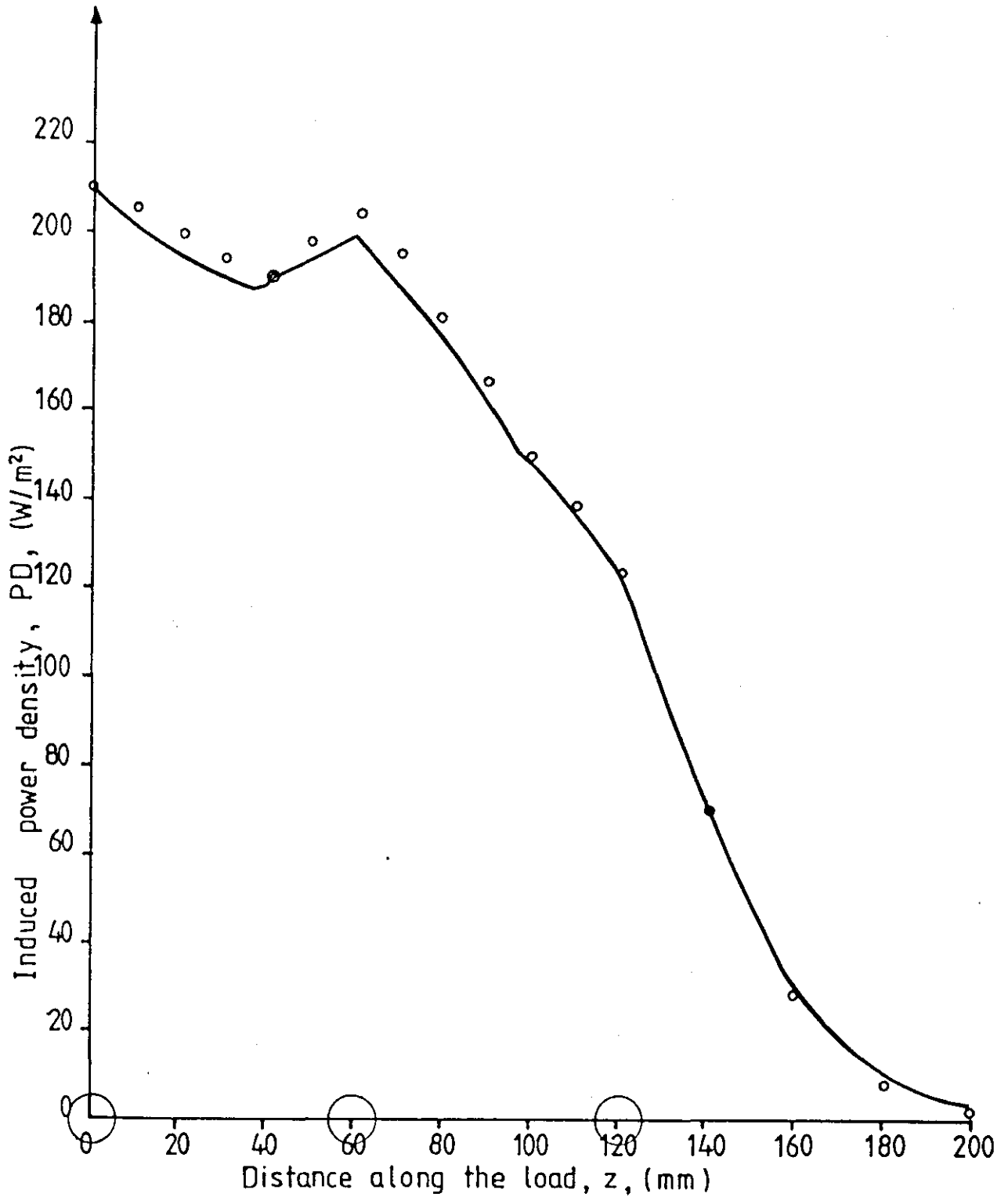


Fig.5.26 Power density distribution along the aluminium slab due to 5 circular conductors at air gap of 39 mm, coil pitch of 60 mm. and a current of 600A/conductor.

———— Prediction by superposition ○ Practical readings.

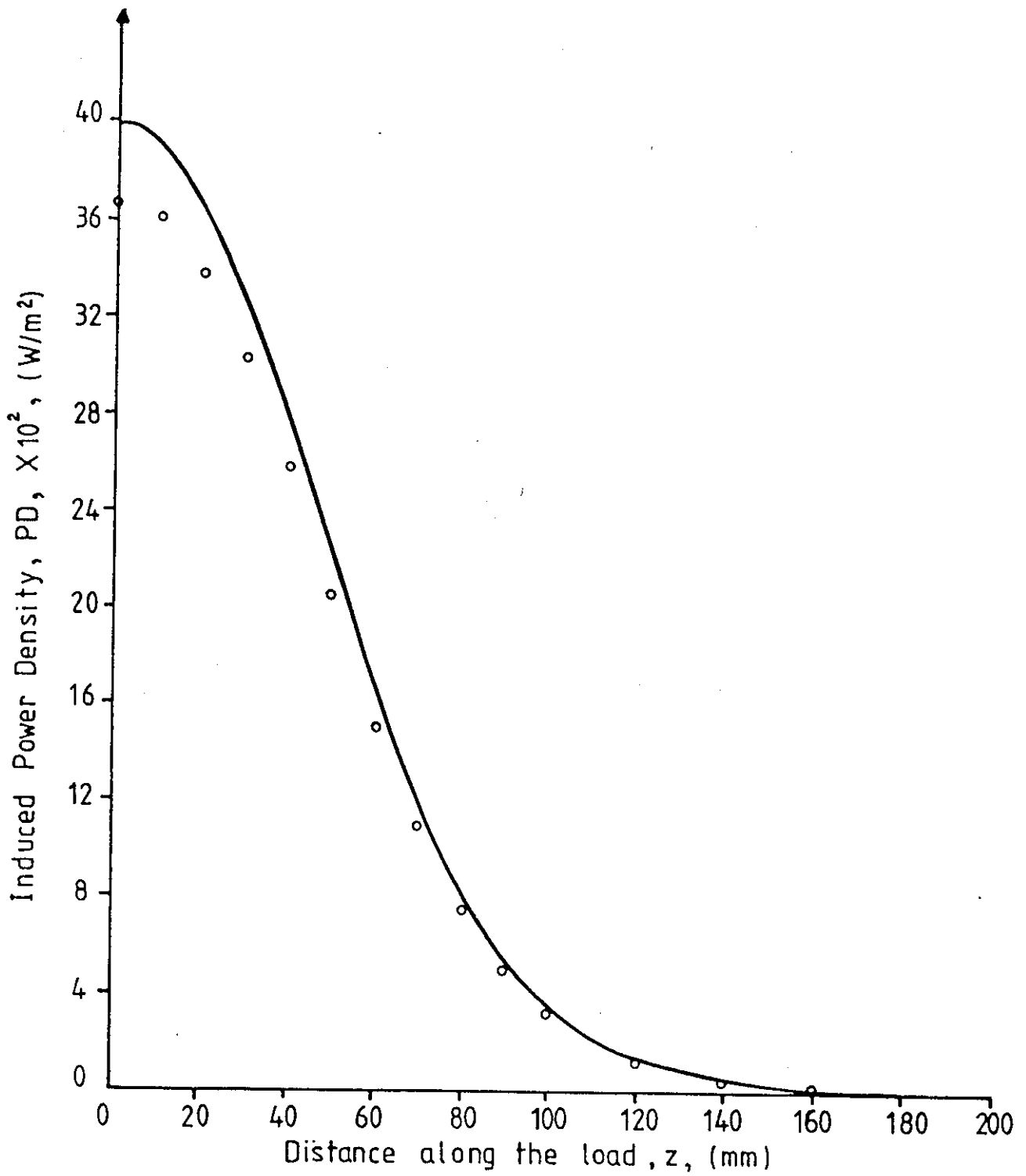


Fig.5.27 Power density distribution along the aluminium slab due to 10 rectangular conductors at air gap of 36mm, coil pitch of 11mm and a current of 300 A/conductor

— Prediction by superposition ○ Practical readings.

conductors, Figs. 5.3 and 5.4, were connected in parallel above the aluminium slab. The air gap between the conductors and the load was varied between 89-23mm. The coil pitch was between 62-11mm, that is down to a 1mm gap between the conductors. In the single conductor experiments, a current of 1000A flowed through the conductor, and a total current of 3000A was used in those involving more than one conductor.

In any of these experiments the difference between the measured induced power density and that calculated by the superposition theory did not exceed 8%. Only one graph is shown, Fig. 5.27, because all the results prove one thing, that is the validity of the superposition theory.

The applicability of the superposition theory was not affected by the surface current distribution of the conductors which was measured for an air gap of 36mm and coil pitch of 11mm. The results, Figs. 5.28 and 5.29, are represented in graphs which match the pattern recorded in Fig. 5.7. The surface current distribution is different on the two conductors as a result of their position amongst other conductors.

5.1.3 Discussion of the Preliminary Investigations

The surface power density distribution on an aluminium slab induced from single conductor was studied. It was found that the equation of a filamentary conductor (4.9) does not represent the actual situation

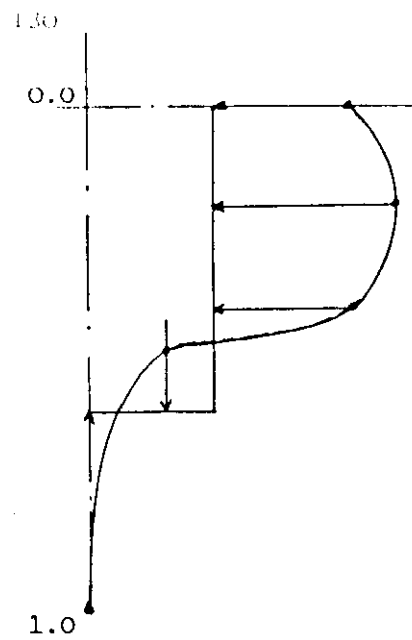


Fig. 5.28 Normalised surface current density on the solid part of a centre rectangular conductor amongst 10 conductors with 11mm coil pitch and at 36mm air gap from aluminium slab

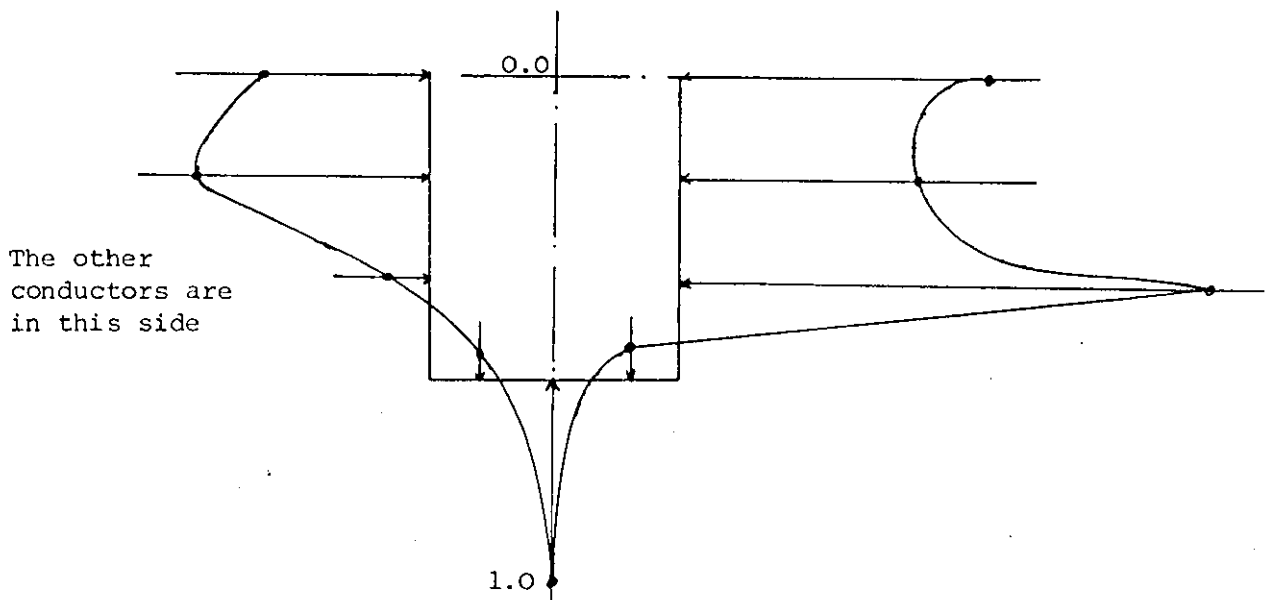


Fig. 5.29 Normalised surface current density on the end conductors of a group of 10 rectangular conductors with 11mm coil pitch and at 36mm air gap from aluminium slab

correctly. A better equation with three new factors was found experimentally. Mathematical expressions were given to calculate these factors.

The investigations on the power density induced on the aluminium slab due to more than one conductor have shown that the superposition method can be applied, regardless of the shape of the conductors or the distribution of the current flowing through them. The proximity effect on the conductors does not prevent the superposition method being applied and nor does the positioning of the conductor alter its effect on the load.

The experimental error was within 8% only and it is due to many factors; among them are the accuracy of the instruments, and the variation in the resistivity of the aluminium slab, due to the change in the ambient temperature. This 8% error in the calculation of the power density implies that the calculation of the voltage and the current induced on the load will be lower than this margin of error as the current is a function of the square root of the power.

The outcome of these experiments proved the theoretical work to be correct, when the coil pitch is large the induced power on the load resembles ripples and is not uniform. The magnitude of these ripples will increase when one decreases the air gap, and when the coil pitch is large they are noticeable.

This preliminary verification on a nonmagnetic load, which is similar to magnetic load subjected to high magnetic field, encourages one to examine the superposition method with ferromagnetic materials and low magnetic field strength. This is to be discussed in the next section.

5.2 Ferromagnetic Materials

Ferromagnetic workpieces are associated with applications within the metal forming industry. These materials are very suitable to induction heating due to their high permeability. Ferromagnetic billets of different sizes are heated efficiently by induction prior to forging, rolling and, to some extent, extrusion [5.3].

Also, the employment of these materials as a flux guide to protect the surrounding metal work from being heated by the stray flux is well known. Packs of low-loss material, such as nickel-iron or silicon-steel, are usually placed on the outside of the coil of a vessel heater or metal melting furnace, so as to confine the magnetic field close to the outside of the coil. This prevents the flux from spreading away from the outside of the coil and linking with surrounding metallic objects.

The aim of this section is to assess the applicability of the superposition method on applications whereby ferromagnetic materials exist in the system, such as off-the-bar forging and vessel heating. This is to be done by dividing this section into two parts. The first part is to study the effect of single conductor on a ferromagnetic load and to examine the superposition method on a workpiece of this kind.

The second part is to investigate the influence of ferromagnetic laminations on the distribution of the power density induced on the load and, also, to inspect the validity of the superposition method with the existence of the laminations.

5.2.1 Ferromagnetic Workpiece

In some induction heating applications such as vessel heating, it is required to heat the mild steel container to a temperature below the Curie point. The main problem associated with these applications is the variable permeability which should be taken into account when designing the installations. Many expressions which attempt to represent the variation of the permeability can be found in the literature [5.4-5.6]. Among these expressions is the Frohlich formula (5.7)

$$\mu = \frac{FA}{FB + H} + FC \quad (5.7)$$

This formula is used extensively in induction heating [5.7-5.9]. It is very accurate at low magnetic field and around the knee of the magnetization curve [5.7] but it gives a saturation flux density of $(FA + FC.H)$, while actual saturation curves never reach such a value.

As far as the Author is aware, there is no published work concerning the design of a coil to heat ferromagnetic load subjected to low magnetic field apart from the empirical formula of Thornton [5.10]

to estimate the coil number of turns. The difficulty in employing this formula lies in the necessity to know the efficiency and the power factor of the system beforehand.

The objective of the experimental work in this section is to assess the possible use of the superposition method on applications containing ferromagnetic workpieces subjected to low magnetic field. This is to be done by investigating the distribution of the power density on a ferromagnetic workpiece subjected to low magnetic field produced by single conductor and examining the superposition method when applied to a load of variable permeability.

The experiments were carried out on an EN3 mild steel slab of (500 x 300 x 25.4)mm dimensions which was taken as a workpiece instead of the aluminium slab on the experimental rig shown in Fig. 5.1. The electrical circuit was similar to that shown in Fig. 5.2.

First, it was necessary to investigate the current distribution on the conductor above such a load, so that the effect of the conductor current flow on the power density induced on the surface of the workpiece can be established. The current distribution on the surfaces of the 28mm diameter circular conductor and the rectangular conductor were measured by the J-probes attached to them. This current distribution was first measured when there was no load beneath the conductor, then the same measurements were repeated with the mild steel slab under the conductor. The results are shown in Figs. 5.30 and 5.31

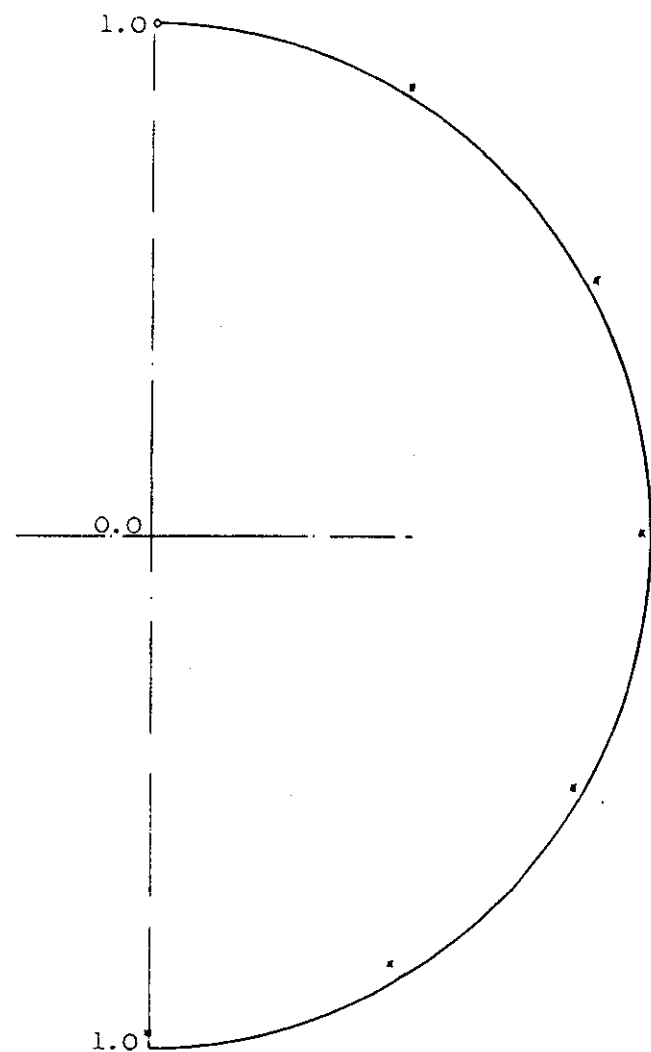


Fig. 5.30 Normalized current density distribution
on the circular conductor

— no load o mild steel slab at $h = 37\text{mm}$

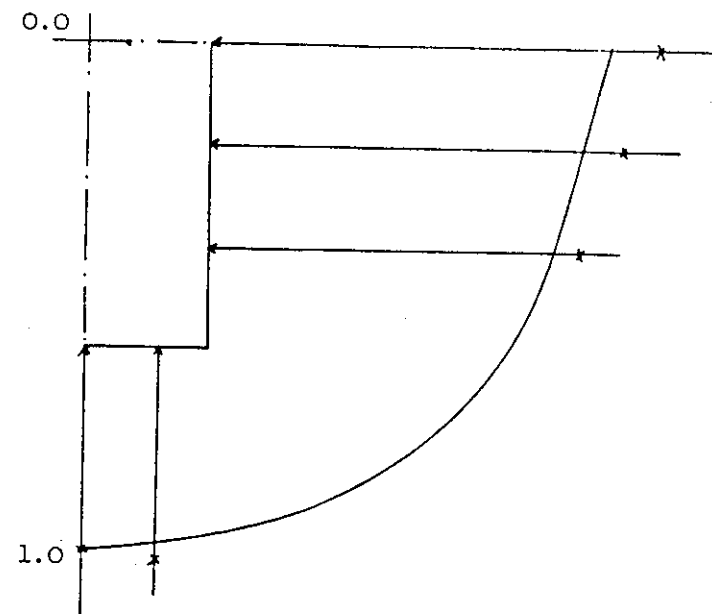


Fig. 5.31 Normalized current density distribution on
the rectangular conductor

— no load x mild steel at $g = 22\text{mm}$

for the circular and rectangular conductors respectively. The difference between the two measurements at any probe on the circular conductor did not exceed 4% when the height h was 37mm, and less than this for a greater height. For the rectangular conductor, this difference was within 8% only for a 22mm air gap and less than this, when the air gap was larger. As the change on the conductor current distribution due to the mild steel slab was small, the influence of this change on the power density induced on the load must be insignificant.

The accurate calculation of the surface power density induced on the mild steel slab necessitates taking the variation of the permeability with the magnetic field strength into account. Frohlich formula was employed in the calculation to represent this variation, as it has the advantages of simplicity and the satisfactory representation of the magnetization curve within the range of the experimental magnetic field strength [5.4, 5.7]. This formula has been used in much of the useful literature [5.7-5.9]. The constants of Frohlich formula for an EN3 mild steel can be found in reference [5.8]. This formula for this material takes the following form:

$$\mu = \frac{2.08}{1380 + H} + (1.257 \times 10^{-6}) \quad (5.8)$$

With the aid of a magnetic field strength probe, H-probe [5.11] the magnetic field was measured on the surface of the workpiece.

The probe consisted of 3 turns of constantan wire wrapped around a (100 x 3)mm perspex sheet of 30mm width. The two ends of the wire were twisted together and connected to the voltmeter by a coaxial wire. By attaching this probe to the surface of the load, the magnetic field strength can be derived from Faraday's law:

$$V_H = NA\omega\mu_o H \quad (5.9)$$

where V_H : The voltage across the H-probe (V)

N : The number of turns of the probe (3 turns)

A : The area of the probe (100 x 3) (mm²)

$\omega = 2\pi f$ (rad/s)

H : The magnetic field strength on the surface (A/m)

In addition, it was necessary to record the current density, by the J-probe, on the surface of the slab induced by different conductors at different heights above the slab.

The following method was taken to measure the power density on the surface of the mild steel slab:

1. The surface magnetic field strength was calculated from the measured voltage across the H-probe V_H as:

$$H = \frac{V_H}{NA\omega\mu_o} \quad (\text{A/m}) \quad (5.10)$$

2. The corresponding permeability was calculated by substituting the value of H from equation (5.10) into equation (5.8)
3. The resistivity of the mild steel was taken as $16.78 \times 10^{-8} \Omega\text{m}$ [5.8]. The penetration depth at each point was calculated from the values of the resistivity and local permeability as:

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_o \mu_r}} \quad (5.11)$$

4. The surface current density was calculated from the measured voltage across the J-probe V_J as:

$$J = \frac{V_J}{\rho \ell} \quad (5.12)$$

5. As the skin depth was small in comparison with the thickness of the slab, the surface power density was calculated by substituting the values of ρ , J and δ into the following equation

$$PD = \frac{\rho J^2 \delta}{2} \quad (5.13)$$

A program "Magnetic Material" was written to a hand calculator to cater for these calculations.

The power density induced by the single conductor on the surface of the mild steel slab is shown in Figs. 5.32-5.35. The involvement of many factors in the calculations makes it difficult to find a simple formula to represent the distribution of the power density induced on the surface of the load. However, it was noticed that the maximum

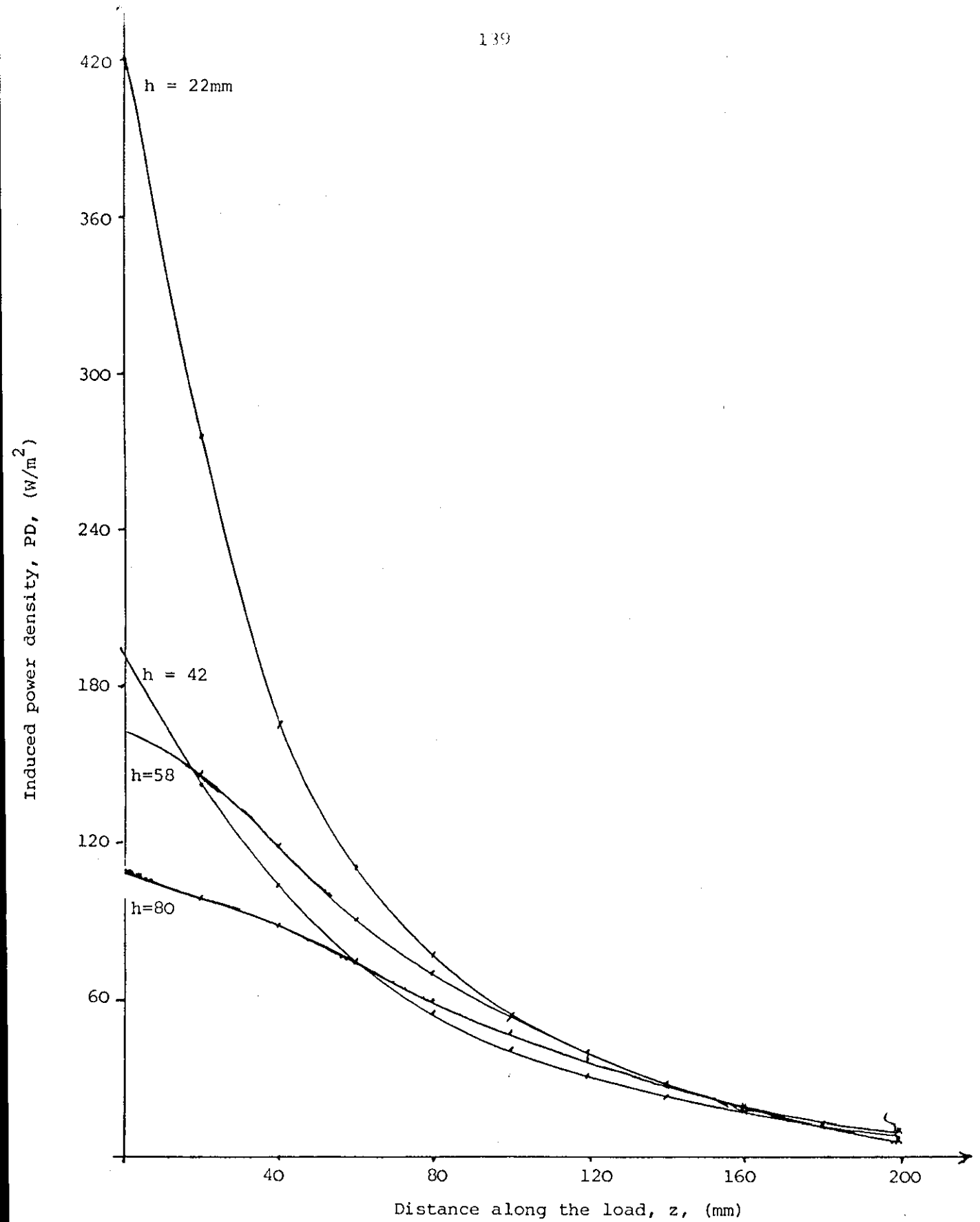


Fig. 5.32 Surface power density distribution along the mild steel slab due to 1000A flowing through circular conductor of 6mm diameter at different heights (h)

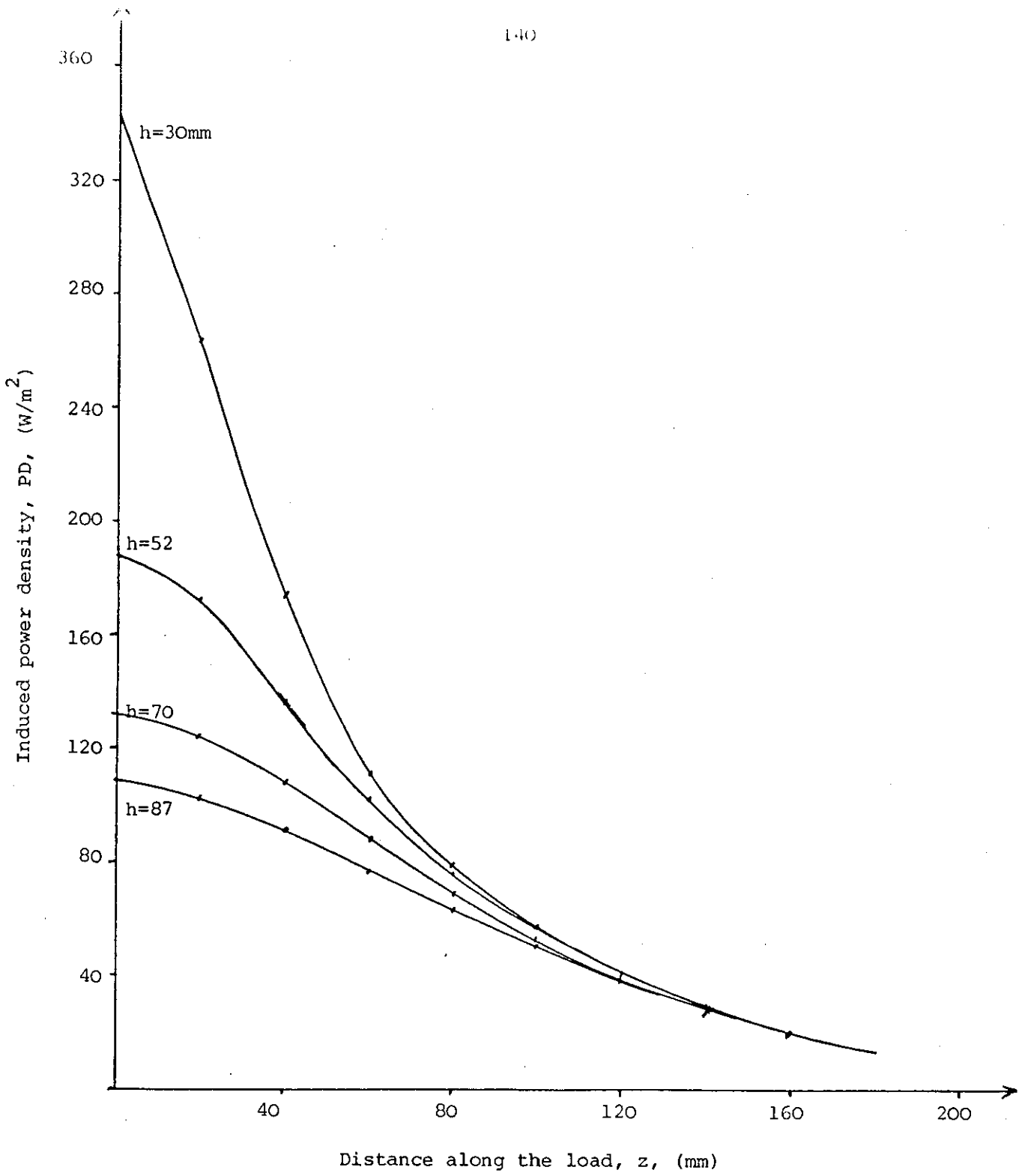


Fig. 5.33 Surface power density distribution along the mild steel slab due to 1000A flowing through circular conductor of 15mm diameter at different heights (h)

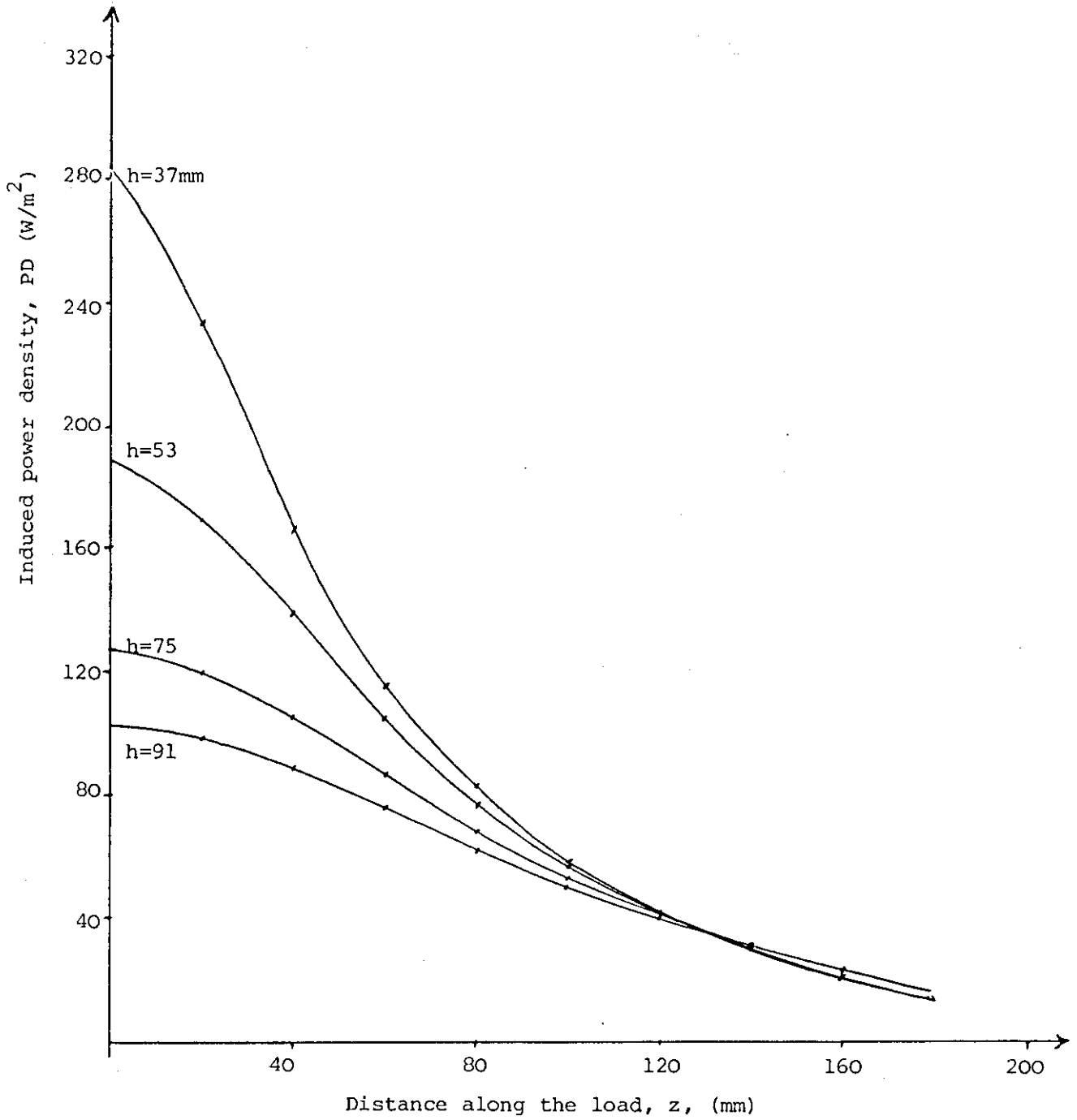


Fig. 5.34 Surface power density distribution along the mild steel slab due to 1000A flowing through circular conductor of 28mm diameter at different heights (h)

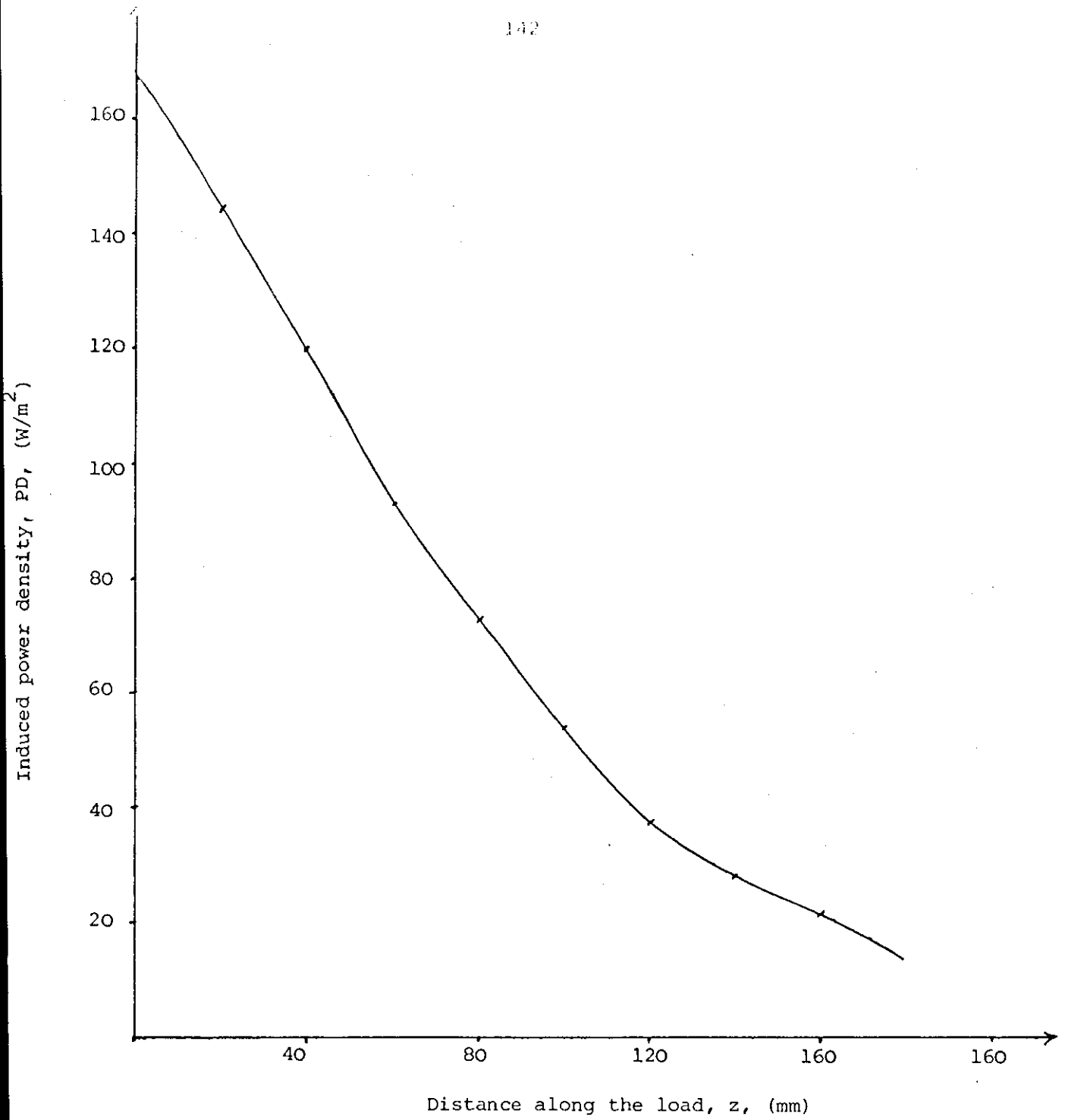


Fig. 5.35 Surface power density distribution along the mild steel slab due to 1000A flowing through rectangular conductor at 41mm air gap.

power density induced on the surface of the load, i.e. beneath the conductor, is a function of the effective height, Fig. 5.36, and takes the following form:

$$PD = \frac{K}{h_e} \quad (5.14)$$

where K is a constant and in this case was found to be 9300.

The factors which affect the power induced on a nonmagnetic load have the same effects for a ferromagnetic load; they are:

1. The induced power density decreases with the increase of both z and h.
2. A conductor of small diameter will concentrate the induced power density beneath the conductor more than would a large diameter conductor.

The next stage was to investigate the applicability of the superposition method on the mild steel slab. This has been done by using 5 circular conductors of 6mm diameter and 5 rectangular conductors respectively.

The voltages across the J-probe and the H-probe induced by a single conductor were combined with the superposition principles to predict the performance of more than one conductor. These calculations were facilitated by the computer program "FITTING1", which was developed for the PRIME 400 system. These values were then substituted into the program "Magnetic Material" to predict the performance of a number of conductors.

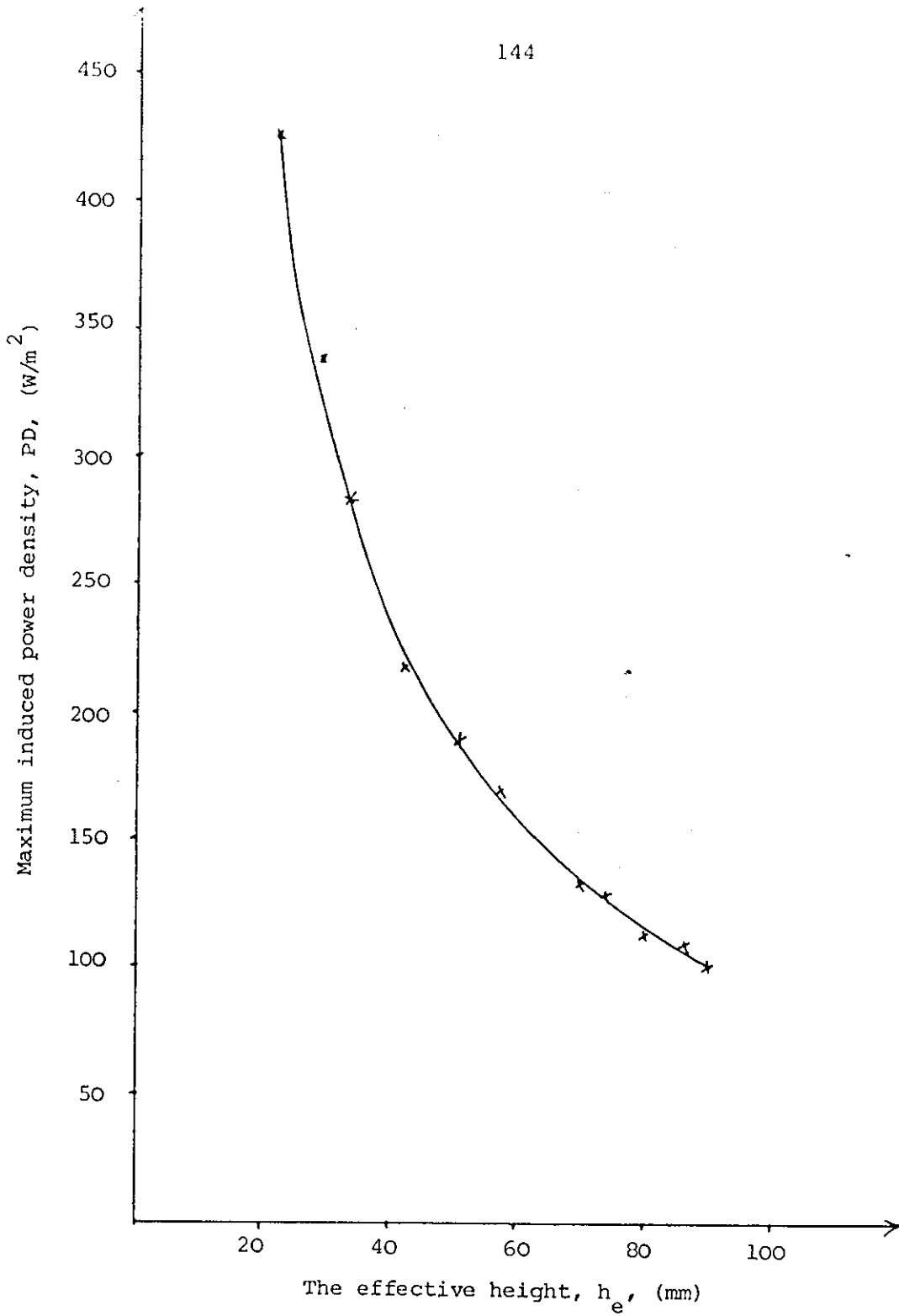
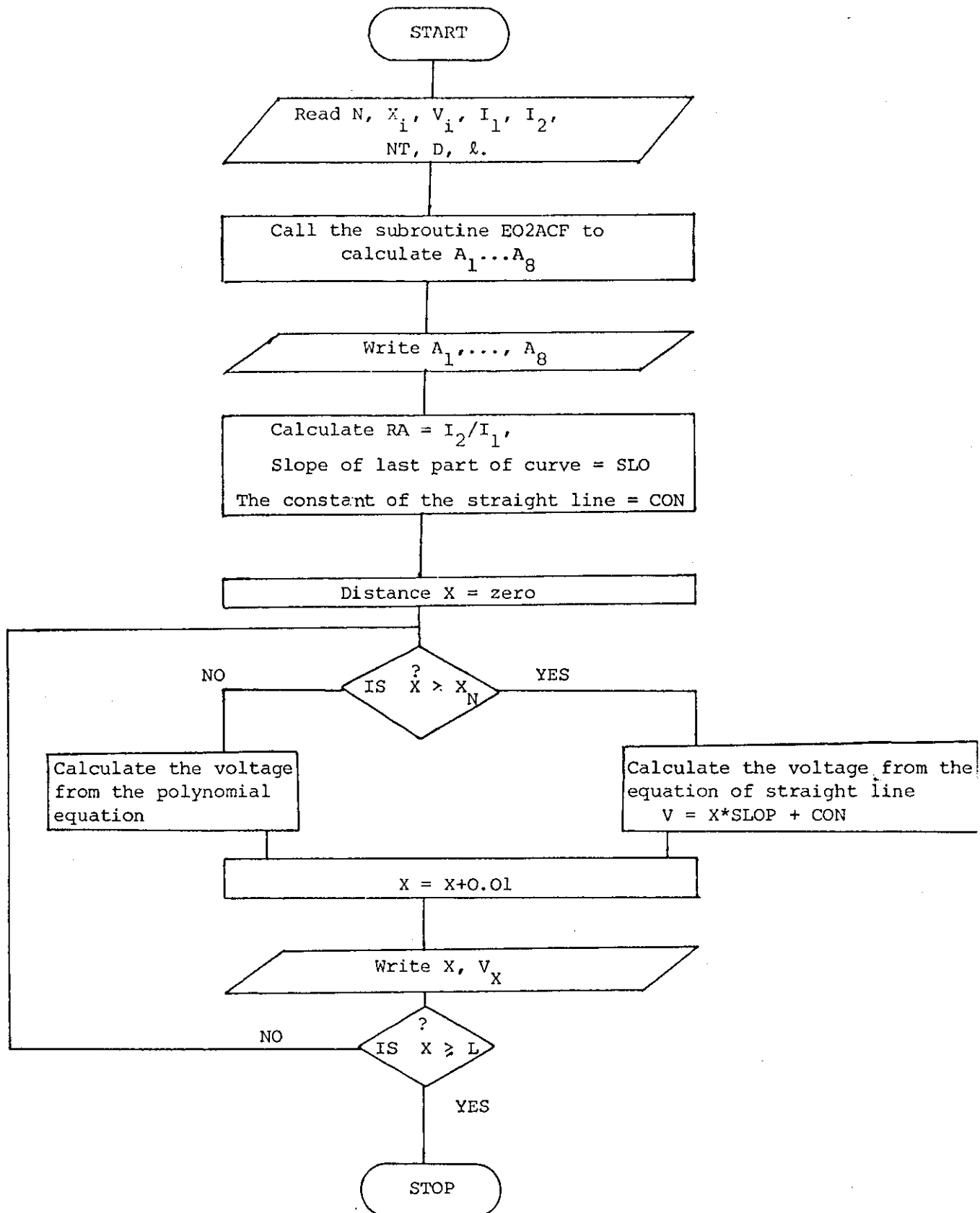


Fig. 5.36 The maximum power density induced directly beneath the conductor as a function of the effective height (h_e)

— calculated from $\frac{K}{h_e}$
 x measurements



The flow chart of the program 'FITTING1'

The measured power density and that predicted by the superposition method are shown in Figs. 5.37 and 5.38 for the circular conductors. The agreement between the two values are very good and the discrepancy did not exceed 6% for air gaps between 80mm and 22mm.

Before examining the superposition method with the rectangular conductors; it was necessary to investigate the current distribution on the conductors so that the effect of this distribution can be assessed. The current density distribution on a rectangular conductor in amongst others and the superposition method with such conductors was investigated.

Despite the nonuniformity and dissimilar conductor current distribution, Figs. 5.39 and 5.40, the superposition still applies with very good accuracy, Fig. 5.41. The difference between the measured and predicted power density was within 6% only. The good agreement between the measured and predicted values for the two kinds of conductors, circular and rectangular, proves that the superposition technique can be applied with ferromagnetic workpiece.

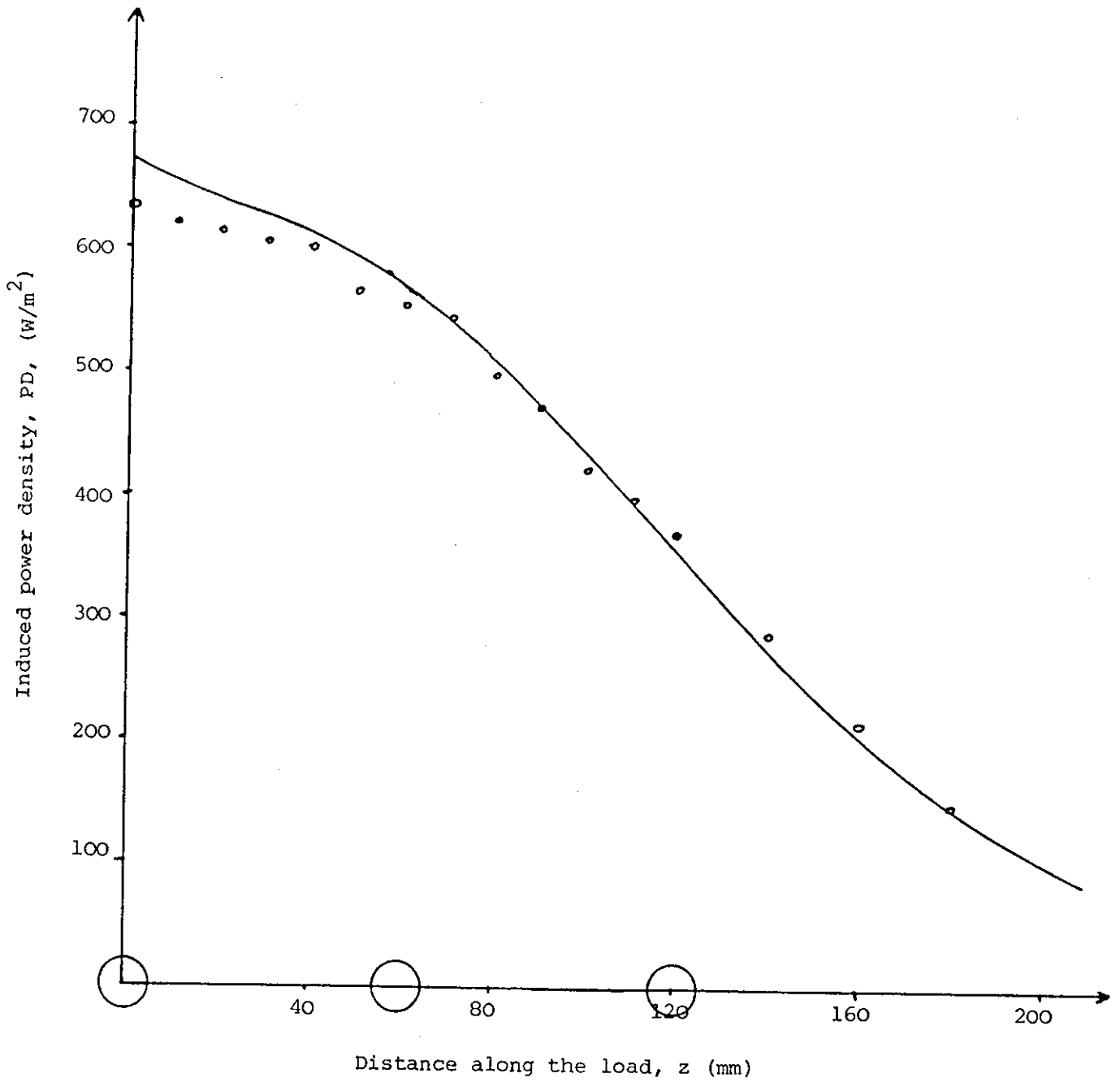


Fig. 5.37 Power density distribution along the mild steel slab due to 5 circular conductors at air gap of 77mm, coil pitch of 60mm, and a current of 600A/conductor.

— Prediction by superposition o Practical readings

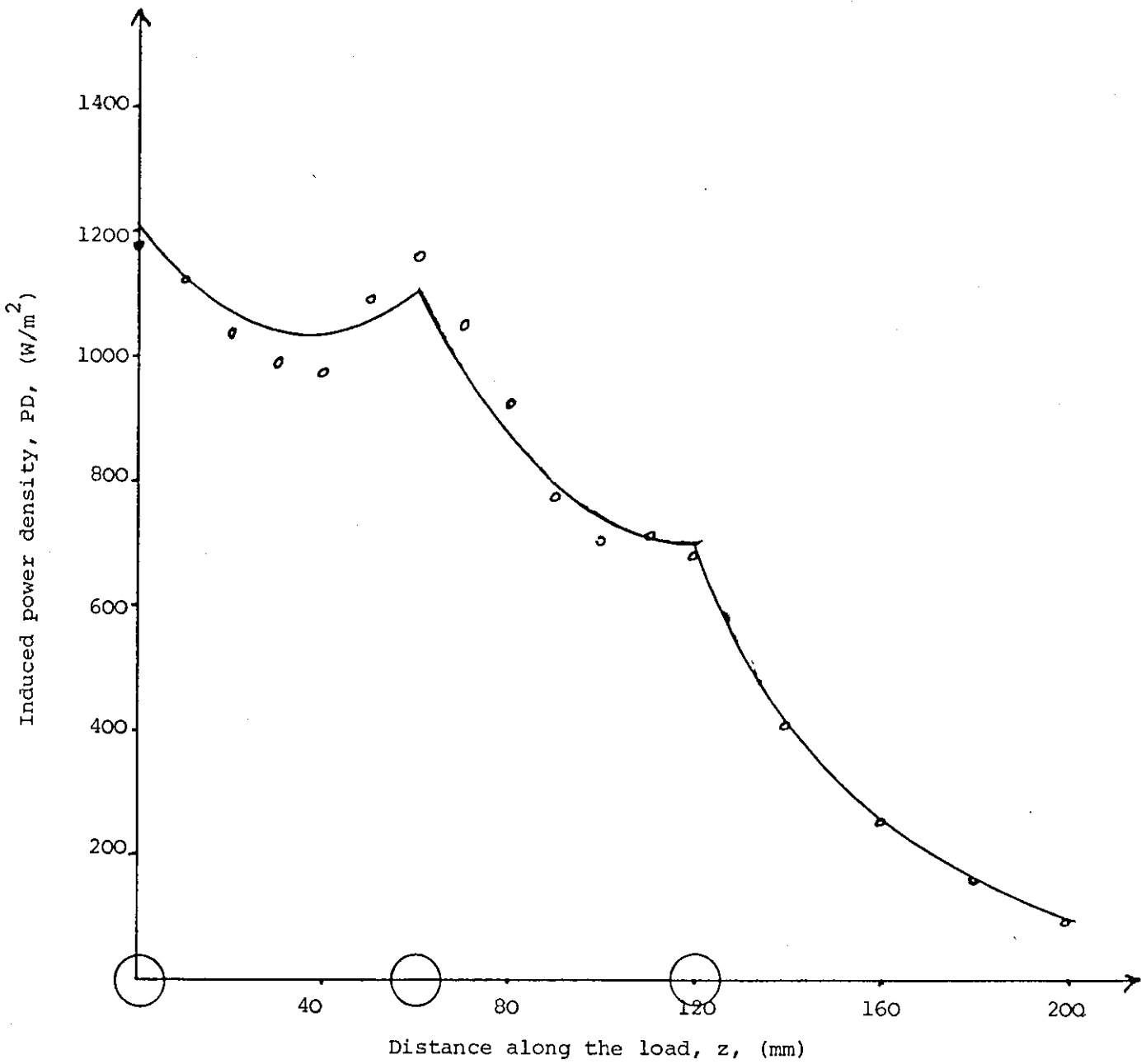


Fig. 5.38 Power density distribution along the mild steel slab due to 5 circular conductors at air gap of 19mm, coil pitch of 60mm and a current of 600A/conductor

— Prediction by superposition o Practical readings

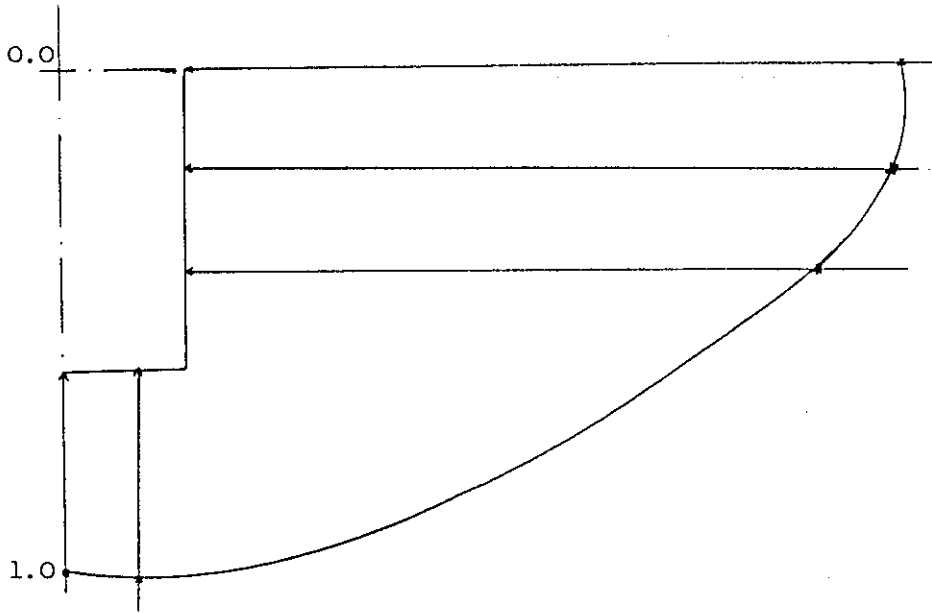


Fig. 5.39 Normalized current distribution on the rectangular conductor when placed centrally among 5 conductors with mild steel slab at $g = 41\text{mm}$ and coil pitch of 31mm

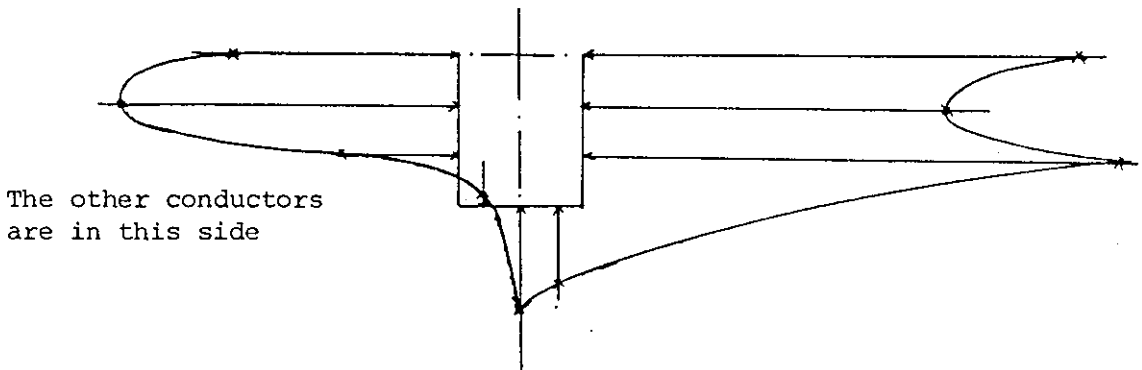


Fig. 5.40 Normalized current distribution on the end rectangular conductor; the others being on one side only with mild steel slab at $g = 41\text{mm}$ and coil pitch of 31mm

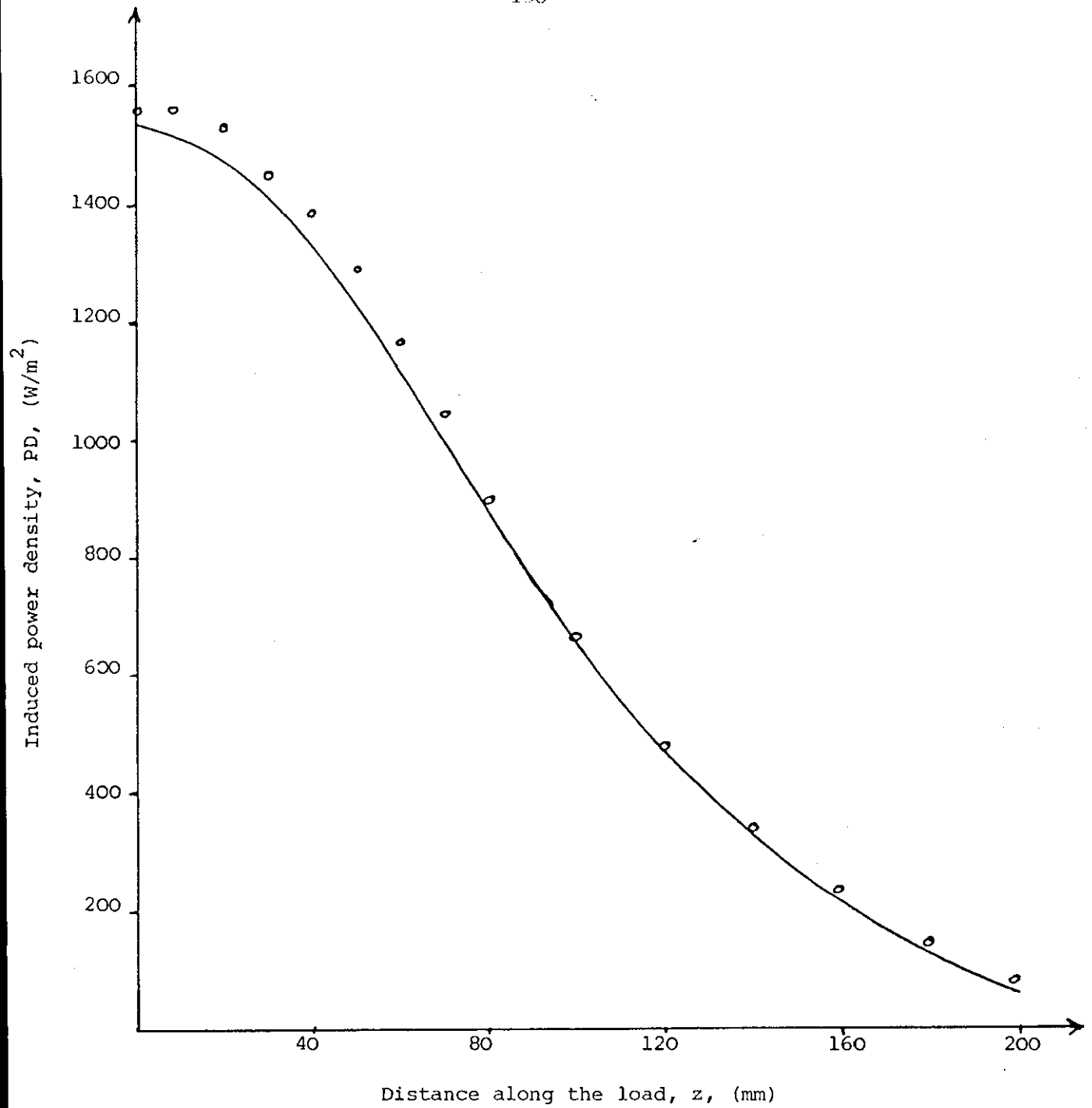


Fig. 5.41 Power density distribution along the mild steel slab due to 5 rectangular conductors at air gap of 41mm, coil pitch of 31mm and a current of 600A/conductor.

— Prediction by superposition o Practical readings

5.2.2 The Lamination Packs

In some induction heating applications, such as vessel heating, lamination packs are installed around the coil. The main reason for using them is to channel the flux in the region outside the coil so that it does not link the surrounding metallic objects, which will heat these objects and, also, might result in sparking. The efficiency will be improved also by the employment of these flux guides due to the reduction on the stray losses and the magnetizing current.

The object of this section is to inspect the applicability of the superposition method in a system containing laminations, before this technique can be recommended as a coil design method to installations involving the laminations. This is to be done by studying the effect of the laminations on the power density induced on the workpiece due to single conductor and number of conductors.

The experiments were carried out on the same aluminium slab employed in Section 5.1 and by using the same experimental rig as shown in Fig. 5.1, with the addition of the laminations. The circuit diagram is that of Fig. 5.2 and the configuration of the system is shown in Fig. 5.42.

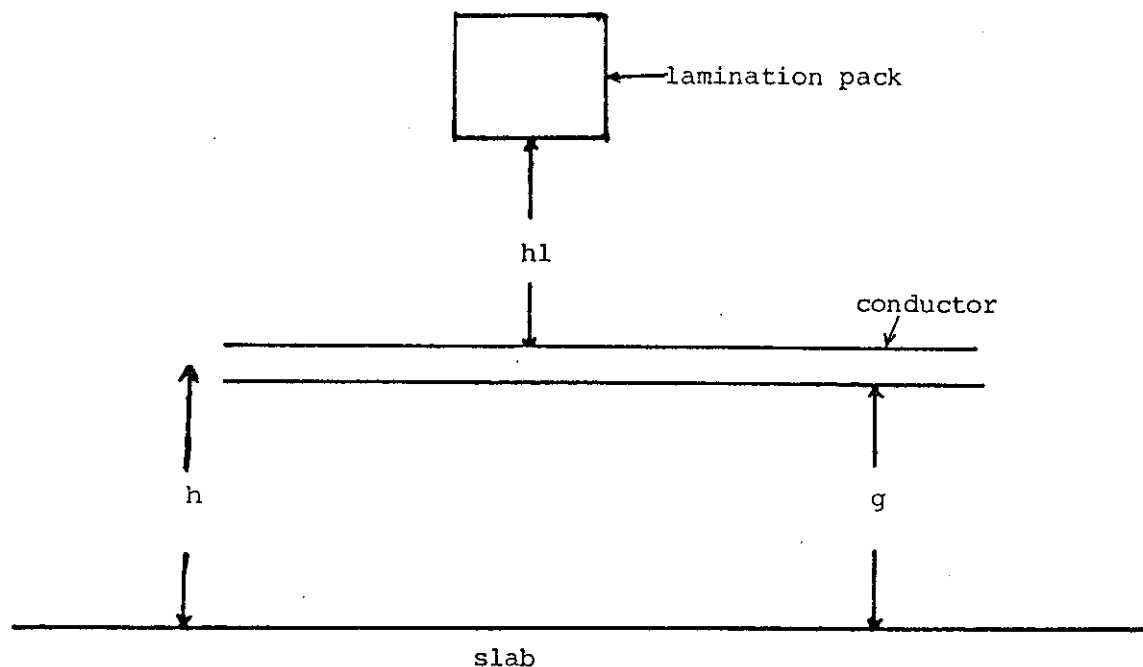


Fig. 5.42 The load, conductor and laminations

A pack of nickel-iron laminations was used in the experiments; the pack contained 250 sheets of the following dimensions 45 x 280 x 0.15mm. It was supported by 2 stainless steel bolts. The pack was then covered with a fibre glass tape to maintain a close proximity between the laminations and to protect their edges.

A circular conductor of 28mm diameter and a rectangular conductor were used in order to investigate the laminations effect on a single conductor. The air gap g between the aluminium slab and the conductor was held constant at 40mm. The current in the conductor was permanently fixed at 1000A. The voltage induced at different points on the load was measured by the current density

probe and the current distribution on the conductors was also recorded, so that the effect of this distribution could be assessed.

When investigating the effect of the laminations on the power density induced on the load a pack of laminations was positioned above the conductor at distance $h_l = 5, 11, 20$ and 35mm respectively. The pack was above the centre of the aluminium slab, where the induced surface power density was measured. Figs. 5.43-5.50 show these measurements together with the calculated values according to equation (5.2). The differences between the two values did not exceed 5%; this proves that the practical equation (5.2) can be used even when there are laminations above the conductor. The constant α is the same as in the case of no laminations, while the other constant β increases when decreasing the distance h_l , see Table 5.1. Decreasing the distance h_l increases the power density induced on the load i.e. increases the coupling.

$h_l (\text{mm})$	β for circular conductor	β for rectangular conductor
∞ (no laminations)	1.13	1.07
35	1.39	1.16
20	1.46	1.19
11	1.51	1.25
5	1.58	1.3

Table 5.1 The variation of β with h_l

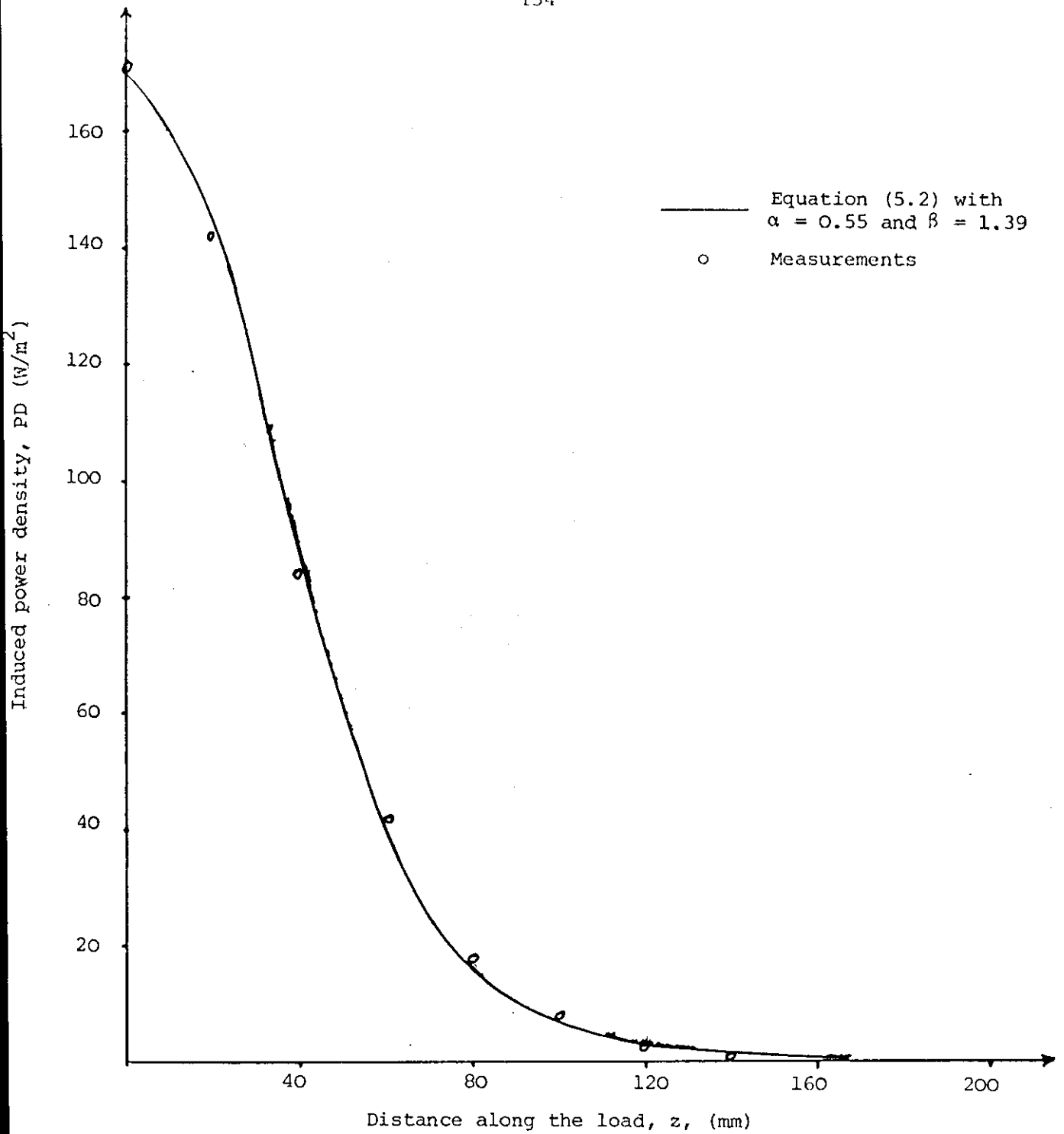


Fig. 5.43 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm at $h = 54\text{mm}$ with lamination pack at $h_1 = 35\text{mm}$

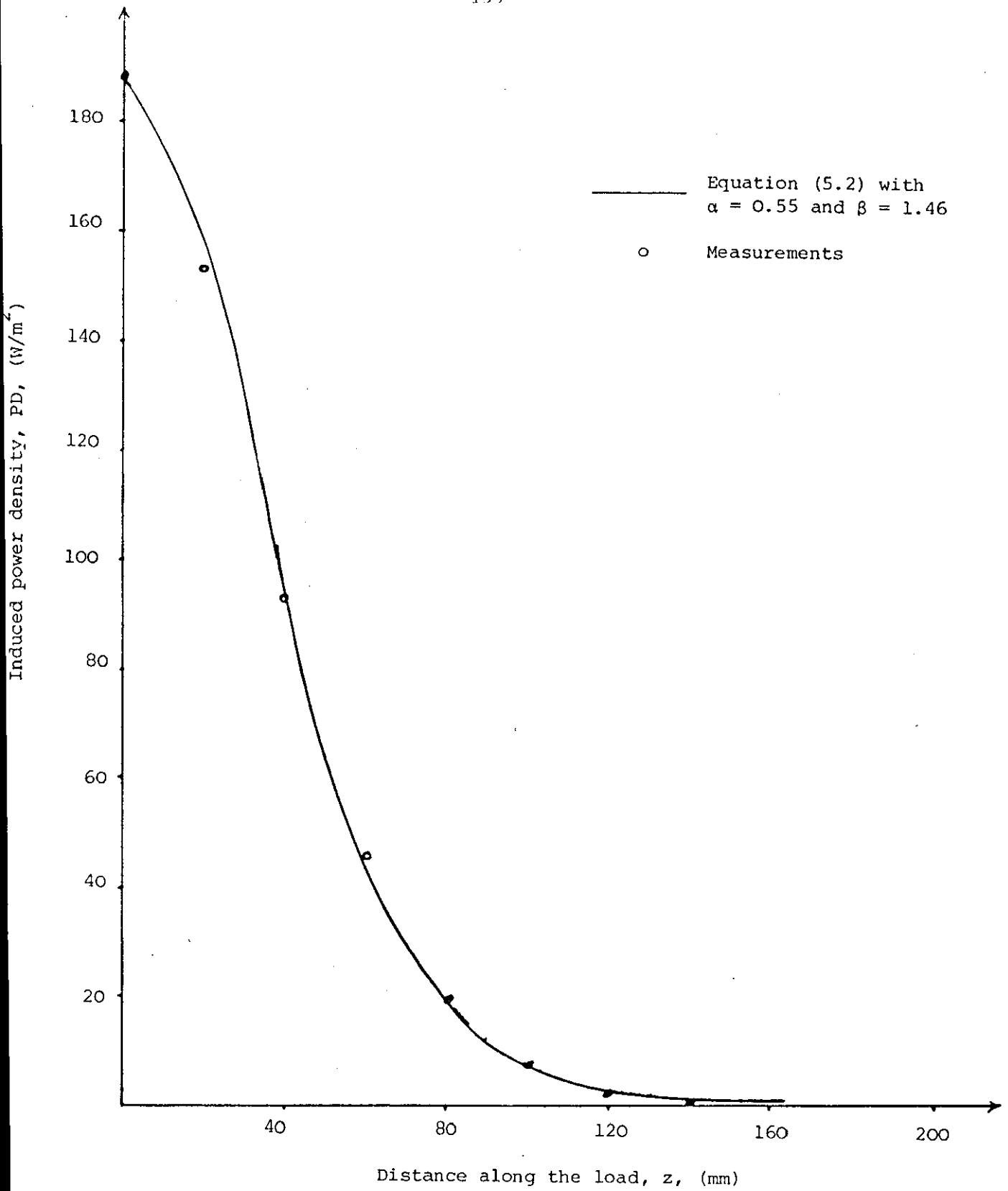


Fig. 5.44 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 54\text{mm}$ with lamination pack at $h_1 = 20\text{mm}$

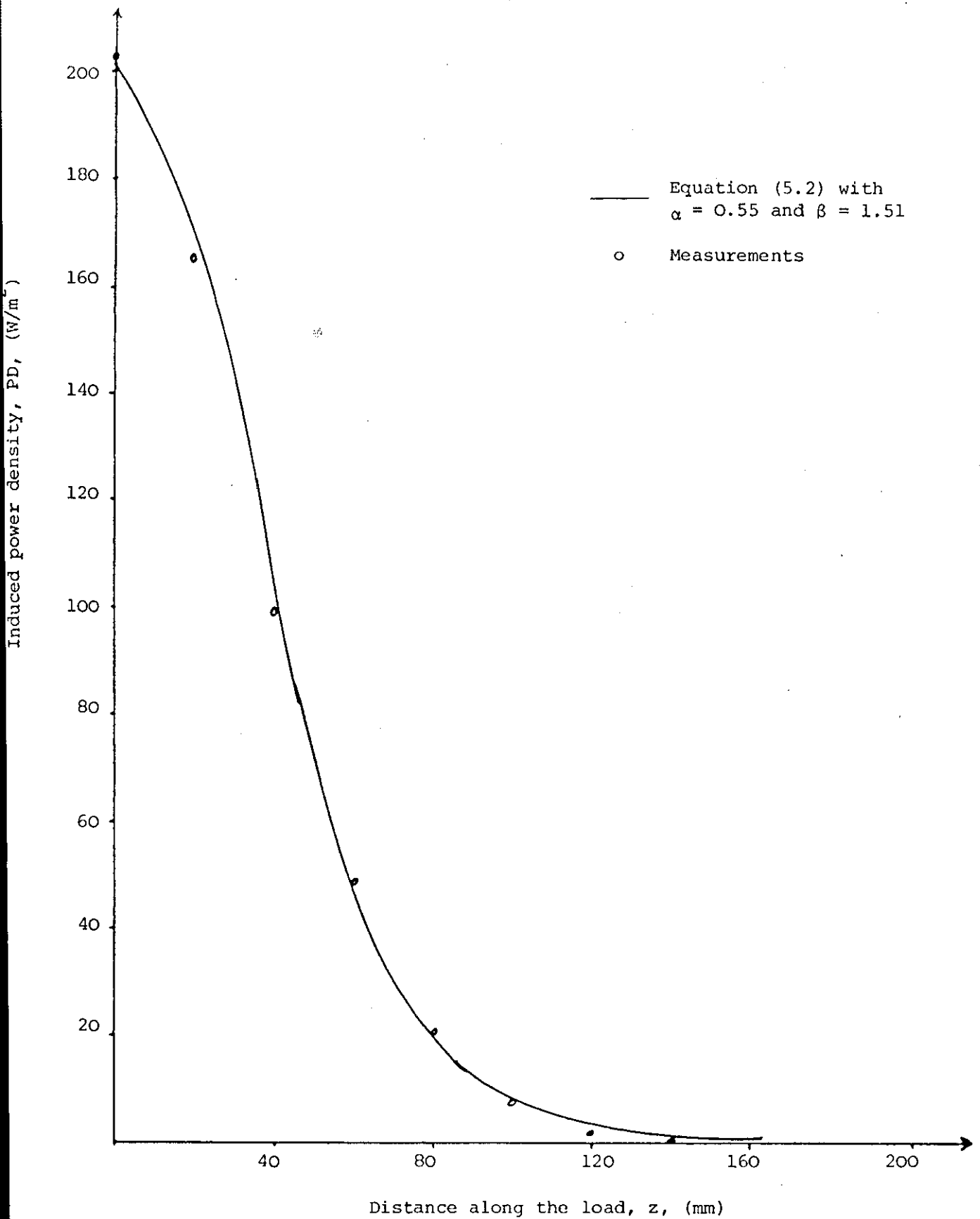


Fig. 5.45 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm at $h = 54\text{mm}$ with lamination pack at $h_1 = 11\text{mm}$

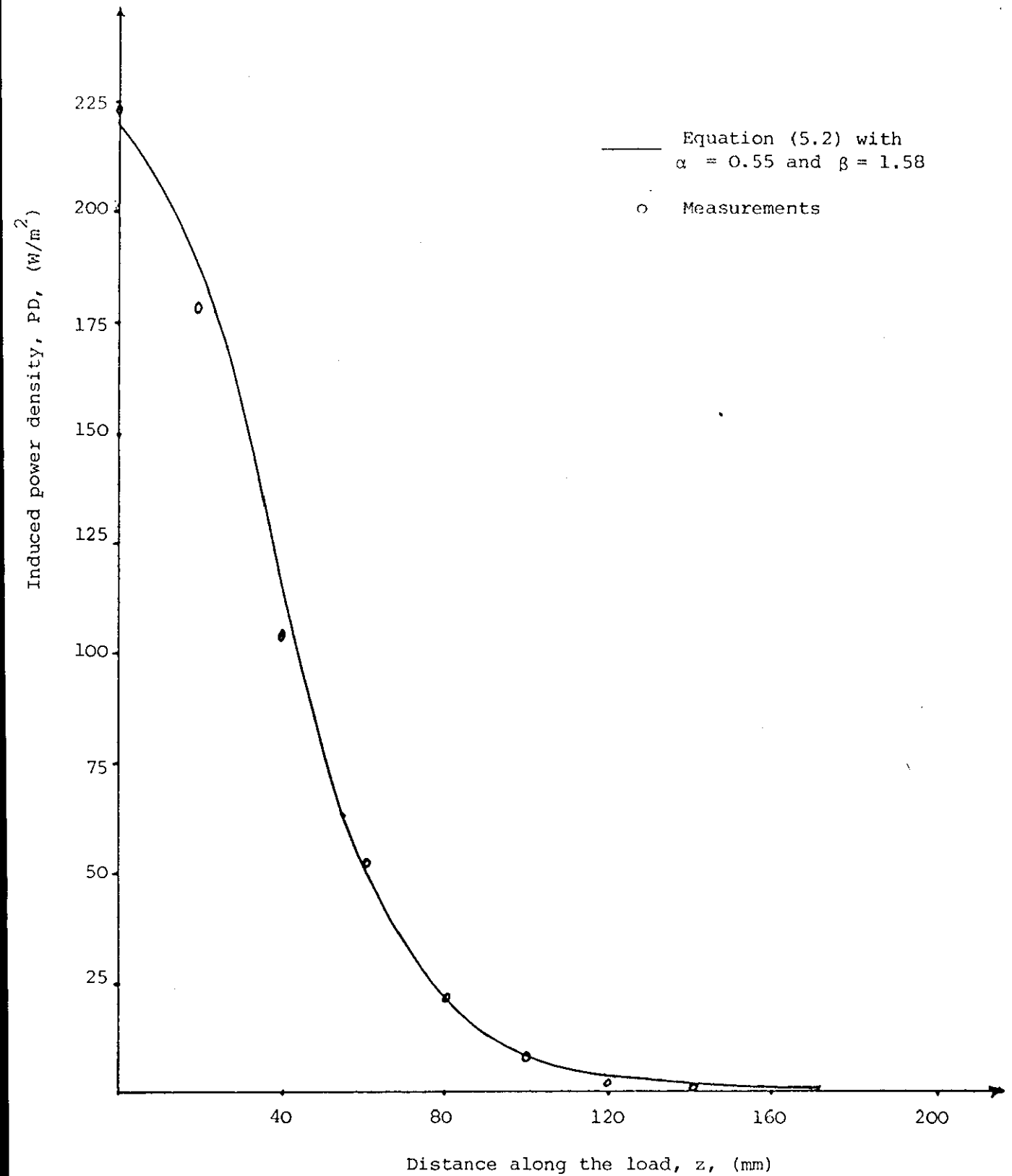


Fig. 5.46 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 54\text{mm}$ with lamination pack at $h_1 = 5\text{mm}$

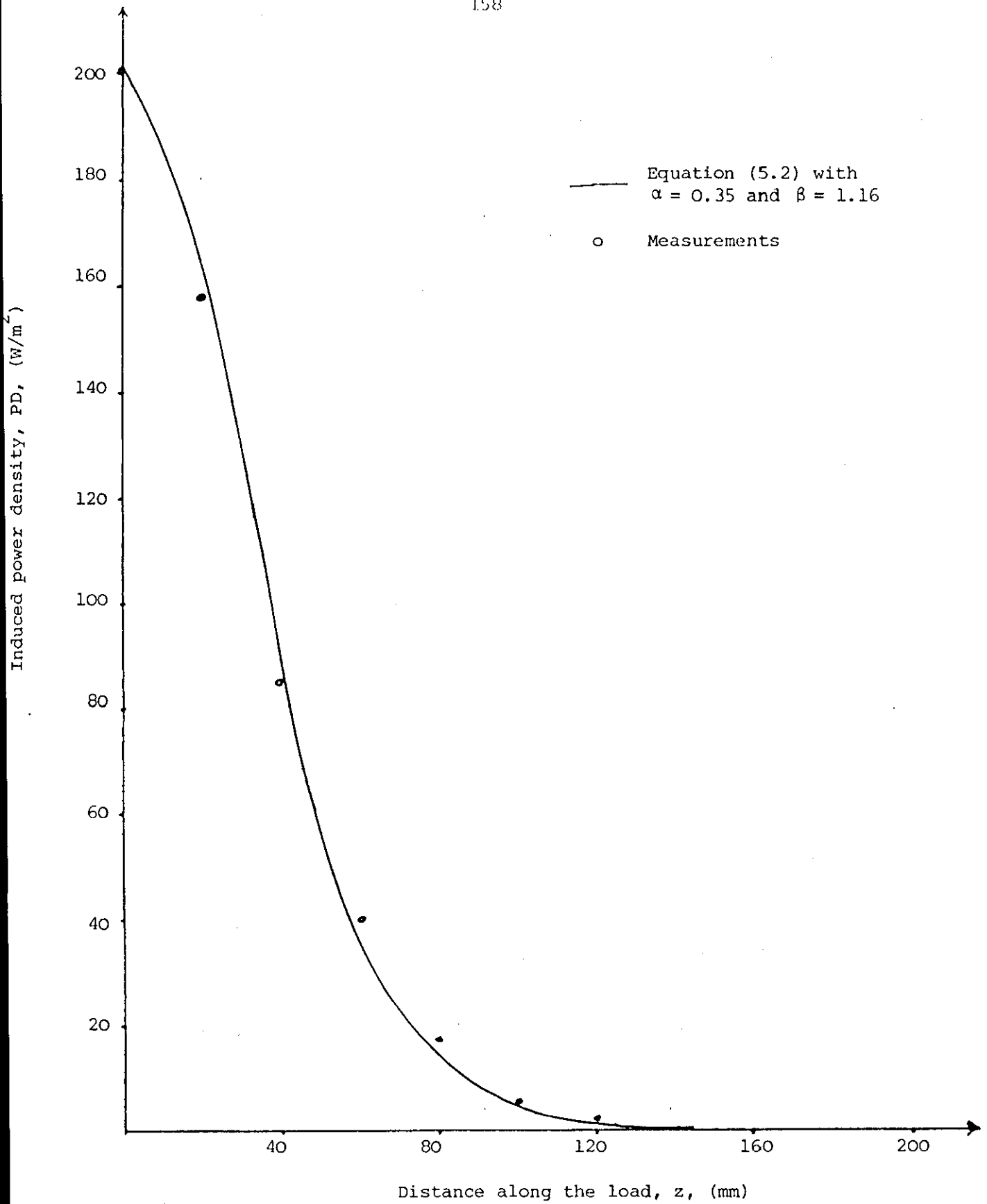


Fig. 5.47 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$ with lamination pack at $h_1 = 35\text{mm}$

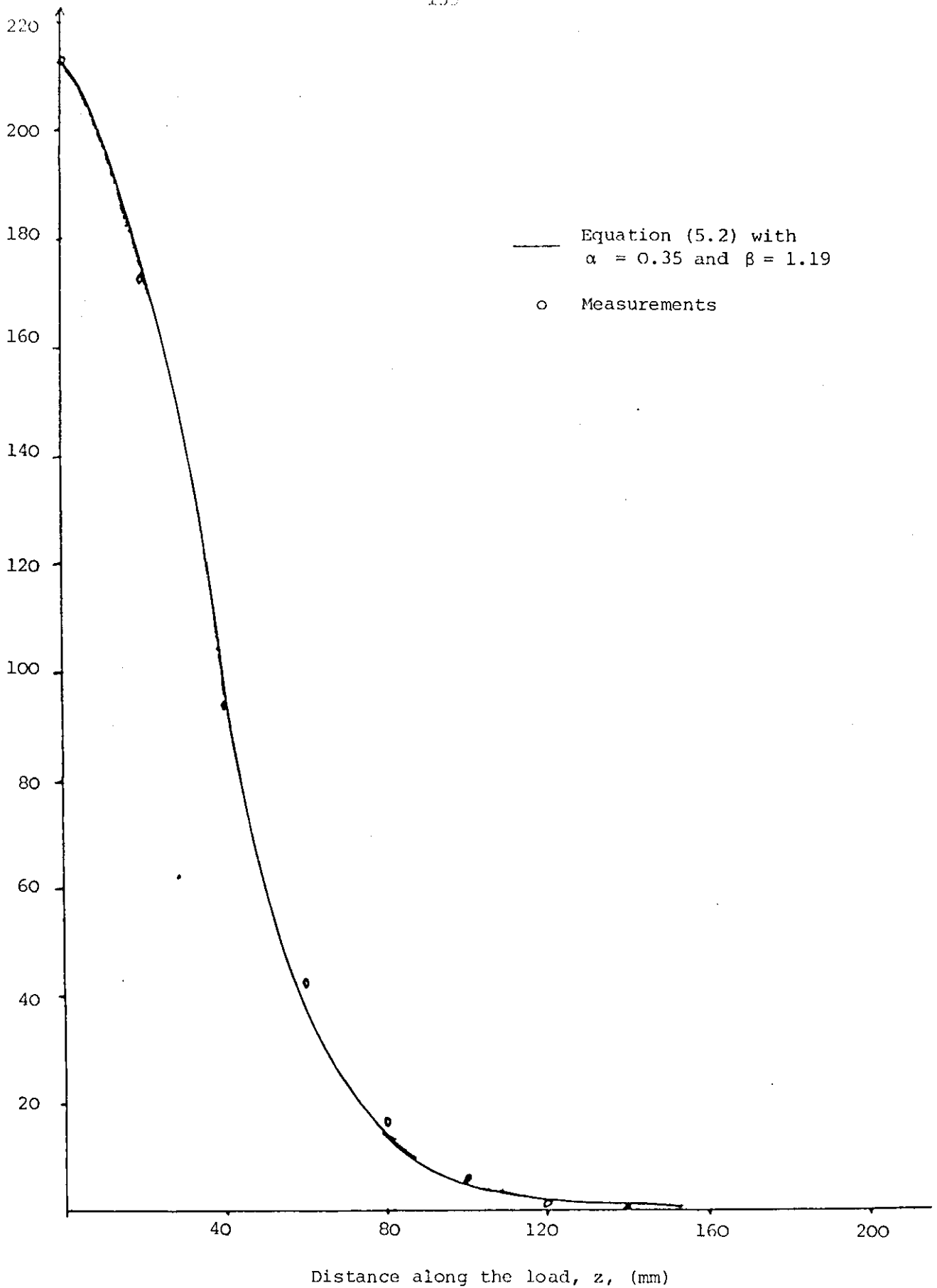


Fig. 5.48 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$ with lamination pack at $h_1 = 20\text{mm}$

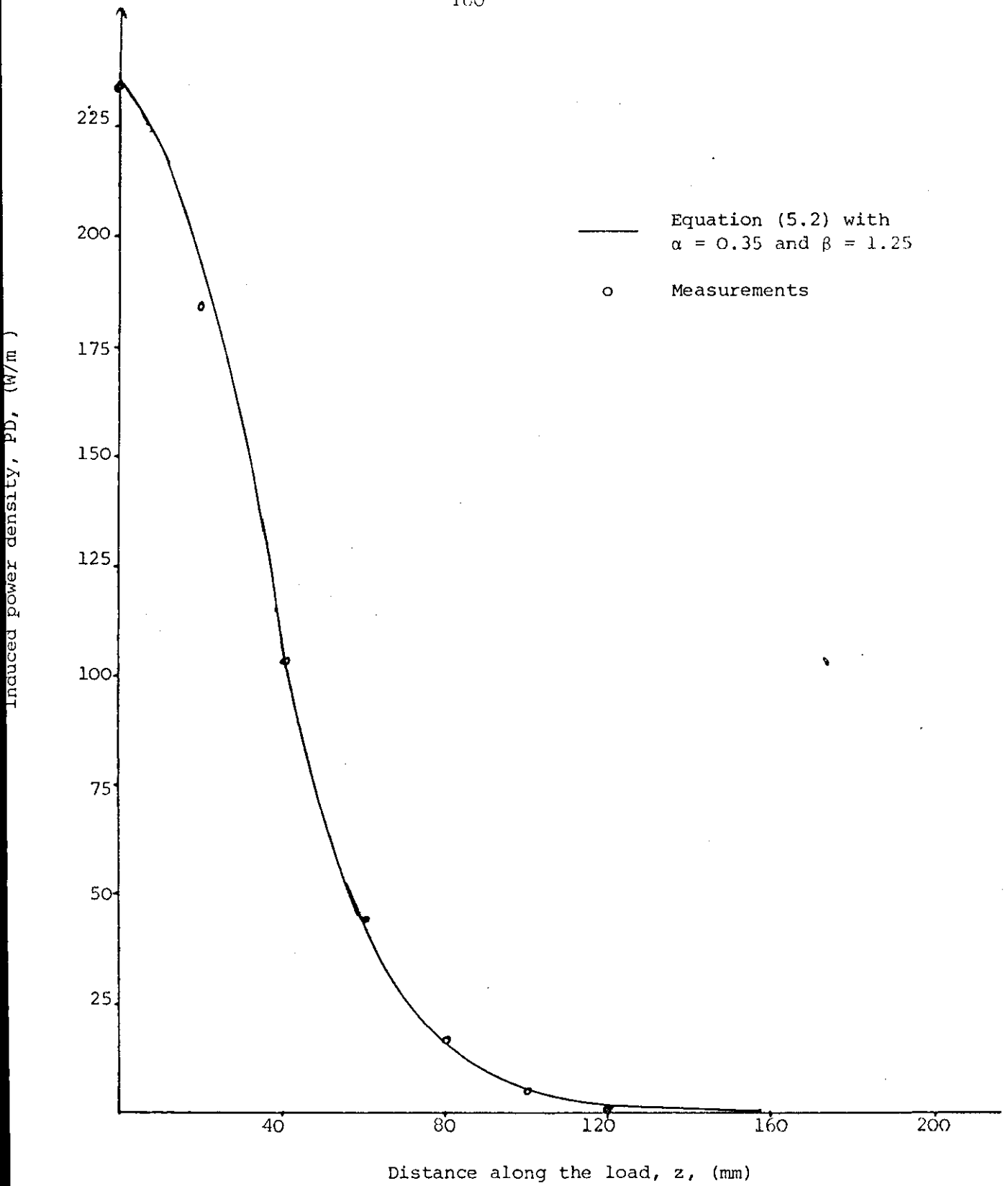


Fig. 5.49 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$ with lamination pack at $h_1 = 11\text{ mm}$

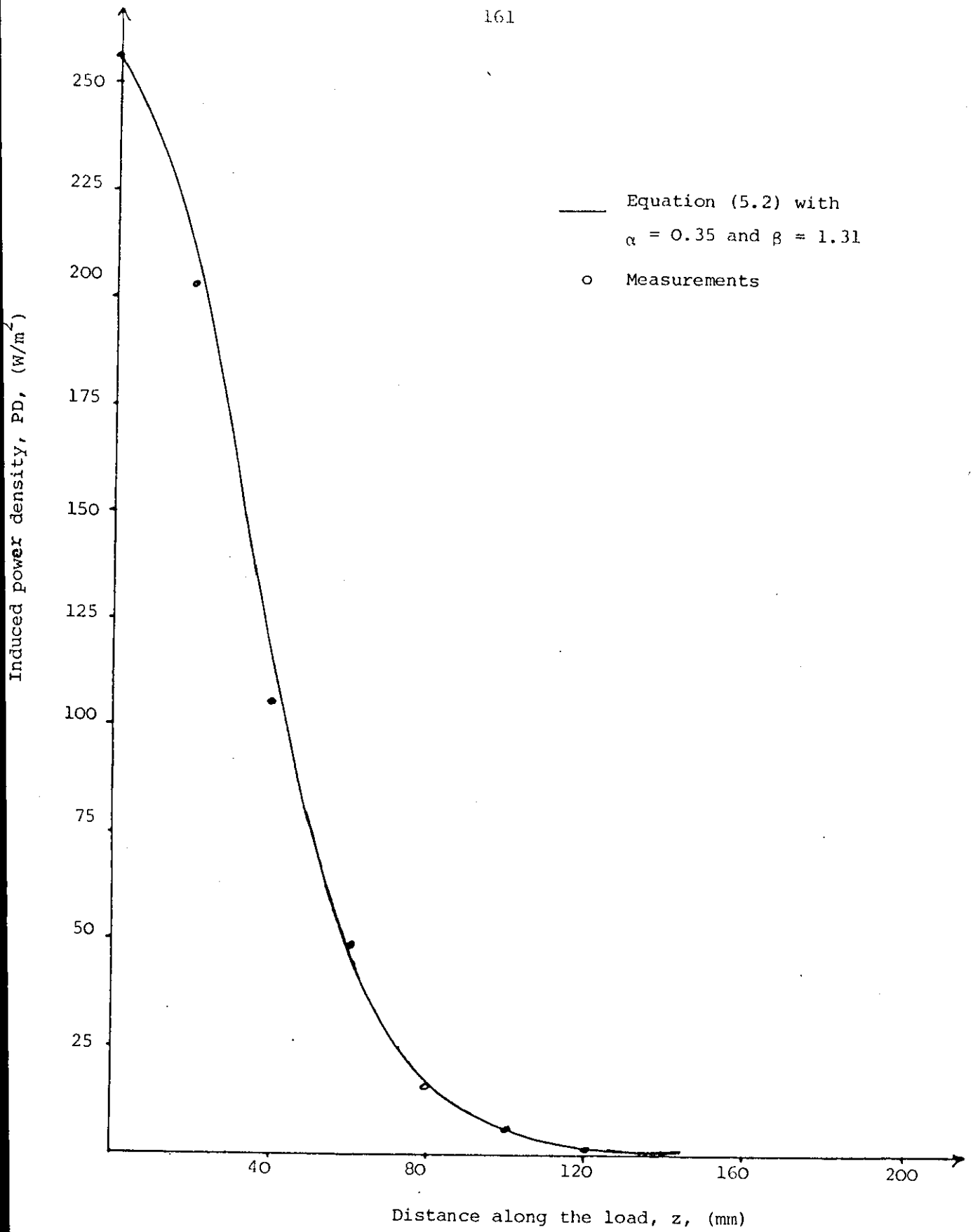


Fig. 5.50 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$ with lamination pack at $h_1 = 5\text{mm}$

hl (mm)	The load surface power density induced from circular conductor (W/m ²)	The load surface power density induced from rectangular conductor (W/m ²)
35	170	200
20	189	213
11	203	233
5	223	256

Table 5.2 The power density induced beneath the conductor

Table 5.2 shows the values of the power density induced on the load directly beneath the conductor for different values of hl.

The next stage was to investigate the current distribution on the conductor itself in order to find out whether the change in the distribution of the induced power density was due to a variation on the conductor current distribution or not. The surface current density distribution on the circular conductor was measured with the lamination pack 5mm above it and the normalized readings are shown in Fig. 5.51.

The same measurements have been taken on the rectangular conductor for hl = 5mm. The readings are normalized with respect to the reading at probe 5, see Fig. 5.3, and the results are shown in Fig. 5.52.

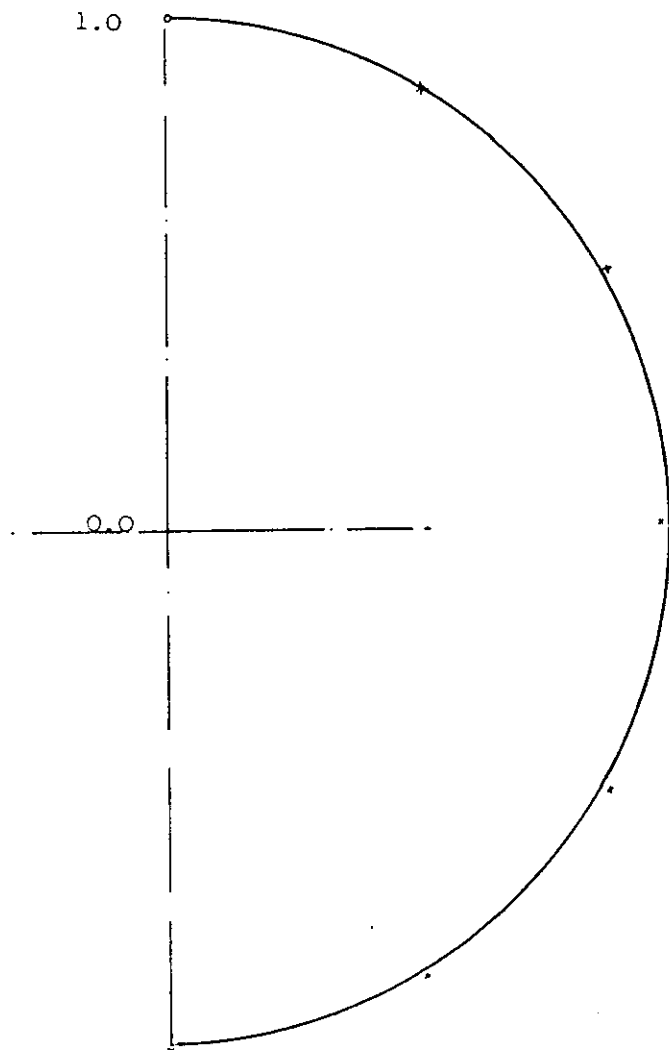


Fig. 5.51 Normalized current density distribution
on the circular conductor at $h = 54\text{mm}$
from an aluminium slab
— no laminations * laminations at $h_1 = 5\text{mm}$

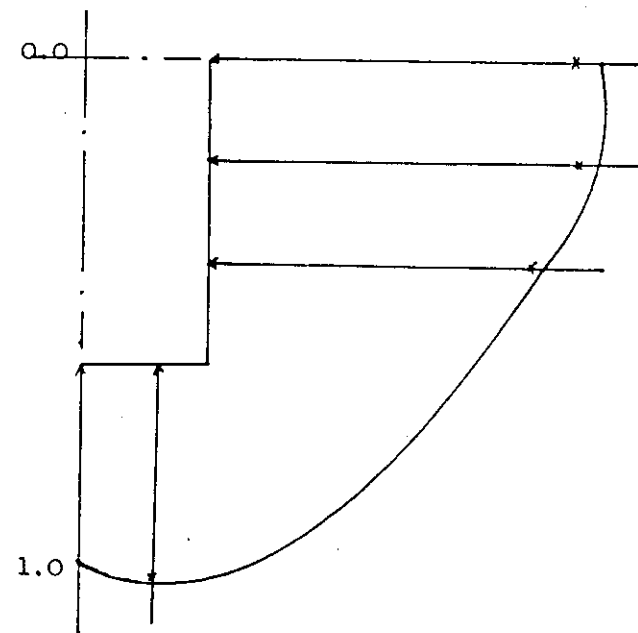


Fig. 5.52 Normalized current density distribution
on the rectangular conductor at
 $g = 40\text{mm}$ from an aluminium slab
— no laminations * laminations at $h_1 = 5\text{mm}$

Figs. 5.51 and 5.52 show that, within the range of the experiments, the laminations do not have an important effect on the distribution of the current on the conductor. This effect did not exceed 5% for the circular conductor and 7% for the rectangular conductor. As the change in the conductor current distribution was small; its effect on the power induced on the load would also be small. Hence the change in the power induced on the load was due mainly to the presence of the laminations.

In induction heating applications, the laminations do not completely surround the coil as most practical designs rely on a number of packets of punchings uniformly distributed around the coil. For this research to be useful in practical applications; it is important to know the effect of the laminations on the parts of the load which are not exactly beneath them, as the power density induced underneath the pack has already been investigated.

One lamination pack was placed above one edge of the load and the induced voltage distribution was measured at both edges of the slab. The distance h_1 was 5, 11, 20 and 35mm respectively. Figs. 5.53-5.56 illustrate the power density distribution for $h_1 = 35\text{mm}$ and $h_1 = 5\text{mm}$ only. The other results are not shown because they lie between the limits of these two cases. These curves show the power density induced on two parts of the load, near the two edges, one is covered with one pack of laminations and the other edge is left exposed.

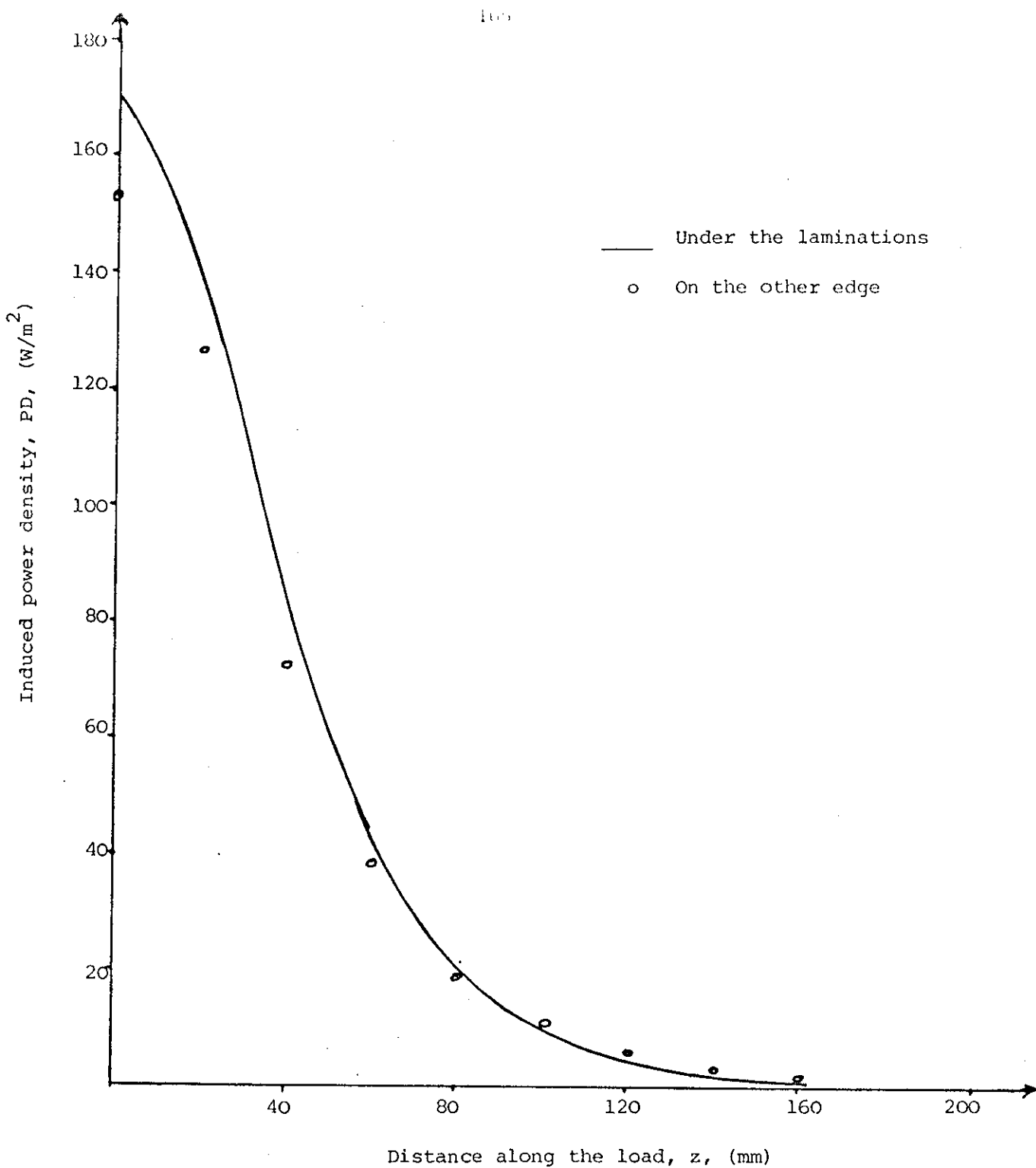


Fig. 5.53 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm at $h = 54\text{mm}$ with lamination pack at $h_1 = 35\text{mm}$ above one edge of the slab

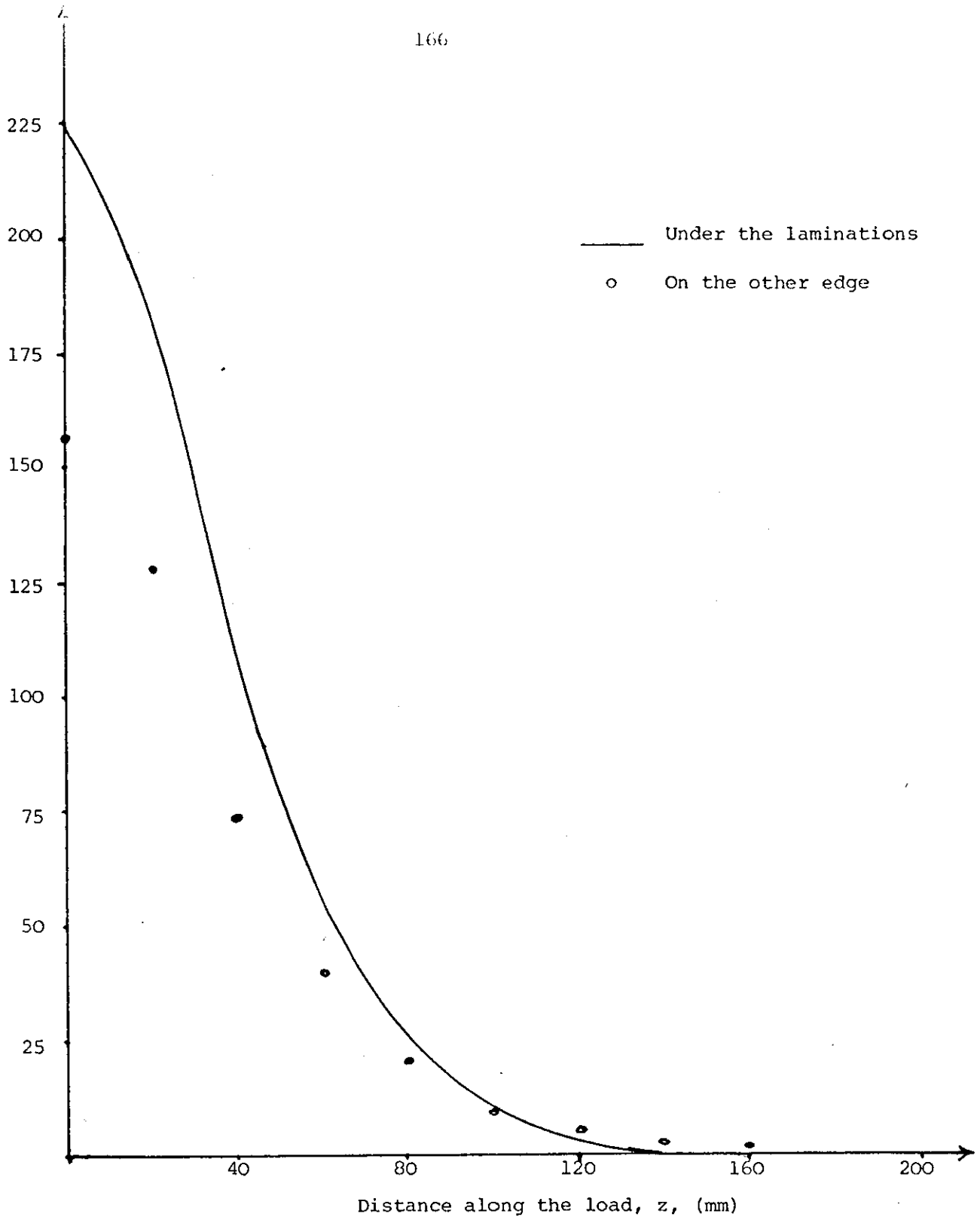


Fig. 5.54 Surface power density distribution along the aluminium slab due to 1000A flowing through circular conductor of 28mm diameter at $h = 54\text{mm}$ with lamination pack at $h_1 = 5\text{ mm}$ above one edge of the slab

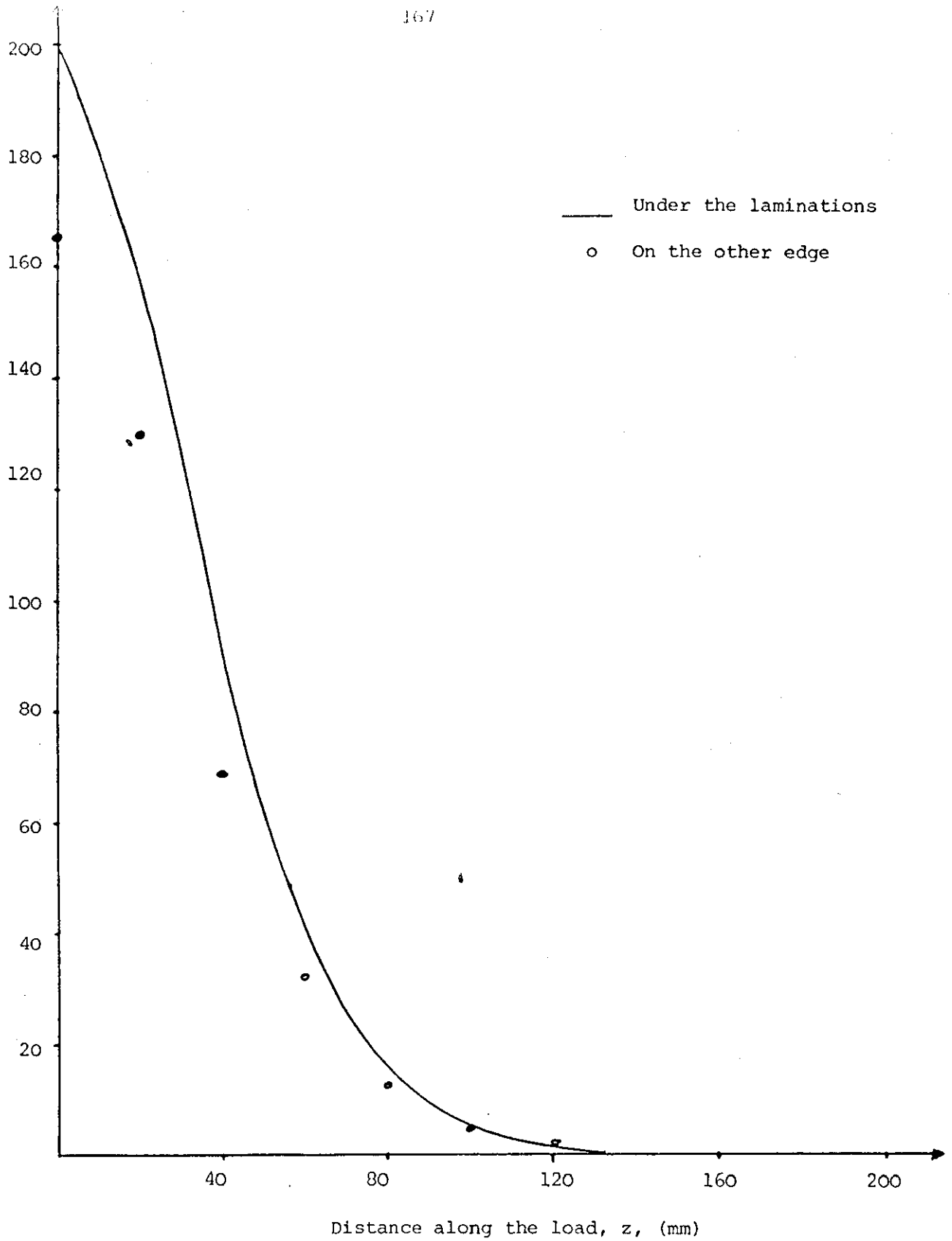
Induced power density, PD, (W/m^2)

Fig. 5.55 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$ with lamination pack at $h_1 = 35\text{mm}$ above one edge of the slab

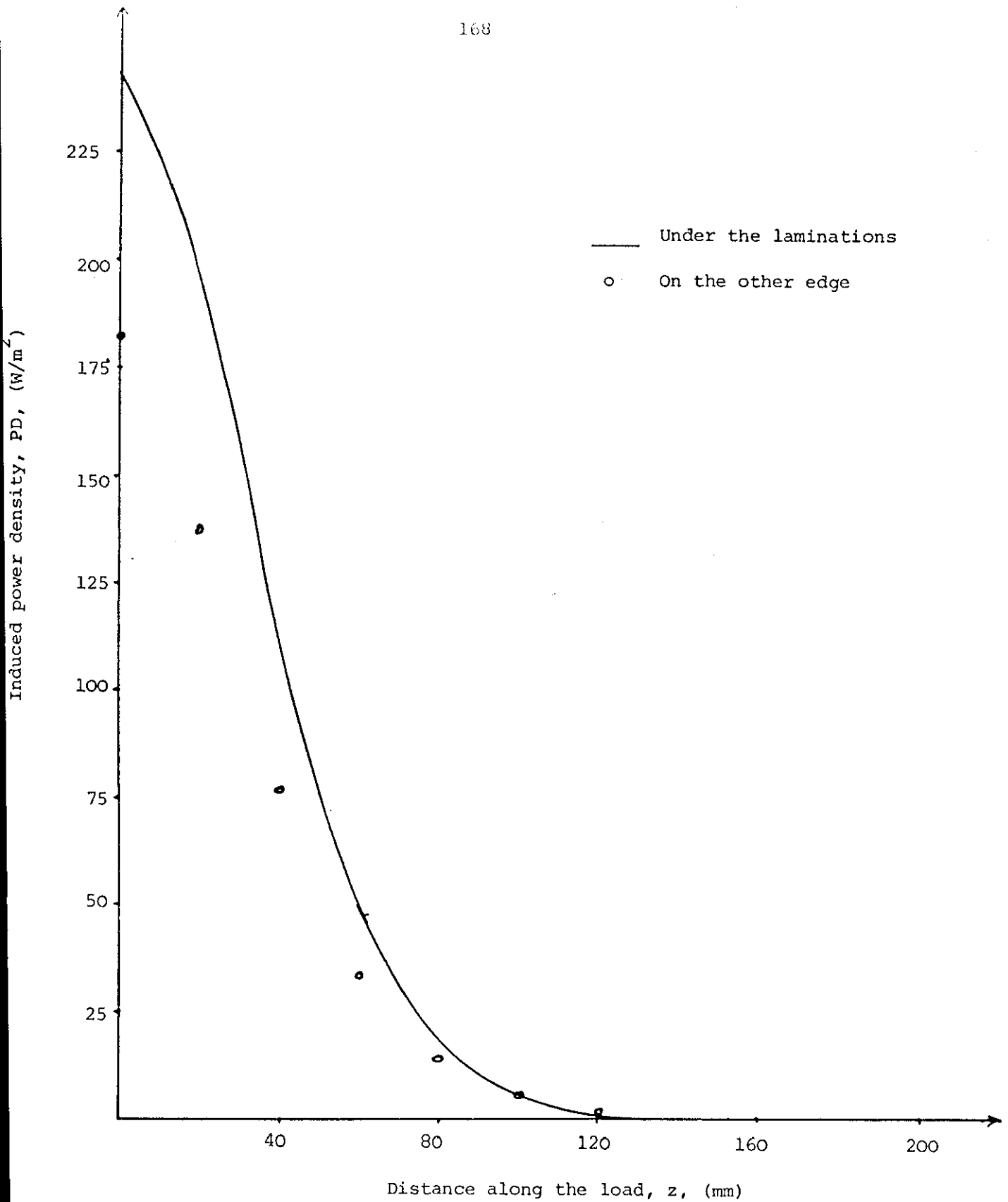


Fig. 5.56 Surface power density distribution along the aluminium slab due to 1000A flowing through rectangular conductor at $g = 40\text{mm}$ with lamination pack at $h_1 = 5\text{mm}$ above one edge of the slab

The power density induced on the first part is higher than that in the second, at given points near the conductor. While the opposite is true at a greater distance from the conductor, it did not, however, appear in the graphs because these values were small in comparison with the maximum value. This point should be taken into account in the applications where laminations were used, otherwise the calculated induced power density distribution might not represent the actual distribution.

Table 5.3 shows the power density induced on the load at those two parts directly beneath the conductor. From this table it can be seen that the power density induced on the other end varies only slightly with h_1 and can thus be regarded as constant.

Distance h_1 (mm)	Circular Conductor			Rectangular Conductor		
	PD under lamin. W/m^2	PD on the other part W/m^2	%Diff	PD under lamin. W/m^2	PD on the other part W/m^2	%Diff
35	170.34	153.45	9.92	199.02	165.25	16.97
20	183.17	157.97	13.76	219.08	179.11	18.25
11	202.87	153.07	24.55	234.53	180.0	23.25
5	223.57	157.21	29.7	242.93	181.95	25.1

Table 5.3 The power density on the load directly beneath the conductor

This phenomenon was investigated over all parts of the load in the next experiment. The power density distribution induced on the load at different distances from the laminations was investigated by assuming the load to comprise of 6 parts, each part being of 50mm width. The lamination pack was situated above the 5th part as shown in Fig. 5.57. The induced voltage distribution was measured on parts 5, 4, 3 and 2 at $h_1 = 20\text{mm}$ and $h_1 = 5\text{mm}$ respectively. The readings are not shown as the changes are small and cannot, therefore, be shown in graph form. The shape of the curves would have been as in Fig. 5.58, so that the smaller the distance h_1 the greater the differences between the different parts.

The main aim of this section is to investigate the superposition method when laminations exist in the system, so that this technique can be used in applications such as vessel heating. When investigating the effect of the laminations on the applicability of the superposition method it proved necessary to measure the voltage distribution, induced on the load, when 1000A flowed through the single rectangular conductor, which was above the slab and below the lamination pack when $h_1 = 5\text{mm}$. These readings were combined with the computer program "W-SC-FIT" to predict the induced power density from 5 similar conductors which were parallel to each other and positioned at 31mm intervals. These 5 conductors operated under identical conditions to those of the single conductor with the exception that a current of 600A/conductor and not 1000A was applied. The results are shown in Fig. 5.59.

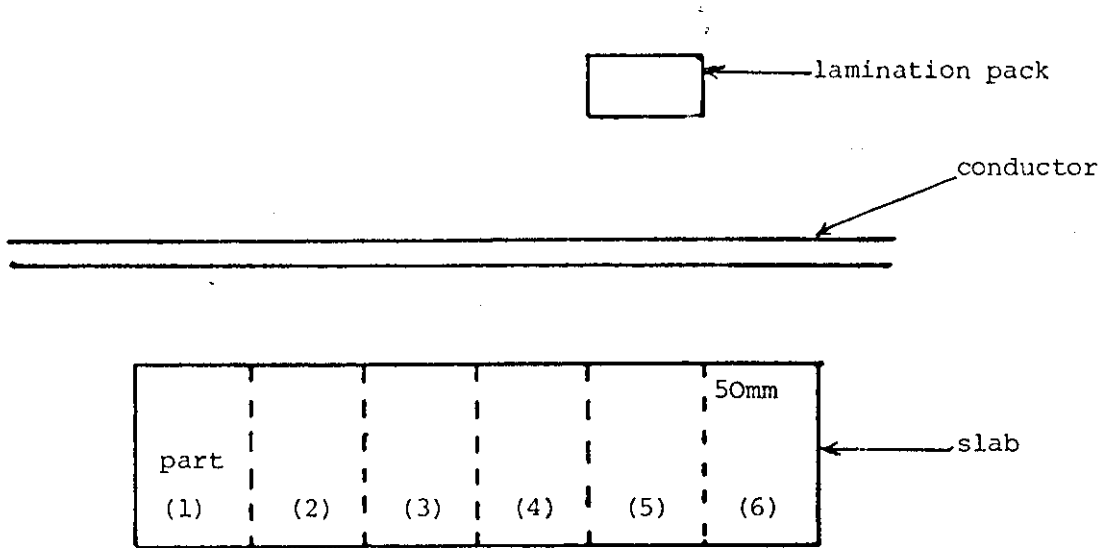


Fig. 5.57 The lamination pack above part (5) of the load

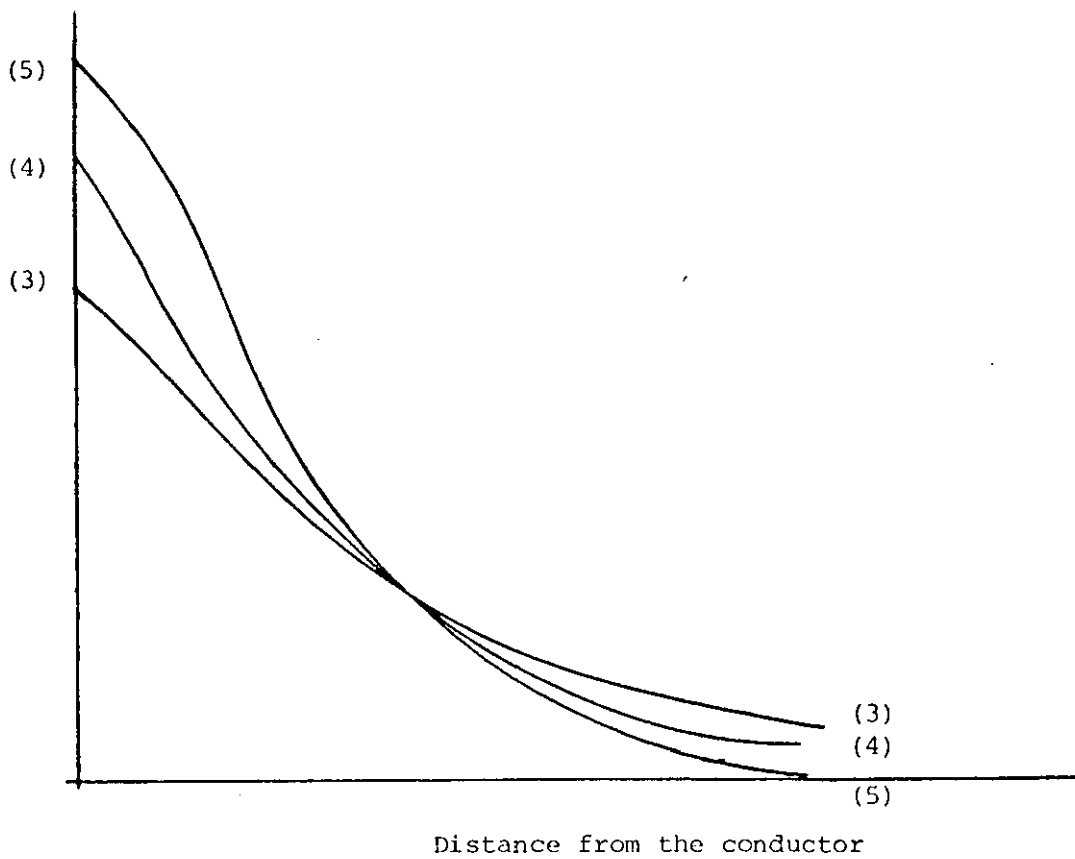


Fig. 5.58 Power density induced on different parts of the load

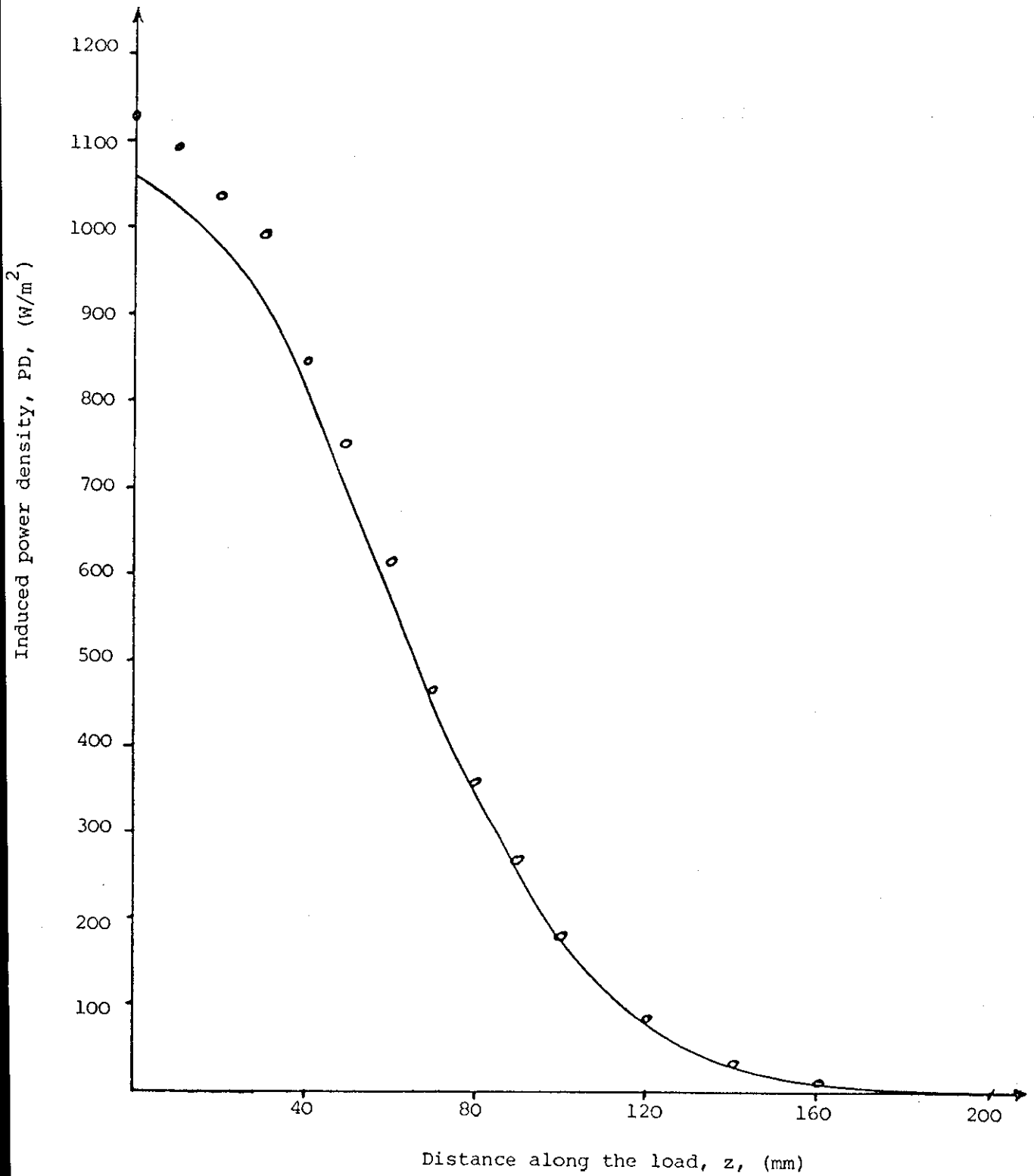


Fig. 5.59 Power density distribution along the aluminium slab due to 5 rectangular conductors at air gap of 40mm, coil pitch of 31mm and a current of 600A/conductor with lamination pack at $h_1 = 5\text{mm}$

— prediction by superposition o practical readings

The existence of the laminations did not affect the applicability of the superposition method.. The discrepancy between the measured power density and that predicted by the superposition method is less than 7%. This small difference is an acceptable experimental error and proves that this technique can be used in the relevant applications.

5.2.3 Conclusions from the Work on Ferromagnetic Materials

Although the numerical values and percentages are for the particular cases under consideration; they do give indications about the effects of the ferromagnetic materials and some conclusions can be drawn.

The superposition method can be applied to a magnetic load and to a system containing laminations. Also, the applicability of this method is not affected by the current distribution on the conductor which depends on the position of the conductor among other conductors.

The use of laminations increases the power density induced on the load beneath the lamination pack. This power density is inversely proportional to the distance between the laminations and the conductor.

The accurate calculation of the power density induced on a ferromagnetic load requires the variation of the permeability to be taken into account.

5.3 Summary of Results and Suggestion

The surface current density distribution induced from a single current carrying conductor on a nonmagnetic slab was investigated and an equation governing this distribution was found. The equation differs from that derived for a filamentary conductor by three factors. These factors can be represented by mathematical expressions.

The superposition method can be employed with magnetic and nonmagnetic loads. The applicability of this method was not affected by the existence of the lamination packs. The permeability variation must be taken into account when using this method with ferromagnetic workpiece. This has been done by adopting a functional representation to the magnetisation curve.

The superposition method can be applied to conductors of different shapes irrespective of the current distribution through them.

Also, neither the positioning of the conductor, amongst others, alters its effect on the workpiece nor the proximity effect on the conductors prevents this method being applied.

The practical verification of the superposition method raises a question concerning the possible use of this method in induction heating applications. From the experiments on aluminium slab, it is evident that this technique can be used with nonmagnetic loads as well as to magnetic loads above the Curie temperature.

Also, it is possible to apply this method to applications requiring relatively low magnetic fields to be induced on a ferromagnetic load such as in vessel heating. This has been shown by the experimental work on the mild steel slab. Finally, the work with lamination packs proves that the superposition method is suitable to applications involving laminations such as metal melting and vessel heating. Hence it is conceivable to employ the superposition method on different applications.

The main reason for developing the superposition method was to design a coil capable of producing a nonuniform power density along the load. This method has not yet been practically proved suitable to this task. Therefore, it is required to test the flexibility of this method and its ability to design such a coil.

Applications such as off-the-bar forging need nonuniform power density along the load. Part of the workpiece in this case will be hot and the other part cold. This problem cannot be tackled in the laboratory due to the high cost of building a heater capable of producing the high power density required for the hot part of the workpiece.

It is reasonable to suggest the aluminium extrusion as the final experimental examination to assess the flexibility of the superposition method practically. This application necessitates nonuniform power density to be induced on the load, and it can be modelled within the laboratory facilities.

CHAPTER 6SUPERPOSITION AS A COIL DESIGN METHOD FOR
HEATING ALUMINIUM PRIOR TO EXTRUSION

The versatility and economic advantages of extrusion as a method of forming aluminium have caused a very wide growth in its use for different products. Superficially, the process appears to be rather simple. It consists of heating a billet, inserting it into the container in a hydraulic press and then applying sufficient pressure to force it through a die.

Unfortunately, maintenance of consistently high quality in aluminium alloy extrusions is difficult. The major problem is that aluminium actually gains heat during extrusion. It is, therefore, desirable to taper-heat the billet. If this is not the case, metallurgical inconsistency in the end product might occur.

In direct extrusion of aluminium, a ram directly pushes a billet through a stationary press container and selected die-shape. The finally extruded section emerges onto the run-out as a correctly shaped product. In this kind of extrusion it is preferred that the extrusion speed of the ram and product temperature through the die be constant throughout the extrusion cycle, thus avoiding continual adjustment of the press. This isothermal effect may be achieved by inducing into the heated billet a temperature at the

front face; that part which enters the die first. In order to appreciate the reason for this isothermal effect it must be explained that additional heat will also be generated during the extrusion operation:-

- a) by heat energy, caused by the shear effect at the restricted die opening,
- b) through the generation of friction, since the hot aluminium billet is not only forced into the die but also moves along the stationary container whilst confined within it's bore.

In indirect extrusion the hot billet remains stationary in relation to the press container, the rear end of the billet becomes progressively cooler as the extrusion proceeds rather than the front end. This, of course, is the exact reversal of the direct extrusion and care must be taken to ensure the correct degree of reverse taper if optimum results are to be achieved.

To demonstrate the usefulness of the design procedure developed in the previous chapters it was decided to design an induction heating coil capable of producing a linearly decreasing temperature taper. This, of course, models the practical requirements met industrially in the induction heating of aluminium prior to extrusion.

It was not possible to build a commercial size billet heater to test the theory because of the capital cost and the limitations of the laboratory power supplies. Megawatt power supplies and large amounts

of power factor correction capacitors are not uncommon in industrial units. However, a laboratory scale model was built to demonstrate the validity of the design procedure.

6.1 The Superposition Method with a Cylindrical Workpiece

Before adopting the superposition method to design the required coil, it is necessary to investigate the applicability of this technique on a cylindrical load. To achieve this, a series of experiments have been carried out to find the effect of single turn and multiturn coil on an aluminium cylinder.

The circuit diagram is the same as that of Fig. 5.2. Instead of a slab, an aluminium cylinder was used as the workpiece. This solid cylinder of 50.8mm radius and 340mm length had its surface machined until smooth. As can be seen, the radius of this cylinder is equal to the thickness of the aluminium slab which was used previously in Chapter 5.

The experimental rig is shown in Fig. 6.1. The single turn and multiturn coils have been constructed from a water cooled copper tube of 6mm diameter. Three single turn coils of 70.8, 80.8 and 100.8mm inner radii were respectively surrounding the aluminium cylinder. The single turn coil was around the middle of the cylinder which was assumed to be the reference point. A constant current of 400A was flowing through the coil. The voltages induced at different distances along the cylinder were measured by the current density probes.

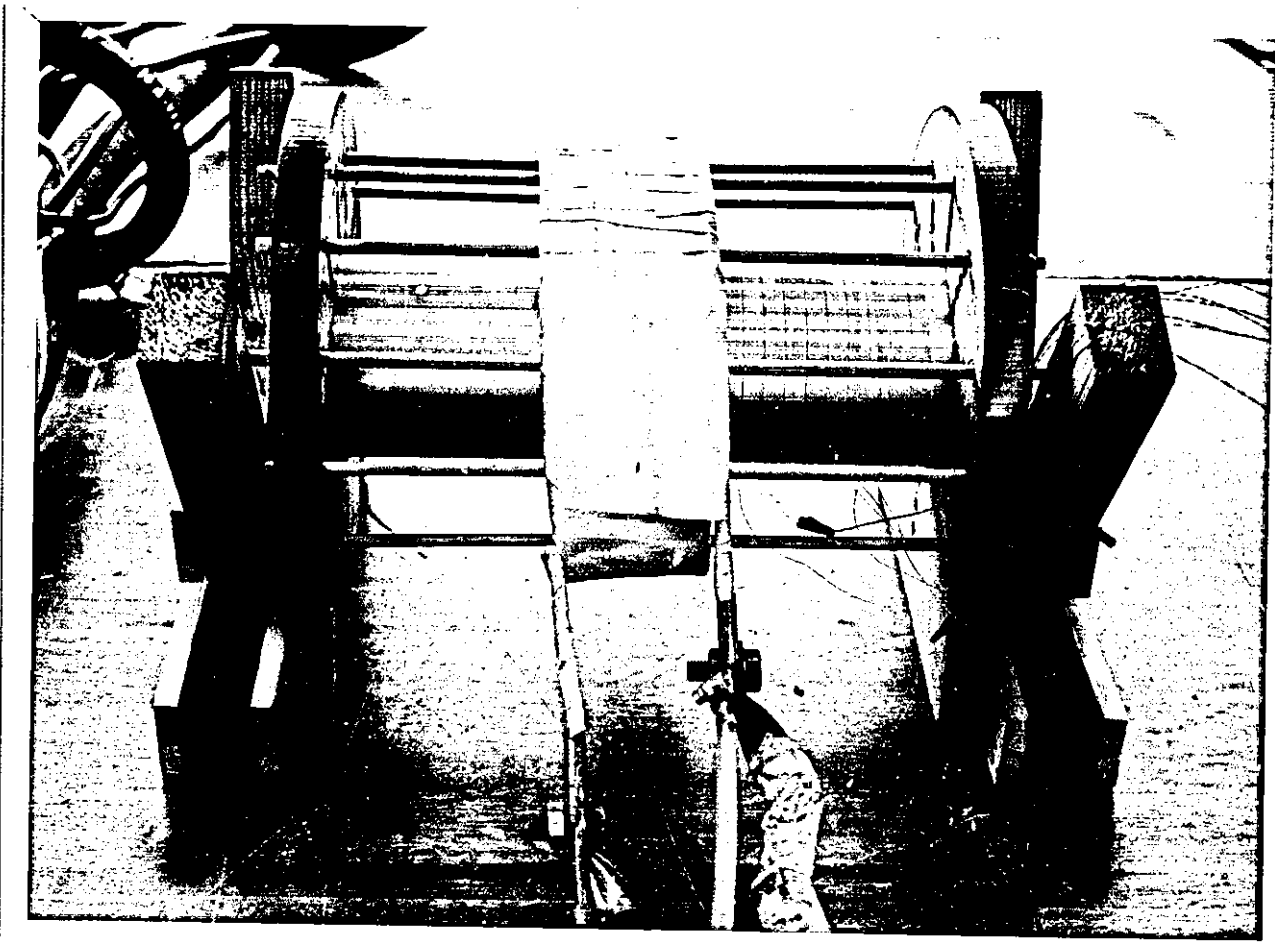


Fig. 6.1 The experimental rig

The probes were made from thin constantan wires wrapped around the cylinder and spaced at 20mm intervals.

The results of the single turn coils are shown in Figs. 6.2-6.4. These graphs show the measured surface power density together with those calculated by using the filamentary conductor theoretical equation (4.9) and the practical equation (5.2). It can be seen that the practical equation (5.2) which was developed for aluminium slab can be applied with the same accuracy to an aluminium cylinder of radius equal to the thickness of the slab.

The calculations of the surface power densities were carried out by substituting the numerical values of the resistivity, $28.24 \times 10^{-9} \Omega\text{m}$, and the measured surface current density into the equations given in Appendix 1.

The results of the single conductor show that the practical equation which was developed for a semi-infinite aluminium slab can be applied for an aluminium cylinder of radius equal to the thickness of the slab. The discrepancy between the measured power density and that calculated by the practical equation (5.2) did not exceed 10% for the range from maximum to 10% of the maximum power density. This accuracy is sufficient for the practical applications. As the power is a function of the square of the current, the error in calculating the current or voltage distribution is lower than that of the power density.

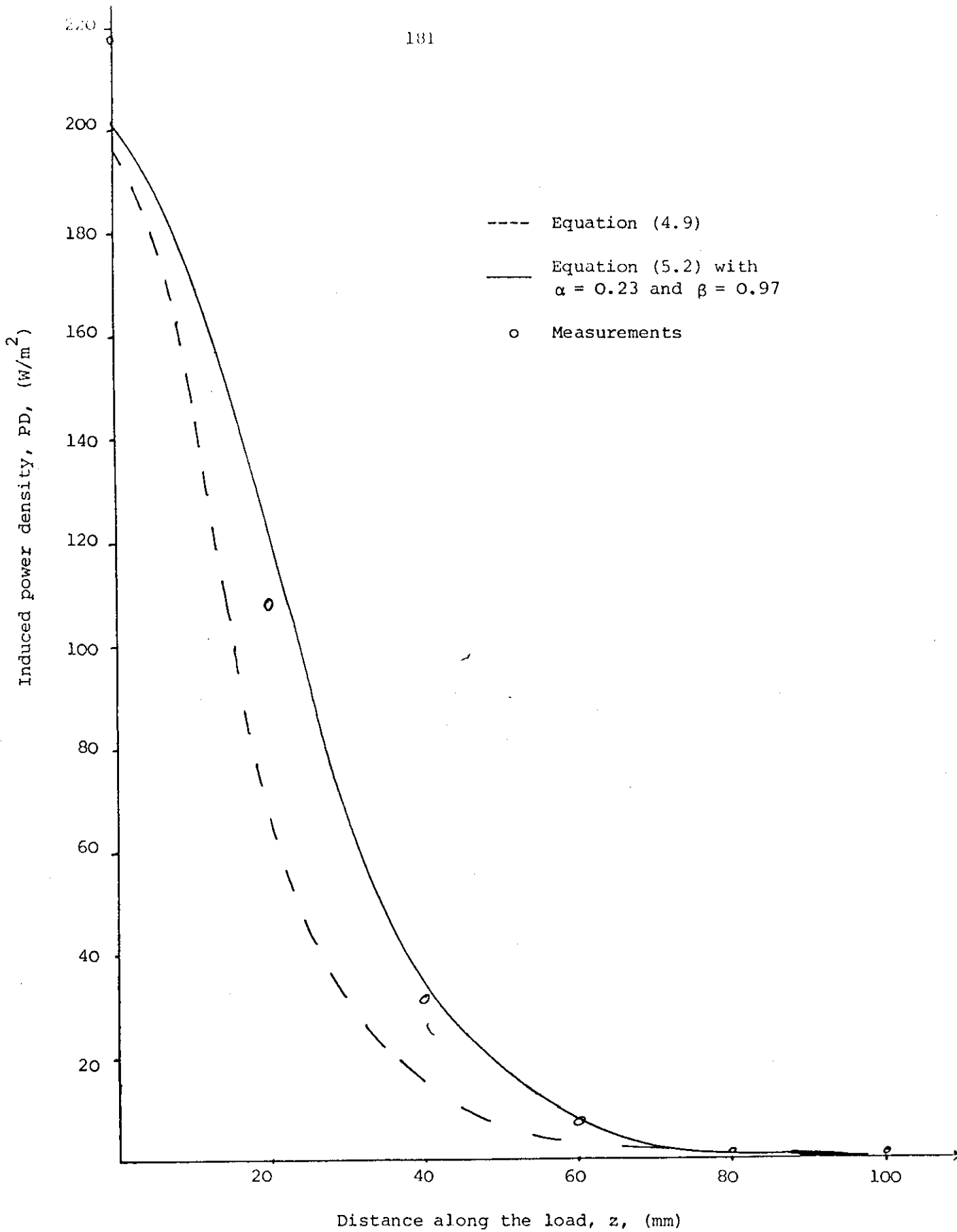


Fig. 6.2 Surface power density distribution along the aluminium cylinder due to 400A flowing through single turn coil at air gap of 20mm

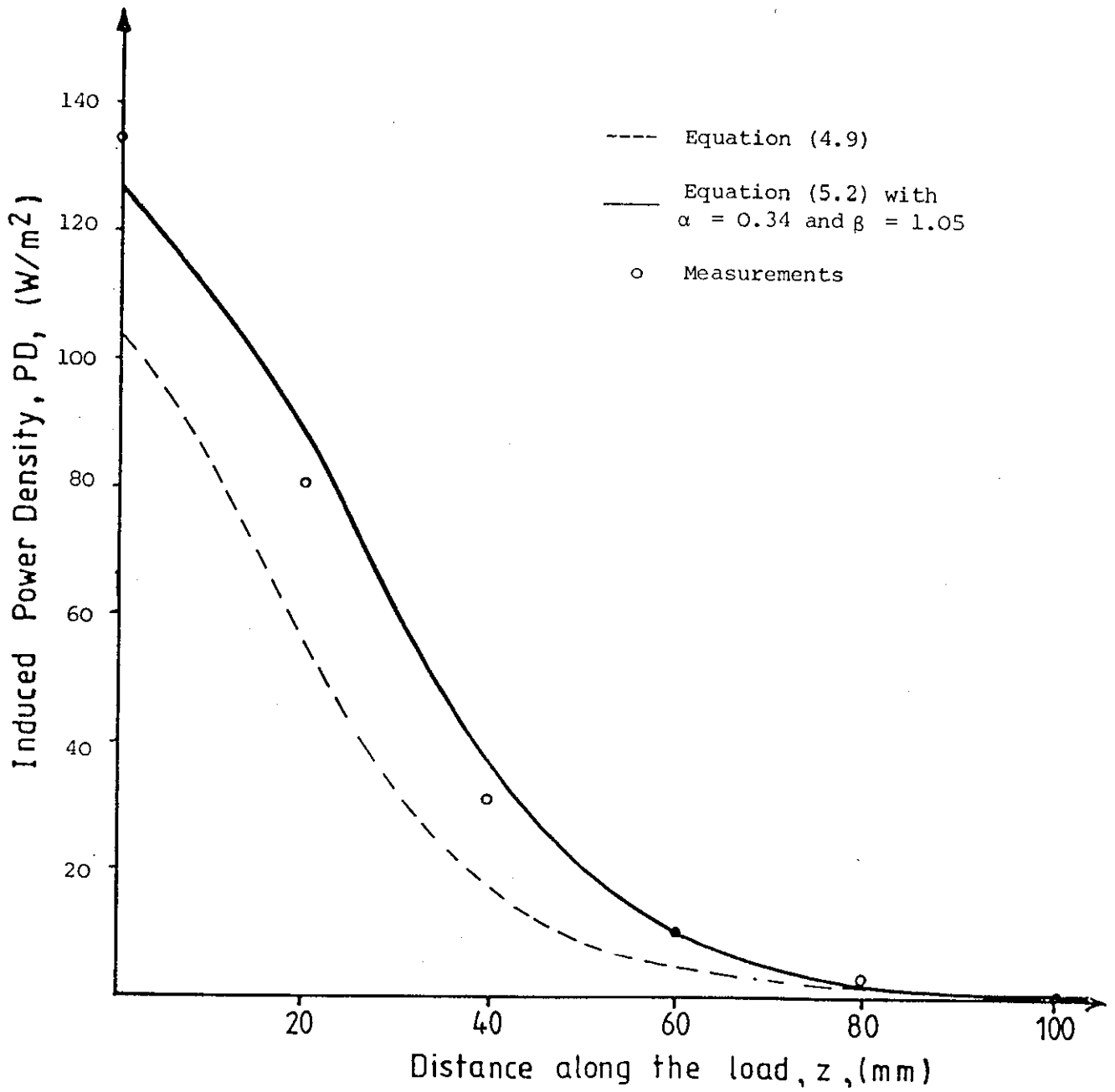


Fig.6.3 Power density distribution along the aluminium cylinder due to 400A flowing in single conductor at air gap of 30 mm.

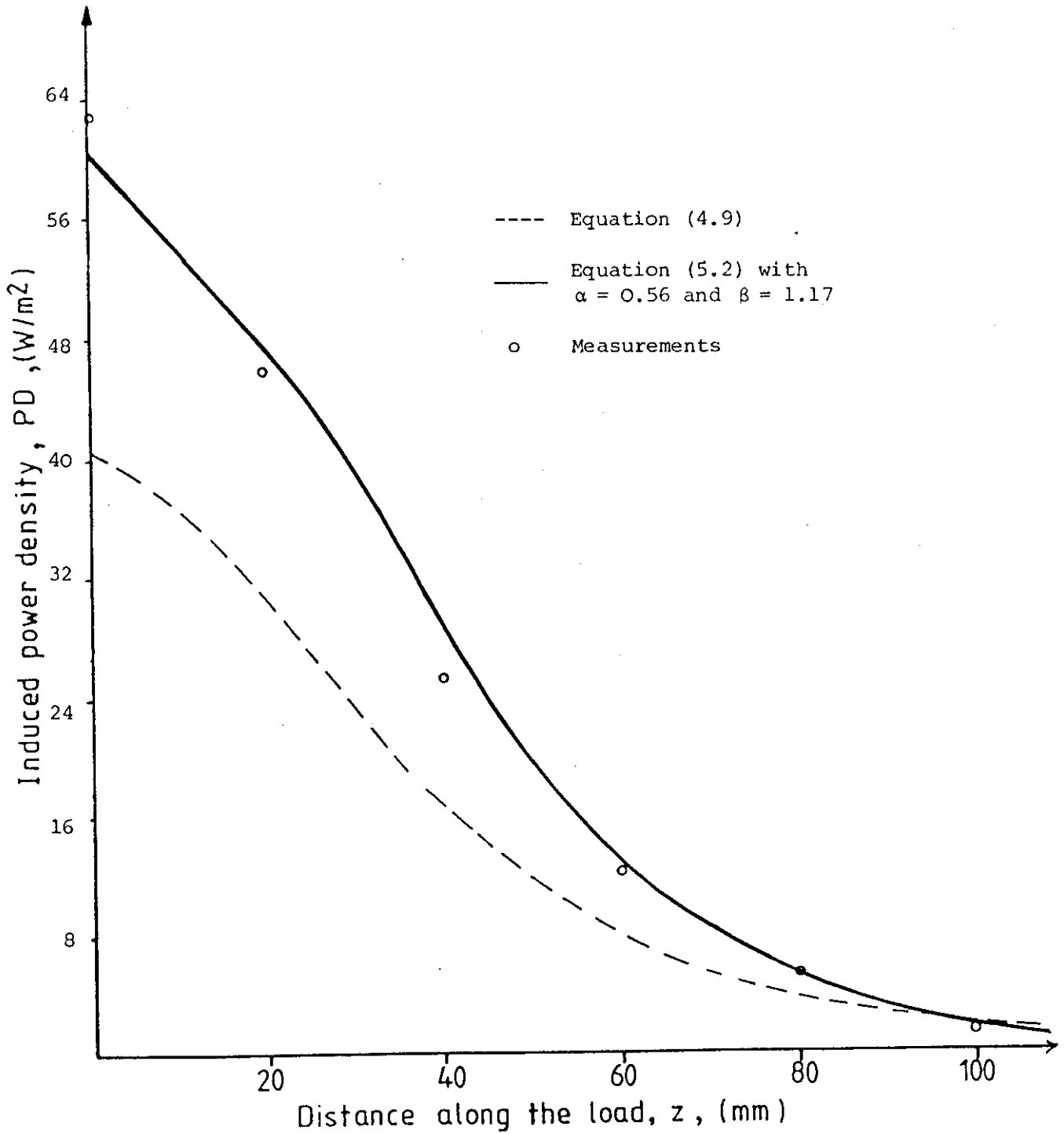


Fig.64 Power density distribution along the aluminium cylinder due to 400A flowing in single conductor at air gap of 50 mm

An 11 turn coil of 100.8mm inner radius and uniform pitch of 7.1mm has been used to verify the superposition method. The centre of the coil was around the middle of the cylinder and this was the reference point. The measured power density distribution induced on the load is shown in Fig. 6.5 together with the predicted values which were derived with the aid of the computer program "W-SC-FIT". This graph shows that the superposition theory can be applied to a cylindrical load and the difference between predicted and measured power densities is within 3%. For large values of distance z ; the error is higher, this is due to the approximation made in calculating the induced voltages at these distances. This is not important practically because the power density induced at these distances is relatively small in comparison with other values.

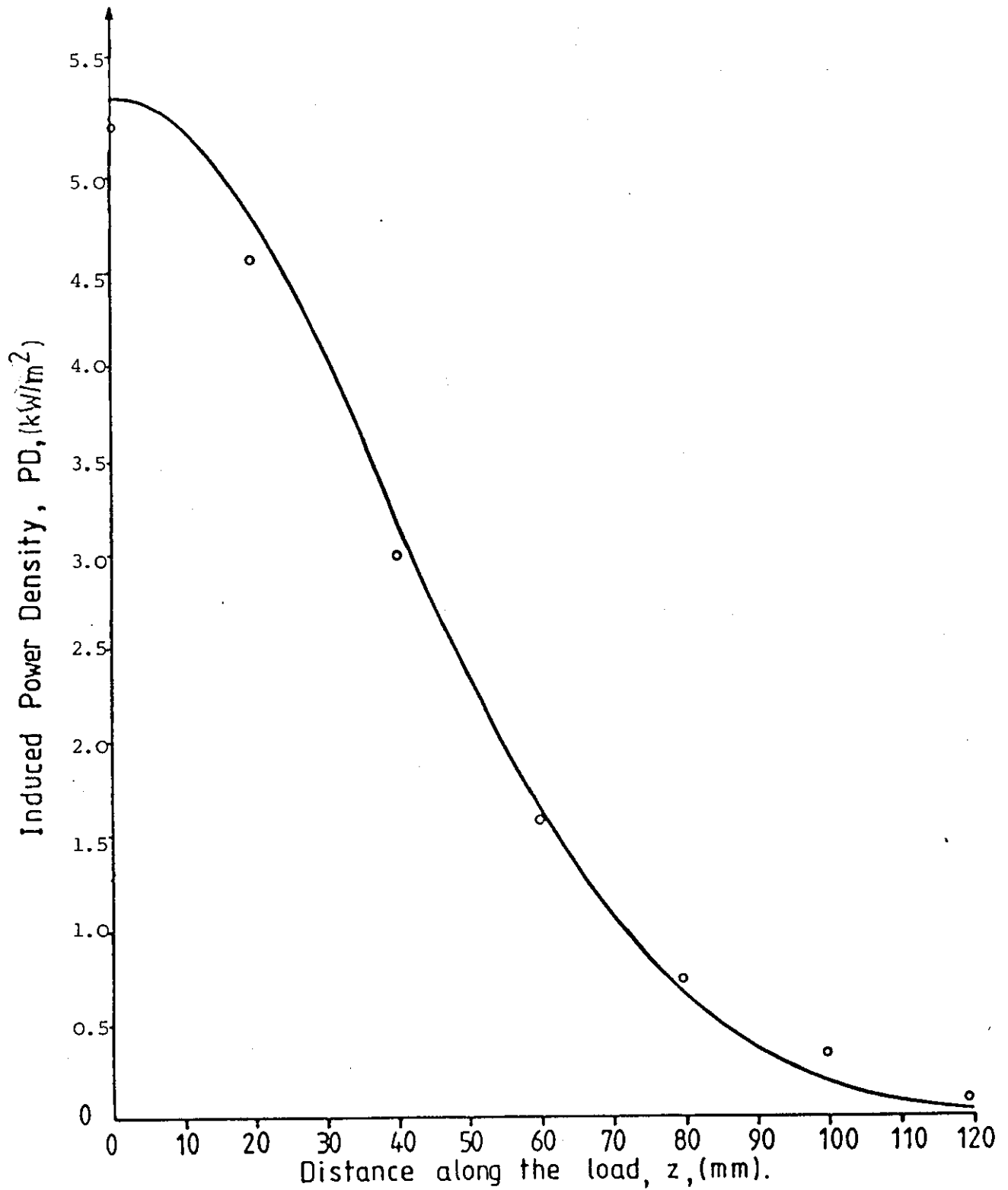


Fig.6.5 Power density distribution along the aluminium cylinder due to 11 turns coil at air gap of 50 mm, coil pitch of 7.1 mm and a current of 400A.

———— Prediction by Superposition ○ Practical Readings

6.2 The Temperature Distribution in a Solid Cylinder

Carslaw and Jaeger [6.1] gave the temperature distribution in a long cylinder initially with uniform temperature when subjected to a constant surface power density with no heat loss from the surface to be:

$$Tr = (PD) \frac{R}{\lambda} \left\{ 2\tau + \frac{r^2}{2R^2} - \frac{1}{4} - 2 \sum_{n=1}^{\infty} e^{-\beta_n^2 \tau} \frac{J_0[\beta_n (r/R)]}{B_n^2 J_0(\beta_n)} \right\} \quad (6.1)$$

where Tr : temperature rise at radius r after time t

$$\tau = \frac{\lambda t}{\gamma C R^2} \quad (\text{dimensionless}) \quad (6.2)$$

λ = thermal conductivity ($W/m^\circ K$)

t = time (s)

γ = density (kg/m^3)

C = specific heat capacity ($J/kg^\circ K$)

R = the radius (m)

$J_0(x)$ is Bessel's function of the first kind and zero order and the β_n 's are the positive roots of $J_1(\beta) = 0$.

At the beginning of the heating cycle, the increase in the temperature of the surface is faster than that of the centre; this is the transient period, which is represented by the summation term in equation (6.1). Baker [6.2] showed that the the transient time

is equivalent to $\tau = 0.25$, at this point the summation term will be very small and as such can be ignored. Once the transient time has elapsed, the temperature of all parts of the cylinder will increase at the same rate and equation (6.1) will be simplified to;

$$T_r = (PD) \frac{R}{\lambda} \left(2\tau + \frac{r^2}{2R^2} - \frac{1}{4} \right) \quad (6.3)$$

The temperature at the surface ($r = R$) is:

$$T_s = (PD) \frac{R}{\lambda} \left(2\tau + \frac{1}{4} \right) \quad (6.4)$$

and at the centre ($r = 0$), the temperature is:

$$T_c = (PD) \frac{R}{\lambda} \left(2\tau - \frac{1}{4} \right) \quad (6.5)$$

Hence the temperature difference between the surface and centre is

$$T_s - T_c = PD \frac{R}{2\lambda} \quad (6.6)$$

By using equations (6.5) and (6.6), equation (6.3) can be rewritten as

$$T_r = T_c + (T_s - T_c) \frac{r^2}{R^2} \quad (6.7)$$

As was stated earlier, the above equations are for a cylinder with power input to the surface which can be applied at high frequencies as the penetration depth is very small when compared with the load diameter. When a finite penetration depth δ exists and there is no heat loss the temperature distribution, with respect to the temperature of the centre, can be calculated [6.3] by

the following equation:

$$T_r - T_c = (PD) \frac{R}{2\lambda} \left\{ \frac{r^2}{R^2} - \frac{X(k_2 \frac{r}{R})^{-1}}{k_2 Z(k_2)} \right\} \quad (6.8)$$

$$\text{where } X(x) = \text{ber}^2 x + \text{bei}^2 x \quad (6.9)$$

$$Z(x) = \text{ber}x\text{ber}'x + \text{beixbei}'x \quad (6.10)$$

$$k_2 = \sqrt{2} \frac{R}{\delta} \quad (6.11)$$

If there is a surface radiation, then the net power input will be less than the total input power [6.4] and equation (6.8) will be:

$$T_r - T_c = P_n \frac{R}{2\lambda} \left\{ \frac{r^2}{R^2} - \frac{PD}{P_n} \left[\frac{X(k_2 \frac{r}{R})^{-1}}{k_2 Z(k_2)} \right] \right\} \quad (6.12)$$

where P_n is the net power density. The radiation loss can be calculated from Stefan-Boltzmann's law:

$$P_R = 5.67 \times 10^{-8} \epsilon (T_s^4 - T_b^4) \quad (6.13)$$

where P_R : the radiated power density (W/m^2)

ϵ : the emissivity coefficient ($\text{W/m}^2 \text{K}^4$) and

T_s, T_b are the surface and the boundary temperatures in $^\circ\text{K}$.

The billet does not enter the die directly after leaving the heater, as there is a handling time or a soaking period which will reduce the temperature difference between different parts of the workpiece.

Carslaw and Jaeger [6.1] showed that a cylinder with initial temperature distribution $T_r = f(r)$ and no heat input will have a

temperature distribution of:

$$T_r = \frac{2}{R^2} \left[\int_0^R r' f(r') dr' + \sum_{n=1}^{\infty} e^{-\alpha_n^2 \tau} \frac{J_0(r \alpha_n / R)}{J_0^2(\alpha_n)} \int_0^R r' f(r') J_0(\alpha_n r' / R) dr' \right] \quad (6.14)$$

where the α_n 's are the positive roots of $J_0'(\alpha R) = J_1(\alpha R) = 0$

The final value of the cylinder's temperature is represented by the first term of equation (6.14). The second term represents the transient period.

From equation (6.7) $f(r)$ is

$$f(r) = T_c + (T_s - T_c) \frac{r^2}{R^2} \quad (6.15)$$

The temperatures of the centre $T_c(t)$ and of the surface $T_s(t)$ during the soaking period as functions of time are given by [6.2] as:

$$T_c(t) = T_c + (T_s - T_c) \left\{ \frac{1}{2} + 4 \sum_{n=1}^{\infty} \frac{e^{-\lambda \beta_n^2 t} \frac{R^2}{\beta_n^2 J_0^2(\beta_n)}}{\beta_n^2 J_0^2(\beta_n)} \right\} \quad (6.16)$$

$$T_s(t) = T_c + (T_s - T_c) \left\{ \frac{1}{2} + \frac{1}{4} \sum_{n=1}^{\infty} \frac{e^{-\lambda \beta_n^2 t} \frac{R^2}{\beta_n^2}}{\beta_n^2} \right\} \quad (6.17)$$

The quantities inside the parentheses are shown in Fig. 6.6.

If the workpiece is of length ℓ and initially with axial temperature distribution of $f(z)$, then during the soaking period this distribution can be represented [6.1] by the following expression;

$$T = \frac{1}{\ell} \int_0^{\ell} f(z') dz' + \frac{2}{\ell} \sum_{n=1}^{\infty} \frac{\cos \frac{n\pi z}{\ell}}{\frac{n^2 \pi^2 \lambda t}{\ell^2 C\gamma}} \int_0^{\ell} f(z') \cos \frac{n\pi z'}{\ell} dz' \quad (6.18)$$

The first term is the steady-state temperature and the second term is the transient. The transient die away with time constants τ' , where

$$\tau' = \frac{1}{e} = \frac{C\gamma \ell^2}{\lambda \pi^2} \quad (6.19)$$

i.e. the temperature difference will drop to 37% of the initial value after a time τ' seconds.

The surface power density distribution required for the extruding of an aluminium billet can be determined by considering the radial and axial temperature distributions during the handling time, i.e. the soaking period.

At a normalized time $\tau = 0.25$, the billet will have an even radial temperature, see Fig. 6.6. This time is equivalent to 16 seconds for an aluminium billet of 70mm radius at a temperature of 500°C [6.4]. This time is far less than the usual handling time which is around 40 seconds [6.4].

If the billet is initially with linear temperature distribution, $f(z) = Cz$, the temperature difference between the two ends $T_{\ell}(t) - T_o(t)$ as a function of time with respect to the initial temperature difference $T_{\ell} - T_o$ is shown in Fig. 6.7 [6.4].

At the end of the handling time, the temperature difference between the two ends will drop to 90% of the initial difference for a billet of 1 metre length and to 58% of the initial difference if the length was 300mm [6.4]. Hence, a billet leaving the heater with axial and radial temperature differences, will enter the die with uniform radial temperature and differential axial temperature.

As the temperature of any point is a linear function of the surface power density on that particular part of the billet, see equation (6.8), hence the temperature distribution of the billet can be controlled by controlling the distribution of the surface power density.

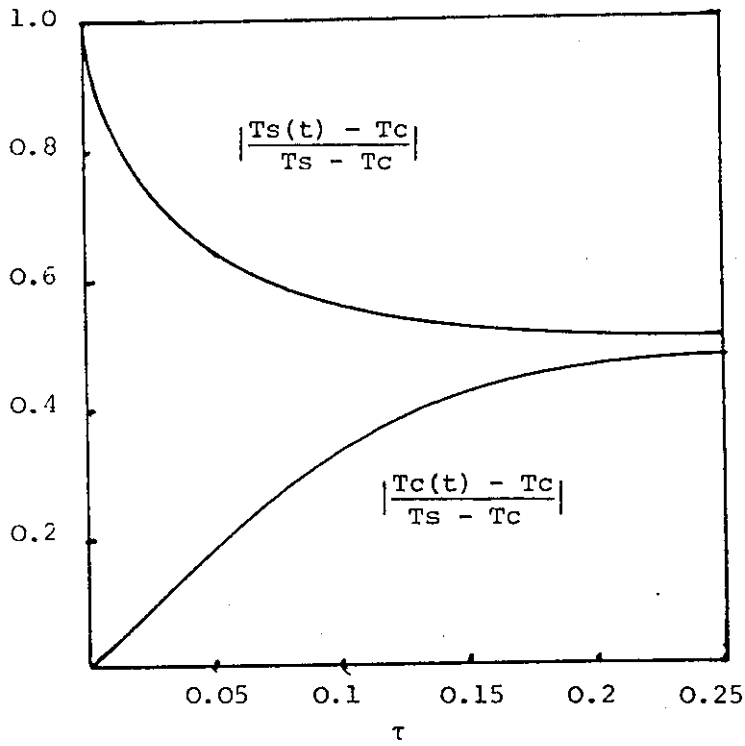


Fig. 6.6 The variation of the surface and centre temperatures, $T_s(t)$ and $T_c(t)$ during the soaking period with respect to the initial temperatures T_s and T_c

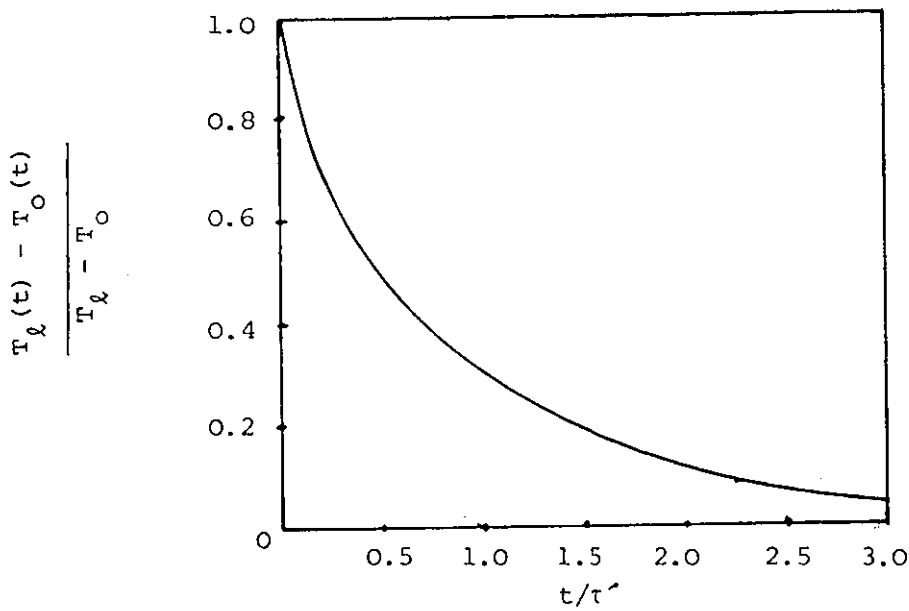


Fig. 6.7 The variation of the temperature difference between the two ends during the soaking period as a function of the initial difference

6.3 A Coil for Power Density Taper

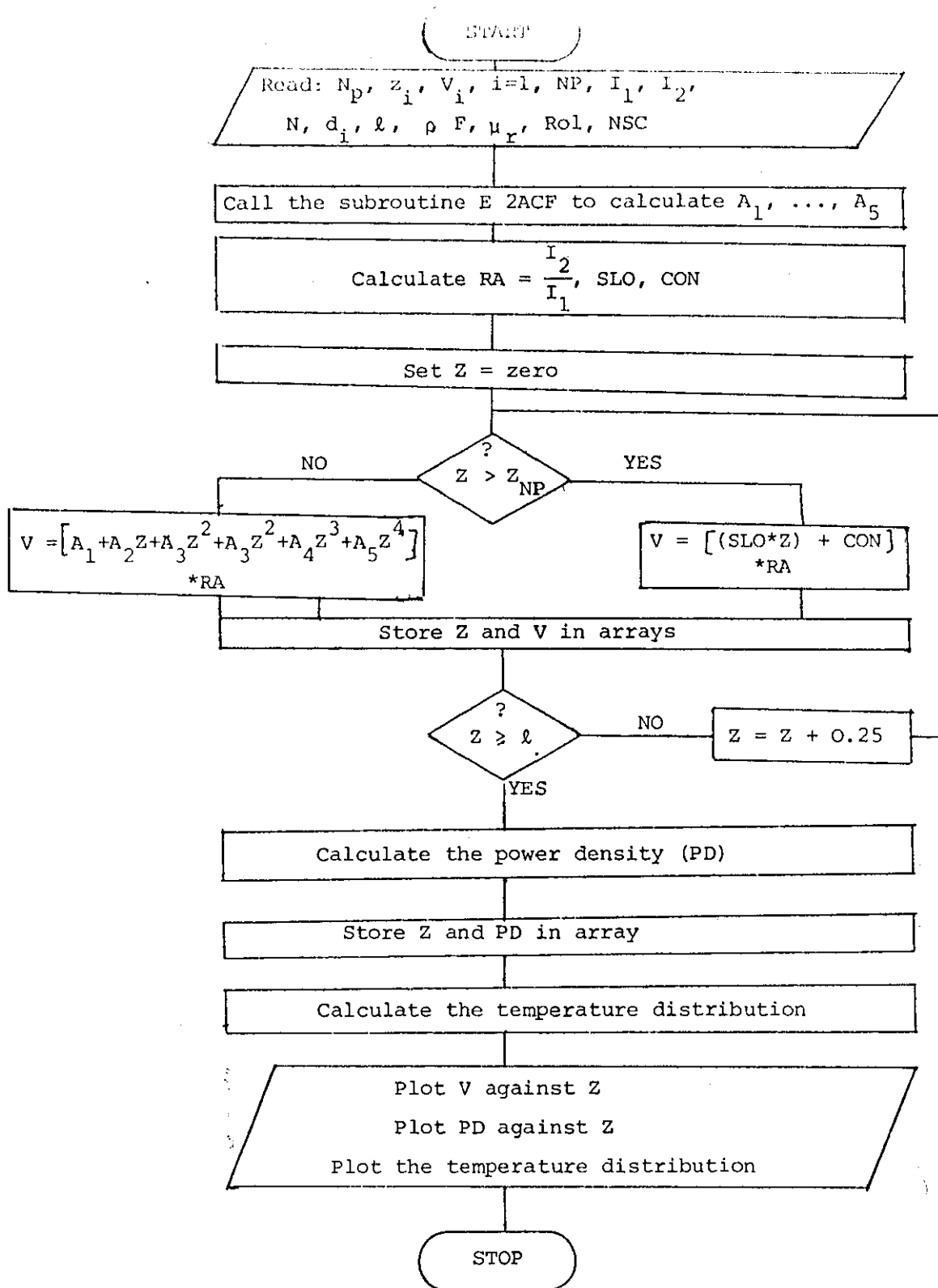
A theory is pointless, unless it is of some practical use in the solution of particular problems. The superposition method to be employed here to design a coil capable of producing linear surface power density distribution along an aluminium cylinder.

A computer program "W-FC-TEMP" was developed to calculate and plot the power density distribution induced on a cylindrical load from a nonuniform coil of unequal pitch . Similar to the uniform coil, the first stage in this program is to find the response of a one turn coil in an equation of the form

$$V(z) = a_0 + a_1z + a_2z^2 + a_3z^3 + a_4z^4 \quad (6.20)$$

It is then necessary to apply the superposition principals to find the total voltage induced at different points on the load. Then using the required Bessel functions to calculate the power density distribution and to plot it in graphical form.

The program, also, calculates the temperature distribution along the length of the cylinder at the moment when it leaves the heater i.e. at the start of the soaking period. The output of this part is a three dimensions graph of $T_r - T_c$ against the length and the radius of the workpiece. The method of calculation in this program is almost identical to that in the program "W-SC-FIT" for a cylindrical load with the exception that different coil pitches instead of one value for the pitch apply and the plotting of the temperature distribution. The flow chart of "W-FC-TEMP" is shown below.



The flow chart of the program "W-FC-TEMP"

The program was used to design a coil capable of producing a linear surface power density on a workpiece. The work coil was made from 6mm diameter water cooled copper tubing which meant that the variable pitch coil could easily be manufactured with the University's facilities. The workpiece was a 200mm long aluminium cylindrical billet of radius 50.8mm and the air gap was taken as 30mm which is in agreement with industrial practice [6.4].

The induced surface current density on the load due to a 400A current flowing through a single turn coil was measured. These values were then used to calculate the resultant surface current density of a number of different work coil configurations using the procedure outlined previously. The design of a work coil capable of generating a linearly decreasing temperature taper on the surface of the billet was then decided upon and details of the work coil pitch are given in Table 6.1. The eleven turn work coil was constructed and the surface current and power densities it produced were measured. The surface power densities induced in the load varied from 3.2 kW/m^2 down to 0.22 kW/m^2 . The practical measurements were compared with those required to produce the linearly decreasing temperature taper and the results are shown in Fig. 6.8 correlation to within an accuracy of 5% can be seen thus demonstrating the validity of this novel method of work coil design.

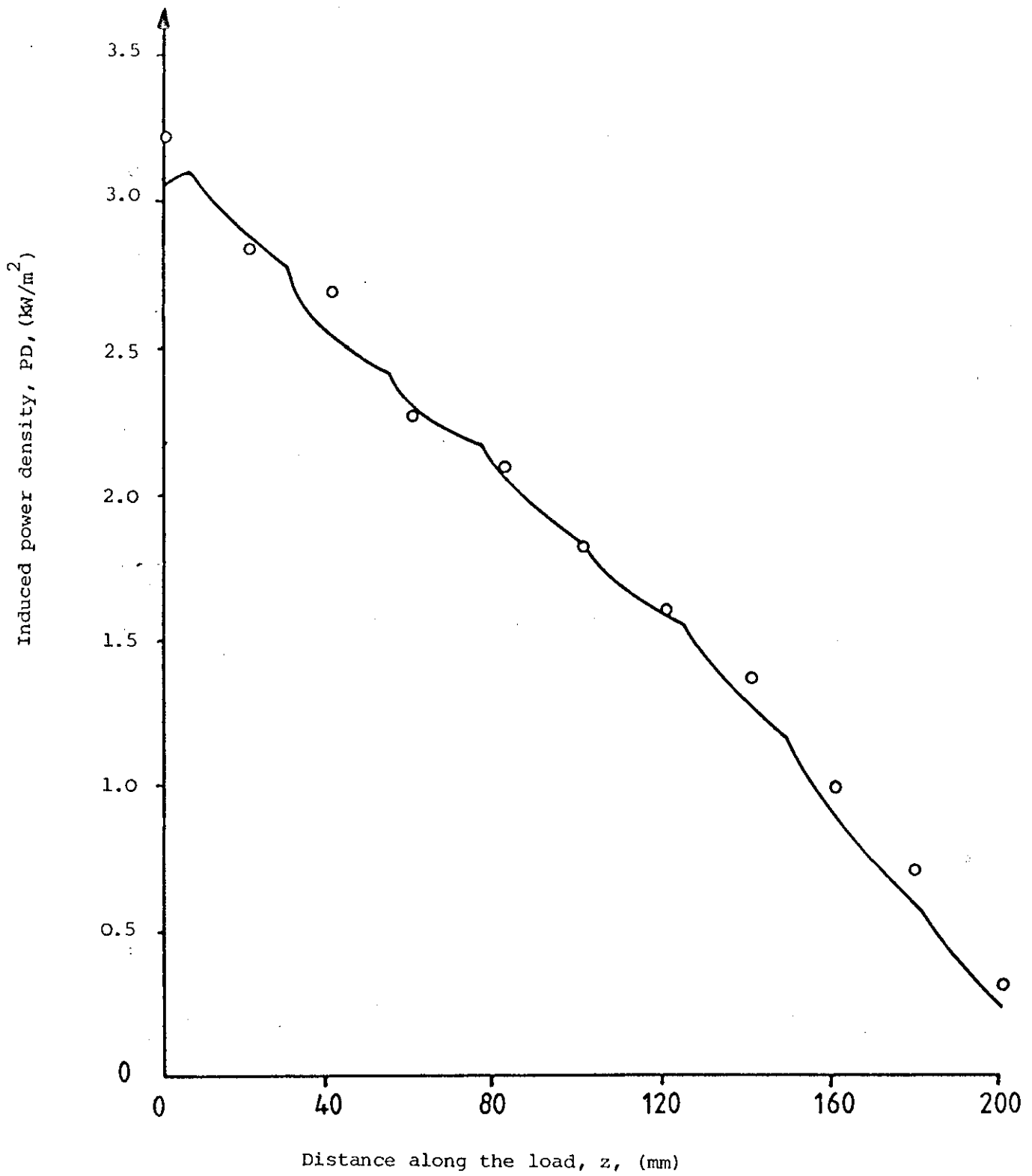


Fig. 6.8 Power density distribution along the aluminium cylinder due to 11 turns coil at air gap of 30mm, nonuniform coil pitch and a current of 400A

— Prediction by Superposition o Practical Reading

Pitch no.	1	2	3	4	5	6	7	8	9	10
mm	7	8	11	23	23	23	25	23	25	32

Table 6.1 The pitch of the coil

The expected temperature distribution in the workpiece exactly after leaving the heater is shown in Fig. 6.9.

6.4 The Construction of a Versatile Coil

To demonstrate fully the usefulness of the superposition method, a versatile coil capable of producing different patterns of surface power density distributions along the aluminium cylinder has been built. A 30 turn, 160mm inner diameter coil with uniform pitch of 10.5mm was made of 10mm diameter copper tube. Each turn of this coil had brazed to it a copper strip for electrical connection so that neighbouring turns could be short circuited to produce a coil with variable inter turn spacings, Fig. 6.10.

The surface power density distribution on the load produced by a single turn of this coil is shown in Fig. 6.11.

The program "W-FC-TEMP" was used to design a coil suitable to produce a linear surface power density distribution along the aluminium cylinder. The coil was of 12 turns with inter turn spacings as

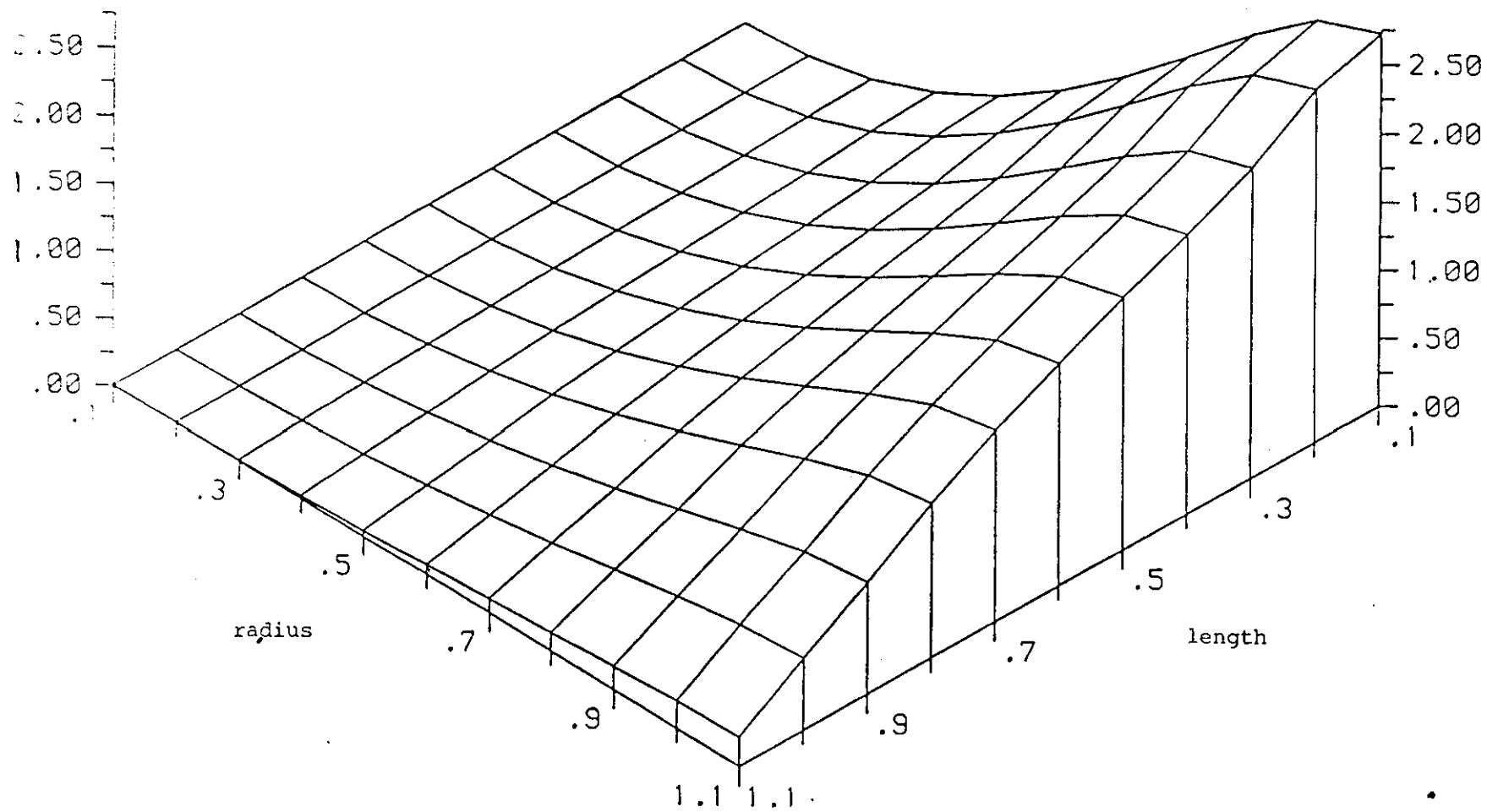


Fig. 6.9 Temperature distribution due to the coil mentioned in Table 6.1

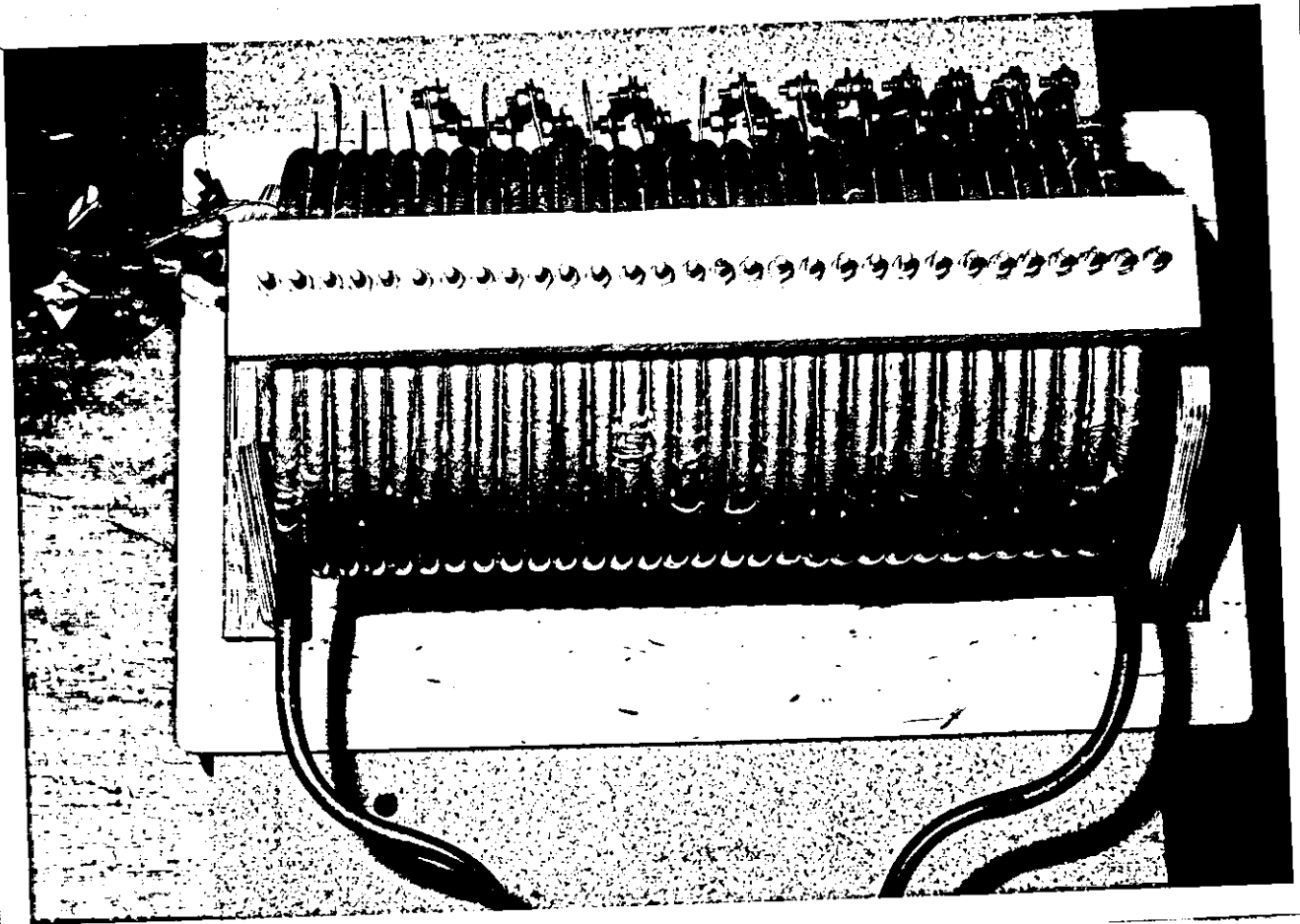


Fig. 6.10 The Versatile Coil

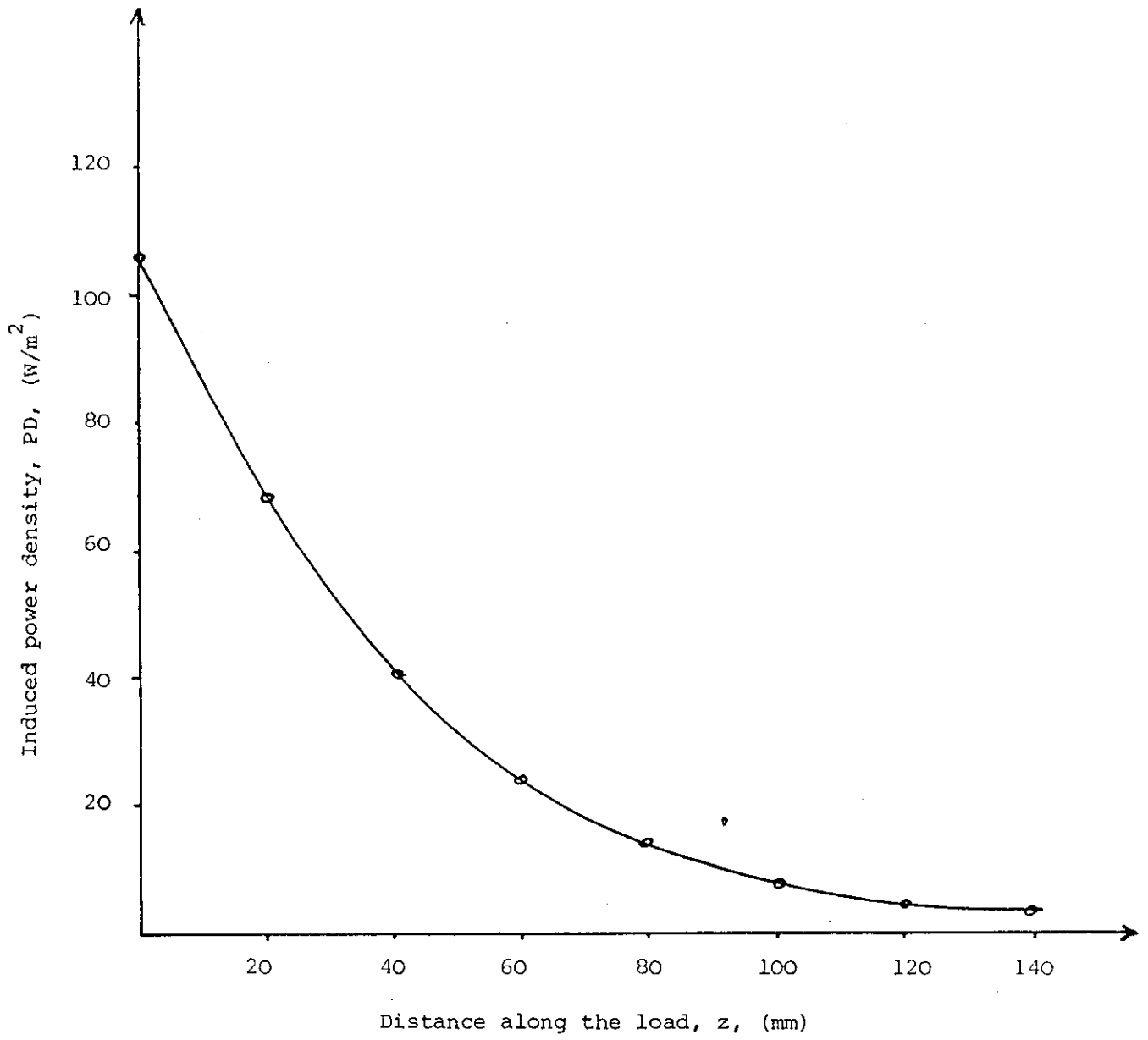


Fig. 6.11 Surface power density distribution along the aluminium cylinder due to single turn of the versatile coil with current of 400A.

shown in Table 6.2 with a coil current of 400A. The practical measurements were compared with the predicted power density distribution and the results are shown in Fig. 6.12.

Pitch no.	1	2	3	4	5	6	7	8	9	10	11
mm	10.5	10.5	10.5	10.5	21	10.5	32.5	10.5	32.5	10.5	32.5

Table 6.2 The inter turn spacings of the coil which produces the surface power density distribution shown in Fig. 6.12

The correlation between the calculated and measured values were very good and the discrepancy did not exceed 10%. The predicted curve is not very linear, this is due to the fact that the spacing between the turns can be varied by 10.5mm, which is the pitch of the coil, or a multiple of this value only. The temperature distribution of the workpiece exactly after leaving the heater, would be that shown in Fig. 6.13.

6.5 A Simplified Coil Design Program

To demonstrate the simplicity of the superposition method as a coil design technique; a nonuniform coil design program "Superposition" was written to be used with a programmable hand calculator.

As the calculator lacks a subroutine, which calculates the best

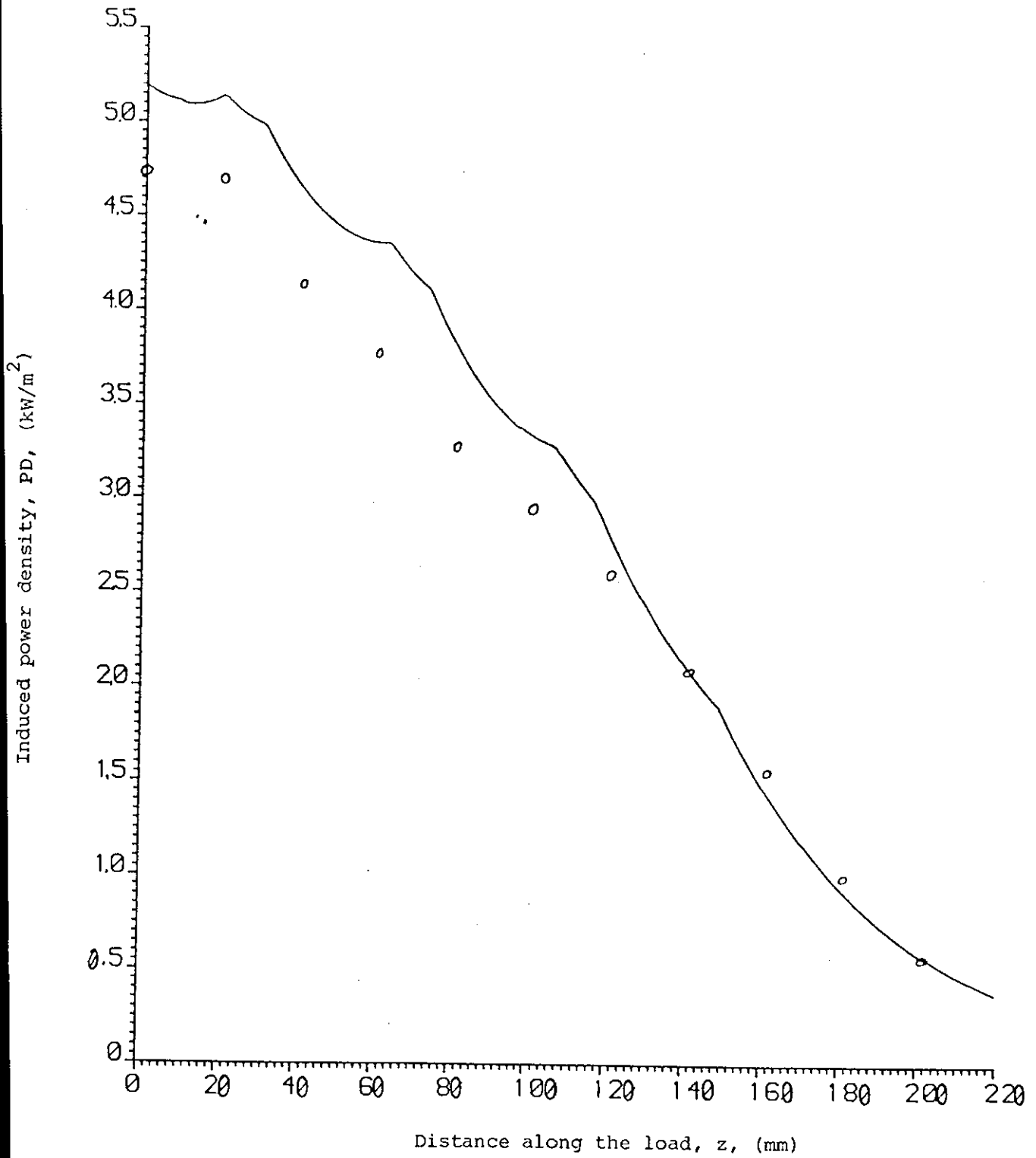


Fig. 6.12 Power density distribution along the aluminium cylinder due to 12 turns versatile coil at air gap of 30mm, nonuniform coil pitch and a current of 400A.

— prediction by superposition o practical reading

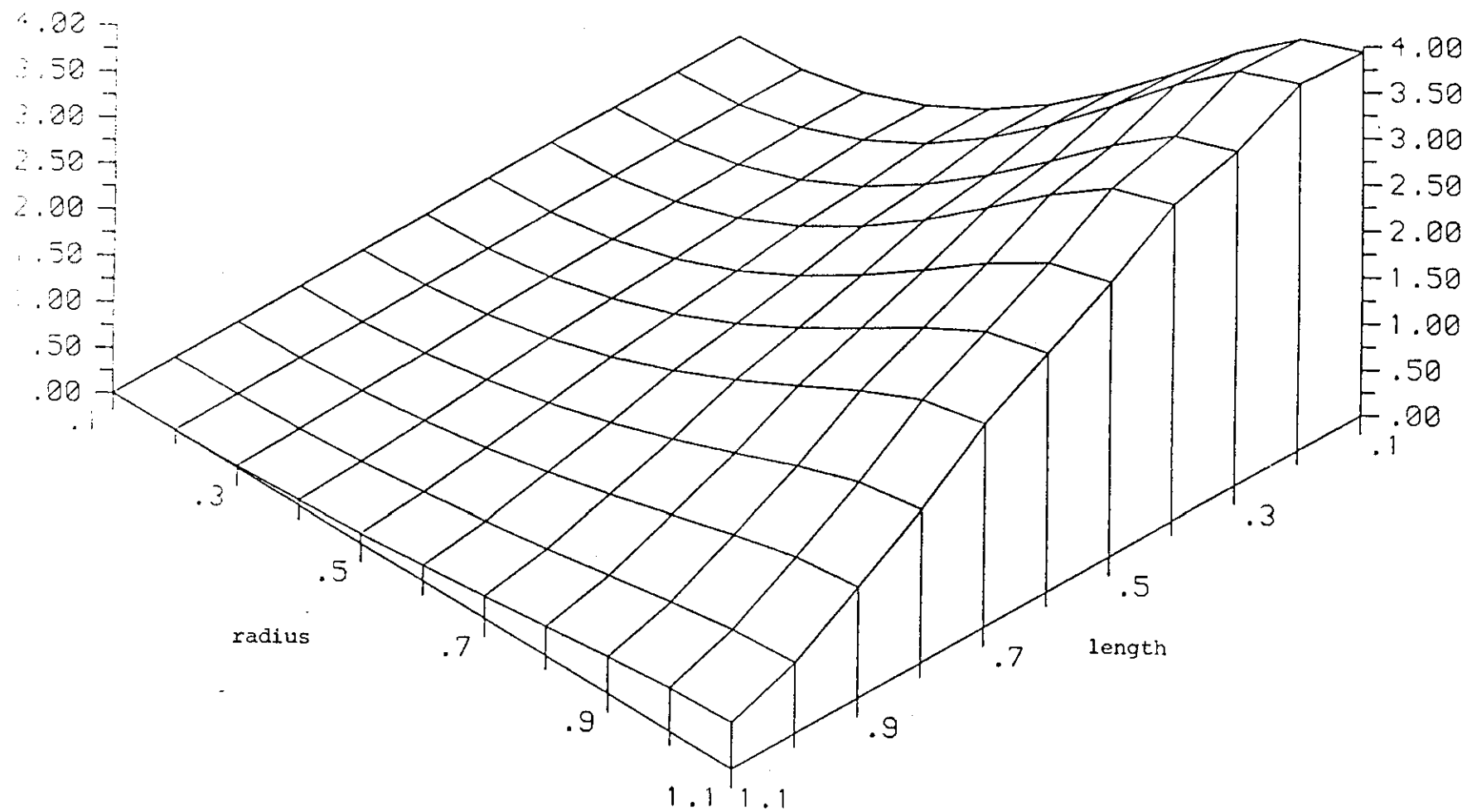


Fig. 6.13 Temperature distribution due to the coil mentioned in Table 6.2

curve to fit the readings of the single conductor, the practical equation (5.2) was taken to predict the surface magnetic field strength, induced on the load, from a single turn around the workpiece. The magnetic field strength induced from all turns are to be calculated by the superposition principles in accordance with the suggested coil pitches. The power density at any point is a function of H^2 ; that is

$$PD = CH^2 \quad (6.21)$$

The value of the constant C should be given, see Appendix 1, as the Bessel functions cannot be computed by the calculator. The suggested coil pitches and the single turn equation parameters were written independently of the program so that they can be varied without affecting the program.

6.6 Conclusion

An equation for the distribution of the surface power density on an aluminium cylinder induced from single turn coil has been found. This equation matches the equation for aluminium slab of thickness equal to the radius of the cylinder.

The practical investigations show that the superposition technique can be applied to a cylindrical load. Applying this method to a practical application i.e. the production of a linear power density taper along the length of a billet, proved highly successful.

The computer program was relatively small, i.e. it did not need a large storage capacity or consume long computing time. A simplified solution suitable for a hand calculator was also given by applying the superposition principles to the single turn equation.

The usefulness of the superposition technique has been fully demonstrated by constructing a versatile coil capable of producing different patterns of surface power density distribution along the cylindrical workpiece.

CHAPTER 7

CONCLUSION AND SUGGESTIONS

FOR FURTHER RESEARCH

The results and their implications are summarised and possible areas of further research are suggested.

7.1 Conclusion

A novel method of work coil design is described which facilitates the design of induction heating work coils capable of inducing a nonuniform power density along a workpiece.

A versatile coil capable of inducing different patterns of power density distribution along the load has been constructed.

A coil producing a linear power density taper along an aluminium cylindrical workpiece has been designed and constructed.

An equation was established for the distribution of the induced surface current density along an aluminium slab and cylinder due to a current in nearby conductor.

Three correction factors were found experimentally to amend the theoretically derived equation of the surface current density distribution along the load induced from a current in a nearby conductor.

Mathematical expressions were found for the three correction factors of the equation of the surface current density distribution along the load induced from a current in a nearby conductor.

The variation of the permeability has been taken into account in the measurement of the power density distribution induced on a magnetic workpiece.

Different types of current density probes have been constructed and used throughout the work.

Magnetic field strength probe has been constructed and used to measure the magnetic field strength on the surface of a ferromagnetic workpiece.

A computer program has been developed to design a coil capable of inducing a predetermined surface power density distribution along the load.

A simplified program has been written for a hand calculator to design a coil to produce a nonuniform power density along the load.

The axial and radial temperature distributions in a cylindrical workpiece inside a nonuniform coil have been predicted and plotted by the use of the computer.

An equivalent circuit based on a coil-multilayer load of different physical properties has been formulated to take the nonuniform physical properties of the workpiece into account.

A computer program has been developed for the coil-multilayer load equivalent circuit and the effect of the load nonuniform physical properties has been assessed.

A two dimensional finite difference computer program has been developed to calculate the magnetic field and power distributions in an inductively heated workpiece.

Computer programs have been developed to aid the understanding of the equivalent circuit technique and to assess the effect of different parameters.

A computer program has been developed to cater for the Bessel functions required for different calculations.

The surface current density distribution on the coil conductor was investigated and the effect of this distribution on the power density induced on the load was assessed.

The investigations included the effect of the laminations on the distribution of the power density induced on the load.

The equivalent circuit method has been explained and different literature concerning this method have been reviewed.

The use of the numerical techniques to solve induction heating problems has been reviewed and number of literature on this subject have been discussed.

7.2 Possible Areas of Further Research

The major achievement of the present work is the development of the superposition method as a design technique for a coil capable of inducing a predetermined nonuniform power density distribution along a load. This method was successfully used to produce a linear power density taper along the aluminium cylinder as required in the extrusion industry.

The applicability of the superposition method to mild steel workpieces was also demonstrated. This technique could therefore be used in applications such as off-the-bar forging where a nonuniform power density is required, and vessel heating where the design of the coil is still very empirical.

In off-the-bar forging, the heater has to deal with a workpiece of differing physical properties along its length as a result of the temperature difference between the cold bar and the partly heated end. Existing techniques allow the workpiece to cool before reheating. This, of course, means a substantial loss of energy and a very inefficient process. The requirement for this application is to induce a nonuniform power density along a workpiece having nonuniform physical properties. It may be possible to use superposition method to deal with these two conditions. The theoretical applications on Chapter 4 showed that this technique can take the nonuniformity of the load's resistivity into account. The experimental work in Section 5.2.1 proved that the variation of the permeability can be dealt with

by the superposition method. Finally, the production of the linear power density taper along the aluminium cylinder, confirmed the ability of this method to design a coil to induce a nonuniform power density along the load.

In certain chemical engineering operations it is necessary to use induction to heat stainless steel or mild steel containers into a low temperature, 200 to 300°C. The main problems associated with vessel heating are that the assumption of a long solenoid cannot be applied as the ratio of the vessel's diameter to its length is relatively large; and the variable permeability of the mild steel vessel should be taken into account as the temperature is well below the Curie point.

The design of the work coils for vessel heaters is a cumbersome task, as the use of the existing coil design methods in this application was not successful. Thornton [7.1] presented empirical formulae to aid on the design of the coils for containers within a specific range, but he stated clearly that these formulae cannot be generalised.

The superposition technique may be suitable as a design method for this application, since this method does not assume the coil as a solenoid and, also, the variation of the permeability can be taken into account. Further research in these areas is recommended.

In the present economic climate abrupt changes in the price and availability of fuels are commonplace, due to social and political effects. Attention should be devoted more to the question of energy conservation, and the newly developed multilayer coil is a good example. Further work to demonstrate the applicability of the superposition theories to the design of multilayer coils is recommended.

The efficiency of the induction heating systems can, also, be improved by the use of the laminations. The use of lamination packs is commonplace in induction melting furnaces, but they are rarely used with induction heating systems. Their employment will improve the efficiency by decreasing the stray losses. Further work to include the use of the laminations in the work coil design procedures is recommended.

Appendix 1

Heating a Cylindrical Workpiece

by Induction

The distribution of the magnetic field in a cylindrical workpiece can be determined by considering a cylinder of radius R , resistivity ρ and permeability μ , when subjected to a surface magnetic field H_R along its length (the symbol H_R is used for the surface magnetic field strength to be consistent with the radius). It is firstly necessary to take a ring of unit length, radius r ($r < R$) and thickness dr .

If H is the magnetic strength and J the current density at radius x , $x < r$, then the total flux within the ring is:

$$\phi_r = \mu \int_0^r 2\pi x H \, dx \quad (A1.1)$$

The emf induced in the ring is:

$$dE = - \frac{\partial \phi_r}{\partial t} \quad (A1.2)$$

$$\begin{aligned} \text{i.e. } dE &= - \frac{\partial}{\partial t} \left\{ \mu \int_0^r 2\pi x H \, dx \right\} \\ &= -2\pi\mu \int_0^r x \frac{\partial H}{\partial t} \, dx \end{aligned} \quad (A1.3)$$

The resistance of the ring is:

$$R_r = \frac{2\pi r \rho}{dr} \quad (A1.4)$$

and the current is:

$$I_r = J \, dr \quad (A1.5)$$

$$\text{also } I_r = \frac{dE}{R_r} \quad (A1.6)$$

$$= \frac{-2\pi\mu \int_0^r x \frac{\partial H}{\partial t} \, dx}{\frac{2\pi r \rho}{dr}} \quad (A1.7)$$

From equations (A1.5) and (A1.7)

$$J = \frac{-\mu}{\rho r} \int_0^r x \frac{\partial H}{\partial t} \, dx \quad (A1.8)$$

Using Maxwell's equation and ignoring the displacement current

$$J = - \frac{dH}{dr} \quad (A1.9)$$

then, by equating (A1.8) and (A1.9) the following results:

$$r \frac{dH}{dr} = \frac{\mu}{\rho} \int_0^r x \frac{\partial H}{\partial t} \, dx \quad (A1.10)$$

If H is a sinusoidal function then:

$$\frac{\partial H}{\partial t} = j\omega H \quad (A1.11)$$

From equations (A1.10) and (A1.11):

$$r \frac{dH}{dr} = \frac{\mu}{\rho} \int_0^r x j\omega H \, dx \quad (A1.12)$$

$$r \frac{dH}{dr} = \frac{j\omega\mu}{\rho} \int_0^r x H dx \quad (A1.13)$$

Differentiating (A1.13) with respect to r and inserting the limits yields:

$$r \frac{d^2 H}{dr^2} + \frac{dH}{dr} = \frac{j\omega\mu}{\rho} Hr \quad (A1.14)$$

$$\text{let } k = \frac{\sqrt{2}}{\delta} \quad (A1.15)$$

$$\text{i.e. } k^2 = \frac{\omega\mu}{\rho} \quad (A1.16)$$

Equation (A1.14) then becomes:

$$\frac{d^2 H}{dr^2} + \frac{1}{r} \frac{dH}{dr} - j k^2 H = 0 \quad (A1.17)$$

This is the diffusion equation, which governs the distribution of the magnetic field strength H . The adoption of Bessel functions facilitates the solution of this equation [A1.1]. The magnetic field strength at radius r , H_r , is:

$$H_r = CI_0(kr\sqrt{j}) + BK_0(kr\sqrt{j}) \quad (A1.18)$$

where C and B are constants

It is known [A1.2] that:

$$K_0(0\sqrt{j}) = \infty \quad (A1.19)$$

As the magnetic field strength in the centre of the cylinder

($r = 0$) is not infinite, then B must be zero. This reduces the solution to

$$H_r = CI_0(kr\sqrt{j}) \quad (\text{A1.20})$$

$$\text{i.e. } H_r = C(\text{ber } kr + j \text{ bei } kr) \quad (\text{A1.21})$$

The magnetic field strength at the surface ($r = R$) is H_R :

$$H_R = C(\text{ber } kR + j \text{ bei } kR) \quad (\text{A1.22})$$

$$\text{i.e. } C = \frac{H_R}{\text{ber } kR + j \text{ bei } kR} \quad (\text{A1.23})$$

From equations (A1.21) and (A1.22):

$$H_r = H_R \left[\frac{\text{ber } kr + j \text{ bei } kr}{\text{ber } kR + j \text{ bei } kR} \right] \quad (\text{A1.24})$$

$$\text{i.e. } H_r = H_R \left[\frac{\text{ber } \frac{\sqrt{2}}{\delta} r + j \text{ bei } \frac{\sqrt{2}}{\delta} r}{\text{ber } \frac{\sqrt{2}}{\delta} R + j \text{ bei } \frac{\sqrt{2}}{\delta} R} \right] \quad (\text{A1.25})$$

From equations (A1.9) and (A1.21):

$$J_r = -Ck(\text{ber}'kr + j \text{ bei}'kr) \quad (\text{A1.26})$$

From equations (A1.23) and (A1.26):

$$J_r = -H_R k \left[\frac{\text{ber}'kr + j \text{bei}'kr}{\text{ber } kR + j \text{bei } kR} \right] \quad (\text{A1.27})$$

$$\text{i.e. } J_r = -\frac{\sqrt{2}}{\delta} H_R \left[\frac{\text{ber}' \frac{\sqrt{2}}{\delta} r + j \text{bei}' \frac{\sqrt{2}}{\delta} r}{\text{ber } \frac{\sqrt{2}}{\delta} R + j \text{bei } \frac{\sqrt{2}}{\delta} R} \right] \quad (\text{A1.28})$$

The total flux lying inside a radius r is given by equation (A1.1).

Inserting equation (A1.21) into equation (A1.1) yields:

$$\phi_r = 2\pi\mu \int_0^r C(\text{ber } kx + j \text{bei } kx) x \, dx \quad (\text{A1.29})$$

The integration of the Bessel functions [A1.2] are

$$\int_0^x x \text{ber } x \, dx = x \text{bei}'x \quad (\text{A1.30})$$

$$\int_0^x x \text{bei } x \, dx = -x \text{ber}'x \quad (\text{A1.31})$$

Substitute equations (A1.30) and (A1.31) into equation (A1.29)

to yield:

$$\phi_r = \frac{2\pi\mu Cr}{k} (\text{bei}'kr - j \text{ber}'kr) \quad (\text{A1.32})$$

Substitute C from equation (A1.23) into equation (A1.32) to produce:

$$\phi_r = \frac{2\pi\mu_H R}{k} \left[\frac{\text{bei}'kR - j \text{ber}'kR}{\text{ber } kR + j \text{bei } kR} \right] \quad (\text{A1.33})$$

The total flux in unit length of the cylinder is;

$$\phi_R = \frac{2\pi\mu_H R}{k} \left[\frac{\text{bei}'kR - j \text{ber}'kR}{\text{ber } kR + j \text{bei } kR} \right] \quad (\text{A1.34})$$

which can be simplified to;

$$\phi_R = \mu_H R A_W (P - jQ) \quad (\text{A1.35})$$

where

$$P = \frac{2}{kR} \frac{\text{ber } kR \text{ bei}'kR - \text{ber}'kR \text{ bei } kR}{\text{ber}^2 kR + \text{bei}^2 kR} \quad (\text{A1.36})$$

and

$$Q = \frac{2}{kR} \frac{\text{bei } kR \text{ bei}'kR + \text{ber } kR \text{ ber}'kR}{\text{ber}^2 kR + \text{bei}^2 kR} \quad (\text{A1.37})$$

P and Q are as shown in Fig. A1.1.

The power dissipated in the ring is:

$$P_r = I_r^2 R_r \quad (\text{A1.38})$$

$$= (Jdr)^2 \frac{2\pi r \rho}{dr} \quad (\text{A1.39})$$

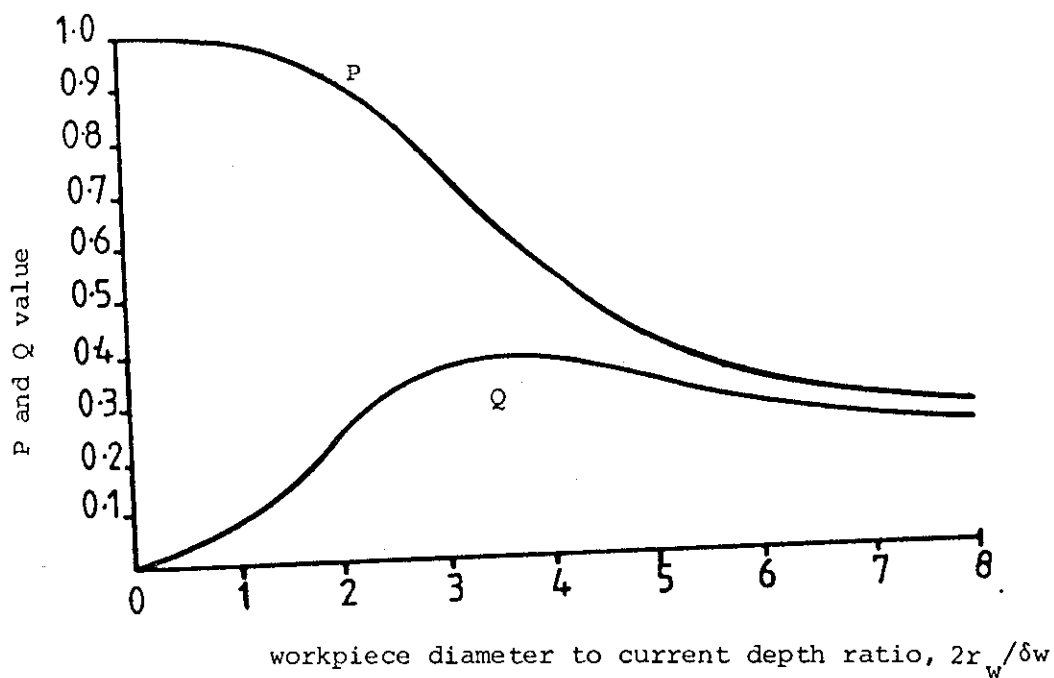


Fig. A1.1 P and Q for a solid cylinder

$$= J^2 2\pi r p \, dr \quad (\text{A1.40})$$

The total power per unit length of the cylinder can be calculated by substituting equation (A1.26) into equation (A1.40) and integrating

$$P_W = \int_0^R 2\pi |C|^2 k^2 \rho (\text{ber}^{-2} kr + j \text{bei}^{-2} kr) r \, dr \quad (\text{A1.41})$$

Using the following integration of Bessel functions [A1.2]

$$\int x (\text{ber}^{-2} x + \text{bei}^{-2} x) dx = x(\text{ber} x \text{ber}' x + \text{beix} \text{bei}' x) \quad (\text{A1.42})$$

Equation (A1.41) becomes;

$$P_W = 2\pi |C|^2 \rho k R (\text{ber } kR \text{ ber}' kR + \text{bei } kR \text{ bei}' kR) \quad (\text{A1.43})$$

Substitute for $|C|^2$ from equation (A1.23) and yield;

$$P_W = 2\pi \rho k R H_R^2 \left[\frac{\text{ber } kR \text{ ber}' kR + \text{bei } kR \text{ bei}' kR}{\text{ber}^2 kR + \text{bei}^2 kR} \right] \quad (\text{A1.44})$$

By using the expression of Q ; equation (A1.44) can be rewritten as:

$$P_W = 2\pi^2 f \mu H_R^2 R^2 Q \quad (\text{A1.45})$$

H_R is the r.m.s. value of the magnetic field strength. If the peak value H_{RM} is to be adopted, then:

$$P_W = \pi^2 f \mu H_{RM}^2 R^2 Q \quad (\text{A1.46})$$

The power density is:

$$PD = \frac{\pi^2 f \mu H_{RM}^2 R^2 Q}{2\pi R} \quad (\text{A1.47})$$

$$\therefore PD = \pi f \mu H_{RM}^2 \frac{R}{2} Q \quad (\text{A1.48})$$

$$\text{or } PD = \pi f \mu H_R^2 R Q \quad (\text{A1.49})$$

Appendix 2

The Reluctances of the Equivalent Circuit

The magnetic circuit of a long coil surrounding a workpiece is shown in Fig. A2.1.

The reluctance R_m of a flux path of length ℓ and cross sectional area A is given by:

$$R_m = \frac{\ell}{\mu_o \mu_r A} \quad (A2.1)$$

According to equation (A2.1) above, the reluctance of the air gap is:

$$R_{mg} = \frac{\ell_c}{\mu_o \pi (r_c^2 - r_w^2)} \quad (A2.2)$$

The reluctance of the cylindrical workpiece is

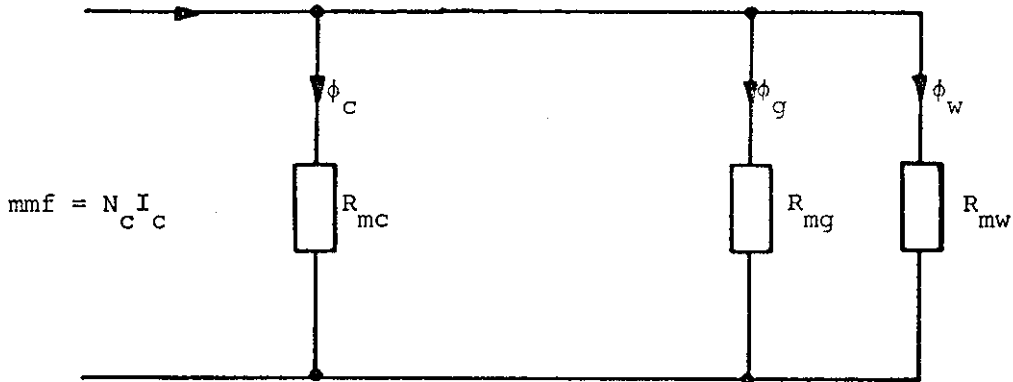
$$R_{mw} = \frac{N I_c}{\phi_w} \quad (A2.3)$$

where ϕ_w represents the total flux within the workpiece, it is given below and is calculated from Appendix 1.

$$\phi_w = \mu_o \mu_r H A_w (P - jQ) \quad (A2.4)$$

The workpiece cross sectional area is,

$$A_w = \pi r_w^2 \quad (A2.5)$$



and the magnetic field strength is

$$H = \frac{N_c I_c}{l_c} \quad (A2.6)$$

Substitute equations (A2.4), (A2.5) and (A2.6) into equation (A2.3) to yield;

$$R_{mw} = \frac{l_c}{\mu_o \mu_r \pi r_w^2 (P-jQ)} \quad (A2.7)$$

The coil reluctance can only be calculated if the spacing between the turns is ignored for the time being. The coil can then be approximated to a semi-infinite slab, as it's curvature is small when compared to the conductor thickness. The latter is greater than the penetration depth [A2.1], the total flux per unit width being:-

$$\phi = \frac{\mu_o \mu_r \delta_c}{2} H(1-j) \quad (A2.8)$$

As the relative permeability of copper is unity, and the width is the coil perimeter, the coil flux can therefore be defined as;

$$\phi_c = \frac{\mu_o \delta_c H}{2} k_r (2\pi r_c) (1-j) \quad (\text{A2.9})$$

where k_r is a correction factor, allowing for the spacing between the turns. Baker [A2.2] suggested that $1.5 \geq k_r \geq 1.0$ with 1.15 being a typical value. Reichert [A2.3] on the other hand employed the following expression for k_r

$$k_r = 0.92 \sqrt{\frac{\ell_c}{N_c t_c}} \quad (\text{A2.10})$$

where the conductor width is t_c .

The coil reluctance is

$$R_{mc} = \frac{N_c I_c}{\phi_c} \quad (\text{A2.11})$$

Substitute equations (A2.6) and (A2.9) into equation (A2.11) so as to yield;

$$R_{mc} = \frac{\ell_c}{\mu_o \pi k_r \delta_c r_c (1-j)} \quad (\text{A2.12})$$

Fig. A2.1 above applies when the return flux is ignored. In induction heating the flux will return along an external path, see Figs. 2.1 and 2.2. The length of the return flux path ℓ_r is a function of: the coil radius; the difference between the coil and the workpiece lengths; and the workpiece penetration depth.

This length was found experimentally [A2.4, A2.5] to be;

$$l_r = l_c - l_w + 2(0.45 + k)r_c \quad (A2.13)$$

where k is: a correction factor; a function of the air gap; the difference between the coil and workpiece lengths; and the workpiece penetration depth. The value of k lies between 0.0 and 0.1 for the return flow through air. If the flux returns through iron, the reluctance will be smaller and, therefore, the range of k will lie between -0.22 and 0.1 [A2.6]. For a coil and workpiece of the same length, equation (A2.13) reduces to

$$l_r = 2r_c(0.45 + k) \quad (A2.14)$$

Hence, according to equation (A2.1), the reluctance of the return path will be:

$$R_{mr} = \frac{2r_c(0.45+k)}{\mu_o \pi r_c^2} = \frac{2(0.45+k)}{\mu_o \pi r_c} \quad (A2.15)$$

Appendix 3

Illustration of Mutually Coupled Circuit Method

To illustrate the use of this method in determining the current distribution in inductively heated workpiece consider the coil-billet configuration shown in Fig. A3.1, [A3.1]. The coil current is known and distributed uniformly along the coil length. The workpiece is divided into n subconductors along current streamlines. The subconductor sizes need to be small near the surface and relatively large in the inside region. The coil is inductively coupled to each subconductor, also each subconductor is coupled to every other one. This will result in the equivalent circuit of Fig. A3.2, where all inductances are mutually coupled to all others.

This circuit can be described by the following matrix equation,

$$[Z_w][I_w] = [Z_{cw}][I_c] \quad (A3.1)$$

where Z_w is the $n \times n$ square matrix of workpiece impedances.

I_w is the column matrix of unknown workpiece currents

I_c is the coil current

Z_{cw} is the column matrix of mutual impedances between the coil and the workpiece subconductors

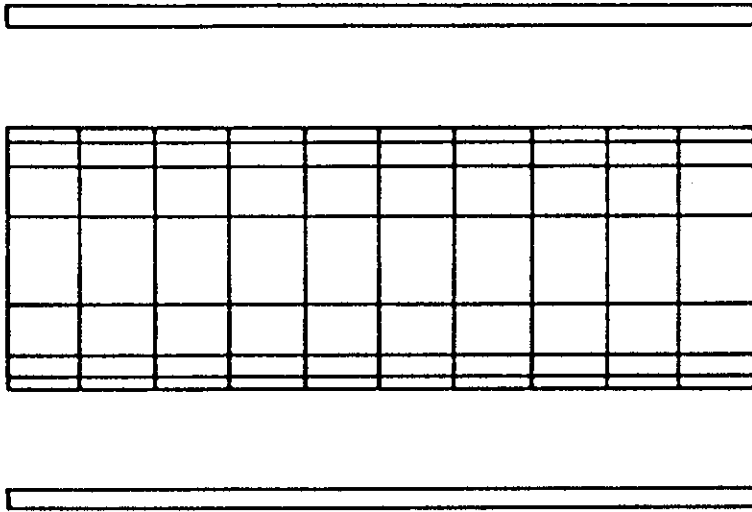


Fig. A3.1 Coil-workpiece

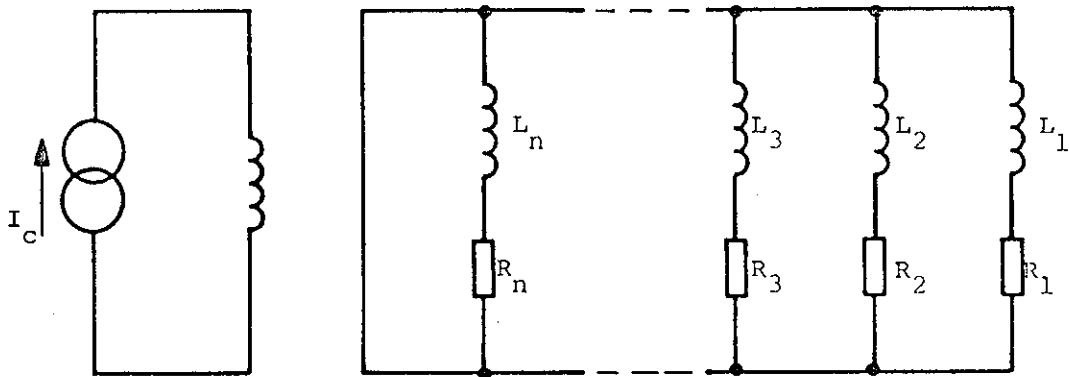


Fig. A3.2 Equivalent circuit for Fig. A3.1

where

$$Z_{wkl} = j\omega L_{kl} \quad \text{for } k \neq l \quad (\text{A3.2})$$

$$Z_{wkk} = j\omega L_{kk} + R_{kk} \quad \text{for } k \cong l \quad (\text{A3.3})$$

$$Z_{cwk} = -j\omega L_{ck} \quad (\text{A3.4})$$

$$k = 1, 2, 3, \dots, n$$

$$l = 1, 2, 3, \dots, n$$

R_{kk} is the resistance of subconductor k , L_{kk} is the self inductance of subconductor k , L_{kl} is the mutual inductance between subconductors k and l . Equation (A3.1) is a set of complex linear algebraic equations and may be solved for the workpiece subconductor currents I_w .

The formulae required to calculate the self and mutual inductances of circular turns of rectangular cross section and the mutual inductance between such a turn and a solenoid are given by Grover [A3.2].

$$L_{kk} = 4\pi r_c (\lambda_1 + \lambda_2) \quad (\text{A3.5})$$

where r_c is the mean radius of the coil

$$\begin{aligned} \lambda_1 = & \ln \frac{8r_c}{c} + \frac{1}{12} - \frac{\pi b}{3c} - \frac{1}{2} \ln \left(1 + \frac{b^2}{c^2}\right) + \frac{c^2}{12b^2} \ln \left(1 + \frac{b^2}{c^2}\right) \\ & + \frac{c^2}{12b^2} \ln \left(1 + \frac{c^2}{b^2}\right) + \frac{2}{3} \left(\frac{b}{c} - \frac{c}{b}\right) \tan^{-1} \frac{b}{c} \end{aligned} \quad (\text{A3.6})$$

$$\begin{aligned}
\lambda_2 = & \frac{c^2}{96r_c^2} \left\{ \left[\ln \frac{8r_c}{c} - \frac{1}{2} \ln \left(1 + \frac{b^2}{c^2} \right) \right] \left(1 + \frac{3b^2}{c^2} \right) \right. \\
& + 3.45 \frac{b^2}{c^2} + \frac{221}{60} - 1.6\pi \frac{b^2}{c^2} + 3.2 \frac{b^2}{c^2} \tan^{-1} \frac{b}{c} \\
& \left. - \frac{c^2}{10b^2} \ln \left(1 + \frac{b^2}{c^2} \right) + \frac{b^4}{2c^4} \ln \left(1 + \frac{c^2}{b^2} \right) \right\} \quad (A3.7)
\end{aligned}$$

and b and c are the breadth and depth of the segment. $\ln$ is the natural logarithm.

The expression of the billet mutual inductances is;

$$L_{12} = 4\pi \sqrt{r_1 r_2} \left[\left(\frac{2}{k} - k \right) F - \frac{2E}{k} \right] \quad (A3.8)$$

where r_1 and r_2 are the radii of the two circles, d is the distance between their centres, and

$$k = \frac{2 \sqrt{r_1 r_2}}{\sqrt{(r_1 + r_2)^2 + d^2}} \quad (A3.9)$$

F and E are integrals to modulus k . The values of these integrals are calculated using polynomial approximations.

The following formula gives the mutual inductance between a solenoid and a coaxial current filament

$$M = \frac{\mu_0 \pi a^2}{2\sqrt{A^2 + h^2}} \left[1 + \sum_{n=0}^{\infty} C_n \left(\frac{a^2 A^2}{(A^2 + h^2)^2} \right)^{n+1} x_{2n} \right] \quad (\text{A3.10})$$

where

$$C_0 = \frac{3}{8} \quad (\text{A3.11})$$

$$C_n = C_{n-2} \left(\frac{2(n+2)-1}{2n} \right) \text{ for } n \geq 1 \quad (\text{A3.12})$$

$$\text{and } x_{2n} = \sum_{p=0}^n K_p \left(-\frac{h^2}{A^2} \right)^{n-p} \quad (\text{A3.13})$$

where $K_0 = 1.0$

$$K_p = K_{p-1} \frac{[2(n+1-p)][2(n+1-p)+1]p}{4p^2(p+1)} \text{ for } p \neq 1 \quad (\text{A3.14})$$

where a is the radius of filament, A is the radius of solenoid, and h is the height of solenoid.

In order to adopt this expression for the mutual inductance between a particular segment and the coil several steps are required.

First, the segment to be divided into two filaments a and b , then the coil is to be divided in the plane of the segment into lengths h_1 and h_2 . The mutual inductance between the whole solenoid and filament a is then:

$$M_a = M_1 \frac{h_1}{h} + M_2 \frac{h_2}{h} \quad (\text{A3.15})$$

where M_1 is the mutual inductance between the filament a and the upper portion of the solenoid, M_2 is the mutual inductance between a and the lower portion of the solenoid. h_1 and h_2 are the lengths of upper portion and lower portion of the solenoid and h is the total length.

This procedure to be followed also for filament b. The mutual inductance between the solenoid and segment k is

$$M_{kc} = \frac{M_a + M_b}{2} \quad (A3.16)$$

Appendix 4

Illustration of Finite Element Methods

To explain the variational formulations procedure in mathematical terms, consider the following linear diffusion equation

$$\frac{1}{\mu} \left[\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} \right] = \frac{j\omega\phi}{\rho} - J \quad (\text{A4.1})$$

and ϕ is specified on the boundary.

The solution of equation (A4.1) can be obtained by solving the equivalent variational problem. This consists of expressing the diffusion equation in terms of an energy functional and minimizing it. The required functional is given by Chari [A4.1] as:

$$\begin{aligned} \text{Fun} = & \frac{1}{2\mu} \int \int_R \left\{ \left(\frac{\partial \phi}{\partial x} \right)^2 + \left(\frac{\partial \phi}{\partial y} \right)^2 \right\} . ds \\ & + \frac{j\omega}{2\rho} \int \int_R \phi^2 . ds - \int \int_R J . \phi ds \end{aligned} \quad (\text{A4.2})$$

Chari has shown, through complicated mathematics, that the Euler equation of the above functional is the linear diffusion equation (A4.1). The field region, R , is divided into subregions of rectangular or triangular shapes. ϕ defined in each subregion as the linear

interpolate of it's vertex values. Functional minimization is then achieved by substituting the value of ϕ in equation (A4.2) and setting it's derivative to zero with respect to each of the vertex potentials. This will lead, after some algebra, to the final matrix equation representing functional minimization to be solved for the required unknowns.

To illustrate the application of the weighted residual procedure to induction heating problems, consider a long ferromagnetic cylinder of arbitrary cross section subjected to a longitudinal magnetic field tangential to the surface, Fig. A4.1 [A4.2].

The value of this magnetic field is;

$$H = H_0 \cos \omega t \quad (A4.3)$$

For ferromagnetic material, the permeability will be a function of H and temperature. While the resistivity is a function of temperature only.

The electromagnetic problem can be solved in terms of the axially directed field strength:

$$H = H(x, y) \hat{k} \quad (A4.4)$$

which must satisfy the following partial differential equation within the cylinder

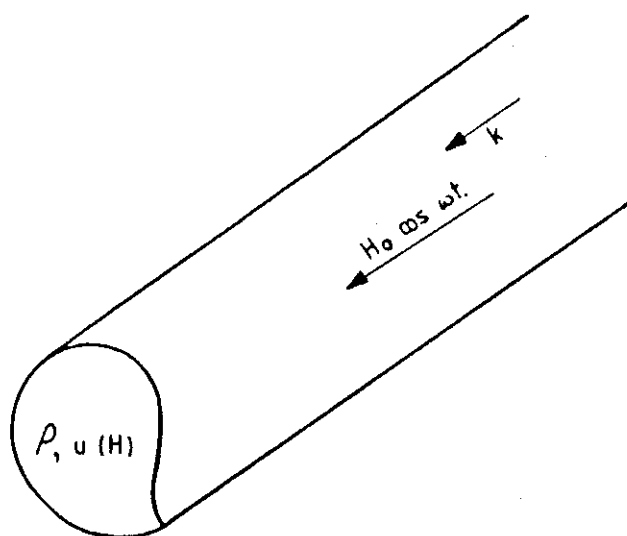


Fig. A4.1 Cylinder subjected to longitudinal magnetic field

$$\frac{\partial}{\partial x} \left(\rho \frac{\partial H}{\partial x} \right) + \frac{\partial}{\partial y} \left(\rho \frac{\partial H}{\partial y} \right) = \frac{\partial B}{\partial t} \quad (\text{A4.5})$$

subject to the boundary condition

$$H = H_0 \cos \omega t \quad (\text{A4.6})$$

on the conductor surface. In the magnetic state, the relation between B and H is governed by

$$B = \mu(H)H \quad (\text{A4.7})$$

The finite element formulation of the system, described by the partial differential equation (A4.5) and the boundary conditions of equation (A4.6) can be obtained by using a Galerkin procedure, which is a weighted residual technique. The application of this

method, usually requires a specialist numerical analyst as it is lengthy and complicated. However, the solution for this problem is known and the resulting matrix equation for the unknown values of H at interior nodes can be taken from Lavers [A4.2]:

$$[[s(\rho)] + j\omega[T][\mu]] \{H\} = 0 \quad (A4.8)$$

where $[s]$ and $[T]$ are the finite element matrices and $[\mu]$ is a diagonal matrix containing nodal permeabilities.

Equation (A4.8) is to be solved by a suitable method, so as to determine the magnetic field and power distributions within the billet.

Appendix 5

Current Density Probes

These probes were used to measure the current density on the surface of a conductor [A5.1-A5.5]. The probe consisted of two thin wires laid along the surface of the conductor.

The entire length of the wires, with the exception of the two extreme ends, was insulated electrically from the conductor.

The two ends were electrically connected to the conductor.

The other two ends of the wires were twisted together and connected to a voltmeter. The probe was aligned with the direction of the current in the conductor. The voltage across the probe was equal to the resistance drop in the element of the conductor immediately adjacent to the wire plus the induced voltage in the probe, i.e.

$$V = V_d + V_i \quad (A5.1)$$

where V : The voltage across the probe

V_d : The resistance drop in the element of the conductor
adjacent to the probe

V_i : The voltage induced in the probe

$$\text{Since } V_d = \rho l J \quad (A5.2)$$

$$V_i = N \frac{d\phi}{dt} \quad (A5.3)$$

$$\text{and } \phi = BA \quad (A5.4)$$

where ρ : The conductor resistivity

ℓ : The probe length

J : The current density on the surface of the conductor

N : The number of turns

ϕ : The flux in the probe-conductor circuit

B : Magnetic flux in the probe-conductor circuit

A : The area of the probe-conductor circuit

Substitute equations (A5.2), (A5.3) and (A5.4) into equation (A5.1), yields

$$V = \rho \ell J + \frac{d}{dt} (BA) \quad (A5.5)$$

The area A is very small as the probe was close to the conductor.

Equation (A5.5) can be written as

$$V = \rho \ell J \quad (A5.6)$$

For the measurements of the induced surface current density on a cylinder, the thin wire was wrapped tightly around the billet so that the emf induced on the surface of the cylinder was equal to that on the probe. As the leads of the probe were twisted together then the voltmeter measured only the voltage on the surface of the billet, which was

$$V = E\ell \quad (A5.7)$$

where E : The electric field intensity, which is

$$E = \rho J \quad (A5.8)$$

and $l = 2\pi R \quad (A5.9)$

substitute equations (A5.8) and (A5.9) into equation (A5.7) to yield,

$$V = 2\pi R \rho J$$

where R was the billet radius.

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The Listing of the Programs and the Publications

i. The Listing of the Programs:

SHORT

LONG

EQUIV

BESSEL

slab

W-P-POWER

PROXIMITY

W-SC-FIT

FITTING1

W-FC-TEMP

ALI

Magnetic Material

Superposition

ii. Publications:

1. "Improvements in the design of induction billet heaters",
19th UPEC, University of Dundee, April 1984.
2. "Methods of analysis for induction heating work coils",
20th UPEC, Huddersfield Polytechnic, April 1985.
3. "Illustrating electromagnetics using an industrial process",
IJEEE, 1986, Vol.23, No.1. (to be published)

SLIST SHORT

REAL PS,V,F,DW,ROWW,URW,ALC,DC,ROWC,URC,AKR,AKE,P,Q
 REAL BER,BEI,KER,KEI,BERD,BEID,KERD,KEID
 COMPLEX THETA,THY,KO,IO,KOD,IOD
 READ(5,*)PS,V,F,DW,ROWW,URW,ALC,DC,ROWC,URC,AKR,AKE
 PI=3.14159265
 PIS=9.8696
 X=10000000.0

DDW=SQRT((ROWW*X)/(4.*PIS*F*URW))

DDC=SQRT((ROWC*X)/(4.*PIS*F*URC))

BA=DW/(1.414214*DDW)

C CALCULATION OF BESSEL FUNCTIONS

IF(BA.GT.8)GO TO 5001

A1=BER(BA)

A2=BEI(BA)

A3=BERD(BA)

A4=BEID(BA)

GO TO 5002

5001 A1=REAL(IO(BA))

A2=AIMAG(IO(BA))

A3=REAL(IOD(BA))

A4=AIMAG(IOD(BA))

5002 CONTINUE

DEN=((A1)*(A1))+((A2)*(A2))

B1=((A1)*(A4))-((A2)*(A3))

B2=((A1)*(A3))+((A2)*(A4))

C1=B1/DEN

C2=B2/DEN

P=2.0*C1/BA

Q=2.0*C2/BA

AK=(8.*PIS*F)/(X*ALC)

AW=PI*(DW)**2/4.

AC=PI*(DC)**2/4.

AG=AC-AW

PC=PI*DC

RW=URW*AW*Q

RC=0.5*AKR*PC*DDC

XC=RC

XW=URW*AW*P

RE=AKE*1.8/PC

XE=ALC/RE

X1=AG+XW+XC

RW1=(RW*(XE)**2)/(RW**2+(X1+XE)**2)

XO=XE*(RW**2+X1**2+XE*X1)/(RW**2+(X1+XE)**2)

RO=RC+RW1

Z=SQRT(RO**2+XO**2)

```

PF=RO/Z
WRITE(1,5)PF
5  FORMAT(1H0,4HPF =,F6.4)
   EFF=(RW1/RO)*100.0
   WRITE(1,6)EFF
6  FORMAT(1H0,13HEFFICIENCY =,F7.4)
   VA=PS/PF
   ZO=Z*AK
   WRITE(1,10)ZO
10  FORMAT(1H0,4HZO =,F12.10)
   EC=SQRT(VA*ZO)
   WRITE(1,9)EC
9   FORMAT(1H0,4HEC =,F9.5)
   AIN=SQRT(VA/ZO)
   WRITE(1,13)AIN
13  FORMAT(1H0,5HAIN =,F12.3)
   TN=V/EC
   WRITE(1,7)TN
7   FORMAT(1H0,4HTN =,F7.2)
   AI=AIN/TN
   WRITE(1,8)AI
8   FORMAT(1H0,9HCURRENT =,F10.4)
   STOP

```

```

END
$INSERT BESSEL

```

OK,

SLIST LONG

REAL PS,V,F,DW,ROWW,URW,ALC,DC,ROWC,URC,AKR,P,Q
REAL BER,BEI,KER,KEI,BERD,BEID,KERD,KEID
COMPLEX THETA,THY,KO,IO,KOD,IOD
READ(5,*)PS,V,F,DW,ROWW,URW,ALC,DC,ROWC,URC,AKR
DDW=503.3*SQRT(ROWW/(F*URW))
DDC=503.3*SQRT(ROWC/(F*URC))
X=1000000.0

BA=DW/(1.414214*DDW)

C CALCULATION OF BESSEL FUNCTIONS

IF(BA.GT.8)GO TO 5001

A1=BER(BA)

A2=BEI(BA)

A3=BERD(BA)

A4=BEID(BA)

GO TO 5002

5001 A1=REAL(IO(BA))

A2=AIMAG(IO(BA))

A3=REAL(IOD(BA))

A4=AIMAG(IOD(BA))

5002 CONTINUE

DEN=((A1)*(A1))+((A2)*(A2))

B1=((A1)*(A4))-((A2)*(A3))

B2=((A1)*(A3))+((A2)*(A4))

C1=B1/DEN

C2=B2/DEN

P=2.0*C1/BA

Q=2.0*C2/BA

SP=(7.896*F)/(ALC*X)

AW=0.785*(DW**2)

AC=0.785*(DC**2)

AG=AC-AW

PC=3.14159265*DC

AP=URW*AW*Q

AA=0.5*AKR*PC*DDC

TB=AA

TF=URW*AW*P

TA=AG

EFF=(AP/(AA+AP))*100.0

WRITE(1,13)EFF

13 FORMAT(1H0,13HEFFICIENICY =,F6.3)

Z=SQRT(((AP+AA)**2)+((TA+TF+TB)**2))

PF=(AP+AA)/Z

WRITE(1,15)PF

15 FORMAT(1H0,4HPF =,F6.4)

VA=PS/PF

TR=Z*SP

E=SQRT(VA*TR)

AIN=SQRT(VA/TR)

TN=(V/E)

WRITE(1,18)TN

18 FORMAT(1H0,4HTN =,F7.2)

AI=AIN/TN

WRITE(1,20)AI

20 FORMAT(1H0,9HCURRENT =,F10.4)

STOP

END

\$INSERT BESSEL

OK,

SLIST EQUIV

```
C *****
C          COIL OF CONCENTRIC CYLINDERS
C    THIS PROGRAMME APPROXIMATES A BILLET AS N CONCENTRIC CYLINDERS
C *****
C
  REAL*4 L,K1,K,N1,I1
  INTEGER S,P
  DIMENSION XPT(100),EFF(100),W1(100),I1(100),Q1(100),
*RES(100),UR(100)
  WRITE(1,800)
800  FORMAT(// 'HOW MANY CONCENTRIC CYLINDERS ARE REQUIRED'//)
  READ(1,*)N
  WRITE(1,15)
15   FORMAT(// ' ENTER 1 IF RANGE OF VALUES REQD'// ' ENTER 2 IF ONE ',
*'VALUE REQD'//)
  READ(1,*)S
  IF(S.EQ.1)WRITE(1,16)
16   FORMAT(// ' WHICH PARAMETER DO YOU WISH TO VARY'// 'ENTER 1 FOR F'
*// '      2 FOR N1'// '      3 FOR L'// '      4 FOR RB'//
*'      5 FOR RC'//)
  IF(S.EQ.1)READ(1,*)M

  WRITE(1,1)
1    FORMAT(// '      **** NCOIL ****'//)
C    ***** INPUT RESISTIVITY PROFILE *****
C
  WRITE(1,803)
803  FORMAT(// 'ENTER RESISTIVITY OF INNER CYLINDER,RES1 '//)
  READ(1,*)RES(1)
  WRITE(1,5000)
5000  FORMAT(// 'ENTER RESISTIVITY OF OUTER CYLINDER,RES2'//)
  READ(1,*)RES(2)
  WRITE(1,776)
776   FORMAT(// 'ENTER INITIAL PERMEABILITY,UR(0) '//)
  READ(1,*)URO
  WRITE(1,777)
777   FORMAT(// 'ENTER FIRST CONSTANT OF PERMEABILITY,AF '//)
  READ(1,*)AAF
  WRITE(1,778)
778   FORMAT(// 'ENTER SECOND CONSTANT OF PERMEABILITY,BF '//)

  READ(1,*)BBF
  WRITE(1,779)
779   FORMAT(// 'ENTER THIRD CONSTANT OF PERMEABILITY,CF '//)
  READ(1,*)CCF
  WRITE(1,2)
2    FORMAT(// 'ENTER VOLTAGE,U1'//)
  READ(1,*)U1
  IF(M.EQ.1)GO TO 100
  WRITE(1,3)
3    FORMAT(// 'ENTER FREQUENCY,F'//)
```

```

READ(1,*)F
  IF(M.EQ.2)GO TO 103
100  WRITE(1,4)
4    FORMAT(//'ENTER POWER,N1'//)
    READ(1,*)N1
    IF(M.EQ.3)GO TO 105
103  WRITE(1,7)
7    FORMAT(//'ENTER LENGTH,L'//)
    READ(1,*)L
    IF(M.EQ.4)GO TO 106
105  WRITE(1,9)
9    FORMAT(//'ENTER BILLET RADIUS,RB'//)
    READ(1,*)RB
106  WRITE(1,10)
10   FORMAT(//'ENTER COIL RESISTIVITY,RESC'//)
    READ(1,*)RESC
    WRITE(1,12)
12   FORMAT(//'ENTER K1'//)
    READ(1,*)K1
    IF(M.EQ.5)GO TO 107
    WRITE(1,13)
13   FORMAT(//'ENTER COIL RADIUS,RC'//)

    READ(1,*)RC
107  WRITE(1,14)
14   FORMAT(//'ENTER K'//)
    READ(1,*)K
    IF(S.EQ.1)WRITE(1,17)
17   FORMAT(//' HOW MANY DATA POINTS ARE THERE ,LESS THAN 100 ONLY'//)
    IF(S.EQ.1)READ(1,*)NN
    IF(S.EQ.1)WRITE(1,18)
18   FORMAT(//' ENTER PARAMETER VALUES,LESS THAN 100 ONLY '//)
    IF(S.EQ.1)READ(1,*)(XPT(I),I=1,NN)
    IF(S.EQ.1)GO TO 19
    CALL SUB(U1,N,RES,UR,F,N1,L,RB,RESC,K1,RC,K,EFF(1),W1(1)
*,I1(1),Q1(1),URO,AAF,BBF,CCF)
    GO TO 21
19   CONTINUE
    DO 20 P=1,NN
    IF(M.EQ.1)CALL SUB(U1,N,RES,UR,XPT(P),N1,L,RB,RESC,K1,RC
*,K,EFF(P),W1(P),I1(P),Q1(P),URO,AAF,BBF,CCF)
    IF(M.EQ.2)CALL SUB(U1,N,RES,UR,F,XPT(P),L,RB,RESC,K1,RC
*,K,EFF(P),W1(P),I1(P),Q1(P),URO,AAF,BBF,CCF)
    IF(M.EQ.3)CALL SUB(U1,N,RES,UR,F,N1,XPT(
P),RESC,K1,RC,K,
*,EFF(P),W1(P),I1(P),Q1(P),URO,AAF,BBF,CCF)
    IF(M.EQ.4)CALL SUB(U1,N,RES,UR,F,N1,L,XPT(P),RESC,K1,RC
*,K,EFF(P),W1(P),I1(P),Q1(P),URO,AAF,BBF,CCF)
    IF(M.EQ.5)CALL SUB(U1,N,RES,UR,F,N1,L,RB,RESC,K1,XPT(P)
*,K,EFF(P),W1(P),I1(P),Q1(

```

```

20  CONTINUE
C    **** RESULTS ****
C
    WRITE(1,22)
22  FORMAT(//'          **** RESULTS ****'//)
C
21  IF(S.EQ.1)GO TO 23
    WRITE(1,24)
24  FORMAT(//'          EFF          W1          I1 ',
*      '          Q1 '//)
    WRITE(1,25)EFF(1),W1(1),I1(1),Q1(1)
25  FORMAT(5E14.5)
    GO TO 500
23  WRITE(1,26)
26  FORMAT(//'          XPT(I)          EFF          W1          I1 ',
*      '          Q1          '//)
    DO 27 I=1,NN
    WRITE(1,28)XPT(I),EFF(I),W1(I),I1(I),Q1(I)
28  FORMAT(6E13.5)
27  CONTINUE
500  CALL EXIT
    END
C
    SUBROUTINE SUB(U1,N,RES,UR,F,N1,
L,RB,RESC,K1,RC,K,EFF,
*W1,I1,Q1,URO,AAF,BBF,CCF)
C
    REAL*4 L,K1,K,N1,KERD,KEID,KER,KEI,KK,I1
    COMPLEX IOT,IODT,KOT,KODT,RF,Z(10),C(10),H(10),JD(10),A,
*ZAA,ZZ,IO,KO,IOD,KOD,AF(10,10)
    INTEGER P,N,M,T
    DIMENSION IOT(10,10),IODT(10,10),KOT(10,10),
*RES(10),UR(10),RR(10),X(10,10),A(20,20)
*,KODT(10,10),R(10),SD(10),YT1(10,10),YT2(10,10)
*,YT3(10,10),YT4(10,10),YT5(10,10),YT6(10,10),
*YT7(10,10),YT8(10,10),XD(10),Y(10)
C
    D=RB/N
    R(1)=D
    IF(N.EQ.1)GO TO 802
    DO 802 I=2,N
    R(I)=R(I-1)+D
802  CONTINUE
    PIE=3.1415926
    UO=4*PIE*1E-7
    W=2*PIE*F
    PC=0.92

    AM=PIE*(RC**2-RB**2)
    DDRR=(RES(2)-RES(1))/(N-1)
    DO 5001 I=2,N
    RES(I)=RES(I-1)+DDRR
5001  CONTINUE
C    ***** CALC. OF SKIN DEPTHS *****
C
    TT

```

```

E=0      UR(1)=URO
          DO 1500 JP=1,N
1510      DO 807 I=1,N
          IF (TTE.EQ.1)SD(I)=SQRT(2.0*RES(I)/(W*UO*UR(I)))
          IF (TTE.NE.1)SD(I)=SQRT(2.0*RES(I)/(W*UO*UR(JP)))
807      CONTINUE
          SDC=SQRT(2.0*RESC/(K1*W*UO))
C          ***** CALC. OF XD(I) *****
C
          DO 808 I=1,N
          XD(I)=-SQRT(2.0)/SD(I)
808      CONTINUE
          RFCR=PC*RESC*RC*2.0*PIE/(K1*L*SDC)
          RFCI=RFCR
          RF=CMPLX(RFCR,RFCI)
          XR=W*UO*PIE*RC/(2.0*(0.45+K))
          XO=W*UO*AM/L
          DO 806 I=1,N
          RR(I)=-RES(I)*2.0*PIE*R(I)/L
806      CONTINUE
C          **

*** CALC. OF X(I,J) ****
C
          X(1,1)=SQRT(2.0)*(R(1)/SD(1))
          IF(N.EQ.1)GO TO 811
          DO 810 I=2,N
          X(I,1)=SQRT(2.0)*(R(I)/SD(I))
          X((I-1),I)=SQRT(2.0)*(R(I-1)/SD(I))
810      CONTINUE
C          ***** CALC. OF Y(I) *****
C
C          *****
M=(N-1)
T=N-2
MM=2*N
MN=(2*N-1)
NM=(2*N-2)
C          *****
          DO 811 I=1,M
          Y(I)=RES(I)/RES(I+1)
811      CONTINUE
C          *****
C

          CALC. OF BESSELL FUNCTIONS
C          *****
          IF(X(1,1).GT.8)GO TO 101
          YT1(1,1)=BER(X(1,1))
          YT2(1,1)=BEI(X(1,1))
          YT3(1,1)=BERD(X(1,1))
          YT4(1,1)=BEID(X(1,1))
          YT5(1,1)=KER(X(1,1))

```

```

YT6(1,1)=KEI(X(1,1))
YT7(1,1)=KERD(X(1,1))
YT8(1,1)=KEID(X(1,1))
IOT(1,1)=CMPLX(YT1(1,1),YT1(1,1))
IODT(1,1)=CMPLX(YT3(1,1),YT4(1,1))
KOT(1,1)=CMPLX(YT5(1,1),YT6(1,1))
KODT(1,1)=CMPLX(YT7(1,1),YT8(1,1))
GO TO 100
101 IOT(1,1)=IO(X(1,1))
IODT(1,1)=IOD(X(1,1))
KOT(1,1)=KO(X(1,1))
KODT(1,1)=KOD(X(1,1))
100 CONTINUE
IF(N.EQ.1)GO TO 620
DO 102 I=2,N
IF(X(I,I).GT.8)GO TO 103
YT1(I,I)=BER(X(I,I))
YT2(I,I)=BEI(X(I,I))
YT3(I,I)=BERD(X(I,I))
YT4(I,I)=BEID(X(I,I))
YT5(I,I)=KER(X(I,I))
YT6(I,I)=KEI(X(I,I))
YT7(I,I)=
KERD(X(I,I))
YT8(I,I)=KEID(X(I,I))
IOT(I,I)=CMPLX(YT1(I,I),YT2(I,I))
IODT(I,I)=CMPLX(YT3(I,I),YT4(I,I))
KOT(I,I)=CMPLX(YT5(I,I),YT6(I,I))
KODT(I,I)=CMPLX(YT7(I,I),YT8(I,I))
GO TO 104
103 IOT(I,I)=IO(X(I,I))
IODT(I,I)=IOD(X(I,I))
KOT(I,I)=KO(X(I,I))
KODT(I,I)=KOD(X(I,I))
104 CONTINUE
IF(X(I-1,I).GT.8)GO TO 105
YT1((I-1),I)=BER(X((I-1),I))
YT2((I-1),I)=BEI(X((I-1),I))
YT3((I-1),I)=BERD(X((I-1),I))
YT4((I-1),I)=BEID(X((I-1),I))
YT5((I-1),I)=KER(X((I-1),I))
YT6((I-1),I)=KEI(X((I-1),I))
YT7((I-1),I)=KERD(X((I-1),I))
YT8((I-1),I)=KEID(X((I-1),I))
IOT((I-1),I)=CMPLX(YT1((I-1),I),Y
T2((I-1),I))
IODT((I-1),I)=CMPLX(YT3((I-1),I),YT4((I-1),I))
KOT((I-1),I)=CMPLX(YT5((I-1),I),YT6((I-1),I))
KODT((I-1),I)=CMPLX(YT7((I-1),I),YT8((I-1),I))
GO TO 102
105 IOT((I-1),I)=IO(X((I-1),I))
IODT((I-1),I)=IOD(X((I-1),I))

```

```

KOT((I-1),I)=KO(X((I-1),I))
KODT((I-1),I)=KOD(X((I-1),I))
102 CONTINUE
C *****
C SOLUTION OF N SIMULTANEOUS EQUATIONS
C *****
C *****
C SETTING MATRIX OF COEFFICIENTS
C *****
C **** SET ARRAY TO ZERO ****
C
DO 900 I=1,MM
DO 901 J=1,MN
A(J,I)=0
901 CONTINUE
900 CONTINUE
A(1,1)=IOT(1,1)
A(1,2)=-IOT(1,2)
A(1,3)=-KOT(1,2)
A(2,1)=IODT(1,1)*Y(1)
A(2,2)=-IODT(1,2)
A(2,3)=-KODT(1,2)
A
(MN,NM)=IOT(N,N)
A(MN,MN)=KOT(N,N)
A(MN,MM)=1
IF(N.LE.2)GO TO 902
DO 902 I=2,M
J=2*I-1
JI=J-1
A(J,JI)=IOT(I,I)
A(J,(JI+1))=KOT(I,I)
A(J,(JI+2))=-IOT(I,(I+1))
A(J,(JI+3))=-KOT(I,(I+1))
A((J+1),JI)=IODT(I,I)*Y(I)
A((J+1),(JI+1))=KODT(I,I)*Y(I)
A((J+1),(JI+2))=-IODT(I,(I+1))
A((J+1),(JI+3))=-KODT(I,(I+1))
902 CONTINUE
DO 700 I=1,MN
DO 612 J=1,MN
C ***** EVALUATE PIVOT COEFFICIENT *****
C
IF(J.EQ.I)GO TO 612
AF(J,I)=A(J,I)/A(I,I)
DO 613 JJ=1,MM
C ***** EVALUATE LINE (J-I) *****
C
A(J,JJ)=A(J,JJ)-AF(J,I)*A(I,JJ)
613 CONTINUE

```

```

612  CONTINUE
      DO 614 J=1,MN
      IF(J.EQ.1)GO TO 614
      A(J,I)=0
C     ***** THIS IS TO REDUCE ROUND-OFF ERROR *****
614  CONTINUE
700  CONTINUE
C     ***** CALC. OF CONSTANTS C(I) *****
C
      DO 620 I=1,MN
      C(I)=A(I,MN)/A(I,I)
620  CONTINUE
      IF(N.NE.1)GO TO 750
C     *****
C     FOR N=1 THERE IS ONLY ONE CONSTANT GIVEN BY:-
C     C(1)=1.0/IOT(1,1)
C     JD(1)=(-SQRT(2.0)*IODT(1,1)*C(1))/SD(1)
C     GO TO 210
C     *****
C
C     *****
C
      CALC. OF JD(I) AND H(I)
C     *****
C
750  H(1)=C(1)*IOT(1,1)
      JD(1)=(-SQRT(2.0)*IODT(1,1)*C(1))/SD(1)
      DO 200 I=2,M
      J=2*I-1
      H(I)=C(J-1)*IOT(I,I)+C(J)*KOT(I,I)
      JD(I)=XD(I)*(C(J-1)*IODT(I,I)+C(J)*KODT(I,I))
200  CONTINUE
      JD(N)=XD(N)*(C(2*N-2)*IODT(N,N)+C(2*N-1)*KODT(N,N))
      UR(JP+1)=((AAF/(BBF+H(JP)))+CCF)/UO
1500  CONTINUE
      IF (TTE.EQ.1)GO TO 1520
      TTE=1
      GO TO 1510
1520  CONTINUE
C     *****
C     CALC. OF IMPEDANCES Z(I)
C     *****
C
      DO 210 I=1,M

210  CONTINUE
      Z(N)=RR(N)*JD(N)
      AA=AIMAG(Z(N))
      B=REAL(Z(N))
      KK=XR/(XR+XO+AA)
      ZB=KK*B
      ZA=(XO+AA)*KK
      ZAA=CMPLX(ZB,ZA)
      ZZ=RF+ZAA
      WRITE(1,1755)ZZ
1755  FORMAT(// 'ZZ= ',

```


2E14.5)

```
ZBR=REAL(ZZ)
ZAI=AIMAG(ZZ)
ZM=CABS(ZZ)
```

C ZM IS MAGNITUDE OF FURNACE IMPEDANCE ZZ

```

C
C      ZAI=IMAGINARY PART OF ZZ
C      ZBR=REAL PART OF ZZ

```

```

W1=(U1/ZM)*(SQRT(ZBR/N1))
I1=U1/((W1**2)*ZM)
Q1=(I1**2)*(W1**2)*ZAI
EFF=100*KK*B/(RFCR+KK*B)
RETURN
END

```

C.
\$INSERT BESSEL

```
COMPLEX FUNCTION DIV(Y,X)
COMPLEX Y,X
REAL MODA,MODB,MODX
```

```
C
C      ****CALC. OF COMPLEX DIVISION *
C
```

C ***** CALC. OF Y/X *****

```
A=REAL(X)
B=AIMAG(X)
MODA=ABS(A)
MODB=ABS(B)
Z=1.0
IF(MODA.GT.MODB)GO TO 500
IF(MODB.LT.0.1)GO TO 501
IF(MODB.LT.1)GO TO 510
X=X/10
MODB=MODB/10
Z=Z*10
IF(MODB.GT.1)GO TO 502
GO TO 510
IF(MODB.LT.0.1)X=X*10
Z=Z/10
MODB=MODB*10
IF(MODB.LT.0.1)GO TO 501
GO TO 510
IF(MODA.LT.0.1)GO TO 503
```

```
504  IF(MODA.LT.1)GO TO 510
      X=X/10
      Z=Z*10
      MODA=MODA/10
      IF(MODA.GT.1)GO TO 504
      GO TO 510
503  IF(MODA.LT.0.1)X=X*10
      Z=Z/10
      MODA=MODA*10
      IF(MODA.LT.0.1)GO TO 503
510  Y=Y/Z
      MODX=CABS(X)
      Y=Y/(MODX**2)
      Y=Y*CONJG(X)
      DIV=Y
      RETURN
      END
```

C
OK,

SLIST BESSEL

REAL*4 FUNCTION BER(X)

X4=(X/8.)**4

Z=1-64*X4

X8=X4*X4

Z=Z+113.77777774*X8

X12=X8*X4

Z=Z-32.36345652*X12

X16=X12*X4

Z=Z+2.64191397*X16

X20=X16*X4

Z=Z-0.08349609*X20

X24=X20*X4

Z=Z+0.00122552*X24

X28=X24*X4

Z=Z-0.00000901*X28

BER=Z

END

C
C

REAL*4 FUNCTION BEI(X)

X2=(X/8.)**2

X4=X2*X2

X6=X2*X4

Z=16*X2-113.77777774*X6

X10=X6*X4

Z=Z+72.81777742*X10

X14=X10*X4

Z=Z-10.56765779*X14

X18=X14*X4

Z=Z+0.52185615*X18

X22=X18*X4

Z=Z-0.01103667*X22

X26=X22*X4

Z=Z+0.00011346*X26

BEI=Z

RETURN

END

C
C

REAL*4 FUNCTION KER(X)

Y=X/2.

Z=-ALOG(Y)*BER(X)+0.25*3.1415926*BEI(X)

X2=(X/8.)**2

X4=X2*X2

Z=Z-0.57721566-59.05819744*X4

X8=X4*X4

Z=Z+171.36272133*X8

X12=X8*X4

Z=Z-60.60977451*X12

X16=X12*X4

Z=Z+5.65539121*X16

X20=X16*X4

```

Z=Z-0.19636347*X20
X24=X20*X4
Z=Z+0.00309699*X24
X28=X24*X4
Z=Z-0.00002458*X28
KER=Z
RETURN
END

```

C
C

```

REAL*4 FUNCTION KEI(X)
Y=X/2.
X2=(X/8.)**2
X4=X2*X2
Z=-ALOG(Y)*BEI(X)-0.25*3.1415926*BER(X)
Z=Z+6.76454936*X2
X6=X4*X2
Z=Z-142.91827687*X6
X10=X6*X4

```

```

Z=Z+124.23569650*X10
X14=X10*X4
Z=Z-21.30060904*X14
X18=X14*X4
Z=Z+1.17509064*X18
X22=X18*X4
Z=Z-0.02695875*X22
X26=X22*X4
Z=Z+0.00029532*X26
KEI=Z
RETURN
END

```

C
C

```

REAL*4 FUNCTION BERD(X)
X2=(X/8.)**2

```

```

X4=X2*X2
X6=X4*X2
X10=X4*X6
X14=X10*X4
X18=X14*X4
X22=X18*X4
X26=X22*X4
Z=-4*X2+14.22222222*X6
Z=Z-6.06814810*X10+0.66047849*X14
Z=Z-0.02609253*X18+0.00045957*X22

```

```

Z=Z-0.00000394*X26
BERD=Z*X
RETURN
END

```

C
C

```

REAL*4 FUNCTION BEID(X)
X2=(X/8.)**2

```

```

X4=X2*X2
X8=X4*X4
X12=X8*X4
X16=X12*X4
X20=X16*X4
X24=X20*X4
Z=0.5-10.66666666*X4
Z=Z+11.37777772*X8-2.31167514*X12
Z=Z+0.14677204*X16-0.00379386*X20
Z=Z+0.00004609*X24
BEID=Z*X
RETURN
END

```

C
C

```

REAL*4 FUNCTION KERD(X)
Y=X/2.
X2=(X/8.)**2
X4=X2*X2

```

```

X6=X2*X4
X10=X6*X4
X14=X10*X4
X18=X14*X4
X22=X18*X4
X26=X22*X4
Z=-ALOG(Y)*BERD(X)-BER(X)/X+0.25*3.1415926*BEID(X)
ZZ=-3.69113734*X2+21.42034017*X6
ZZ=ZZ-11.36433272*X10+1.41384780*X14
ZZ=ZZ-0.06136358*X18+0.00116137*X22
ZZ=ZZ-0.00001075*X26
ZZ=ZZ*X
KERD=Z+ZZ
RETURN
END

```

C
C

```

REAL*4 FUNCTION KEID(X)

```

```

Y=X/2.
X2=(X/8.)**2
X4=X2*X2
X8=X4*X4
X12=X4*X8
X16=X12*X4
X20=X16*X4
X24=X20*X4
Z=-ALOG(Y)*BEID(X)-BEI(X)/X-0.25*3.1415926*BERD(X)
ZZ=0.21139217-13.39858846*X4
ZZ=ZZ+19.41182758*X8-4.65950823*X12

```

ZZ=ZZ+0.33049424*X16-0.00926707*X20
ZZ=ZZ+0.00011997*X24
ZZ=ZZ*X
KEID=Z+ZZ
RETURN
END

C
C

COMPLEX FUNCTION THETA(X)
REAL*4 K,J
Y=8./X
Y2=Y*Y
Y3=Y2*Y
Y4=Y3*Y
Y5=Y4*Y
Y6=Y5*Y

K=0.0110486*Y
K=K-0.0000906*Y3
K=K-0.0000252*Y4
K=K-0.0000034*Y5
K=K+0.0000006*Y6
J=-0.0110485*Y-0.3926991
J=J-0.0000901*Y3-0.0009765*Y2
J=J+0.0000051*Y5+0.0000019*Y6
THETA=CMPLX(K,J)
RETURN
END

C
C

COMPLEX FUNCTION THY(X)
REAL*4 K,J

Y2=Y*Y
Y3=Y2*Y
Y4=Y3*Y
Y5=Y4*Y
Y6=Y5*Y
K=0.7071068-0.0625001*Y
K=K-0.0013813*Y2+0.0000005*Y3
K=K+0.0000346*Y4+0.0000117*Y5
K=K+0.0000016*Y6
J=0.7071068-0.0000001*Y
J=J+0.0013811*Y2+0.0002452*Y3
J=J+0.0000338*Y4-0.0000024*Y5
J=J-0.0000032*Y6
THY=CMPLX(K,J)
RETURN
END

C
C

```

COMPLEX FUNCTION KO(X)
COMPLEX CN,Z2,F,K,THETA
Y=SQRT(3.1415926/(2*X))
W=SQRT(1./2.)
Z=W
CN=CMPLX(-W,-Z)
K=THETA(-X)
Z2=CN*X+THETA(-X)
F=CEXP(Z2)
KO=Y*F
RETURN
END

```

C
C

```

COMPLEX FUNCTION IO(X)
COMPLEX CN,ZX,F,FY,B,D,DF,THETA,KO
REAL M,N
Y=1/(SQRT(2*3.1415926*X))
W=SQRT(1./2.)

```

```

CN=CMPLX(W,Z)
ZX=CN*X+THETA(X)
F=CEXP(ZX)
FY=F*Y
B=KO(X)
M=AIMAG(B)
N=REAL(B)
D=CMPLX(-M,N)
DF=D/3.1415926
IO=FY+DF
RETURN
END

```

C
C

```

COMPLEX FUNCTION KOD(X)
COMPLEX CN,ZX,F,FY,FX,THETA,THY
Y=SQRT(3.1415926/(2*X))

```

```

W=1/SQRT(2.0)
Z=W
CN=CMPLX(-W,-Z)
ZX=CN*X+THETA(-X)
F=CEXP(ZX)
FY=F*Y
FX=-FY*THY(-X)
KOD=FX
RETURN
END

```

C
C

```

COMPLEX FUNCTION IOD(X)
COMPLEX CN,ZX,F,FY,FX,B,D,DX,THETA,THY,KOD
REAL*4 M,N

```

```

Y=SQRT(1/(2*3.1415926*X))
W=SQRT(1./2.)
Z=W
CN=CMPLX(W,Z)
ZX=CN*X+THETA(X)
F=CEXP(ZX)
FY=F*Y
FX=FY*THY(X)
B=KOD(X)
M=AIMAG(B)
N=REAL(B)
D=CMPLX(-M,N)
DX=D/3.1415926
IOD=FX+DX
RETURN
END

```

OK,

pr slab.fortran

slab.fortran

03/26/85 1716.9 gmt Tue

c THE SLAB PROGRAM- this has been written by Ali Al-Shaikhli on his Ph.D
c project.It calculates the magnetic field ,power
c density and power distribution in a
c non magnetic rectangular slab.It also calculates the
c total power induced in the slab.

```
print'(a,/)', 'This program carries out finite difference analysis'  
print'(a,/)', 'on a non magnetic rectangular slab.'  
print'(a,/)', 'It calculates the magnetic, power density and power  
1 distribution in the slab.'  
print'(a,/)', 'It also calculates the total power induced in the  
1 non magnetic slab.'  
print'(a,/)', 'To run the slab program ,type slab and press return'  
  
print'(a,/)', 'You will be asked to input values for HAG(the surface  
1 magnetic field strength'
```

```
1 magnetic field strength'  
print'(a,/)', 'and FREQ(the supply frequency).'  
print'(a,/)', 'You will also be asked to input a value for q(7).If  
1 q(7) is set to 0.0'  
print'(a,/)', 'then you will get a detailed output including arrays  
1 ROXA,ROX,ROY and MU.'  
print'(a,/)', 'To use the program for your own example you will hav  
1 e to do some simple editing'  
print'(a,/)', 'This is outlined in the EDDY report.'  
print'(a,/)', 'Also included in the report is a section which tells  
1 you how to get a print-'  
print'(a,/)', 'out of the results at the computer centre.'
```

c PARAMETER STATEMENTS SETTING UP THE MESH GEOMETRY

c the number of horizontal meshes in the quarter section of the slab
parameter(mx=9)

c the number of vertical meshes in the quarter section of the slab
parameter(my=10)

c
the number of refractory meshes

parameter(mr=4)

c the total number of mesh nodes

parameter(m=(mx+1)*(my+1)+mr+1)

c the total number of mesh elements in the quater section

slab

parameter(n=mx*my)

c mac will be used to dimension array ac which is used in
c subroutine gauss

parameter(mac=mx**2)

c nbc should be greater than mx*mx*(my-1) for the subroutine gauss to be
c succesfull

parameter(nbc=1000)

c mq will be used to dimension general array q

parameter(mq=15)

c DECLARATION OF REAL AND COMPLEX ARRAYS

real hx(mx),hy(my),mu(n),rotab,roxa(n)
real rox

(n),roy(n),muo,pg(m),b(m),q(mq)
real p(m),mg(n),hag,freq,max
complex h(n),ac(mac),bc(nbc),ca,cb,cc

c pringo is a logical variable which determines the amount of output
c the user will get

logical pringo

c SETTING UP THE GEOMETRY OF THE SLAB

c setting up the horizontal mesh spacings

data hx/.075,.07,.039,.025,.018,.009,.006,.005,.003/

c setting up the vertical mesh spacings

data hy/.01,.009,.007,.005,.004,.003,.003,.003,.003/

c slbthx is the length of the cross-sectional face of the slab
c which is being used for the finite difference analysis

data sl

bthx/.5/

c slbthy is the width of the cross-sectional face of the slab
c which is being used for the finite difference analysis

data slbthy/.1/

c the resistivity of non magnetic aluminium

data rotab/2.824e-8/

c HAG is the surface magnetic field strength(A/m)

c FREQ is

the supply frequency (Hz)

```
print*, 'INPUT VALUES OF HAG AND FREQ'  
read*, hag, freq
```

```
print*, 'INPUT VALUE OF Q(7)'  
read *, q(7)
```

c if q(7) is 0.0 user will get a fuller output, otherwise the
c user will only get the power density, power generated, and
c the magnetic field distributions in addition to the total power
c induced in the slab

```
pringo=(q(7).eq.0.0)
```

c THE FOLLOWING SECTION CHECKS FOR MESH CONSISTENCY

c checks if one or more of the horizontal meshes is less than 1e-10.
c if so the user is informed And the program stops

```
100 xa=0.0  
do 1001 i=1, mx  
xa=xa
```

```
+hx(i)  
if (hx(i).lt.1e-10)then  
print*, 'HORIZONTAL MESH(es) IS/ARE TOO SMALL(less than 1e-10 m)'  
stop  
endif  
1001 continue
```

c check if twice the sum of the horizontal mesh spacings is consistent
c with the dimension slbthx

```
if(abs(xa+xa-slbthx).gt.slbthx*1e-5)then  
print*, 'TWICE THE SUM OF HORIZONTAL MESHES IS NOT CONSISTENT WITH  
1 DIMENSION SLBTHX'  
stop  
endif
```

c check if one or more of the vertical meshes is smaller than 1e-10

```
xa=0.0  
do 10015 i=1, my  
xa=xa+hy(i)  
if (hy(i).lt.1e-10)then  
print*, 'VERTIC
```

AL MESH(es) IS/ARE TOO SMALL (less than 1e-10 m)'

```
stop  
endif
```

```
10015 continue
```

c check if twice the sum of the vertical mesh spacings is consistent
c with the dimension slbthy

```
if(abs(xa+xa-slbthy).gt.slbthy*1e-5)then  
pri
```

```

nt*, 'twice the sum of vertical meshes not consistent with
1 dimension slbthy'
print*, 'TWICE THE SUM OF VERTICAL MESHES NOT CONSISTENT
1 WITH DIMENSION SLBTHY'
stop
endif

c CALLING SUBROUTINE SETUP TO USE THE FINITE DIFFERENCE EQUATIONS
c DEFINED AND SETUP THE MATRIX

call setup(pg,b,h,hag,hx,hy,rox,roy,mu,ac,bc,
1rotab,roxa,mg,pringo,max,
2freq,q,mq,m,mx,my,mac,nbc,n,p)
stop
end

c *****START OF SUBROUTINE SETUP *****

subroutine setup(pg,b,h,hag,hx,hy,rox,roy,mu,ac,bc,
1rotab,roxa,mg,pringo,max,
2freq,q,mq,m,mx,my,mac,nbc,n,p)

c
declaration of real and complex arrays

real hx(mx),hy(my),mu(n),rotab,roxa(n)
real rox(n),roy(n),muo,pg(m),b(m),q(m)
real p(m),mg(n),hag,freq,max
complex h(n),ac(mac),bc(nbc),ca,cb,cc

logical pringo

10 format(10f7.2,6x)
15 format(9f8.0,2x)
20 format(9f8.0,x)

mxpi=mx+1
mslab=mxpi*my+mxpi

c checking for any array bound irregularities in subroutine setup

if(m.gt.mslab.
1 and.mac.ge.mx*mx)
2 goto 101

print*, '

THERE ARE ARRAY BOUND IRREGULARITIES IN SUBROUTINE SETUP
1 PLEASE CHECK'

print'(4(a,i6/))',
1 'mx=',mx
2, 'my=',my
3, 'm=',m
4, 'mac =',mac

100 return

```

101 mxmi=mx-1

c the following sets muo to be equal to $4\pi \cdot 10^{-7}$

```
xa=8*atan(1.0)
omega=xa*freq
muo=xa*2e-7
```

c THE FOLLOWING SECTION CALCULATES THE MESH AREA*RESISTIVITY
C FOR EACH MESH ELEMENT AND ALSO SETS THE PERMEABILITY OF EACH
c MESH TO MUO

```
do 104 j=1,my
  ia=mx*(j-1)
  do 103 i=1,mx
    ia=ia+1
    mu(ia)=muo
    roxa(ia)=rotab*hx(i)*hy(j)
103  continue
104  continue
```

```
if(.not.pringo)goto 1051
print*, 'TH
```

E MESH AREA X RESISTIVITY FOR EACH MESH ELEMENT'
print'(4e16.7)',(roxa(j),j=1,n)

c THE FOLLOWING SECTION CALCULATES THE RESISTANCE VECTORS ROX AND
c ROY AT EACH NODE

```
1051 xa=2/omega
do 1055 j=my,1,-1
  ia=j*mx
  xb=xa/(hy(j)*hy(j))
  do 1054 i=mx,1,-1
    xc=xa/(hx(i)*hx(i))
    if(ia.eq.1)then
      roy(ia)=xb*rox(ia)
    else
      roy(ia)=xb*(roxa(ia)+roxa(ia-1))
    endif
    if(i.eq.1)roy(ia)=xb*roxa(ia)
    if(j.eq.1) goto 1052
    rox(ia)=xc*(roxa(ia)+roxa(ia-mx))
    goto 1053
1052 rox(ia)=xc*roxa(ia)
1053 ia=ia
```

```
-1
1054 continue
1055 continue
```

```
if(.not.pringo)goto 1084
print*, 'THE RESISTANCE VECTORS ROX AT EACH MODE (array rox)'
print'(4e16.7)',(rox(j),j=1,n)
print*, 'THE RESISTANCE VECTORS ROY AT EACH NO
```

```
DE (array roy)'
  print'(4e16.7)',(roy(j),j=1,n)
```

c THE FOLLOWING SECTION SETS UP THE RIGHT HAND SIDE
c OF THE MATRIX EQUATION IN THE ARRAY H

```
1084 j=n-mx
      do 1087 i=1,j
        h(i)=0.0
1087 continue
      do 1088 i=mx,j,mx
        h(i)=rox(i)*hag
1088 continue
      j=j+1
      do 1089 i=j,n
        h(i)=roy(i)*hag
1089 continue
      h(n)=h(n)+rox(n)*hag
```

c

```
      do 1092 j=1,my
        do 1091 i=1,mx
          mu(j*mx-mx+i)=mu(j*mx-mx+i)*hx(i)*hy(j)
1091 continue
1092 continue
        do 1094 j=my,2,-1
          ia=j*mx
          ib=ia-mx+2
          do 1093 i=ia,ib,-1
            mu(i)=mu(i)+mu(i-1)+mu(i-mx)+mu(i-mx-1)
1093 continue
          mu(ib-1)=mu(ib-1)+mu(ib-1-mx)
1094 continue
          do 1095 i=mx,2,-1
            mu(i)=mu(i)+mu(i-1)
1095 continue

      if(.not.pringo)goto 2000
      print*, 'THE NODE ARRAY MU'
      print'(4e16.7)',(mu(j),j=1,n)
```

c THE SUBROUTINE GAUSS IS NOW CAL

LED UPON TO SOLVE THE FINITE

c DIFFERENCE MATRIX USING THE METHOD OF GAUSSIAN ELIMINATION

```
2000 call gauss(rox,roy,mu,h,n,ac,mac,bc,nbc,mx,my)
```

c THE MAGNETIC FIELD STRENGTHS OBTAINED BY SUBROUTINE GAUSS WILL
c NOW BE USED TO CALCULATE

c a) the power at each node

c b) the power density over each mesh element
 c c) the total power in the slab from the power generated at
 c each node

```

121  ic=my-1
      do 130 j=1,ic
        ia=j*mx
        ib=ia-mx
        do 125 i=1,mxmi
          ca=h(ia+i+1)-h(ib+i)
          cb=h(ia+i)-h(ib+i+1)
          cc=(ca+cb)/hy(j)
          cb=(ca-cb)/hx(i)
          rox(ib+i)=roxa(ib+i)*(cc*conjg(cc)+cb*conjg(cb))*0.03125
125  continue
        ca=hag-h(ia)
        cb=h(ia+mx)-hag
        cc=(ca+cb)/hy(j)
        cb=(ca-cb)/hx(mx)
        rox(ia)=roxa(ia)*(cc*conjg(cc)+cb*conjg(cb))*0.03125
130  continue
        ia=mx*my-mx

      do 140 i=1,mxmi
        ca=hag-h(ia+i)
        cb=hag-h(ia+i+1)
        cc=(ca+cb)/hy(my)
        cb=(ca-cb)/hx(i)
        rox(ia+i)=roxa(ia+i)*(cc*conjg(cc)+cb*conjg(cb))*0.03125
140  continue
        cc=(hag-h(mx*my))/hy(my)
        cb=(hag-h(mx*my))/hx(mx)
        rox(mx*my)=roxa(mx*my)*(cc*conjg(cc)+cb*conjg(cb))*0.03125

c    the following section calculates the power generated at each
c    node of the quarter section slab mesh

      pg(1)=rox(1)
      pg(mxpi)=rox(mx)
      pg(mxpi*my+1)=rox(mx*my-mxmi)
      pg(mslab)=rox(mx*my)
      do 150 i=2,mx
        pg(i)=rox(i-1)+rox(i)
        pg(mslab-mxpi+i)=rox(mx*my-mx+i-1)+rox(mx*my-mx+i)
150  continue

      do 160 j=2,my
        ia=mxpi*j-mxpi
        ib=mx*j-mx
        pg(ia+1)=rox(ib+1)+rox(ib+1-mx)
        pg(ia+mxpi)=rox(ib+mx)+rox(ib)
        do 155 i=2,mx
          pg(ia+i)=rox(ib+i)+rox(ib+i-1)+rox(ib+i-mx)+rox(ib+i-mx-1)
155  continue

```

160 continue

c setting the power at the refractory nodes to zero

```
ia=mslab+1
do 161 i=ia,m
pg(i)=0.0
```

161 continue

c the following section calculates the power density within each mesh
c element of the quarter section slab

```
ia=0
ib=0
do 164 j=1,my
do 162 i=1,mx
b(ia+i)=4/hy(j)/hx(i)*rox(ib+i)
```

162 continue

c setting the power densities at the surface to zero

```
ia=ia+mxpi
ib=ib+mx
b(ia)=0.0
```

164 continue

```
ia=mslab-mx
do 165 i=ia,mslab
b(i)=0.0
```

165 continue

print'(a,/)', 'THE POWER DENSITY(W/m3) IN EACH ELEMENT OF THE MESH'

```
print 15,(b(j),j=1,9)
print 15,(b(j),j=11,19)
print 15,(b(j),j=21,29)
print 15,(b(j),j=31,39)
print 15,(b(j),j=41,49)
print 15,(b(j),j=51,59)
print 15,(b(j),j=61,69)
```

```
print 15,(b(j),j=71,79)
print 15,(b(j),j=81,89)
print 15,(b(j),j=91,99)
```

c adding the powers generated at each node and then multiplying by
c four to give the total power in the slab

```
xa=0.0
do 1665 i=1,mslab
xa=xa+pg(i)
```

1665 continue

xa=4*xa

print'(/,a,f15.2,/)', 'TOTAL POWER(W/m) IN THE SLAB=', xa

print'(a,/)', 'POWER GENERATED (W) AT EACH NODE

OF THE MESH'

```
print 10,(pg(j),j=1,10)
print 10,(pg(j),j=11,20)
print 10,(pg(j),j=21,30)
print 10,(pg(j),j=31,40)
print 10,(pg(j),j=41,50)
print 10,(pg(j),j=51,60)
print 10,(pg(j),j=61,70)
print 10,(pg(j),j=71,80)
print 10,(pg(j),j=81,90)
print 10,(pg(j),j=91,100)
print 10,(pg(j),j=101,110)
```

c expressing the power densities in each mesh as a percentage of the
c largest mesh power density

c the largest mesh power density being denoted by max

c finding the largest mesh power density

```
max=b(1)
do 166 k=1,m
  if(b(k).gt.max)then
    max=b(k)
```

c expressing the power densities as a percentage of the largest
c mesh power density mxa

```
do 167 k=1,m
  p(k)=(b(k)/max)*100
167 continue
```

print'(/,a,/)', 'THE POWER DENSITY IN EACH ELEMENT OF THE MESH
1 EXPRESSED AS A PERCENTAGE OF THE LARGEST MESH POWER DENSITY'

```
print 10,(p(j),j=1,9)
print 10,(p(j),j=11,19)
print 10,(p(j),j=21,29)
print 10,(p(j),j=31,39)
print 10,(p(j),j=41,49)
print 10,(p(j),j=51,59)
print 10,(p(j),j=61,69)
print 10,(p(j),j=71,79)
print 10,(p(j),j=81,89)
print 10,(p(j),j=91,99)
```

print'(/,a,f15.2,/)', 'THE LARGEST MESH POWER DENSITY(w/m3)=' ,max

```
if(.not.pringo)goto 169
print*, 'MAGNETIC FIELD STRENGTHS WRITTEN AS COMPLEX NUMBERS'
print' (4d16.6)', (h(j),j=1,n)
```

c mg is the array of the modulus of the complex magnetic field
c strengths

```

169   ia=0
      do 170 i=1,n
        mg(i)=cabs(h(i))
170   continue

      print'(a,/)', 'MAGNETIC FIELD STRENGTHS AT EACH NODE OF THE MESH'

      print 20,(mg(j),j=1,9)
      print 20,(mg(j),j=10,18)
      print 20,(mg(j),j=19,27)
      print 20,(mg(j),j=28,36)
      print 20,(mg(j),j=37,45)
      print 20,(mg(j),j=46,54)
      print 20,(mg(j),j=55,63)
      print 20,(mg(j),j=64,72)
      print 20,(mg(j),j=73,81)
      print 20,(mg(j),j=82,90)

      print'(/,a,f10.2,/)', 'THE SURFACE MAGNETIC FIELD STRENGTH=',hag
      return
      end

```

c *****START OF SUBROUTINE GAUSS T

0 SOLVE THE MATRIX *****

```

      subroutine gauss(rx,ry,mu,h,nxy,a,na,b,nb,mx,my)
      complex h(nxy),a(na),b(nb),ca
      real rx(nxy),ry(nxy),mu(nxy)

```

```

      logical ntgo

```

```

      mxmi=mx-1
      mxmx=mx*mx
      mxy=mx*my

```

c checks if there are any array bound errors in subroutine gauss

```

      if(na.ge.mxmx.
1 and.nxy.ge.mxy.
2 and.mx.gt.2.
3 and.my.gt.2.)goto 100

```

```

      print'(a,/)', 'THERE ARE ARRAY BOUND ERRORS IN SUBROUTINE GAUSS'

```

```

      print'(6(a,i6/))',
1 'mx='

```

,mx

```

2, 'my=',my
3,'mxy=',mxy
4,'nxy=',nxy
5,'mxmx=',mxmx
6,'na=',na
      return

```

```

100   mxpi=mx+1
      mxt=2*mx
      mxmxmm=mxmx-mx

```

c i

f ntgo is true then the subroutine gauss will not give the
c right solution because the array b is not large enough

```
ntgo=nb.lt.mxm*(my-1)
if(ntgo)then
  print*,'NBC IN THE MAIN PROGRAM IS NOT LARGE ENOUGH FOR THE MATRIX
1  TO BE SOLVED'
  print*,'nbc=',nb
  stop
endif
kb=mx
if (ntgo) kb=0
ka=-kb
```

c setting up the elements in array a

```
a(1)=cmplx(rx(1)+ry(1),mu(1))
a(2)=-rx(1)
do 200 i=3,mxm
a(i)=0.0
200 continue
ia=mx-2
do 240 i=1,ia

ib=i*mxpi
a(ib+1)=cmplx(rx(i)+rx(i+1)+ry(i+1),mu(i+1))
a(ib+2)=-rx(i+1)
a(ib)=-rx(i)
240 continue
a(ib+mx+1)=-rx(mx-1)
a(ib+mx+2)=cmplx(rx(mx-1)+rx(mx)+ry(mx),mu(mx))
ie=mxy-mxt+1
```

c start of the gaussian elimination process

```
do 600 i=1,ie,mx
  ia=1
  ka=ka+kb
  ca=1/a(1)
  do 410 j=2,mx
    b(ka+j-1)=ca*a(j)
410 continue
    b(ka+mx)=-ca*ry(i)
    h(i)=h(i)*ca
    ib=i
    do 430 j=mxpi,mxm,mx
      do 420 k=1,

mxmi
a(k+j-mx-1)=a(k+j)-a(j)*b(ka+k)
420 continue
a(j-1)=-a(j)*b(ka+mx)
ib=ib+1
h(ib)=h(ib)-a(j)*h(i)
430 continue
```

```

do 440 k=1,mx
  a(mxmxmm+k)=ry(i)*b(ka+k)
c  print*,(h(j),j=1,90)
440  continue
  a(mxmx)=a(mxmx)+cplx(ry(i)+rx(i+mx)+ry(i+mx),mu(i+mx))
  h(i+mx)=h(i+mx)+ry(i)*h(i)
  ib=i+1
  ic=i+mxmi
  do 550 ia=ib,ic
    ka=ka+kb
    ca=1/a(1)
506  do 510 j=2,mx
    b(ka+j-1)=ca*a(j)
510  continue
    b(ka+mx)=-ca*ry(ia)
    h(ia)=h(ia)*ca
    id=ia
    do 530 j=mxpi,mxmx,mx
    do 520 k=1,mxmi
      a(k+j-mx-1)=a(k+j)-a(j)*b(ka+k)
520  continue
      a(j-1)=-a(j)*b(ka+mx)
      id
=id+1
    h(id)=h(id)-a(j)*h(ia)
530  continue
    do 540 k=1,mx
      a(mxmxmm+k)=ry(ia)*b(ka+k)
540  continue
      a(mxmxmm)=a(mxmxmm)-rx(ia+mx-1)
      a(mxmx-1)=a(mxmx-1)-rx(ia+mx-1)
      a(mxmx)=a(mxmx)+cplx(ry(ia)+rx(ia+mx-1)+rx(ia+mx)+ry(ia+mx)
1,mu(ia+mx))
545  h(mx+ia)=h(mx+ia)+ry(ia)*h(ia)
550  continue
600  continue
    ja=0
    i=2
    ic=mxpi
    id=mxy-1
    ie=mxy-mxmi
    do 640 ia=ie,id
      ca=1/a(ja+i-1)
      do 610 j=i,mx
        a(j+ja)=ca*a(j+ja)
610  continue
        h(ia)=ca*h(ia)
    )
    ib=ia
    do 630 j=ic,mxmx,mx
    do 620 k=i,mx
      a(k+j-i+1)=a(k+j-i+1)-a(j)*a(ja+k)
620  continue
      ib=ib+1
      h(ib)=h(ib)-a(j)*h(ia)
630  continue
      ja=ja+mx

```

```

h(mxy)=h(mxy)/a(mxmx)
i=mx
ja=mxmx-mxt
do 700 ia=id,ie,-1
do 690 j=1,mx
h(ia)=h(ia)-h(ia-i+1+j)*a(ja+j)
690 continue
i=i-1
ja=ja-mx
700 continue
id=mxy-mx
810 do 850 ia=id,1,-1
do 830 j=1,mx
h(ia)=h(ia)-h(ia+j)*b(ka+j)
830 continue
ka=ka-kb
850 continue
return
end

```

r 17:38 0.878 76 level 7

SLIST W-P-POWER

```

      DIMENSION A(6),H(6),AL(6),F(6),ROW1(6),UR(6),AIC(6),AAP(999)
1,N(6),LH(6),B(6),C(6),D(6),AF(6),DD(6),AXZ(999),AAJ(999),ROW2(6)
      COMMON XZ(6,999),AJ(6,999),AP(6,999)
      READ(5,*)LM,(A(I),H(I),AL(I),F(I),ROW1(I),ROW2(I),UR(I),
1AIC(I),N(I),LH(I),I=1,LM)
      DO 1000 LS=1,LM
      B(LS)=AIC(LS)*H(LS)*(SQRT(1.0-(A(LS)/H(LS))**2))/3.14159265
      C(LS)=H(LS)**2-A(LS)**2
      ROW=ROW1(LS)
      DR=(ROW2(LS)-ROW1(LS))/700.0
C      WRITE(1,20)DD
C20  FORMAT(1H,4HDD =,F15.6)
C      WRITE(1,4)A
C4   FORMAT(1H,3HA =,F15.9)
C      WRITE(1,8)H
C8   FORMAT(1H,3HH =,F15.9)
C      WRITE(1,11)AL
C11  FORMAT(1H,8HLENGTH =,F15.9)

C      WRITE(1,12)N
C12  FORMAT(1H,3HN =,I8)
C      WRITE(1,61)F
C61  FORMAT(1H,11HFREQUENCY =,F15.4)
C      WRITE(1,90)AIC
C90  FORMAT(1H,14HCOIL CURRENT =,F9.3,4X,'AMP')
C      WRITE(1,62)ROW
C62  FORMAT(1H,5HROW =,E15.7,4X,'OHM-M')
C      WRITE(1,63)UR
C63  FORMAT(1H,4HUR =,F15.3)
C      WRITE(1,9)
C9   FORMAT(1H,2X,'Z',13X,'CURRENT DENSITY',10X,'POWER DENSITY')
C      WRITE(1,7)
C7   FORMAT(1H,1X,'---',12X,'-----',10X,'-----')
      IND=0
      D(LS)=AL(LS)/(N(LS)-1)
      AF(LS)=AL(LS)/700.0

      Z=0.0
5     IND=IND+1
      XZ(LS,IND)=Z
      SUM=0.0
      NL=N(LS)
      DO 10 K=1,NL
      I=K-1
      SUM1=1./(C(LS)+Z**2+2.0*Z*D(LS)*I+D(LS)*D(LS)*I*I)
      SUM=SUM+SUM1
10    CONTINUE

```

```

PI=3.14159265
UO=4.0*PI/10000000.0
AXA=UR(LS)*F(LS)
XAX=PI*UO*AXA
DD(LS)=(ROW/XAX)**0.5
AJ(LS,IND)=B(LS)*SUM
AP(LS,IND)=ROW*AJ(LS,IND)*AJ(LS,IND)/(DD(LS))
DIS=ABS(Z)
C WRITE(1,6)DIS,AJ(LS,IND),AP(LS,IND)
C6 FORMAT(1H ,F6.3,5X,F15.6,14X,E10.4)
    ROW=ROW+DR
    Z=Z-AF(LS)
    E=Z+AL(LS)
    XZ(LS,IND)=ABS(XZ(LS,IND))
    IF (E.GE.0.0)GO TO 5
1000    CONTINUE

AL1=0.0
    YB=0.0
    YET=-9999.0
    YEV=-9999.0
    DO 1500 MM=1,LM
    DO 2000 J=1,IND
    IF (YET.LT.AP(MM,J))YET=AP(MM,J)
    IF (YEV.LT.AJ(MM,J))YEV=AJ(MM,J)
    IF (AL1.LT.AL(MM))AL1=AL(MM)
2000    CONTINUE
1500    CONTINUE
    IF (LH(LS).EQ.3)CALL C1051N
    IF (LH(LS).EQ.1)CALL TREND
    IF (LH(LS).EQ.2)CALL S5600
    CALL DEVPAP(210.0,297.0,1)
    CALL MOVTO2(0.0,0.0)
    CALL LINTO2(0.0,297.0)
    CALL MOVTO2(210.0,0.0)
    CALL LINTO2(210.0,297.0)
    CALL WINDOW(2)
    CALL ERRMAX(100)
    CALL AXIPOS(1,40.0,90.0,150.0,1)
    CALL AXIPOS(1,40.0,90.0,180.0,2)

    CALL AXISCA(1,9,0.0,AL1,1)
    CALL AXISCA(1,9,0.0,YEV,2)
    CALL AXIDRA(2,1,1)
    CALL AXIDRA(-2,-1,2)
    CALL MOVTO2(25.0,285.0)
    CALL CHAHOL('A (A/m)*.')
```

CALL MOVTO2(190.0,78.0)

```

    CALL CHAHOL('Z(m)*.')
```

CALL MOVTO2(40.0,70.0)

```

CALL CHAHOL('Fig.4. . CURRENT PER UNIT LENGTH ALONG THE LOAD *.')
```

```

CALL MOVTO2(40.0,54.0)
CALL CHAHOL('N=      Turns      d=      mm      h=      mm *.' )
CALL MOVTO2(40.0,46.0)
CALL CHAHOL('a=      mm      I=      Amp *.' )
      DO 1700 II=1,LM
        DO 1750 I=1,IND
          AXZ(I)=XZ(II,I)
          AAJ(I)=AJ(II,I)
1750      CONTINUE
        CALL GRACUR(AXZ,AAJ,IND)
1700      CONTINUE
CALL PICCLE
CALL DEVEND
IF(LH(LS).EQ.3)CALL C1051N
IF(LH(LS).EQ.2)CALL S5600
IF(LH(LS).EQ.1)CALL TREND

CALL DEVPAP(210.0,297.0,1)
CALL MOVTO2(0.0,0.0)
CALL LINTO2(0.0,297.0)
CALL MOVTO2(210.0,0.0)
CALL LINTO2(210.0,297.0)
CALL WINDOW(2)
CALL ERRMAX(100)
CALL AXIPOS(1,40.0,90.0,150.0,1)
CALL AXIPOS(1,40.0,90.0,180.0,2)
CALL AXISCA(1,9,0.0,AL1,1)
CALL AXISCA(1,9,0.0,YET,2)
CALL AXIDRA(2,1,1)
CALL AXIDRA(-2,-1,2)
CALL MOVTO2(25.0,285.0)
      CALL CHAHOL('PD(W/SQ m)*.' )
CALL MOVTO2(190.0,78.0)
CALL CHAHOL('Z(m)*.' )
      CALL MOVTO2(40.0,70.0)
CALL CHAHOL('Fig.4. . POWER DENSITY DISTRIBUTION ALONG THE LOAD *.' )
CALL MOVTO2(95.0,62.0)

CALL CHAHOL('NONMAGNETIC LOAD*.' )
CALL MOVTO2(40.0,54.0)
CALL CHAHOL('INITIAL RESISTIVITY      =      Ohm.m *.' )
CALL MOVTO2(40.0,46.0)
CALL CHAHOL('FINAL RESISTIVITY      =      Ohm.m*.' )
CALL MOVTO2(40.0,38.0)
CALL CHAHOL('RELATIVE PERMEABILITY=*.' )

```


CALL MOVTO2(40.0,30.0)

CALL CHAHOL('N= Turns d= mm h= mm *.')

CALL MOVTO2(40.0,22.0)

CALL CHAHOL('a= mm I= Amp F= Hz*.')

DO 1800 II=1,LM

DO 1850 I=1,IND

AXZ(I)=XZ(II,I)

AAP(I)=AP(II,I)

1850 CONTINUE

CALL GRACUR(AXZ,AAP,IND)

1800 CONTINUE

CALL PICCLE

IF(LH(LS).NE.2)CALL DEVEND

STOP

END

OK,

```

SLIST PROXIMITY
  READ(5,*)A,H,AL,F,ROW,UR,AIC,PW
  N=2
  B=AIC*H*(SQRT(1.0-(A/H)**2))/3.14159265
  C=H**2-A**2
  DD=503.3*SQRT(ROW/(F*UR))
C   WRITE(1,20)DD
C20  FORMAT(1H ,4HDD =,F15.12)
    WRITE(1,4)A
  4   FORMAT(1H ,17HINDUCTOR RADIUS =,F5.4,4X,'m')
    WRITE(1,8)H
  8   FORMAT(1H ,9HAIR GAP =,F5.4,4X,'m')
    WRITE(1,11)AL
  11  FORMAT(1H ,8HLENGTH =,F6.3,4X,'m')
    WRITE(1,44)PW
  44  FORMAT(1H ,16HPOWER REQUIRED =,F15.3,4X,'Watts/meter')
    WRITE(1,61)F
  61  FORMAT(1H ,11HFREQUENCY =,F15.1,4X,'Hz')

  WRITE(1,66)AIC
  66  FORMAT(1H ,14HCOIL CURRENT =,F15.3,4X,'Amp')
    WRITE(1,62)ROW
  62  FORMAT(1H ,18HLOAD RESISTIVITY =,E10.4,4X,'Ohm.m')
    WRITE(1,63)UR
  63  FORMAT(1H ,14HPERMEABILITY =,F10.1)
    AF=AL/100
  16  ASS=0.0
    D=AL/(N-1)
    Z=0.0
  5   SUM=0.0
    DO 10 K=1,N
      I=K-1
      SUM1=1./(C+Z**2+2.0*Z*D*I+D*D*I*I)
      SUM=SUM+SUM1
  10  CONTINUE
    AJ=B*SUM

  AP=ROW*AJ*AJ/(2.0*DD)
  AAP=AP*AF
  AAP=ASS+AAP
  Z=Z-AF
  E=Z+AL
  IF (E.GE.0.0)GO TO 5
  IF (AAP-PW)28,25,25
  28  N=N+1
    GO TO 16
  25  WRITE(1,22)N
  22  FORMAT(1H ,17HNUMBER OF TURNS =,I10,4X,'Turns')
    WRITE(1,7)AAP
  7   FORMAT(1H ,22HPOWER PER UNIT WIDTH =,F15.3,4X,'Watts/meter')
    ER=100.0*(AAP-PW)/PW
    WRITE(1,53)ER
  53  FORMAT(1H ,12HDIFFERENCE =,F5.2,'%')
    WRITE(1,77)
  77  FORMAT(1H ,'------')
    STOP
    END

```

SLIST W-SC-FIT

C THE SYMBOLS ARE

C N NUMBER OF POINTS
C X DISTANCE FROM SINGLE TURN (MM)
C Y VOLTAGES FROM SINGLE TURN (UV)
C AI1 CURRENT FOR SINGLE TURN (AMP)
C AI2 CURRENT FOR THE COIL (AMP)
C NT NUMBER OF TURNS
C D COIL PITCH (MM)
C AL LOAD LENGTH (MM)
C ROW LOAD RESISTIVITY (OHM.M)
C ROL LOAD RADIUS OR SLAB'S PROBE LENGTH (MM)
C UR LOAD RELATIVE PERMEABILITY
C F FREQUENCY (Hz)
C LH CHOICE OF THE DEVICE, TREND=1, S5600=2, C1051N=3
C NSC CHOICE OF SLAB=1, CYLINDER=2
C

REAL BER,BEI,KER,KEI,BERD,BEID,KERD,KEID
COMPLEX THETA,THY,KO,IO,KOD,IOD
DIMENSION AZ(999),AV(999),PD(999),AJ(999)
1,SAJ(999),HRM(999)

REAL*8 X(50),Y(50),A(50),REF
INTEGER N,NT
READ(5,*)N,(X(I),Y(I),I=1,N),AI1,AI2,NT,D,AL,ROW,ROL,F,UR,NSC,LH
CALL E02ACF(X,Y,N,A,5,REF)
C WRITE(1,30)(A(I),I=1,5)
C30 FORMAT(1H,5X,E12.4)
SLO=(Y(N)-Y(N-1))/(X(N)-X(N-1))
CON=Y(N)-(SLO*X(N))
CL=D*(NT-1)
R=-(CL/2.0)
C WRITE(1,9)
C9 FORMAT(1H,'DISTANCE',7X,'VOLTAGE')
C WRITE(1,90)
C90 FORMAT(1H,'-----',7X,'_____')
ML=1
1000 SUM=0.0
RA=AI2/AI1
DO 200 J=1,NT
I=J-1
ZA=R+(I*D)
Z=ABS(ZA)

IF(Z.GT.X(N))GO TO 300
SUM0=A(1)+(A(2)*Z)+(A(3)*(Z**2))+(A(4)*(Z**3))+(A(5)*(Z**4))
SAM0=0.0
SUM1=SUM0+SAM0
GO TO 400
300 SUM1=(SLO*Z)+CON
400 IF(SUM1.LT.0.0)SUM1=0.0
SUM=(SUM+SUM1)
200 CONTINUE
SUM2=SUM*RA

```

ZZ=R+(CL/2.0)
AZ(ML)=ZZ
AV(ML)=SUM2
ML=ML+1
R=R+0.25
IF(ZZ.LE.AL)GO TO 1000
K=ML-1
DO 500 I=1,K
500  AV(I)=AV(I)/1000000.0
    CONTINUE
    IF(NSC.EQ.1)ROL=ROL/1000.0
    IF(NSC.EQ.2)R=ROL
PIY=3.141592654
U0=4.0*PIY/10000000.0
U=U0*UR
W=2.0*PIY*F
R=R/1000.0
PER=2.0*R*PIY
RL=ROW*PER
D2=(2.0*ROW)/(W*U)
D1=SQRT(D2)

BA=R*SQRT(2.0)/D1
C    CALCULATION OF BESSEL FUNCTIONS
    IF(BA.GT.8)GO TO 5001
    A1=BER(BA)
    A2=BEI(BA)
    A3=BERD(BA)
    A4=BEID(BA)
    GO TO 5002
5001  A1=REAL(IO(BA))
    A2=AIMAG(IO(BA))
    A3=REAL(IOD(BA))
    A4=AIMAG(IOD(BA))
5002  CONTINUE
    DEN=((A1)*(A1))+((A2)*(A2))
    B1=((A3)*(A3))+((A4)*(A4))
    B2=((A1)*(A3))+((A2)*(A4))
    C1=B1/DEN
    C2=B2/DEN

IF(NSC.EQ.2)GO TO 550
    DP=D1/(2.0*ROW*ROL*ROL)
    DO 510 I=1,K
    510  PD(I)=DP*AV(I)*AV(I)
        CONTINUE
        GO TO 999
550    CONTINUE
    CPD=U*PIY*F*R*C2/2.0

```

```

DO 110 I=1,K
  AJ(I)=AV(I)/RL
  SAJ(I)=AJ(I)*AJ(I)
  HRM(I)=D2*SAJ(I)/C1
  PD(I)=CPD*HRM(I)
110  CONTINUE
999  CONTINUE
C    WRITE(1,1001)
C1001  FORMAT(1H,2X,'DISTANCE',7X,'VOLTAGE',6X,'POWER DENSITY')
C    WRITE(1,1100)
C1100  FORMAT(1H,2X,'-----',7X,'-----',6X,'-----')
C    DO 120 I=1,K
C    WRITE(1,130)AZ(I),AV(I),PD(I)
C130  FORMAT(1H,3X,F6.1,5X,E12.5,5X,E12.5)
C120  CONTINUE

YB=0.0
  YET=-9999.0
  YEV=-9999.0
  DO 2000 J=1,K
    IF(YET.LT.PD(J))YET=PD(J)
    IF(YEV.LT.AV(J))YEV=AV(J)
2000  CONTINUE
      YET=1.1*YET
      YEV=1.1*YEV
      IF(LH.EQ.3)CALL C1051N
      IF(LH.EQ.1)CALL TREND
      IF(LH.EQ.2)CALL S5600
      CALL DEVPA(210.0,297.0,1)
      CALL MOVTO2(0.0,0.0)
      CALL LINTO2(0.0,297.0)

CALL MOVTO2(210.0,0.0)
  CALL LINTO2(210.0,297.0)
  CALL WINDOW(2)
    CALL ERRMAX(100)
    CALL AXIPOS(1,40.0,90.0,150.0,1)
    CALL AXIPOS(1,40.0,90.0,180.0,2)
    CALL AXISCA(1,9,0.0,AL,1)
    CALL AXISCA(1,9,0.0,YEV,2)
    CALL AXIDRA(2,1,1)
    CALL AXIDRA(-2,-1,2)
    CALL MOVTO2(25.0,285.0)
    CALL CHAHOL('V (VOLT)*.')
    CALL MOVTO2(182.0,78.0)
    CALL CHAHOL('Z(mm)*.')

CALL GRACUR(AZ,AV,K)
  CALL MOVTO2(50.0,70.0)
  CALL CHAHOL('SURFACE VOLTAGE DISTRIBUTION ALONG THE LOAD *.')
  CALL MOVTO2(95.0,60.0)
  IF(NSC.EQ.1)CALL CHAHOL('ALUMINUM SLAB*.')
  IF(NSC.EQ.2)CALL CHAHOL('ALUMINUM CYLINDER*.')
  CALL MOVTO2(40.0,50.0)
  CALL CHAHOL('RESISTIVITY=2.8E-8 Ohm.m *.')
  CALL MOVTO2(140.0,50.0)

```

```

CALL CHAHOL('RELATIVE PERMEABILITY=1*.')
CALL MOVTO2(40.0,40.0)
CALL CHAHOL('N=      Turns      d=      mm      g=      mm *.')
CALL MOVTO2(113.0,30.0)
CALL LINTO2(125.0,30.0)
CALL MOVTO2(126.0,30.0)
CALL CHAHOL('PREDICTION BY SUPERPOSITION*.')
CALL MOVTO2(145.0,40.0)
CALL CHAHOL('o PRACTICAL READINGS*.')
CALL MOVTO2(40.0,30.0)
CALL CHAHOL('I=      Amp      F=      Hz*.')
CALL PICCLE
CALL DEVEND
IF(LH.EQ.3)CALL C1051N

IF(LH.EQ.2)CALL S5600
IF(LH.EQ.1)CALL TREND
CALL DEVPAP(210.0,297.0,1)
CALL MOVTO2(0.0,0.0)
CALL LINTO2(0.0,297.0)
CALL MOVTO2(210.0,0.0)
CALL LINTO2(210.0,297.0)
CALL WINDOW(2)
CALL ERRMAX(100)
CALL AXIPOS(1,40.0,90.0,150.0,1)
CALL AXIPOS(1,40.0,90.0,180.0,2)
CALL AXISCA(1,9,0.0,AL,1)
CALL AXISCA(1,9,0.0,YET,2)
CALL AXIDRA(2,1,1)

CALL AXIDRA(-2,-1,2)
CALL MOVTO2(25.0,285.0)
CALL CHAHOL('PD(W/SQ m)*.')
CALL MOVTO2(190.0,78.0)
CALL CHAHOL('Z(mm)*.')
CALL GRACUR(AZ,PD,K)
CALL MOVTO2(50.0,70.0)
CALL CHAHOL('POWER DENSITY DISTRIBUTION ALONG THE LOAD *.')
CALL MOVTO2(95.0,60.0)
IF(NSC.EQ.1)CALL CHAHOL('ALUMINUM SLAB*.')

```

```
IF(NSC.EQ.2)CALL CHAHOL('ALUMINUM CYLINDER*.')
CALL MOVTO2(40.0,50.0)
CALL CHAHOL('RESISTIVITY=2.8E-8 Ohm.m *.')
CALL MOVTO2(140.0,50.0)
CALL CHAHOL('RELATIVE PERMEABILITY=1*.')

```

```
CALL MOVTO2(113.0,30.0)
CALL LINTO2(125.0,30.0)
CALL MOVTO2(126.0,30.0)
CALL CHAHOL('PREDICTION BY SUPERPOSITION*.')
CALL MOVTO2(145.0,40.0)
CALL CHAHOL('o PRACTICAL READINGS*.')
CALL MOVTO2(40.0,40.0)
CALL CHAHOL('N=      Turns    d=      mm    g=      mm *.')
CALL MOVTO2(40.0,30.0)
CALL CHAHOL('I=      Amp    F=      Hz*.')
CALL PICCLE
IF(LH.NE.2)CALL DEVEND
STOP
END
$INSERT BESSEL
OK,
```

SLIST FITTING1

```

      REAL*8 X(50),Y(50),A(50),REF,D,AI1,AI2,AL
      INTEGER N,NT
      READ(5,*)N,(X(I),Y(I),I=1,N),AI1,AI2,NT,D,AL
      CALL E02ACF(X,Y,N,A,5,REF)
      WRITE(1,30)(A(I),I=1,5)
30    FORMAT(1H ,5X,E12.4)
      SLO=(Y(N)-Y(N-1))/(X(N)-X(N-1))
      CON=Y(N)-(SLO*X(N))
      CL=D*(NT-1)
      R=-(CL/2.0)
      WRITE(1,9)
9     FORMAT(1H , 'DISTANCE',7X, 'VOLTAGE' )
      WRITE(1,90)
90    FORMAT(1H , '-----',7X, '_____')
1000   SUM=0.0
      RA=AI2/AI1
      DO 200 J=1,NT
      I=J-1

      Z=ABS(ZA)
      IF(Z.GT.X(N))GO TO 300
      SUM0=A(1)+(A(2)*Z)+(A(3)*(Z**2))+(A(4)*(Z**3))+(A(5)*(Z**4))
      SAM0=0.0
      SUM1=SUM0+SAM0
      GO TO 400
300    SUM1=(SLO*Z)+CON
400    IF(SUM1.LT.0.0)SUM1=0.0
      SUM=(SUM+SUM1)
200    CONTINUE
      SUM2=SUM*RA
      ZZ=R+(CL/2.0)
      WRITE(1,6)ZZ,SUM2
6     FORMAT(1H ,F6.1,10X,E15.7)
      R=R+10.0
      IF(ZZ.LE.AL)GO TO 1000
      CALL EXIT
      STOP
      END

```

OK,

SLIST W-FC-TEMP

NON UNIFORM COIL

THE SYMBOLS ARE

CD CONDUCTOR DIAMETER (MM)
AG AIR GAP (MM)
N NUMBER OF POINTS
X DISTANCE FROM SINGLE TURN (MM)
Y VOLTAGES FROM SINGLE TURN (UV)
AI1 CURRENT FOR SINGLE TURN (AMP)
AI2 CURRENT FOR THE COIL (AMP)

NT NUMBER OF TURNS
D(I) COIL PITCHES (MM)
AL LOAD LENGTH (MM)
ROW LOAD RESISTIVITY (OHM.M)
R LOAD RADIUS
UR LOAD RELATIVE PERMEABILITY
F FREQUENCY (Hz)
THC THERMAL CONDUCTIVITY (W/M K)
ALL NUMBER OF CYLINDERS
SSS NUMBER OF DISCS

REAL BER,BEI,KER,KEI,BERD,BEID,KERD,KEID
COMPLEX THETA,THY,KO,IO,KOD,IOD
DIMENSION Z(120),AV(120),PD(120),D(50),AX(120),AJ(120)
1,SAJ(120),HRM(120),AR(120),HR(120),SRR(120),BR(120),ARM(120)
2,BI(120),SBR(120),SBI(120),XHR(120),XHRZ(120),SXZ(120),ITIT(80)

REAL*8 X(9),Y(9),A(9),REF,D
1,AX,AZ,V,V1,RA
REAL T
COMMON T(11,11),WX(242)
DIMENSION WXL(4),WXH(4),WYL(4),WYH(4)
DATA WXL(1),WXL(2),WXL(3),WXL(4)/20.0,140.0,20.0,140.0/
DATA WXH(1),WXH(2),WXH(3),WXH(4)/140.0,260.0,140.0,260.0/
DATA WYL(1),WYL(2),WYL(3),WYL(4)/110.0,110.0,20.0,20.0/
DATA WYH(1),WYH(2),WYH(3),WYH(4)/200.0,200.0,110.0,110.0/
READ(5,*)CD,AG,N,(X(I),Y(I),I=1,N),AI1,AI2,NT,AL,(D(I),I=1,NT)
1,ROW,R,UR,F,THC,ALL,SSS

WRITE(1,3000)

3000 FORMAT(1H,34X,'W-PDC')

WRITE(1,3001)

3001 FORMAT(1H,34X,'*****')

WRITE(1,3002)

3002 FORMAT(1H,72X,'COIL AND LOAD SPECIFICATIONS')

WRITE(1,3003)

3003 FORMAT(1H,21X,'-----')

WRITE(1,6001)CD,R

6001 FORMAT(1H,5X,'CONDUCTOR DIAMETER = ',F6.1,1X,'MM',
120X,'LOAD RADIUS = ',F4.1,1X,'MM')

```

WRITE(1,6002)AG,ROW
6002  FORMAT(1H ,16X,'AIR GAP = ',F6.1,1X,'MM'
      1,7X,'LOAD RESISTIVITY = ',E9.4,1X,'OHM.M')
      WRITE(1,6003)NT,UR
6003  FORMAT(1H ,8X,'NUMBER OF TURNS = ',I5,1X,'TURNS'
      1,15X,'LOAD PERMEABILITY = ',F6.0)
      WRITE(1,6004)AI2,F
6004  FORMAT(1H ,11X,'COIL CURRENT = ',F6.0,1X,'AMP',20X,'FREQUENCY = '
      1,F6.0,1X,'Hz')
      WRITE(1,4650)THC
4650  FORMAT(1H ,3X,'THERMAL CONDUCTIVITY = ',F5.1,1X,'W/M K')
      WRITE(1,6005)
6005  FORMAT(1H ,//30X,'THE RESULTS')
      WRITE(1,6006)
6006  FORMAT(1H ,29X,'-----',/3X)
      CALL EO2ACF(X,Y,N,A,5,REF)
      WRITE(1,10)
10    FORMAT(1H ,14X,'THE CONSTANTS OF THE SINGLE TURN EQUATION')
      WRITE(1,77)
77    FORMAT(1H ,14X,'-----')
      WRITE(1,30)A(1),A(2),A(3),A(4),A(5)

30    FORMAT(1H ,5(E10.3,4X))
      SLO=(Y(N)-Y(N-1))/(X(N)-X(N-1))
      CON=Y(N)-(SLO*X(N))
      RA=AI2/AI1
      S=AL/SSS
      AK=AL/S
      K=AK+1
      PIY=3.141592654
      UO=4.0*PIY/10000000.0
      U=UO*UR
      W=2.0*PIY*F
      R=R/1000.0
      PER=2.0*R*PIY
      RL=ROW*PER
      D2=(2.0*ROW)/(W*U)

D1=SQRT(D2)
      BA=R*SQRT(2.0)/D1
C    CALCULATION OF BESSEL FUNCTIONS
      IF(BA.GT.8)GO TO 5001
      A1=BER(BA)
      A2=BEI(BA)
      A3=BERD(BA)
      A4=BEID(BA)
      GO TO 5002
5001  A1=REAL(IO(BA))
      A2=AIMAG(IO(BA))

      A3=REAL(IOD(BA))
      A4=AIMAG(IOD(BA))
5002  CONTINUE
      DEN=((A1)*(A1))+((A2)*(A2))
      B1=((A3)*(A3))+((A4)*(A4))
      B2=((A1)*(A3))+((A2)*(A4))
      C1=B1/DEN
      C2=B2/DEN

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      AX(1)=D(1)
      DO 80 I=2,NT
      AX(I)=AX(I-1)+D(I)
80    CONTINUE
      WRITE(1,9000)
9000  FORMAT(1H ,//28X,'THE COIL PITCHES')
      WRITE(1,9001)
9001  FORMAT(1H ,27X,'-----')
      WRITE(1,40)
40    FORMAT(1H ,20X,'DISTANCE',10X,'THE COIL PITCHES')
      DO 20 I=1,NT
      WRITE(1,60)AX(I),I,D(I)
60    FORMAT(1H ,20X,F6.1,14X,2HD(,I2,3H) =,F6.1)
20    CONTINUE

      Z(1)=0.0
      DO 70 I=2,K
      Z(I)=Z(I-1)+S
70    CONTINUE
      DO 90 I=1,K
      V1=0.0
      DO 100 II=1,NT
      ABC=Z(I)-AX(II)
      AZ=ABS(ABC)
      IF(AZ-X(N))200,200,300
200   V=A(1)+(A(2)*AZ)+(A(3)*AZ**2)+(A(4)*AZ**3)
      *+(A(5)*AZ**4)
      GO TO 400
300   V=(SLO*AZ)+CON
400   IF(V.LT.0.0)V=0.0
      V1=V1+V
100   CONTINUE

      AV(I)=V1*RA/1000000.0
90    CONTINUE
      CPD=U*PIY*F*R*C2/2.0
      DO 110 I=1,K
      AJ(I)=AV(I)/RL
      SAJ(I)=AJ(I)*AJ(I)
      HRM(I)=D2*SAJ(I)/C1
      PD(I)=CPD*HRM(I)
110   CONTINUE
C     WRITE(1,9003)
C9003  FORMAT(1H ,//23X,'VOLTAGE AND POWER DENSITY')
C     WRITE(1,9004)
C9004  FORMAT(1H ,22X,'-----')
C     WRITE(1,1000)
C1000  FORMAT(1H ,2X,'DISTANCE',7X,'VOLTAGE',6X,'POWER DENSITY')
C

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C1100  FORMAT(1H ,2X,'-----',7X,'-----',6X,'-----')
C      DO 120 I=1,K
C      WRITE(1,130)Z(I),AV(I),PD(I)
C130   FORMAT(1H ,3X,F6.1,5X,E12.5,5X,E12.5)
C120   CONTINUE
      YB=0.0
      YET=-9999.0
      YEV=-9999.0
      DO 2000 J=1,K
      IF(YET.LT.PD(J))YET=PD(J)
      IF(YEV.LT.AV(J))YEV=AV(J)
2000   CONTINUE
      LLL=ALL+1
      DR=R/ALL
      H1=R/(2.0*THC)
      H2=BA/R
      H3=B2*BA
      AR(1)=0.0

      HR(1)=0.0
      SRR(1)=0.0
      DO 4100 I=2,LLL
      AR(I)=AR(I-1)+DR
      HR(I)=H2*AR(I)
      SRR(I)=(AR(I)*AR(I))/(R*R)
4100   CONTINUE
C      CALCULATIONS OF BESSEL FUNCTIONS
      DO 4200 I=1,LLL
      IF(HR(I).GT.8)GO TO 4250
      BR(I)=BER(HR(I))
      BI(I)=BEI(HR(I))
      GO TO 4200
4250   BR(I)=REAL(IO(HR(I)))
      BI(I)=AIMAG(IO(HR(I)))
4200   CONTINUE
      DO 4400 I=1,LLL
      SBR(I)=BR(I)*BR(I)
      SBI(I)=BI(I)*BI(I)

      XHRZ(I)=XHR(I)/H3
      SXZ(I)=H1*(SRR(I)-XHRZ(I))
4400   CONTINUE
      DO 4050 I=1,LLL
      ARM(I)=1000.0*AR(I)
4050   CONTINUE
C      WRITE(1,4000)
C4000  FORMAT(1H ,//23X,'TEMPERATURE DISTRIBUTION')
C      WRITE(1,4350)
C4350  FORMAT(1H ,22X,'-----')
      DO 4500 J=1,K

      DO 4600 L=1,LLL
      T(J,L)=SXZ(L)*PD(J)
C      WRITE(1,4450)J,L,T(J,L)
C4450  FORMAT(1H ,22X,'T(',I3,',',I3,',') =',E12.5)
4600   CONTINUE
4500   CONTINUE
      PRINT 44
44     FORMAT('WRITE TITLE OF PLOT')

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READ(1,49)ITIT
49  FORMAT(80A1)
    PRINT 42
42  FORMAT('TYPE 1=S5600 , 2=C1051N')
    READ(1,*)KP
    NW=2*K*LLL
    ANX=K
    ANY=LLL
63  PRINT 45
45  FORMAT('ENTER IVIEW')
    READ(1,*)IVIEW
    IF(KP.EQ.1)CALL S5600
    IF(KP.EQ.2)CALL C1051N

CALL MOVTO2(20.0,15.0)
    CALL CHAARR(ITIT,80,1)
    CALL DATE
    CALL ISOFRA(0)
    IF(IVIEW.EQ.4)GO TO 85
    CALL ISOPRJ(K,1.0,ANX,LLL,1.0,ANY,T,IVIEW,NW,WX)
89  CALL PICCLE
    PRINT 82
82  FORMAT('1=PLOT AGAIN , 2=STOP')
    READ(1,*)IPLOT
    IF(IPLOT.EQ.1)GO TO 63
    IF(IPLOT.EQ.2)GO TO 99
85  DO 11 I=1,4
    IVIEW=I-1
    CALL WINDO2(WXL(I),WXH(I),WYL(I),WYH(I))
11  CALL ISOPRJ(K,1.0,ANX,LLL,1.0,ANY,T,IVIEW,NW,WX)

GO TO 89
99  CALL DEVEND
    STOP
    END
    SUBROUTINE DATE
    INTEGER ADATE(8)
    CALL DATE$A(ADATE)
    CALL MOVTO2(240.0,200.0)
    CALL CHAARR(ADATE,8,2)
    RETURN
    END
$INSERT BESSEL
OK,

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ALI

PROGRAMME
PROGRAMMILIN
PROGRAMMIER

A.K.M. Al-Shaikhli

PAGE
DATE
DATUM
DATE

NO. 1000
REVISION
REVISION
REVISION

LOC ADR	CODE KODE	KEY TASTE	COMMENTS BEMERKUNGEN	LOC ADR	CODE KODE	KEY TASTE	COMMENTS BEMERKUNGEN	LOC ADR	CODE KODE	KEY TASTE	COMMENTS BEMERKUNGEN
0		LRN		5	0003	Ø 3	I				
1	76	2ND LbI		5	65	X					
2	11	A		5	03	3					
3	42	STO		5	93	.					
4	00 01	Ø 1	Z	5	02	2					
5	91	R/S		6	04	4					
6	43	RCL		6	01	1					
7	00 04	Ø 4		6	06	6					
8	33	X ²	h ²	6	52	EE					
9	85	+	+	6	05	5					
10	43	RCL		6	94	+/-					
11	00 01	Ø 1		6	95	=					
12	33	X ²	z ²	6	33	X ²					
13	95	=	=	6	65	X					
14	34	√x	√h ² + z ¹	6	02	2					
15	65	X	X	7	93	.					
16	43	RCL		7	02	2					
17	00 06	Ø 6	α	7	07	7					
18	55	÷	÷	7	06	6					
19	43	RCL		7	05	5					
20	00 04	Ø 4	h	7	02	2					
21	65	X	X	7	08	8					
22	43	RCL	α/cosα	7	06	6					
23	00 01	Ø 1		7	52	EE					
24	33	X ²		7	04	4					
25	95	=	Z ² α/cosα	8	94	+/-					
26	85	+	+	8	95	=					
27	43	RCL		8	41	R/S					
28	00 04	Ø 4	h ²			LRN					
29	33	X ²				β	The values				
30	75	-	-			STO	of the con-				
31	43	RCL				Ø 2	stants and				
32	00 05	Ø 5	a ²			I	current to				
33	33	X ²				STO	be stored				
34	95	=	h ² -a ² + $\frac{\alpha}{\cos\alpha}z^2$			Ø 3	here, outside				
35	35	1/X	1/---			h	the program,				
36	42	STO				STO	so that it				
37	00 07	Ø 7	1/(h ² + $\frac{\alpha z^2}{\cos\alpha}$)			Ø 4	does not				
38	43	RCL				a	affect the				
39	00 04	Ø 4				STO	program when				
40	33	X ²	h ²			Ø 5	the changed.				
41	75	-	-			α					
42	43	RCL				STO					
43	00 05	Ø 5				Ø 6					
44	33	X ²	a ²								
45	95	=	h ²								
46	34	√x	h _e								
47	65	X									
48	43	RCL									
49	00 07	Ø 7	1(h _e ² + $\frac{\alpha z^2}{\cos\alpha}$)								
50	65	X	X								
51	43	RCL									
52	00 02	Ø 2	β								
53	65	X	X								
54	43	RCL									

62	STO	72	STO	83	STO
63	RCL	73	RCL	84	RCL
64	SUM	74	SUM	92	INV

TEXAS INSTRUMENTS

PROGRAMMER
PROGRAMMEUR
A.K.M. Al-Shaikhli

PAGE
DATE
DATE

CODING FORM
KODEFORM
FEUILLE DE PROGRAMMATION

LOC ADR ADR	CODE KODE CODE	KEY TASTE TOUCHE	COMMENTS BEMERKUNGEN COMMENTAIRES	LOC ADR ADR	CODE KODE CODE	KEY TASTE TOUCHE	COMMENTS BEMERKUNGEN COMMENTAIRES	LOC ADR ADR	CODE KODE CODE	KEY TASTE TOUCHE	COMMENTS BEMERKUNGEN COMMENTAIRES
0		LRN		50	43	RCL		11	76	2ND Lbl	
1	76	2ND Lbl	V	51	02 07	27		11	15	E	PD
2	16	2ND A'	J	52	95	=		11	43	RCL	
3	42	STO		53	42	STO		11	01 04	14	
4	00 00	0 0		54	01 02	12	μ	11	91	R/S	
5	91	R/S		55	05	5				LRN	
6	76	2ND Lbl	V	56	00	0					
7	17	2ND B'	H	57	65	X				AF	The constants
8	42	STO		58	89	2ND π				STO	of Frohlich
9	00 01	0 1		59	65	X				29	formula and
10	91	R/S		60	43	RCL				BF	the resistivi
11	02	2		61	01 02	12				STO	of the materi
12	43	.		62	95	=				28	to be store
13	08	8		63	35	1/X				CF	outside the
14	01	1		64	65	X				STO	program
15	04	4		65	43	RCL				27	
16	04	4		66	02 05	26				ρ	
17	07	7		67	95	=				STO	
18	07	7		68	34	√X				26	
19	03	3		69	42	STO					
20	65	X		70	01 03	13	δ				
21	43	RCL		71	43	RCL					
22	00 01	0 1		72	01 01	11					
23	95	=		73	33	x ²					
24	42	STO		74	85	X					
25	01 00	1 0	H	75	43	RCL					
26	03	3		76	02 05	26					
27	00	0		77	65	X					
28	93	.		78	43	RCL					
29	05	5		79	01 03	13					
30	52	EE		80	55	÷					
31	03	3		81	02	2					
32	65	X		82	95	=					
33	43	RCL		83	42	STO					
34	02 05	26		84	01 04	14	PD				
35	95	=		85	76	2ND Lbl					
36	35	1/X		86	11	A	H				
37	65	X		87	43	RCL					
38	43	RCL		88	01 00	10					
39	00 00	0 0		89	91	R/S					
40	95	=		90	76	2ND Lbl					
41	42	STO		91	12	B	J				
42	01 01	11	J	92	43	RCL					
43	43	RCL		93	01 01	11					
44	01 00	1 0		94	91	R/S					
45	85	+		95	76	2ND Lbl					
46	43	RCL		96	13	C	μ				
47	02 08	28		97	43	RCL					
48	95	=		98	01 02	12					
49	35	1/X		99	91	R/S					
50	65	X		100	76	2ND Lbl					
51	43	RCL		101	13	C					
52	02 09	29		102	43	RCL					
53	95	=		103	01 02	12					
54	85	+		104	91	R/S					
				105	76	2ND Lbl					
				106	14	D	δ				
				107	43	RCL					
				108	01 03	13					
				109	91	R/S					

MERGED CODES
KOMBINATIONEN-KODES
TOUCHES COMBINEES

62	72	83
63	73	84
64	74	92

TEXAS INSTRUMENTS

Superposition

PROGRAMMER
PROGRAMMIERER
PROGRAMMEUR

A.K.M. Al-Shaikhli

DATE
DATUM
DATE

CODE FORM
KODEFORM
FEUILLE DE PROGRAMMATION

LOC ADR	CODE KODE CODE	KEY TASTE TOUCHE	COMMENTS BEMERKUNGEN COMMENTAIRES	LOC ADR	CODE KODE CODE	KEY TASTE TOUCHE	COMMENTS BEMERKUNGEN COMMENTAIRES	LOC ADR	CODE KODE CODE	KEY TASTE TOUCHE	COMMENTS BEMERKUNGEN COMMENTAIRES
0		LRN		5 2	71	SBR		11	95	=	
1	76	2ND Lbl		5 6	25	CLR		11	33	x ²	
2	11	A		5 7	43	RCL		11	34	√x	
3	42	STO		5 8	00 01	∅ 1		11	42	STO	
4	00 01	∅ 1		5 9	75	-		11	01 05	15	
5	91	R/S		6 0	43	RCL		11	71	SBR	
6	43	RCL		6 1	02 01	21		11	25	CLR	
7	00 04	∅ 4		6 2	95	=		11	43	RCL	
8	33	x ²		6 3	33	x ²		11	00 01	∅ 1	
9	75	-		6 4	34	√x		11	75	-	
10	43	RCL		6 5	42	STO		12	43	RCL	
11	00 05	∅ 5		6 6	01 05	15		12	02 06	26	
12	33	x ²		6 7	71	SBR		12	95	=	
13	95	=		6 8	25	CLR		12	33	x ²	
14	42	STO		6 9	43	RCL		12	34	√x	
15	00 00	∅ ∅		7 0	00 01	∅ 1		12	42	STO	
16	34	√x		7 1	75	-		12	01 05	15	
17	42	STO		7 2	43	RCL		12	71	SBR	
18	00 07	∅ 7		7 3	02 02	22		12	25	CLR	
19	65	X		7 4	95	=		12	43	RCL	
20	43	RCL		7 5	33	x ²		13	00 01	∅ 1	
21	00 02	∅ 2		7 6	34	√x		13	75	-	
22	65	X		7 7	42	STO		13	43	RCL	
23	43	RCL		7 8	01 05	15		13	02 07	27	
24	00 03	∅ 3		7 9	71	SBR		13	95	=	
25	55	÷		8 0	25	CLR		13	33	x ²	
26	89	2ND π		8 1	43	RCL		13	34	√x	
27	95	=		8 2	00 01	∅ 1		13	42	STO	
28	42	STO		8 3	75	-		13	01 05	15	
29	00 08	∅ 8		8 4	25	RCL		13	71	SBR	
30	00	∅		8 5	02 03	23		14	25	CLR	
31	42	STO		8 6	95	=		14	43	RCL	
32	01 04	14		8 7	33	x ²		14	00 01	∅ 1	
33	43	RCL		8 8	34	√x		14	75	-	
34	00 01	∅ 1		8 9	42	STO		14	43	RCL	
35	75	-		9 0	01 05	15		14	02 08	28	
36	43	RCL		9 1	71	SBR		14	45	=	
37	01 09	19		9 2	25	CLR		14	33	x ²	
38	95	=		9 3	43	RCL		14	34	√x	
39	33	x ²		9 4	00 01	∅ 1		14	42	STO	
40	34	√x		9 5	75	-		15	01 05	15	
41	42	STO		9 6	25	RCL		15	71	SBR	
42	01 05	15		9 7	02 04	24		15	25	CLR	
43	71	SBR		9 8	95	=		15	43	RCL	
44	25	CLR		9 9	33	x ²		15	00 01	∅ 1	
45	43	RCL		10 0	34	√x		155	75	-	
46	00 01	∅ 1		10 1	42	STO		15	25	RCL	
47	75	-		10 2	01 05	15		15	02 09	29	
48	43	RCL		10 3	71	SBR		15	95	=	
49	02 00	20		10 4	25	CLR		15	33	x ²	
50	95	=		10 5	43	RCL		MERGED CODES KOMBINATIONEN-KODES TOUCHES COMBINEES 62 72 83 63 73 84 64 74 92			
51	33	x ²		10 6	00 01	∅ 1					
52	34	√x		10 7	75	-					
53	42	STO		10 8	43	RCL					
54	01 05	15		10 9	02 05	25		TEXAS INSTRUMENTS			

A.K.M. Al-Shaikhli

CODING FORM

KODEFORM

FEUILLE DE PROGRAMMATION

LOC ADR	CODE KODE	KEY TASTE	COMMENTS BEMERKUNGEN	LOC ADR	CODE KODE	KEY TASTE	COMMENTS BEMERKUNGEN	LOC ADR	CODE KODE	KEY TASTE	COMMENTS BEMERKUNGEN
ADR	CODE	TOUCHE	COMMENTAIRES	ADR	CODE	TOUCHE	COMMENTAIRES	ADR	CODE	TOUCHE	COMMENTAIRES
16 ⁰	34	$\sqrt{x}$		21 ²	22 71	INV LRN					
16 ¹	42	STO				LRN					
16 ²	01 05	15									
16 ³	71	SBR				β	Constant				
16 ⁴	25	CLR				STO					
16 ⁵	43	RCL				$\emptyset 2$					
16 ⁶	01 04	14				I	Current				
16 ⁷	33	x^2				STO					
16 ⁸	65	X				$\emptyset 3$					
16 ⁹	43	RCL				h	Height				
17 ⁰	01 00	10				STO					
17 ¹	95	=				$\emptyset 4$					
17 ²	42	STO				a	Cond. Radius				
17 ³	01 01	11				STO					
17 ⁴	43	RCL				$\emptyset 5$					
17 ⁵	01 04	14				α	Constant				
17 ⁶	91	R/S	H			STO					
17 ⁷	76	2ND Lbl				$\emptyset 6$					
17 ⁸	12	B	PD			PP	The constant				
17 ⁹	43	RCL				STO	of the PD				
18 ⁰	01 01	11				11					
18 ¹	91	R/S				Z1	The distances				
18 ²	76	2ND Lbl	The starting			STO	of the				
18 ³	25	CLR	of the			19	turns from				
18 ⁴	43	RCL	subroutine			22	the zero				
18 ⁵	00 04	$\emptyset 4$				STO	point on				
18 ⁶	33	x^2				20	the load				
18 ⁷	85	+				Z3					
18 ⁸	43	RCL				STO					
18 ⁹	01 05	15				21					
19 ⁰	33	x^2				Z4					
19 ¹	95	=				STO					
19 ²	34	$\sqrt{x}$				22					
19 ³	65	X				25					
19 ⁴	25	RCL				STO					
19 ⁵	00 06	$\emptyset 6$				23					
19 ⁶	65	X				Z6					
19 ⁷	25	RCL				STO					
19 ⁸	01 05	15				24					
19 ⁹	33	x^2				Z7					
20 ⁰	55	+				STO					
20 ¹	25	RCL				25					
20 ²	00 04	$\emptyset 4$				28					
20 ³	95	=				STO					
20 ⁴	85	+				26					
20 ⁵	25	RCL				Z9					
20 ⁶	00	$\emptyset \emptyset$				STO					
20 ⁷	95	=				27					
20 ⁸	35	1/x				Z10					
20 ⁹	65	X				STO					
21 ⁰	25	RCL				28					
21 ¹	00 08	$\emptyset 8$				Z11					
21 ²	95	=				STO					
21 ³	44	SUM				29					
21 ⁴	01 04	14									

MERGED CODES
KOMBINATIONEN KODES

TOUCHES COMBINEES

62  72  83  63  73  84  64  74  92  

TEXAS INSTRUMENTS

IMPROVEMENTS IN THE DESIGN OF INDUCTION BILLET HEATERS.

A K M Al-Shaikhli and L Hobson

Loughborough University of Technology, UK.

1 INTRODUCTION

Induction heating of metals is well established throughout the metal forming industry for heating metals prior to forging, rolling and extrusion. Although the unit cost of electricity is high compared with fossil fuels, induction heating techniques offer many advantages including high efficiency, high power densities, and improved working environment. These process advantages have ensured that induction heating has achieved a substantial market share of the heating installations within certain parts of the metal forming industry. Other areas however have so far remained virtually untouched, especially those requiring a non-uniform surface power density along the length of the workpiece or those requiring many different workpieces diameters to be heated during the working day.

In industrial practice the design of induction billet heaters is largely based on the equivalent circuit method devised by Baker [1,2] and developed by Reichert [3] and Vaughan and Williamson [4,5]. The method assumes a uniform magnetic field strength along the length of the workpiece and many empirical factors are incorporated to take into account such things as a shortness of coil and spacing between turns. Numerical solutions using finite difference and finite element techniques are in existence [6,7] but they also assume a uniform magnetic field strength and hence a surface power density along the length of the workpiece. Even though they are at best two dimensional solutions they require specialist knowledge and powerful computational facilities not normally available to U.K. induction heating manufacturers.

In order to deal realistically with problems involving a non-uniform surface power density along the length of a workpiece, the surface current density distribution produced by a single conductor has been investigated. The superposition principle to find the power density produced by a number of conductors has been verified and a work coil design to produce a particular temperature distribution in a billet has been carried out. Investigations have also included the effect of conductor size, operating frequency, magnetic permeability and shape of workpiece, and current distribution within the work conductor itself.

2 SYMBOLS

$a_0, a_1, \dots, a_n$	= Constants to be found for each case
b	= Thickness of slab (m)
c	= Specific heat ($J K^{-1} kg^{-1}$)
f	= Frequency (Hz)
$J_0(x)$	= Bessel function of the first kind and zero order
$J(z)$	= The surface current density on the load ($A m^{-2}$)
r	= A radius in the cylinder (m)
R	= The radius of the cylinder (m)

t	= Time (s)
x	= Depth in the slab measured from the surface (m)
z	= The distance along the load measured from a point directly beneath the conductor (m)
β	= Positive roots of $J_1(\beta) = 0$
γ	= Density ($kg m^{-3}$)
$\delta = \sqrt{\frac{\rho}{\pi f \mu}}$	= Skin depth (m)
ρ	= Resistivity (Ωm)
μ	= Permeability (Hm^{-1})
σ	= Temperature ($^{\circ}K$)

3 THE SUPERPOSITION METHOD

The temperature of a load is a function of the surface power density. By controlling the current density distribution on the load the temperature distribution along its length can be controlled. To achieve this aim, the surface current density distribution in the load produced by a single conductor was investigated and it was found that the distribution could be represented by the following equation:

$$J(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots + a_n z^n \quad (1)$$

If there are a number of similar conductors then the surface current density at any point on the load can be found by applying superposition principles. Consider Fig. 1 which represents N identical conductors parallel to each other at unequal distances from each other but at a specific air gap from a metallic slab. If a current I flows through each conductor then the surface current density (J_p) at point P is given by equation (2).

$$J_p = J_{p1} + J_{p2} + \dots + J_{pN} \quad (2)$$

where $J_{p1}, J_{p2}, \dots, J_{pN}$ are the surface current densities produced by the individual conductors 1, 2, ..., N at point P and they are given by:

$$J_{p1} = a_0 + a_1 z_1 + a_2 z_1^2 + \dots + a_n z_1^n \quad (3)$$

$$J_{p2} = a_0 + a_1 z_2 + a_2 z_2^2 + \dots + a_n z_2^n \quad (4)$$

$$J_{pN} = a_0 + a_1 z_N + a_2 z_N^2 + \dots + a_n z_N^n \quad (5)$$

The same equations can apply if it is required to produce a specific current density J_p at point P by changing the distances $z_1, z_2, \dots, z_N$.

By using this method, the induced surface current density and hence the power density and the temperature distributions along the load can be controlled.

It should be mentioned here that the relation between the current in the conductor and the induced surface current density in the load is a linear relation and it should be taken into account if the current used in multi-turn coil is different from that of single conductor.

4 THE EXPERIMENTAL WORK

To verify the superposition method practically, a series of tests were carried on an aluminium slab of 50.8 mm thickness and a cylinder of 50.8 mm radius. In all the experiments a single current carrying conductor was placed above the slab or around the cylinder at a specific air gap. The surface current density distribution induced in the load was measured by a current density probe [8]. Using the results of the single turn coil the performance of a multi-turn coil can be predicted based on the calculation of the current density by the superposition method and the power density based on the equations given in the Appendix.

Two types of water cooled conductors were used in the experiments with the slab. The first type was a circular conductor of 6 mm diameter, in these conductors the current flows uniformly. The second type was a rectangular conductor, Fig. 2, which are used in mains frequency induction heating installations. The current does not flow uniformly in this kind of conductor.

In the experiments on the cylindrical load only water cooled circular conductors of 6 mm diameter were used.

All the experimental work was carried out using a mains frequency supply.

Fig. 3 shows the surface current density distribution along the length of the slab due to current of 600 A flowing through a single circular conductor with a 39 mm air gap. The corresponding power density distribution induced from 5 conductors carrying 600 A/conductor at 60 mm coil pitch is shown in Fig. 4. The agreement between the practical readings and the calculation values is 5% i.e. within the range of experimental error.

The surface current density induced in the slab due to single rectangular conductor carrying 600 A with an air gap of 36 mm is shown in Fig. 5. The superposition theorem was verified by using 10 conductors very close to each other, coil pitch of 11 mm, carrying a current of 600 A/conductor as shown in Fig. 6. The discrepancy in this case is less than 9%.

Figs. 7 and 8 show the results obtained from the experiments using a cylindrical load. The current in the single turn and the 11 turns coil was 600 A and the air gap was 50 mm. The coil pitch of the 11 turns coil was 7.1 mm. The difference between practical readings and calculated values is less than 3% for power densities down to 5% of the maximum value. At power densities below this the accuracy was not as good but because of the magnitude of the discrepancies was small this was neglected.

5 PRODUCTION OF A POWER DENSITY TAPER

There are applications in induction heating, such as off-the-bar forging or the extrusion of aluminium which require a non-uniform power density to be produced along the bar or billet.

In direct extrusion of aluminium, a ram directly pushes a billet through a stationary press container and selected die-shape. The finally extruded section emerges onto the run-out as a correctly shaped product. In this kind of extrusion it is preferred that the extrusion speed of the ram and product temperature through the die be constant throughout the extrusion cycle, thus avoiding continual adjustment of the press. This isothermal effect may be

achieved by inducing into the heated billet a temperature at the front face; that part which enters the die first. In order to appreciate the reason for this isothermal effect it must be explained that additional heat will also be generated during the extrusion operation:-

- By heat energy, caused by the shear effect at the restricted die opening,
- Through the generation of friction, since the hot aluminium billet is not only forced into the die but also moves along the stationary container whilst confined within its bore.

In indirect extrusion the hot billet remains stationary in relation to the press container, the rear end of the billet becomes progressively cooler as the extrusion proceeds rather than the front end. This, of course, is the exact reversal of the direct extrusion and care must be taken to ensure the correct degree of reverse taper if optimum results are to be achieved.

To demonstrate the usefulness of the design procedure developed in the previous sections it was decided to design an induction heating coil capable of producing a linearly decreasing temperature taper. This, of course, models the practical requirements met industrially in the induction heating of aluminium prior to extrusion.

It was not possible to build a commercial size billet heater to test the theory because of the capital cost and the limitations of the laboratory power supplies. Megawatt power supplies and large amounts of power factor correction capacitors are not uncommon in industrial units. However, a laboratory scale model was built to demonstrate the validity of the design procedure. The work coil was made from 6 mm diameter water cooled copper tubing which meant that the variable pitch coil could easily be manufactured with the University's facilities. The workpiece was a 200 mm long aluminium cylindrical billet of radius 50.8 mm and the air gap was taken as 30 mm which is in agreement with industrial practice [10].

The induced surface current density on the load due to a 600 A current flowing through a single turn coil was measured and the results shown in Fig. 9. These values were then used to calculate the resultant surface current density of a number of different work coil configurations using the procedure outlined previously. The design of a work coil capable of generating a linearly decreasing temperature taper on the surface of the billet was then decided upon and details of the work coil pitch are given in table 1. The eleven turn work coil was constructed and the surface current and power densities it produced were measured. The surface power densities induced in the load varied from 7 kW/m² down to 0.5 kW/m². The practical measurements were compared with those required to produce the linearly decreasing temperature taper and the results are shown in Fig. 10 correlation to within an accuracy of 5% can be seen thus demonstrating the validity of this novel method of work coil design.

Pitch No.	1	2	3	4	5	6	7	8	9	10
mm	7	8	11	23	23	23	25	23	25	32

Table 1: The pitches of the coil

6 CONCLUSION

The practical investigations show that the superposition method can be applied on loads of different shapes. The non-uniform flow of current through the conductors did not affect the applicability of this method within the range of the test carried out.

Applying this method to a practical application, i.e. the production of a linear power density taper along the length of a billet, proved highly successful.

The computer program is relatively small, i.e. it does not need a large storage capacity or consume long computing time. This method is therefore suitable for desk top computers which are cheap and readily available within even the smallest company.

Although the results are shown for tests on mains frequency and non-magnetic loads, further work now being carried out to assess the effect of frequency, proximity of laminations packs and non-uniform current distribution in conductor. Work also was carried out on magnetic material to assess the effect of the permeability and the possible use of this method in the off-the-bar forging applications.

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APPENDIX

The Relationship Between Surface Current and Power Densities and the Temperature of the Load

In induction heating, the heat flows almost in the radial direction only because of the high power usually used in this kind of heating. The one dimension solution to the problem of heat flow in a slab and cylinder heated by induction is shown below.

a) Slab heated from one side

The temperature (σ) at a depth (x) in a slab of total thickness (b) heated on one side at a constant power density (PD) is given by [9].

$$\sigma = \frac{(PD)t}{bc\gamma} + \frac{(PD)b}{2k} \left[(1 - x/b)^2 - \frac{1}{3} - \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \exp(-n^2 \pi^2 \tau) \cos n\pi(1 - x/b) \right] \quad (1A)$$

$$\text{where } \tau = \frac{kt}{\gamma c b^2}$$

The first term is the final temperature of the slab. The summation term is the transient temperature rise and can be ignored for values of $\tau \geq 0.25$. The transient period represents a small part of the heating cycle, for example it corresponds to 30 seconds for aluminium slab of 100 mm thickness at 473° K. Equation (1A) then simplifies to

$$\sigma = \frac{(PD)b\tau}{k} + \frac{(PD)b}{2k} \left[(1 - x/b)^2 - \frac{1}{3} \right] \quad (2A)$$

$$\therefore \sigma = \frac{b}{2k} \left\{ 2\tau + \left[(1 - x/b)^2 - \frac{1}{3} \right] \right\} PD \quad (3A)$$

i.e. the temperature at any depth (x) is a function of the surface power density which is in turn a function of the surface current density [10] and it is given by equation (4A) below

$$PD = \rho J^2 \frac{\delta}{2} \quad (4A)$$

This shows that the temperature at any depth is a function of the surface current density.

b) Cylinder of radius R

The temperature (σ) at a radius (x) in a cylinder of radius (R) heated by induction is given by [9]:

$$\sigma = (PD) \frac{R}{k} \left\{ 2\tau + \frac{x^2}{2R^2} - \frac{1}{4} - 2 \sum_{n=1}^{\infty} e^{-\beta_n^2 \tau} \frac{J_0 \left[\beta_n (x/R) \right]}{\beta_n^2 J_0(\beta_n)} \right\} \quad (5A)$$

$$\text{where } \tau = \frac{kt}{\gamma c R^2}$$

as in the case of slab the summation term can be ignored when $\tau = 0.25$ and equation (5A) can then be simplified to

$$\sigma = PD \frac{R}{k} \left(2\tau + \frac{k^2}{2R^2} - \frac{1}{4} \right) \quad (6A)$$

hence the temperature at any radius is a function of the surface power density which is, as in slab, a function of the surface current density [10]

$$PD = \mu \pi f \frac{R}{2} J^2$$

$$\left[\frac{\text{ber}(\alpha) \text{ber}'(\alpha) + \text{bei}(\alpha) \text{bei}'(\alpha)}{\text{ber}^2(\alpha) + \text{bei}^2(\alpha)} \right] \quad (7A)$$

$$\text{where } \alpha = \frac{\sqrt{2}}{\delta} R$$

As in the case of slab the temperature at any radius is a function of the surface current density.

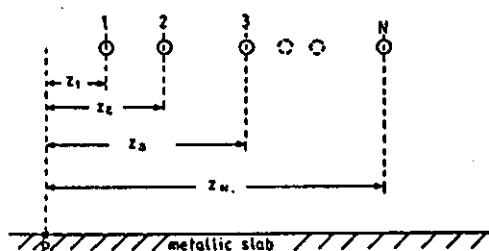


Fig 1. Number of identical conductors at a specific air gap from a metallic load.

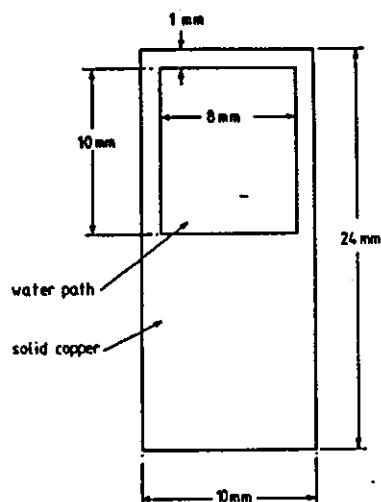


Fig. 2. Rectangular conductor used in mains frequency induction heating.

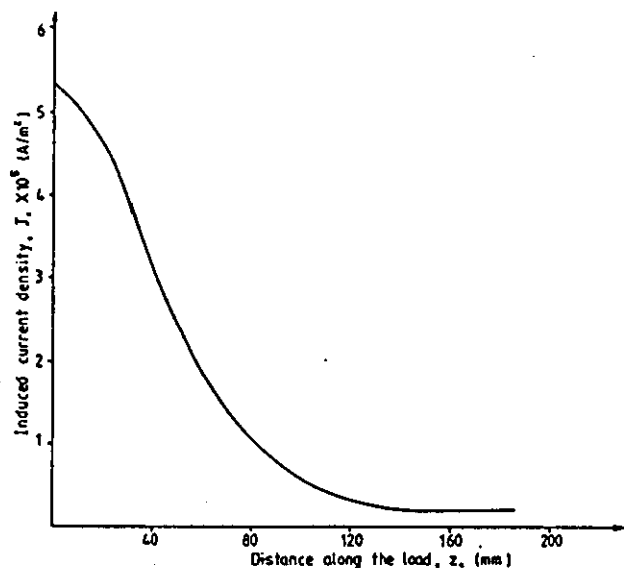


Fig. 3. Current density distribution along the aluminium slab due to 600A flowing in single circular conductor at air gap of 39 mm.

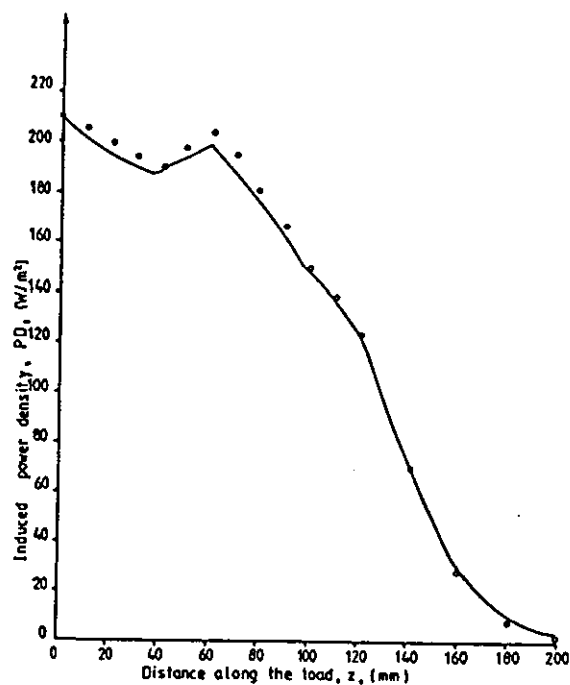


Fig. 4. Power density distribution along the aluminium slab due to 5 circular conductors at air gap of 39 mm, coil pitch of 60 mm, and a current of 600A/conductor.

— Prediction by superposition o Practical readings.

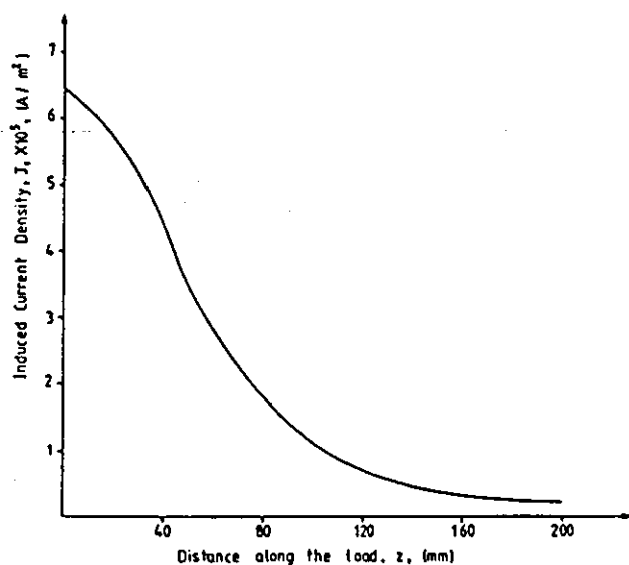


Fig. 5. Current density distribution along the aluminium slab due to 600A flowing in single rectangular conductor at air gap of 36 mm.

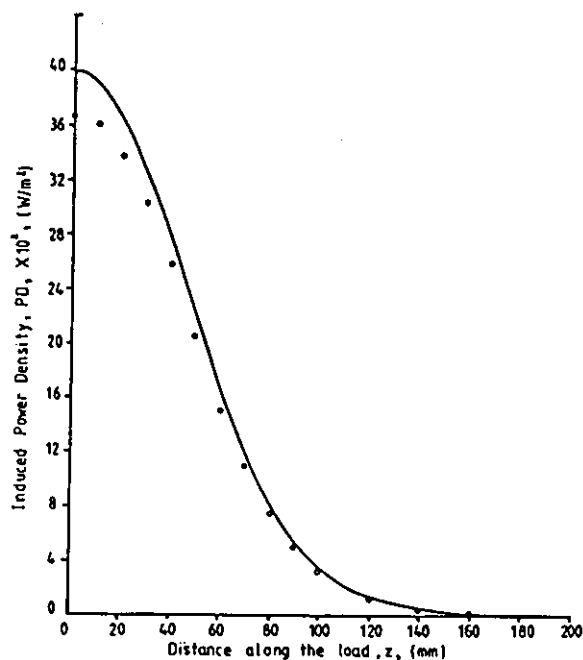


Fig. 6. Power density distribution along the aluminium slab due to 10 rectangular conductors at air gap of 36 mm, coil pitch of 11 mm and a current of 600 A/conductor.

— Prediction by superposition o Practical readings.

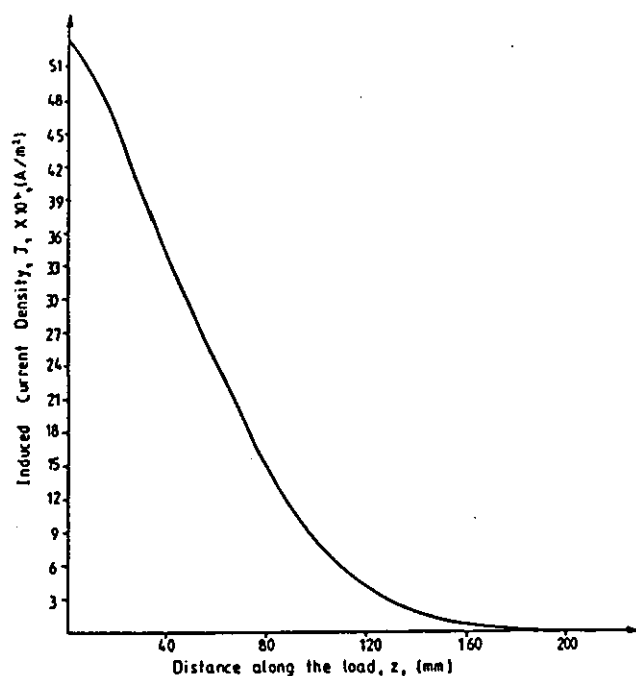


Fig. 7. Current density distribution along the aluminium cylinder due to 600A flowing in single circular conductor at air gap of 50 mm.

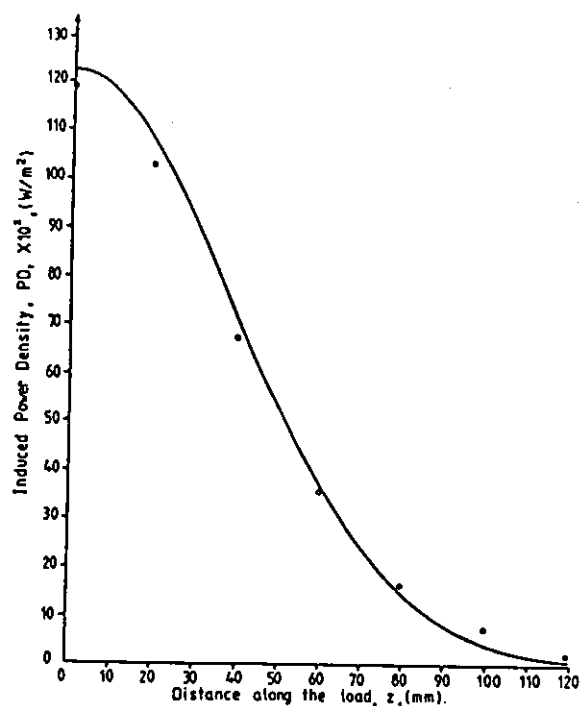


Fig. 8. Power density distribution along the aluminium cylinder due to 11 turns coil at air gap of 50 mm, coil pitch of 7.1 mm and a current of 600A.

— Prediction by Superposition o Practical Readings

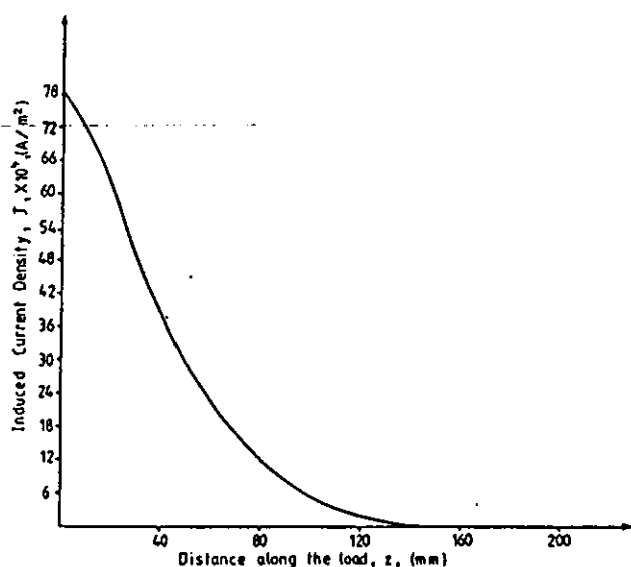


Fig. 9. Current density distribution along the aluminium cylinder due to 600A flowing in single conductor at air gap of 30 mm.

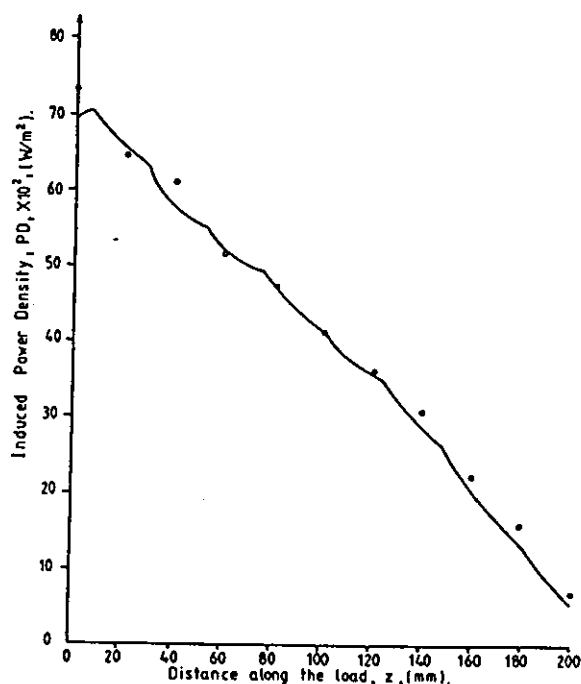


Fig. 10. Power density distribution along the aluminium cylinder due to 11 turns coil at air gap of 30 mm, nonuniform coil pitch and a current of 600A.

— Prediction by Superpositions. o Practical Readings

METHODS OF ANALYSIS FOR INDUCTION HEATING WORK COILS

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ABSTRACT

Industrial designers of induction heating work coils depend largely on relatively simple equivalent circuit techniques supplemented by empirical data accumulated over many years. On the other hand academics and other research workers have developed highly sophisticated numerical techniques to solve particular problems. This paper explains the principles behind each form of work coil design and assesses the range of their applicability. Suggestions are made to improve the existing work coil design techniques and brief details are given of a new method of design more readily applied to certain industrial applications of major present day interest, i.e. aluminium extrusion and off-the-bar forging.

1. INTRODUCTION

Induction heating has a number of inherent advantages over its fuel fired competitive processes; including fast heating rates, precise temperature control, suitability for incorporation in fully automated processes, small standby losses, and good working environment with little extraneous heating and a low noise level. These process advantages have ensured that induction heating has achieved a substantial market share of the heating installations within certain parts of the metal forming industry (1). Other areas, however, have so far remained virtually untouched, especially those requiring a non-uniform surface power density along the length of the workpiece (2).

Industrial designers of induction heating work-coils depend largely on relatively simple equivalent circuit techniques supplemented by empirical data accumulated over many years. On the other hand academics and other research workers have developed highly sophisticated numerical techniques to solve particular problems. Existing methods of work-coil design almost universally assume that the load is subjected to a uniform magnetic field.

It is very difficult, therefore, to adapt these methods to solve the problems of non-uniform power density distribution along a workpiece.

This paper explains the principles behind each form of work-coil design and assesses the range of their applicability. Suggestions are made to improve the existing work coil design techniques and details are given of a new method of design more readily applied when non-uniform surface power density distributions are required.

2. EQUIVALENT CIRCUIT METHOD

The basic assembly of induction billet heater consists of a water cooled coil surrounding a metallic workpiece. A relatively large air gap between the coil and the workpiece is required so that the billet can move freely through the heater and, in addition, thermal insulation is provided to reduce the radiated heat losses from the billet to the coil. The magnetic flux produced by the work coil has alternative parallel paths through either the workpiece, the coil or the air gap as shown in Fig.1.

An expression for the total ampere-turn requirement

is derived using the reluctance of each of the magnetic circuits and Ampere's law (3,4).

The work coil voltage, V_c , is related to the total flux by Faraday's law and hence an expression for the total circuit impedance, Z , can be obtained. An electrical equivalent circuit is therefore derived and shown in Fig.2 where

$$X_r = \omega \mu_0 \pi N_c^2 \frac{r_c}{2(0.45+k)}$$

$$X_g = \omega \mu_0 \pi N_c^2 \frac{(r_c^2 - r_w^2)}{l_c}$$

$$Z_w = \omega \mu_0 \pi N_c^2 \frac{\mu_r r_w^2}{l_w} (Q+jP) = R_w + jX_w$$

$$Z_c = \omega \mu_0 \pi N_c^2 \frac{k_c \delta_c r_c}{l_c} (1+j) = R_c + jX_c$$

where N_c is the number of turns. r_c , r_w , l_c and l_w are the radii and lengths of the coil and the workpiece respectively. Q and P are dimensionless constants, δ_c is the coil skin depth and k and k_c are empirical factors.

The equivalent circuit can be solved by circuit analysis to give work coil turns, current carrying capacity, efficiency and power factor.

It should be mentioned here that other forms of the equivalent circuit may be used by choosing different positions for X_r in the circuit (3,5,6).

The effect of the different parameters in the equivalent circuit depends on the case under consideration. To assess the influence of the external reactance X_r and the empirical factors k and k_r , three computer programs were developed based on three different equivalent circuits. A specific work coil/nonmagnetic billet configuration was assumed and calculations were carried out for frequencies of 50 Hz and 3 kHz. The required number of turns obtained from the three circuits is within 8% difference from each other for the 50 Hz frequency, this difference increases to become 10% when the frequency is 3 kHz. Varying k between -0.22 and 0.1 changed the required number of turns by up to 10%. The maximum effect of changing k_r from 1.0 to 1.5 was 20% difference in the required number of turns.

The equivalent circuit method suffers, mainly, from three imperfections.

(1) The need of empirical data. The values of these factors are not necessarily known for every application.

(2) The assumption of uniform material properties which is not correct as the resistivity and permeability of the workpiece change with the temperature and magnetic field strength. This deficit can partially be overcome by considering the workpiece as consisting of a number of concentric cylinders as it will be seen in the next section.

(3) The assumption of uniform magnetic field along the length of the workpiece hinders this method from being applied to applications which require nonuniform power densities in the load. This

problem can be solved by employing the relatively new superposition method (2).

2.1. Improvement on the Equivalent Circuit Method

One of the limitations of the equivalent circuit method is that it does not take the nonuniformity of the workpiece into account. This can be overcome by assuming the workpiece to consist of more than one concentric cylinder, each with different physical properties.

Fig. 3 shows a typical billet heating application in which the coil of radius r_c surrounds a billet of radius r_w . The billet is shown divided into n concentric cylinders. The impedance of the composite load can be determined by employing the complex Poynting's vector (4) to derive an expression for the impedance, Z_x , on radius x within a cylinder

$$Z_x = \frac{-2\pi x p}{I_w} \frac{J_x}{H_0} \frac{H_x^*}{H_0^*}$$

then solving Maxwell's equations on the boundaries between the concentric cylinders.

A computer program was developed for a work coil/multicylinder billet. Two cases were considered. The first is; a load of 8 concentric cylinders with different physical properties. The second is for a uniform load with physical properties equal to the average properties of the first case. The difference between the required number of turns for the two loads was found to be 23%. This shows the importance of the load's nonuniformity in the calculations.

3. NUMERICAL METHODS OF WORK COIL DESIGN

Three distinct numerical approaches have been applied to induction heating problems: finite difference, finite element and mutually coupled circuit methods. In the finite difference and finite element methods the workpiece is divided into subregions and the relevant non-linear partial differential equations are replaced by a set of algebraic equations to be solved in each subregion by means of iterative procedures. The mutually coupled circuit method (7) divides the system into a number of subconductors mutually coupled with each other and applies Kirchhoff's Laws to obtain a system of linear equations describing the problem which in turn are solved by procedures similar to those used in other numerical methods. The application of these numerical methods to induction heating problems will now be discussed in detail.

3.1. Mutually Coupled Circuit Method

This method was developed to solve problems in induction heating and melting applications by Kolbe and Reiss (7) and later developed by Dudley and Burke (8). The region to be investigated is subdivided into subconductors and using known expressions (9) the resistance and self inductance of each subconductor, together with the mutual inductances between each subconductor are determined. The expressions are usually presented in the form of an impedance matrix and the application of Kirchhoff's Laws lead to a system of linear equations with complex coefficients of the form

$$[Z][I] = [V]$$

where $[Z]$ is the square matrix comprising all of the coil and billet self and mutual impedances, $[I]$ is the column matrix of unknown coil and billet segment currents and $[V]$ is the column matrix of driving voltages.

The mutually coupled circuit method can be used to

determine the induced currents, power distribution and mechanical forces produced by any two dimensional, linear and axisymmetric induction heating problem. The limitations of the method are that magnetic materials cannot be included in the problem geometry and that the number of elements into which the conductors can be divided is limited by the size of the matrix which may be solved. This number is of the order of 300 unknowns for the IBM 3033 computer (10) which has a memory capability of approximately one megabyte. Difficulties also arise when dealing with curved surfaces and when the penetration depth is small in comparison with the load dimensions. The number of unknowns can partly be reduced by finely subdividing only the region of greatest interest, typically between a distance 0.1 to 0.33 times the skin depth δ (10) from the surface and using larger subdivisions elsewhere. This, of course, requires more complicated software.

3.3. Finite Element Method

The finite element method has become an important and practical numerical analysis tool. It has found application in almost all areas of engineering and applied mathematics.

In this method, the field region is subdivided into a finite number of discrete sized subregions or finite elements. The unknown quantities at each element are presented by suitable interpolation functions that contain the node values of each element.

From the mathematical point of view the finite element method is based on integral formulations. By way of comparison the older finite difference methods are usually based on differential formulations. Finite element integral formulations are obtained by two different procedures; Variational formulations (11, 12) and weighted residual formulations (13). Both of these techniques generate the final assembly of algebraic equations that must be solved for the unknown nodal parameters.

It must be borne in mind that when employing the finite element methods to solve induction heating problems, one must (10)

(i) Include an "exterior element" to represent the region outside the coil.

(ii) Use a sparse matrix solution routine.

Typically, induction heating problems involving the computation of the electromagnetic and thermal fields at 500 nodes requires approximately 0.75 megabytes of computer memory (10) even when using a sparse matrix solution routine. Problems involving 500 to 2000 unknowns can be run at a reasonable cost on an IBM 3033 which compares very favourably with the mutually coupled circuit method mentioned previously.

3.3. Finite Differences

This was the first numerical technique applied to induction heating problems (14) and is still in use today. The technique provides a relatively straightforward means of formulation and solving two dimensional problems.

The basis of the finite difference method is the conversion of the governing differential equations of the problem into a set of algebraic equations by the use of the Taylor series approximations. The equations are solved at every point on a grid constructed over the required space domain. The mathematical operations are performed not upon continuous function but rather in terms of equations about one discrete point. By this approach operations such as differentiation and

integration may be reduced to simple arithmetic forms and can then be conveniently solved using digital computers.

A program has been written to predict the magnetic field and power density distributions in a homogeneous nonmagnetic rectangular workpiece subjected to a uniform surface magnetic field strength.

The magnetic field and power density distribution within an aluminium workpiece of cross-section 500 mm by 150 mm subjected to a surface magnetic field strength of 10^5 Am^{-1} at a frequency of 50 Hz are shown in Figs. 4 and 5 respectively.

The program was relatively simple to implement and gave satisfactory results when applied to a simple two dimensional problem.

4. CONCLUSIONS AND RECOMMENDATIONS

An accurate knowledge of the power density and temperature distribution in an inductively heated load is of great importance. Analytical expressions are available for simple one dimensional geometries such as a cylinder or a slab (15) but for the solution of even the simplest two dimensional geometries numerical techniques are required.

In industrial practice the design of induction billet heaters is largely based on the equivalent circuit method which assumes a uniform magnetic field strength along the length of the workpiece and many empirical factors are incorporated to take into account such things as a shortness of coil and spacing between turns. Numerical solutions using finite difference, finite element, and mutually coupled circuit techniques are in existence but they also assume a uniform magnetic field strength and hence a surface power density along the length of the workpiece. Even though they are at best two dimensional solutions they require specialist knowledge and powerful computational facilities not normally available to U.K. induction heating manufacturers.

Induction billet heating is a well established industrial process for preheating prior to forging, rolling and extrusion. There are areas, however, in which induction heating has not achieved the market share that perhaps one might expect e.g. off-the-bar forging or the extrusion of aluminium. In both applications the ability to create a non-uniform surface power density along the billet or bar is required. Existing methods of work coil design which are based on the assumption of an infinitely long solenoid and/or the use of a great amount of empirical data are not easily adapted to the design of non-uniform surface power densities. Further investigations to develop a new technique for work coil design is required which can be adapted to solve problems requiring a non-uniform surface power density. In order to deal realistically with problems involving a non-uniform surface power density along the length of a workpiece, the surface current density distribution produced by a single conductor has been investigated. The superposition principle to find the power density produced by a number of conductors has been verified and a work coil design to produce a particular temperature distribution in a billet has been carried out. (2, 16). Investigations

have also included the effect of conductor size, operating frequency, magnetic permeability and shape of workpiece, and current distribution within the work conductor itself.

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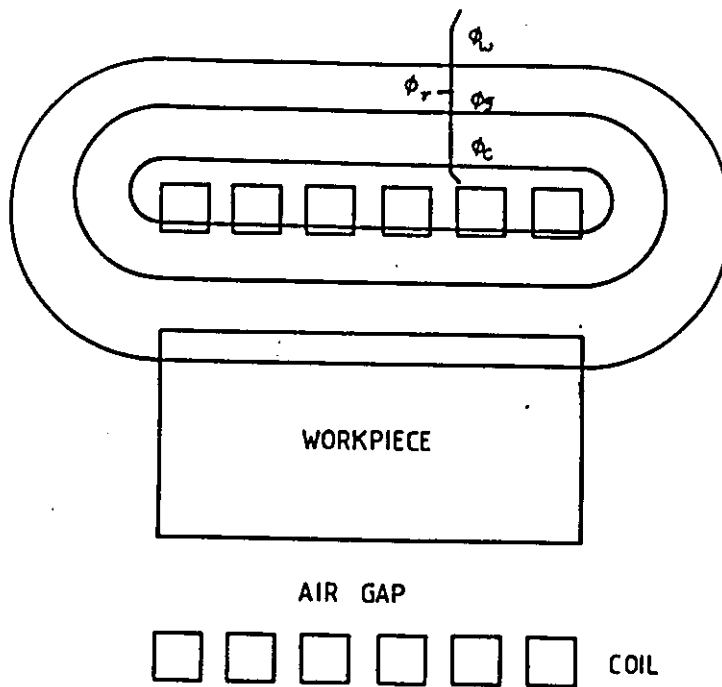


Fig. 1 The flux paths in induction heating system

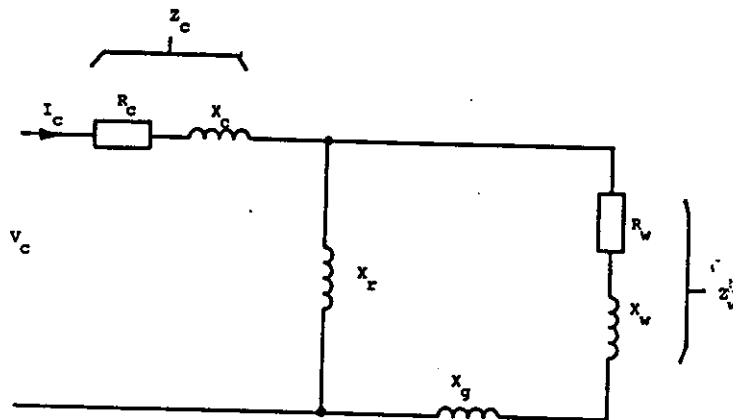


Fig. 2 The equivalent electrical circuit

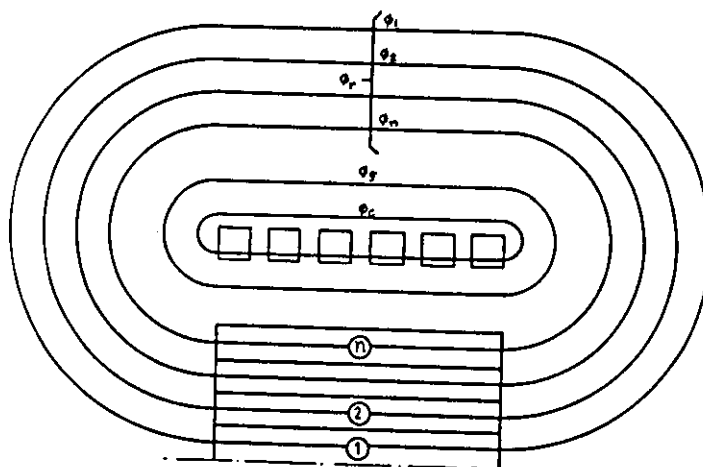


Fig. 3 The flux paths in a load of concentric cylinders

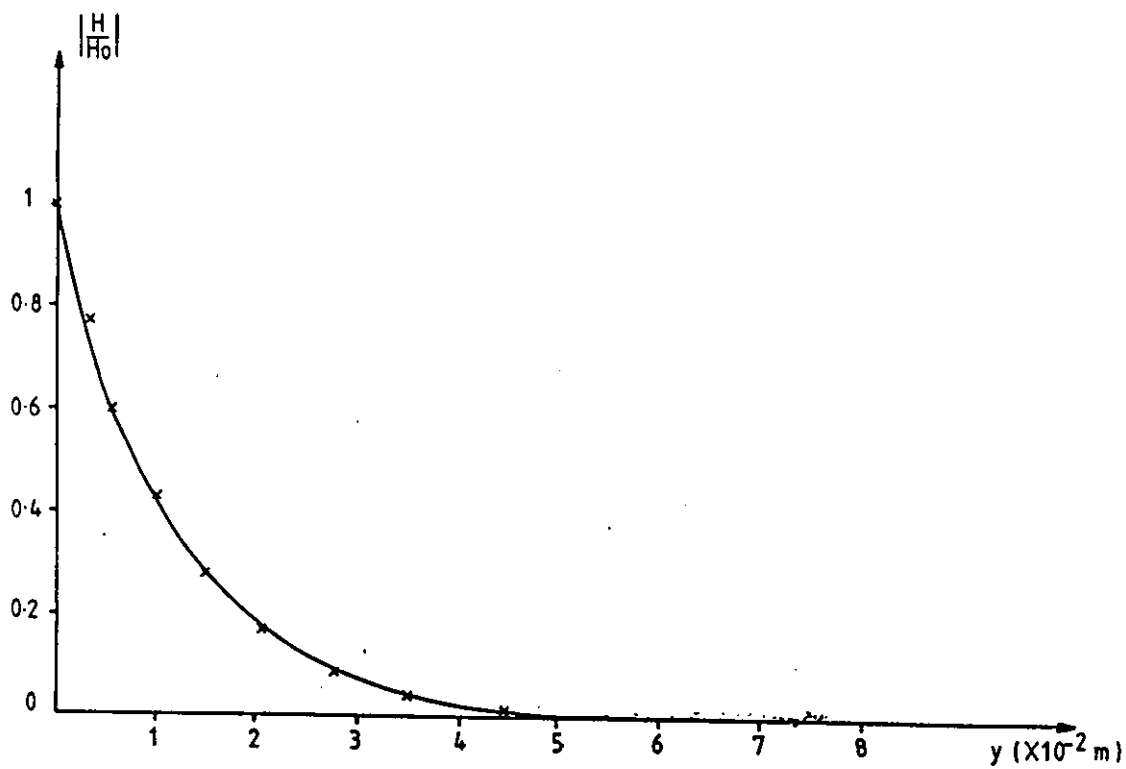


Fig. 4 Variation of magnetic field distribution (as a fraction of the surface field) with depth y .

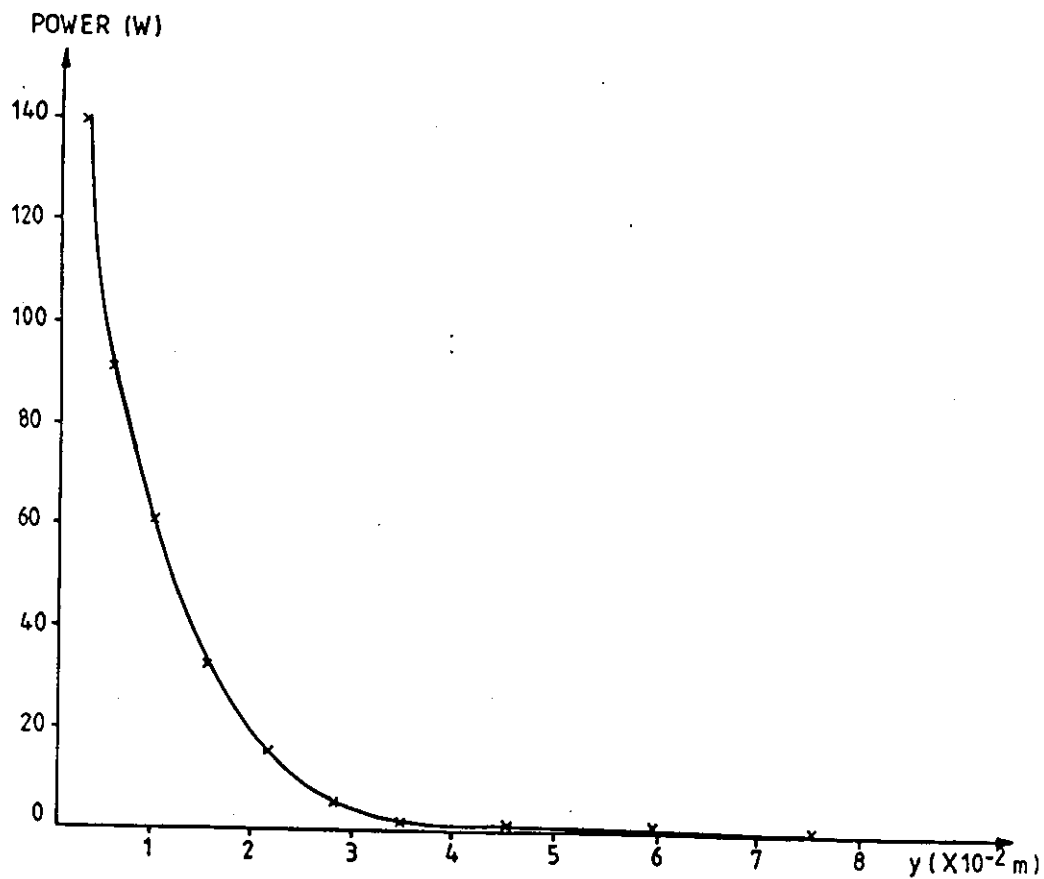


Fig. 5 Power distribution down the centre line of the cross-sectional face

ILLUSTRATING ELECTROMAGNETICS USING AN INDUSTRIAL PROCESS

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ABSTRACT

The authors have attempted to illustrate basic electromagnetic phenomena using readily available laboratory equipment and unsophisticated current density measuring probes. A series of experiments are described explaining induction heating theory and relating it to the requirements of the industrial process of aluminium extrusion.

ILLUSTRATING ELECTROMAGNETICS USING AN INDUSTRIAL PROCESS

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SYMBOLS

a	=	The radius of the conductor (m)
A	=	Constant found from practical measurement (dimensionless)
B	=	" " " " " "
f	=	Frequency (Hz)
h	=	Perpendicular distance between centre of conductor or filament and surface of the slab or cylinder (m)
h_e	=	$\sqrt{h^2 - a^2}$ = Effective height (m)
H	=	Magnetic field strength (Am^{-1})
H_p	=	Magnetic field strength at point P (Am^{-1})
H_r	=	Radial component of magnetic field strength (Am^{-1})
H_z	=	Axial component of magnetic field strength (Am^{-1})
I	=	Current (A)
J	=	The surface current density (Am^{-2})
J_p	=	The surface current density at point P (Am^{-2})
ℓ	=	The probe length = The perimeter of the cylinder (m)
V	=	The voltage across the probe terminals (V)
W_p	=	The power density induced on point P (Wm^{-2})
z	=	The distance of point P from plane of filament or conductor (m)
δ	=	$\sqrt{\frac{\rho}{\pi f \mu}}$ = Skin depth (m)
ρ	=	Resistivity (Ωm)
μ	=	Permeability (Hm^{-1})

1. INTRODUCTION

Electromagnetic field theory is often taught in an extremely abstract manner which does not appeal to the vast majority of students. Laboratory experiments designed to illustrate the fundamental concepts of electromagnetics are also often contrived to avoid practical measurement problems and over simplified to such an extent that their industrial relevance is not obvious. In an attempt to reverse this situation the authors have suggested a series of laboratory experiments which illustrate basic electromagnetic phenomena and relate it to a particular industrial process.

When a current carrying conductor passes in close proximity to a metallic structure then eddy currents are induced in the metal which can give rise to heating effects or even excessive mechanical forces within the structure. The proximity effect is a fundamental feature of electromagnetics and is put to use industrially in the induction heating of metallic objects prior to rolling, forging or extrusion. Induction billet heaters are made with many turns of water cooled copper tubing and the design of these coils requires knowledge of the induced currents from not one conductor but many in close proximity to each other and to the workpiece⁽¹⁾. One of the most common industrial processes using induction billet heating is the heating of aluminium prior to extrusion. Aluminium workpieces heated prior to extrusion require a temperature taper⁽²⁾ and hence a surface power density taper along their length. The exact requirements depend on the exact extrusion process being used.

The use of a long filament close to a semi-infinite slab of metal would be the easiest example which could be analysed theoretically.

However this situation is far from those seen in practice. The interaction of a finite sized conductor close to a circular billet is of greater industrial relevance but theoretical analysis of this situation is not easy and graphical results with experimentally verified correction factors are preferred.

The series of experiments put forward consists of an investigation of the interaction of a single current carrying conductor and a metallic workpiece; the illustration of the validity of the method of superposition, i.e. the prediction of the total surface current density induced by a number of conductors in close proximity; and finally the design of a simple induction billet heater which can produce a non-uniform current density distribution often required in the industrial process of induction heating of aluminium prior to extrusion.

2. SINGLE CONDUCTOR

Fig. 1 shows a filament carrying current I at a distance h from the surface of a semi-infinite metal slab. Given that the slab is a good conductor the magnetic field strength at any point P at distance z along the surface of the slab can be calculated by assuming an imaginary filament within the slab at a distance h from the surface and carrying current $-I$. Both the conductor and the image will produce a magnetic field strength H , according to Ampere's law, equal to

$$H = \frac{I}{2\pi\sqrt{h^2 + z^2}} \quad (1)$$

The axial components H_z of the conductor's magnetic field strength and its image will aid each other, while the radial components H_r will cancel each other. The resultant field on point P will therefore be:-

$$H_p = 2 H_z \quad (2)$$

$$\text{but } H_z = H \cos \theta = H \frac{h}{\sqrt{h^2 + z^2}} \quad (3)$$

Hence

$$H_p = \frac{Ih}{\pi [h^2 + z^2]} \quad (\text{Am}^{-1}) \quad (4)$$

If the filament were replaced by a circular conductor of finite radius a , then the magnetic field at P is modified to the following expression⁽³⁾:-

$$H_p = \frac{I \sqrt{h^2 - a^2}}{\pi [h^2 - a^2 + z^2]} \quad (5)$$

The surface current density on the semi-infinite slab in a direction opposite to that of the conductor current also given by⁽⁴⁾:

$$J_p = H_p \frac{\sqrt{2}}{\delta} \quad (\text{Am}^{-1}) \quad (6)$$

and the surface power density is:-

$$W_p = \frac{\rho \delta}{2} J_p^2 \quad (\text{Wm}^{-2}) \quad (7)$$

If the metallic workpiece is a long cylinder then it can, to a first approximation, be modelled by the semi-infinite slab configuration providing the radius of the cylinder is many times the electromagnetic skin depth within the material⁽⁴⁾.

A laboratory experiment has been designed in which a single circular turn of 6mm diameter water cooled copper tubing was wrapped around an aluminium cylindrical workpiece of 50.8mm radius. With air gaps between the coil and workpiece of 30mm and 50mm, and a coil current of 600 A at 50 Hz, the current density at 10mm intervals along the surface of the billet was measured. The measurements were made using surface current density probes. A number of surface current density probes were positioned along the length of the billet, Fig. 2. Each probe consisted of a single turn of constantan wire electrically insulated from the billet. The output of the probe was measured by a high impedance voltmeter shielded from the high magnetic field. The voltage reading was directly related to the current density within the surface of the workpiece^{(5) (6)}, as:-

$$J = \frac{V}{\rho l} \quad (8)$$

Fig. 3 shows the results obtained and whilst the shape of the curves measured was consistent with equation (6) a more accurate expression for the surface current density is given by

$$J_P = \frac{\sqrt{2}BI}{\pi\delta} \frac{\sqrt{h^2 - a^2}}{h^2 - a^2 + (AZ \frac{2\sqrt{h^2 + z^2}}{h})} \quad (9)$$

where A and B are constants found from practical measurement.

They are functions of the effective height h_e . A is a straight line represented by

$$A = 10^{-2} h_e \quad (10)$$

The application of the closest fit computer routine was found to produce the equation of the other constant

$$B = (-2.642 \times 10^{-1}) + (7.36 \times 10^{-5})h_e - (1.539 \times 10^{-9})h_e^3 \quad (11)$$

The variation of A and B with h_e is shown on Fig. 4. For simplicity and within the accuracy levels required the use of graphical results given in Fig. 3 are preferred.

A schematic diagram of the electrical circuit used is shown in Fig. 5. A 415 V, 50 Hz variable voltage transformer was connected to the input of a 415/5 V transformer which gave an output current capability to the work coil in excess of 2000 A. A current transformer on the high current output was used to measure the work coil current in the majority of experiments.

3. MULTIPLE CONDUCTORS

Induction billet heaters have, of course, many turns and the second part of the proposed laboratory experiment investigated the interaction of multiple conductors round a cylindrical aluminium work-piece. The results verify the principle of superposition associated with the surface current density distribution along the length of the billet.

The same aluminium billet used in the first experiment was

surrounded by an 11 turn work coil made from 6mm diameter water cooled copper tubing. The air gap used was 50mm and the coil pitch was constant at 7.1mm between each turn. The surface current density distribution along the billet was measured and the results correlated with those predicted from theory. The summation of the effects of all the conductors could be done using the rather complicated expression of equation (9). To simplify the procedure it is suggested that the resultant surface current density at any point P be calculated using the closest fit curve of the practically measured values of Fig. 3. Correlating the results in this manner gives an error of less than 5% as shown in Fig. 6.

4. TEMPERATURE TAPER

Research work at Loughborough⁽¹⁾ has produced a novel computer aided technique for the design of induction billet heating work coils which can give any particular non-uniform surface power density or temperature taper required.

To demonstrate the usefulness of the design procedure it was decided to design an induction heating coil capable of producing a linearly decreasing temperature taper. This, of course, models the practical requirements met industrially in the induction heating of aluminium prior to extrusion.

A computer program was developed to calculate the power density at any point P on the surface of the workpiece when heated by multiple turn work coil of arbitrary pitch. The values of the surface current density of 30mm air gap single turn coil, Fig. 3, were used to calculate the resultant surface current density of a number of

different work coil configurations using the procedure outlined previously.

A work coil was constructed again using 6mm diameter copper tubing but with variable coil pitch as detailed in Table 1. A work coil with the spacings given in Table 1 should produce a linear temperature distribution⁽¹⁾ usually associated with aluminium extrusion. In the third part of the laboratory exercise the practical measurements of surface current density for the coil should be correlated with the theoretical predictions. The results are shown in Fig. 7.

Pitch	1	2	3	4	5	6	7	8	9	10
Spacing (mm)	7	8	11	23	23	23	25	25	25	32

Table 1: Coil pitches to give linear temperature taper.

It was not possible to build a commercial size billet heater to test the theory because of the capital cost and the limitations of the laboratory power supplies. Megawatt power supplies and large amounts of power factor correction capacitors are not uncommon in industrial units. However, the laboratory scale model does demonstrate the validity of the design procedure.

The type of laboratory experiment has been used to illustrate the basic electromagnetic phenomena associated with induction heating and the essentials of taper heating associated specifically with induction heating of aluminium prior to extrusion. It has been used with final year undergraduates undertaking an electroheat option and to technical

staff trainees of the Electricity Supply Industry involved in specialist short courses on Industrial Applications of Electricity⁽⁷⁾.

5. CONCLUSIONS

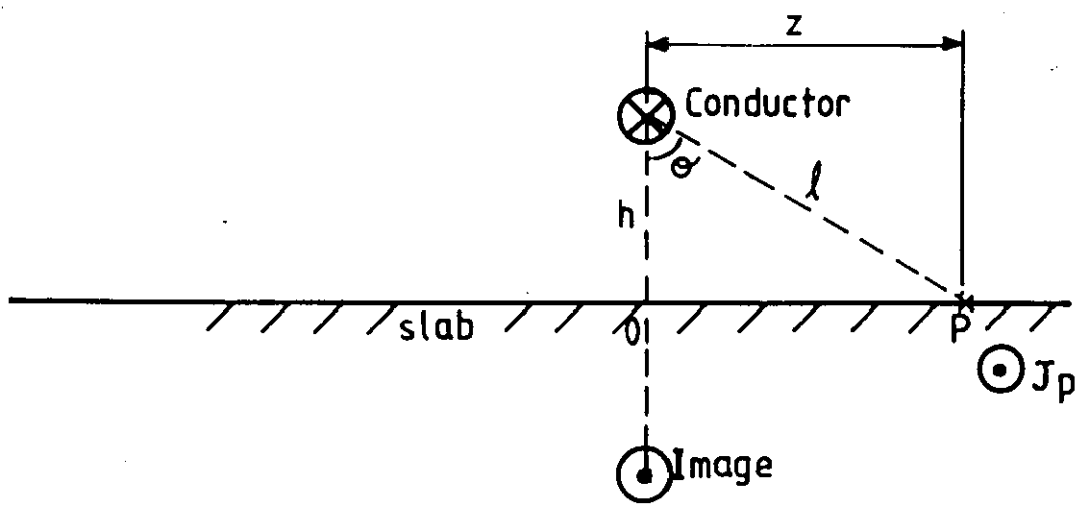
The authors have attempted to illustrate the basic electromagnetic phenomena of the proximity effect using readily available laboratory equipment and unsophisticated current density measuring probes. The temptation to use unrealistic workpiece/coil configurations has been avoided and by the use of graphical superposition techniques complex mathematics have been eliminated. Throughout the series of experiments the work has been related to induction heating theory and finally the requirements of the highly relevant industrial process of aluminium extrusion have been explained.

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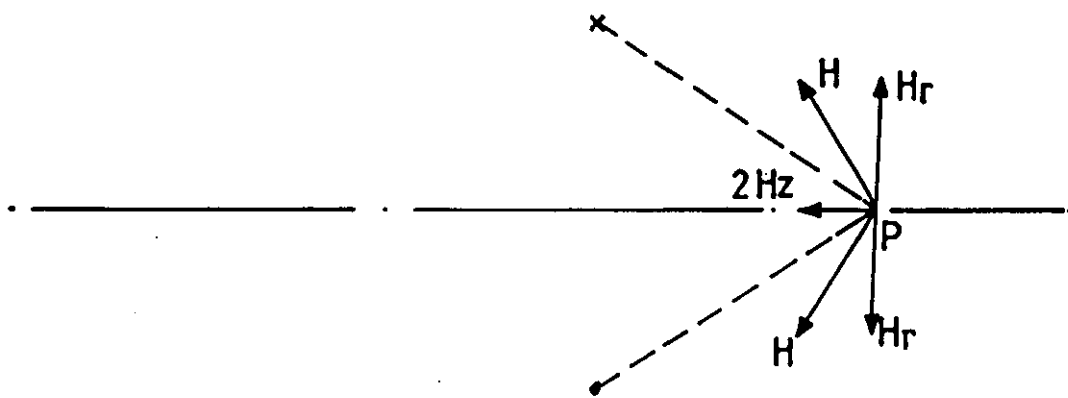
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(a)



(b)

